Lecture Notes in Networks and Systems 667

Serhiy Shkarlet · Anatoliy Morozov ·
Alexander Palagin · Dmitri Vinnikov ·
Nikolai Stoianov · Mark Zhelezniak ·
Volodymyr Kazymyr *Editors*

Mathematical Modeling and Simulation of Systems

Selected Papers of 17th International Conference, MODS, November 14–16, 2022, Chernihiv, Ukraine

Springer

Lecture Notes in Networks and Systems

Volume 667

The series “Lecture Notes in Networks and Systems” publishes the latest developments in Networks and Systems—quickly, informally and with high quality. Original research reported in proceedings and post-proceedings represents the core of LNNS.

Volumes published in LNNS embrace all aspects and subfields of, as well as new challenges in, Networks and Systems.

The series contains proceedings and edited volumes in systems and networks, spanning the areas of Cyber-Physical Systems, Autonomous Systems, Sensor Networks, Control Systems, Energy Systems, Automotive Systems, Biological Systems, Vehicular Networking and Connected Vehicles, Aerospace Systems, Automation, Manufacturing, Smart Grids, Nonlinear Systems, Power Systems, Robotics, Social Systems, Economic Systems and other. Of particular value to both the contributors and the readership are the short publication timeframe and the world-wide distribution and exposure which enable both a wide and rapid dissemination of research output.

The series covers the theory, applications, and perspectives on the state of the art and future developments relevant to systems and networks, decision making, control, complex processes and related areas, as embedded in the fields of interdisciplinary and applied sciences, engineering, computer science, physics, economics, social, and life sciences, as well as the paradigms and methodologies behind them.

Indexed by SCOPUS, INSPEC, WTI Frankfurt eG, zbMATH, SCImago.

All books published in the series are submitted for consideration in Web of Science.

For proposals from Asia please contact Aninda Bose (aninda.bose@springer.com).

Serhiy Shkarlet · Anatoliy Morozov ·
Alexander Palagin · Dmitri Vinnikov ·
Nikolai Stoianov · Mark Zhelezniak ·
Volodymyr Kazymyr
Editors

Mathematical Modeling and Simulation of Systems

Selected Papers of 17th International Conference, MODS, November 14–16, 2022, Chernihiv, Ukraine

Springer

Editors
Serhiy Shkarlet
The Ministry of Education and Science
Kyiv, Ukraine

Alexander Palagin
Academician of NAS of Ukraine
V.M. Glushkov Institute of Cybernetics
Kyiv, Ukraine

Nikolai Stoianov
Bulgarian Defence Institute
Sofia, Bulgaria

Volodymyr Kazymyr
Chernihiv Polytechnic National University
Chernihiv, Ukraine

Anatoliy Morozov
Academician of NAS of Ukraine
Institute of Mathematical Machines
and Systems Problems
Kyiv, Ukraine

Dmitri Vinnikov
Tallinn University of Technology
Tallinn, Estonia

Mark Zheleznіak
Institute of Environmental Radioactivity
Fukushima University
Fukushima, Japan

ISSN 2367-3370 ISSN 2367-3389 (electronic)
Lecture Notes in Networks and Systems
ISBN 978-3-031-30250-3 ISBN 978-3-031-30251-0 (eBook)
https://doi.org/10.1007/978-3-031-30251-0

This Springer imprint is published by the registered company Springer Nature Switzerland AG
The registered company address is: Gewerbestrasse 11, 6330 Cham, Switzerland

Preface

The International Conference "Mathematical Modeling and Simulation of Systems" (MODS) was formed to bring together outstanding researchers and practitioners in the field of mathematical modeling and simulation from all over the world to share their experience and expertise.

The conference MODS was established by the Institute of Mathematical Machines and Systems Problems of the National Academy of Sciences of Ukraine (NASU) in 2006. MODS is now an annual international conference held by Chernihiv Polytechnic National University with the assistance of the Ministry of Education and Science of Ukraine, the NASU, and the State Research Institute for Testing and Certification of Arms and Military Equipment, universities and research organizations from UK, Japan, Sweden, Bulgaria, Poland, Estonia, and Ukraine participating as co-organizers of the conference.

The XVIIth International Conference MODS'2022 was held in Chernihiv, Ukraine, during November 14–16, 2022. MODS'2022 received 48 paper submissions from different countries. All papers went through a rigorous peer-review procedure including pre-review and formal review. Based on the review reports, the Program Committee finally selected 24 high-quality papers for presentation on MODS'2022, which are included in "Lecture Notes in Networks and Systems" series.

This book contains papers devoted to relevant topics including tools and methods of mathematical modeling and simulation in ecology and environment, manufacturing and energetics, information technology, modeling, analysis and tools of safety in distributed information systems, mathematical modeling and simulation of special-purpose equipment samples, and cyber-physical systems. All of these offer us plenty of valuable information and would be of great benefit to the experience exchange among scientists in modeling and simulation.

The organizers of MODS'2022 made great efforts to ensure the success of this conference despite the active military operations on the territory of Ukraine. We hereby would like to thank all the members of MODS'2022 Advisory Committee for their guidance and advice, the members of Program Committee and Organizing Committee, the referees for their effort in reviewing and soliciting the papers, and all authors for their contribution to the formation of a common intellectual environment for solving relevant scientific problems.

Also, we are grateful to Springer-Verlag and Janusz Kacprzyk as the editor responsible for the series "Lecture Notes in Networks and Systems" for their great support in publishing these selected papers.

Serhiy Shkarlet
Alexander Palagin
Anatoliy Morozov
Dmitri Vinnikov
Nikolai Stoianov
Mark Zhelezniak
Volodymyr Kazymyr

Organization

Organizers

Ministry of Education and Science of Ukraine
The National Academy of Sciences of Ukraine
Academy of Technological Sciences of Ukraine
Engineering Academy of Ukraine
State Scientific Research Institute of Armament and Military Equipment Testing and Certification, Ukraine
Glyndwr University, Wrexham, UK
Defence Institute of Tsvetan Lazarov, Bulgaria
Lodz University of Technology, Poland
Riga Technical University, Latvia
Tallinn University of Technology, Estonia
University of Extremadura, Badajoz, Spain
Institute of Mathematical Machines and Systems Problems of the NASU, Ukraine
Institute for Safety Problems of Nuclear Power Plants of the NASU, Ukraine
National Technical University of Ukraine "Kyiv Polytechnic Institute," Ukraine
National University «Yuri Kondratyuk Poltava Polytechnic», Ukraine
Cherkasy State Technological University, Ukraine
Chernihiv Polytechnic National University, Ukraine

Chairs

Alexander Palagin	Academician of NASU, Ukraine, V.M. Glushkov Institute of Cybernetics of the NASU, Ukraine
Ireneusz Zbiciński	Lodz University of Technology, Poland

Anatoliy Morozov	Academician of the NASU, Institute of Mathematical Machines and Systems Problems of the NASU, Ukraine
V. Onishchenko	National University «Yuri Kondratyuk Poltava Polytechnic», Ukraine
Enrique Romero-Cadaval	University of Extremadura, Badajoz, Spain
Serhiy Shkarlet	Ministry of Education and Science of Ukraine, Ukraine
Volodymyr Bashynskyi	State Scientific Research Institute of Armament and Military Equipment Testing and Certification, Ukraine
A. Vasiliev	A. N. Podgorny Institute for Mechanical Engineering Problems of the NASU, Ukraine
Dmitri Vinnikov	Tallinn University of Technology, Estonia
Ilya Galkin	Riga Technical University, Latvia
John Davies	Glyndwr University, Wrexham, UK

Program Committee

V. Adamchuk	The National Academy of Agrarian Sciences of Ukraine, Ukraine
O. Azarov	Vinnytsia National Technical University, Ukraine
Serhiy Bushuev	Kyiv National University of Construction and Architecture, Ukraine
Volodymyr Dmytriiev	State Scientific Research Institute of Armament and Military Equipment Testing and Certification, Ukraine
Mariia Dorosh	Chernihiv Polytechnic National University, Ukraine
Serhiy Holub	Cherkasy State Technological University, Ukraine
V. Gryshko	National University «Yuri Kondratyuk Poltava Polytechnic», Ukraine
Janis Zakis	Riga Technical University, Latvia
Volodymyr Kazymyr	Chernihiv Polytechnic National University, Ukraine
V. Kharchenko	National Aerospace University named after N.E. Zhukovsky "Kharkiv Aviation Institute," Ukraine

V. Klimenko	Institute of Mathematical Machines and Systems Problems of the NASU, Ukraine
Ivan Kovalets	Institute of Mathematical Machines and Systems Problems of the NASU, Ukraine
V. Kraskevych	Kyiv National University of Trade and Economics, Ukraine
Dmytro Lande	Institute for Information Recording of NAS of Ukraine, Ukraine
Olexandr Liakhov	National University «Yuri Kondratyuk Poltava Polytechnic», Ukraine
V. Maderych	Institute of Mathematical Machines and Systems Problems of the NASU, Ukraine
V. Myronenko	Member of the NAASU, The National Academy of Agrarian Sciences of Ukraine, Ukraine
A. Nosovskyi	Corresponding Member of the NASU, Institute for Safety Problems of Nuclear Power Plants of the NASU, Ukraine
V. Snytiuk	Taras Shevchenko National University of Kyiv, Ukraine
Inna V. Stetsenko	National Technical University of Ukraine "Igor Sikorsky Kiev Polytechnic Institute," Ukraine
V. Tarasenko	National Technical University of Ukraine "Igor Sikorsky Kiev Polytechnic Institute," Ukraine
A. Verlan	National Technical University of Ukraine "Igor Sikorsky Kiev Polytechnic Institute," Ukraine
O. Voloshyn	Taras Shevchenko National University of Kyiv, Ukraine
V. Zatserkovnyi	Taras Shevchenko National University of Kyiv, Ukraine
Mark Zhelezniak	Institute of Environmental Radioactivity, Fukushima University, Japan

Steering Chairs

Volodymyr Kazymyr	Chernihiv Polytechnic National University, Ukraine
Igor Brovchenko	Institute of Mathematical Machines and Systems Problems of the NASU, Ukraine
Iryna Bilous	Chernihiv Polytechnic National University, Ukraine

Local Organizing Committee

Iryna Bilous	Chernihiv Polytechnic National University, Ukraine
Oleksii Trunov	Chernihiv Polytechnic National University, Ukraine

EasyChair Support

Mariia Voitsekhovska	Chernihiv Polytechnic National University, Ukraine

Sponsor

«S&T Ukraine», Kyiv, Ukraine

Contents

List of Contributors

Andrii Akymenko Chernihiv Polytechnic National University, Chernihiv, Ukraine

Roman Bezhenar Institute of Mathematical Machine and System Problems, Kyiv, Ukraine

Iryna Bilous Chernihiv Polytechnic National University, Chernihiv, Ukraine

Vadim Bodunov Chernihiv Polytechnic National University, Chernihiv, Ukraine

Oleksandr Bondarenko National Technical University of Ukraine "Igor Sikorsky Kyiv Polytechnic Institute", Kyiv, Ukraine

Igor Brovchenko Institute of Mathematical Machines and System Problems NAS of Ukraine, Kyiv, Ukraine

Yuriy Denisov Chernihiv Polytechnic National University, Chernihiv, Ukraine

Serhii Denysov Taras Shevchenko National University of Kyiv, Kyiv, Ukraine

Volodymyr Dmytriiev State Scientific Research Institute of Armament and Military Equipment Testing and Certification, Chernihiv, Ukraine

Iryna Dohtieva Vinnytsia National Technical University, Vinnytsia, Ukraine

Anatoliy Doroshenko Institute of Software Systems of National Academy of Sciences of Ukraine, Kyiv, Ukraine

Oleksandr Drozd Chernihiv Polytechnic National University. 95, Chernihiv, Ukraine

Eugene Fedorov Cherkasy State Technological University, Cherkasy, Ukraine

Alexander Gai National University of Life and Environmental Sciences of Ukraine, Kyiv, Ukraine

Vadim Garin National Aerospace University "Kharkiv Aviation Institute", Kharkiv, Ukraine

Viktor Grechaninov Academy of Science, Kyiv, Ukraine

Tetyana Herasymenko National University of Food Technologies, Kiev, Ukraine

Oleksandr Hlushko Chernihiv Polytechnic National University, Chernihiv, Ukraine

Dmytro Horval Chernihiv Polytechnic National University, Chernihiv, Ukraine

Serhii Hrybkov National University of Food Technologies, Kiev, Ukraine

Borys Horlynskyi Institute of Telecommunications and Global Information Space of the National Academy of Sciences of Ukraine, Kyiv, Ukraine

Pavlo Ivanenko Institute of Software Systems of National Academy of Sciences of Ukraine, Kyiv, Ukraine

Ihor Ivanov Institute of Mathematical Machines and System Problems NAS of Ukraine, Kyiv, Ukraine;
S. P. Timoshenko Institute of Mechanics NAS of Ukraine, Kyiv, Ukraine

Dmytro Kamak State Scientific Research Institute of Armament and Military Equipment Testing and Certification, Chernihiv, Ukraine

Yevgen Kaplun National Academy of National Guard of Ukraine, Kharkiv, Ukraine

Oleksandra Kapinus The Ukrainian-American Joint Venture “KODA”, Kharkiv, Ukraine

Volodymyr Kazymyr Chernihiv Polytechnic National University, Chernihiv, Ukraine

Natalia Khalimon National Aviation University, Kyiv, Ukraine

Oleksandr Khoshaba Academy of Science, Kyiv, Ukraine;
Vinnytsia National Technical University, Vinnytsia, Ukraine

Sergii Kivva Institute of Mathematical Machines and System, Problems, Kyiv, Ukraine

Andrii Kondratiev O. M. Beketov National University of Urban Economy in Kharkiv, Kharkiv, Ukraine

Ihor Korniienko Chernihiv Polytechnic National University, Chernihiv, Ukraine

Svitlana Korniienko Chernihiv Polytechnic National University, Chernihiv, Ukraine

Daria Kosareva Chernihiv Polytechnic National University, Chernihiv, Ukraine

Aleksandr Kosmach Chernihiv Polytechnic National University, Chernihiv, Ukraine

Ivan Kovalets Institute of Mathematical Machine and System Problems, Kyiv, Ukraine

Anna A. Kozlova Scientific Centre for Aerospace Research of the Earth, NAS of Ukraine, Kiev, Ukraine

Dmytro Kucherov National Aviation University, Kyiv, Ukraine

Sergiy Kukobko State Scientific Research Institute of Armament and Military Equipment Testing and Certification, Chernigiv, Ukraine

Tetiana Kulko Chernihiv Polytechnic National University, Chernihiv, Ukraine

Pavel Kurbet Institute of Telecommunications and Global Information Space of the National Academy of Sciences of Ukraine, Kyiv, Ukraine

Anatoliy Lopushanskyi Academy of Science, Kyiv, Ukraine

Nataliia Lutska National University of Food Technologies, Kiev, Ukraine

Viacheslav Mamchurovskyi Chernihiv Polytechnic National University, Chernihiv, Ukraine

Kateryna Maiorova National Aerospace University "KhAI", Kharkiv, Ukraine

Dmytro Mekhed Chernihiv Polytechnic National University, Chernihiv, Ukraine

Tetiana Molodetska Vinnytsia National Technical University, Vinnytsia, Ukraine

Dmytro Mykhalyk Ternopil Ivan Puluj National Technical University, Ternopil, Ukraine

Svitlana Myronova National Aerospace University "KhAI", Kharkiv, Ukraine

Tetyana Nabokina National Aerospace University "Kharkiv Aviation Institute", Kharkiv, Ukraine

Anastasiia Neskorodieva Vasyl' Stus Donetsk National University, Vinnytsia, Ukraine

Tetiana Neskorodieva Vasyl' Stus Donetsk National University, Vinnytsia, Ukraine

Anatolii Pavlenko State Scientific Research Institute of Armament and Military Equipment Testing and Certification, Chernihiv, Ukraine

Mykhaylo Petryk Ternopil Ivan Puluj National Technical University, Ternopil, Ukraine

Sergiy Plankovskyy O.M. Beketov National University of Urban Economy in Kharkiv, Kharkiv, Ukraine

Daryna Pryschepa Chernihiv Polytechnic National University, Chernihiv, Ukraine

Anatoliy Prystupa Chernihiv Polytechnic National University, Chernihiv, Ukraine

Evgeniy Roshchupkin Ivan Kozhedub Kharkiv National Air Force University, Kharkiv, Ukraine

Yevhen Ryndych Chernihiv Polytechnic National University, Chernihiv, Ukraine

Yana Savytska National University of Life and Environmental Science of Ukraine, Kiev, Ukraine

Vladimir Semenov Taras Shevchenko National University of Kyiv, Kyiv, Ukraine

Mykhailo Shelest Chernihiv Polytechnic National University, Chernihiv, Ukraine

Vitalii Shelestovskii National University of Life and Environmental Science of Ukraine, Kiev, Ukraine

Maryna Shevtsova National Aerospace University "Kharkiv Aviation Institute", Kharkiv, Ukraine

Anatolii Shyian Vinnytsia National Technical University, Vinnytsia, Ukraine

Olga Shypul National Aerospace University "Kharkiv Aviation Institute", Kharkiv, Ukraine

Tetiana Sichko Vasyl' Stus Donetsk National University, Vinnytsia, Ukraine

Marina Sinenko Chernihiv Polytechnic National University, Chernihiv, Ukraine

Victor Smolii National University of Life and Environmental Science of Ukraine, Kiev, Ukraine

Natalia Sokorynska Chernihiv Polytechnic National University, Chernihiv, Ukraine

Maksym Solodchuk State Scientific Research Institute of Armament and Military Equipment Testing and Certification, Chernihiv, Ukraine

Sergey A. Stankevich Scientific Centre for Aerospace Research of the Earth, NAS of Ukraine, Kiev, Ukraine

Serhii Stepenko Chernihiv Polytechnic National University, Chernihiv, Ukraine

Yuliia Tkach Chernihiv Polytechnic National University, Chernihiv, Ukraine

Denys Tkachenko National Aerospace University "Kharkiv Aviation Institute", Kharkiv, Ukraine

Elena Trunova Chernihiv Polytechnic National University, Chernihiv, Ukraine

Oleg Tryfonov National Aerospace University "Kharkiv Aviation Institute", Kharkiv, Ukraine

Anton Tsaritsynskyi National Aerospace University "Kharkiv Aviation Institute", Kharkiv, 61070 Ukraine

Sergiy Tymchenko National Academy of National Guard of Ukraine, Kharkiv, Ukraine

Tetyana Utkina Cherkasy State Technological University, Cherkasy, Ukraine

Oleksii Vambol National Aerospace University “Kharkiv Aviation Institute”, Kharkiv, Ukraine

Lidiia Vlasenko State University of Trade and Economics, Kiev, Ukraine

Volodymyr Voytenko Lund University, Lund, Sweden;
Chernihiv Polytechnic National University, Chernihiv, Ukraine

Olena Yatsenko Ternopil Ivan Puluj National Technical University, Ternopil, Ukraine

Roman Yershov Chernihiv Polytechnic National University, Chernihiv, Ukraine

Anatolijs Zabašta Riga Technical University, Riga, Latvia

Sergei Zaitsev Chernihiv Polytechnic National University, Chernihiv, Ukraine;
University of Technology (Politechnika Świętokrzyska), Kielce, Poland;
Institute of Telecommunications and Global Information Space of the National Academy of Sciences of Ukraine, Kyiv, Ukraine

Liliia Zaitseva Institute of Telecommunications and Global Information Space of the National Academy of Sciences of Ukraine, Kyiv, Ukraine

Sergiy Zaklinskyy National Aerospace University “Kharkiv Aviation Institute”, Kharkiv, Ukraine

Kostiantyn Zavertailo Academy of Science, Kyiv, Ukraine

Mathematical Modeling and Simulation of Systems in Ecology

Modeling of Radioactive Contamination of the Black Sea and Its Impact on Humans Due to Hypothetical Heavy Accident at the Zaporizhzhia NPP

Roman Bezhenar and Ivan Kovalets

Abstract The paper describes one of the many possible scenarios of the hypothetical accident at the Zaporizhzhia Nuclear Power Plant (ZNPP), located in southern part of Ukraine not far from the Black Sea coast. The special scenario with atmospheric release of a large amount of activity with the subsequent transport to the south and predominant deposition on the Black Sea is considered. In the study, the atmospheric dispersion model, marine dispersion model and dose models are applied within the EU nuclear emergency response system JRODOS. According to model results, under the selected conditions the whole western part of the Black Sea will be radioactively contaminated. The values of atmospheric deposition will reach 400 kBq/m^2 at distances up to 290 km from ZNPP. The maximum concentrations of radionuclides in the surface water could exceed 1000 Bq/m^3 immediately after the deposition, while 20 years later the concentration of ^{137}Cs in the Black Sea will be around 25 Bq/m^3, which is several times higher than the current concentration. The concentration of ^{137}Cs in fish could exceed 100 Bq/kg. The doses from terrestrial and marine pathways could reach considerable levels in large territories. For example, the isoline of 1 mSv of the calculated total effective dose received by 1-yr children from terrestrial exposure pathways during 1 year after the accident covers large parts of the territories of southern Ukraine, Romania and Bulgaria. The calculated annual individual dose for adults due to seafood consumption will also exceed 1 mSv for large areas of the Black Sea.

Keywords Black sea · Radionuclides · Zaporizhzhia NPP · Doses to human

1 Introduction

Black Sea is the semi-enclosed deep sea in the Eastern Europe. It is connected with the Mediterranean Sea by the narrow straits Bosporus and Dardanelles separated by the Marmara Sea, as well as with the See of Azov via the Kerch Strait. Due to

R. Bezhenar (✉) · I. Kovalets
Institute of Mathematical Machine and System Problems, Kyiv, Ukraine
e-mail: romanbezhenar@gmail.com

S. Shkarlet et al. (eds.), *Mathematical Modeling and Simulation of Systems*, Lecture Notes in Networks and Systems 667,
https://doi.org/10.1007/978-3-031-30251-0_1

limited water exchange with adjacent seas, any contamination that enters the Black Sea remains there for a long time. For example, according to estimations based on results of modeling and measurement data, concentration of radioactive isotope of cesium ^{137}Cs in surface water that fell into the Black Sea in result of the accident on the Chornobyl NPP decreased two times every 14 years, according to the long-term measurement data [1]. The long retention time of contaminants in the sea and the existence of significant sources of radionuclides in the past made the Black Sea one of the most radioactively contaminated seas in the world [2].

Main sources of radionuclides in the Black Sea were global fallout from nuclear weapons tests in the middle of XX century and atmospheric deposition due to the Chornobyl accident in 1986 [3]. The fact is that there are 13 NPPs in operation on the watershed of the Black Sea. This means that any accident at one of these NPPs can significantly contaminate the Black Sea by different radionuclides. The probability of such an accident increases in case of hostilities, which are ongoing near the NPP.

The aim of the current study is to consider one of the possible scenarios for the hypothetical accident at the Zaporizhzhia NPP, which is captured by military forces of Russian Federation, with the predominant deposition of radionuclides on the sea surface and estimate its impact on humans. The special atmospheric conditions were chosen to obtain needed deposition. Atmospheric dispersion model, marine dispersion model and dose model, which are components of the EU nuclear emergency response system JRODOS, were applied for simulations that was done for the first time for the Black Sea. Results of simulations allow to estimate the possible damage to the ecosystem of the Black Sea, the fishing industry, and the long-term impact on humans living in the coastal areas due to probable accident on the ZNPP.

2 Models Description and Their Customization to the Black Sea

The EU real-time on-line nuclear emergency response system RODOS and it's redesigned version JRODOS (https://resy5.iket.kit.edu/JRODOS/) is widely used in Europe and over the world for assessment of real and potential consequences of radiation accidents [4]. It includes simulation tools that allow for assessment of all stages of the accident from the early phase to late consequences that cover many years after the accident. The atmospheric transport module of JRODOS includes three local-scale atmospheric dispersion models DIPCOT, LASAT and RIMPUFF, that could be used for assessment of atmospheric transport of radionuclides up to 800 km from the point of release. In this study RIMPUFF model was used that combines computational efficiency with the advanced algorithms for assessment of puff's growth at medium-range time and spatial intervals of atmospheric transport [5]. The numerical weather prediction (NWP) data are calculated by WRF-ARW mesoscale meteorological model (https://www2.mmm.ucar.edu/wrf/users/) operated by the Ukrainian Hydrometeorological Center (UHMC) in the frame of WRF-Ukraine NWP system.

The WRF model covers territory of Ukraine and partly of neighbor countries and Black Sea with spatial resolution 0.15°. It uses operational numerical weather prediction data of Global Forecasting System (GFS) of the US National Center of Environmental Prediction [6]. The 96-h forecasts are produced by WRF-Ukraine system in UHMC with time resolution of 1 h and forecasts are updated every 6 h. In calculations of atmospheric transport by the JRODOS system the option 'switch to newer forecasts' was used, so that WRF forecasts were effectively used as reanalysis data. When atmospheric transport and deposition were calculated, the JRODOS Food Dose Model Terrestrial (FDMT) [7] was then used for the assessment of long-term doses from different exposure pathways. The JRODOS system contains default parameters of FDMT model and necessary maps of soil, land use and agriculture production for the territory of Europe.

For the long-term simulation of the transfer of radionuclides in the marine environment (water, sediments, and biota), the compartment model POSEIDON-R [8] is used in the JRODOS system. In the model, the area of interest is considered as a system of compartments. Each compartment includes water column, bottom sediments and marine organisms forming the food web. Water column may be divided by several vertical layers. In addition, water column in each compartment contains the constant concentration of suspended sediments, which are continuously settling down from upper to lower layers and finally to the bottom. Radionuclides that enter in the compartment system are transferred between compartments in three dimensions according to average currents in the modeling area. Certain part of every radionuclide that depends on the value of distribution coefficient K_d are adsorbed by suspended sediments and gradually settle to the bottom forming the contamination of bottom sediments. There is also exchange of activity between water and bottom sediments due to diffusion and bioturbation processes. Marine organisms in the model are organized in the food web. They uptake radionuclides directly from water and by ingesting organisms from lower trophic levels. Model equations, method of their solution and parameters of the food web are given in details in [8]. Obtained in the model concentration of radionuclides in biota, as well as average annual consumption rates of different types of marine organisms are used for calculation the internal doses of radiation to human from the seafood consumption.

The POSEIDON-R model was previously customized for the Black Sea as it is described in [3]. The system of boxes (Fig. 1) consists of 45 boxes in the Black Sea, Dnipro-Buh Estuary (box 49), Sea of Azov (box 50), Marmara Sea (box 3), Aegean Sea (box 2), Mediterranean Sea (box 1), Dnipro River and Danube River. Compartments in the Black Sea are vertically divided to four layers: 1st layer from surface to 25 m depth, 2nd layer from 25 to 100 m, 3rd layer from 100 to 600 m, and 4th layer from 600 m to the bottom. Amount of layers in each compartment depends on the depth of that compartment. Fluxes of water between boxes were calculated by averaging 3D currents from the reanalysis data for the period 2006–2015 years, which are available online [9]. Climatic data for flows of water from Dnipro and Danube rivers were also taken into account. The model correctly reproduces the long-term changes of ^{137}Cs concentration in water and marine organisms that was shown in

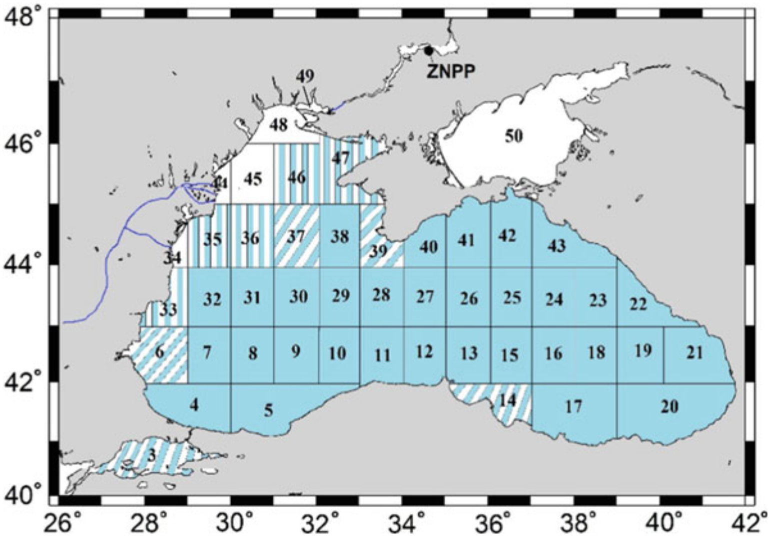


Fig. 1 The system of compartments for the Black Sea in the POSEIDON-R model. Compartments that have four vertical layers are shaded, diagonal and vertical strips indicate three-layer and two-layer compartments, respectively, one-layer boxes are white. Location of Zaporizhzhia NPP (ZNPP) is shown by the black dot

[3], where model results agree well with measurement data for the period 1960–2012 years that covers main sources of ^{137}Cs such as global fallout from nuclear weapon tests and Chornobyl accident.

POSEIDON-R model together with atmospheric dispersion model and dose model are integrated in the JRODOS system. This means that calculated deposition density of radionuclides on the sea surface from atmospheric dispersion model is automatically converted to the source term for the POSEIDON-R model. For this aim, the total deposition in each node of the atmospheric dispersion model computational grid is distributed between Black Sea compartments according to geographical coordinates.

3 Scenario of Hypothetical Accident at the ZNPP

The source term for the hypothetical accident at the ZNPP was taken from the official assessments of the Ukrainian National Energy Generating Company (NAEC) "Energoatom" [10]. In the cited study reassessment of potential heavy accidents at ZNPP had been performed with taking into account Fukushima lessons. One of the heaviest scenarios considered in this work is 'Full loss of electric power supply together with loss of coolant for Block 1 of Zaporizhzhia NPP with taking into account actions on reducing hydrogen concentration'. In this scenario, the following fractions of the inventory of main radionuclides are released during 25 h after start of the accident: 3.4% for ^{137}Cs and ^{134}Cs, and 1.3% for ^{90}Sr. Releases of ^{106}Ru, ^{238}Pu and ^{244}Cm are also considered. Here we mentioned only the long-lived radionuclides ($T_{1/2}$ > 1 year) from the total of 26 radionuclides that were simulated by JRODOS. The total ^{131}I equivalent released inventory of all radionuclides in the considered scenario according to International Nuclear and Radiological Event Scale (INES) [11] is $2.261 \cdot 10^{18}$ Bq. It is less by the factor of 2.3 than the respective value for Chornobyl accident and greater by the factor of almost 3 than the respective value for Fukushima accident [11].

Since the aim of this study was to evaluate the consequences of hypothetical scenario at ZNPP for the Black Sea, the meteorological scenario was intentionally selected to provide high deposition of radionuclides on the surface of the Black Sea. The amounts of deposited materials usually highly increase when transport of cloud over the area of interest is accompanied by rain, leading to wet removal of radionuclides by precipitation. The influence of wet removal may be illustrated by the effective deposition velocity v_d [m·s^{-1}] defined as the ratio of deposition flux F [Bq·m^{-2} s^{-1}] to concentration in air c [Bq·m^{-3}] at a given point. Under the influence of wet removal, the effective deposition velocities were increased in some points by order of magnitude and more—from $v_d = 5 \cdot 10^{-4}$ m·s^{-1}, a representative for dry deposition on the water surface in JRODOS, up to $v_d = 0.04$ m·s^{-1} in areas with high wet deposition. Therefore the weather maps for the period of 01 May–31 July built on the basis of GFS final analysis data [6] where manually processed to select the corresponding weather situations when atmospheric transport in the direction from ZNPP to the Black Sea is accompanied by intense rains over the Black Sea. Ten such meteorological scenarios were selected and calculation of the atmospheric transport was then performed. The largest amounts of deposition were obtained for the scenario of release happening on 27 June 2022:01 h UTC, as described in more detail in the next section. The weather situation in the region of ZNPP and Black Sea during that date was dominated by the cyclone, with its pressure minimum located in the south-central part of the Black Sea. The area of the most intense precipitation (up to 7 mm/h) covered the central part of Black Sea together with shoreline regions including the southern part of Crimea Peninsula. During the period of release (27 June 22:01 h–28 June 22:02 h UTC) rains happened over the whole western part of Black Sea. At the same time, high atmospheric pressure was located in the northern part of Belarus. Therefore high pressure gradient existed in the region of ZNPP leading to the fresh wind (6–7 m/s) blowing to the south-west and transporting radionuclides in the direction of the Black Sea where it was then washout by rains.

4 Results of Modeling

4.1 Calculated Deposition Density of Radionuclides

Figure 2 presents the calculated by JRODOS system areal density of total deposition (dry+wet) for the hypothetical accident scenario at ZNPP described above. The release started on 27 June 2022, 01 h UTC and simulation was performed for the next 72 h. The cloud was transported over the Black Sea according to the meteorological scenario described above and the high deposition values were mainly created by wet removal by rain. The isoline is plotted for the value of 400 kBq/m^2. This value is considered by National Regulation on Radiation Safety of Ukraine [12] as the lowest possible value, justifying permanent relocation of people, i.e. relocation from the corresponding territories could be considered. The maximum distance from ZNPP to this isoline reaches the value of 290 km. Similar maps were calculated for other radionuclides listed in Sect. 3 and passed to POSEIDON-R model for subsequent assessments of sea water and fish radioactive contamination and respective dose assessments.

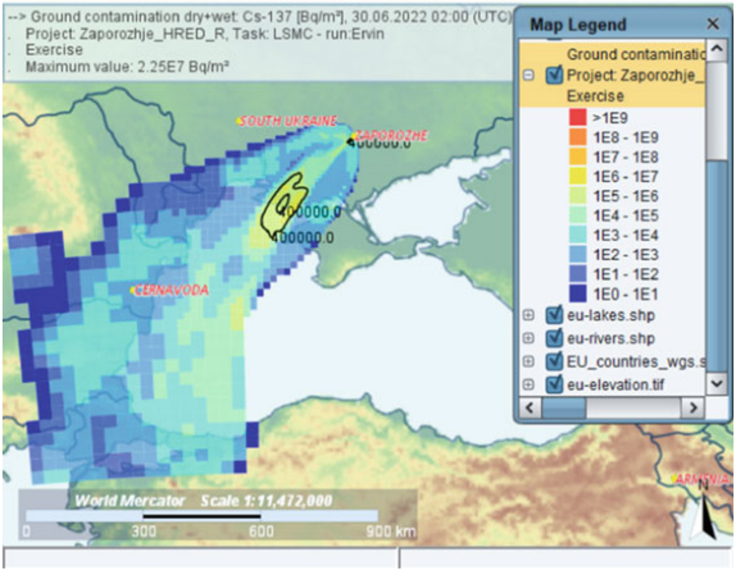


Fig. 2 Map of total (dry+wet) deposition density of Cs-137 simulated by JRODOS system for the hypothetical accident scenario at ZNPP started on 27 June 2022, 01 h UTC. Isoline is plotted for the value 400 kBq/m^2

4.2 *Concentration of Radionuclides in Water and Marine Organisms*

Since the maximum deposition of radionuclides under the selected meteorological conditions will be in the western part of the Black Sea, the maximum concentrations of radionuclides in water and marine organisms will also occur there. Figure 3 shows how the concentration of various radionuclides in the water will change in compartment No 47 (see Fig. 1) where the maximum concentrations of radionuclides were obtained as a result of modeling. In a few years, the contamination will spread to the entire sea that will lead to a decrease in the concentration of radionuclides in the most contaminated areas and an increase in the areas that were less contaminated immediately after the accident. The concentration of radionuclides will also decrease due to radioactive decay, especially for ^{106}Ru and ^{134}Cs (blue and green lines in Fig. 3), whose half-lives are 1.01 and 2.06 years, respectively. Other radionuclides are characterized by a much longer half-life (28.8 years for ^{90}Sr; 30.2 years for ^{137}Cs; 14.4 years for ^{241}Pu), so their concentration in the Black Sea will remain at a quite high level for much longer time. For example, 20 years after the considered accident the concentration of ^{137}Cs in the Black Sea (yellow line in Fig. 3) will be around 25 Bq/m^3, which is several times higher than the current concentration [13].

The maximum concentrations of radionuclides in different components of the marine environment will be reached at different time moments. In surface water, it will be reached immediately after deposition, a little later—in prey fish due to the time needed for assimilation of radionuclides by organism, even later—in predatory fish due to the time of accumulation of radionuclides in the food chain. According to model results, concentration of ^{137}Cs in fish could exceed 100 Bq/kg (Fig. 4 and 5).

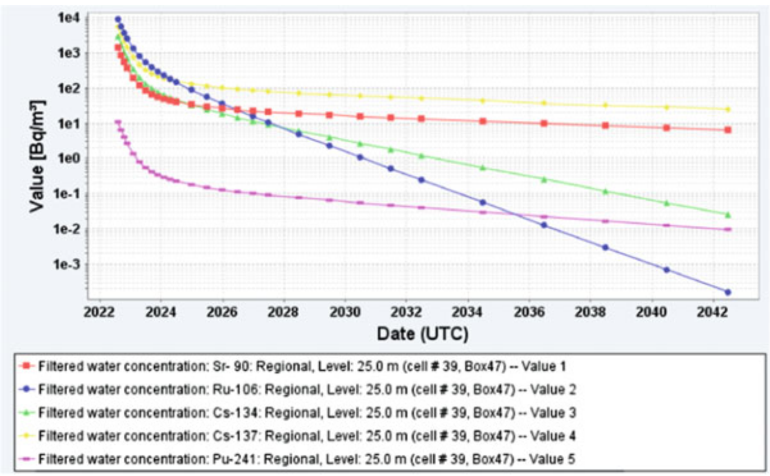


Fig. 3 Calculated concentrations of radionuclides in water in the compartment No 47

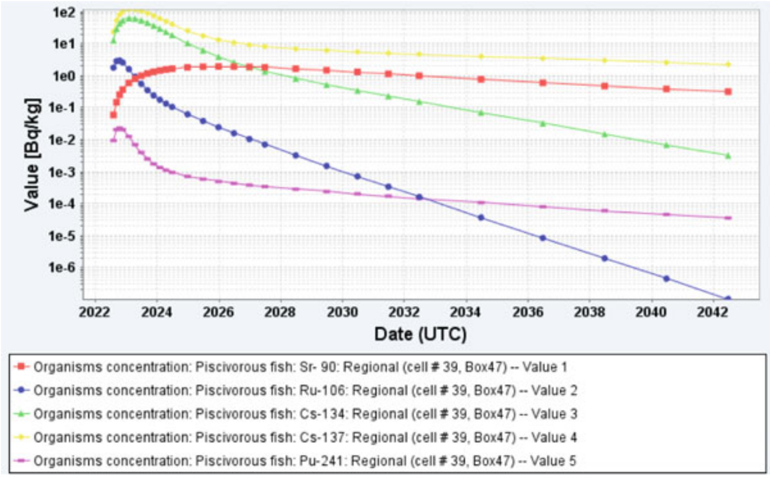


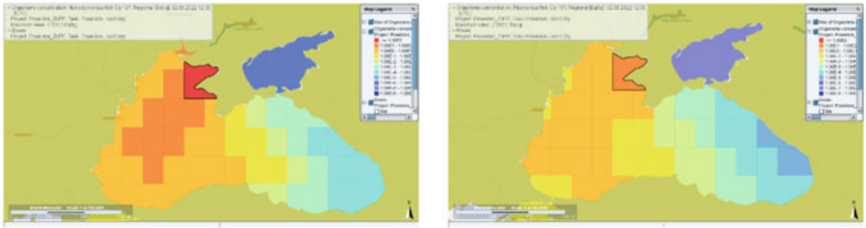
Fig. 4 Calculated concentrations of radionuclides in predatory fish in the compartment No 47

The model also takes into account that radionuclides accumulate in body of fish non-uniformly: different radionuclides mainly accumulate in different tissues [14]. For example, among considered radionuclides, the radionuclide with the longest retention time in organisms is ^{90}Sr because it accumulates in bones, which are characterized by a very slow renewal of living cells. ^{106}Ru and ^{241}Pu have the shortest retention time due to the fast renewal of living cells in organs where they mostly accumulate. Isotopes of cesium (^{134}Cs, ^{137}Cs) mostly accumulate in muscles, which take the dominant part of the organism. Therefore, we can expect the largest concentrations of ^{137}Cs and ^{134}Cs in fish after the accident.

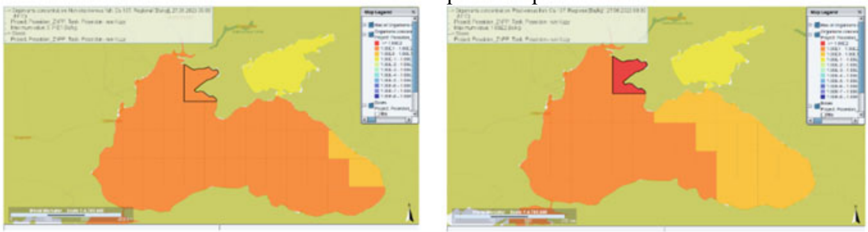
Figure 5 shows how the concentration of ^{137}Cs in prey and predatory fish will change with time. Initially, the concentration of ^{137}Cs in prey fish will be higher with maximum in regions where the highest deposition density took place. Over time, the concentration of ^{137}Cs in predatory fish will become higher due to the ability of cesium for biomagnification in the food chain [15]. Note that red color on the Fig. 5 indicates areas where concentration of ^{137}Cs in fish will exceed 100 Bq/kg; in dark orange areas, the concentration will be in the range from 10 to 100 Bq/kg; in orange areas—from 1 to 10 Bq/kg and so on.

4.3 Estimated Doses to Humans

Figure 6 presents map of total effective dose received by 1-yr children from all terrestrial exposure pathways except ingestion for all 26 simulated radionuclides during 1 year after the hypothetical accident simulated by JRODOS-FDMT. As it could be seen from the figure, areas within isoline 1 mSv cover large parts of Southern



1 month after atmospheric deposition



1 year after atmospheric deposition

Fig. 5 Calculated concentrations of ^{137}Cs in prey fish (left side) and predatory fish (right side) in the Black Sea at different times after the atmospheric deposition

Ukraine, Romania and Bulgaria. Moreover the values of 10 mSv are reached in some places at distances up to 840 km from ZNPP. For example the maximum dose obtained at the territory of Bulgaria for the considered scenario reaches the value of 14.6 mSv. As was expected, the largest doses are reached at the territory of Ukraine. Note that deterministic effects were not predicted by the JRODOS-EMERSIM model for this scenario. The ingestion doses were not presented in this work because exact estimates for the territories outside Ukraine require additional efforts on JRODOS customization. As shown by preliminary tests, the ingestion of food products would increase the presented in Fig. 6 values by about the order of magnitude at the territories of Ukraine and by several times at larger distances from ZNPP.

Doses for humans from the seafood consumption were calculated based on the average annual consumption rates of marine organisms in European countries [16] and concentrations of radionuclides in marine organisms obtained in the current modeling study. Here the impact of all long-lived radionuclides were taken into account. Calculations were made for each compartment separately supposing that all consumption was from this compartment. As we can see in the Fig. 7, the annual individual dose to humans in the half of the see will exceed 1 mSv (orange color) that is the maximal allowable dose to public according to International Atomic Energy Agency (IAEA) recommendations [17]. In some areas, the individual dose could exceed 10 mSv per year (dark orange color). This means that described heavy accident at the Zaporizhzhia NPP could significantly and for a long time contaminate the Black See by radioactive materials.

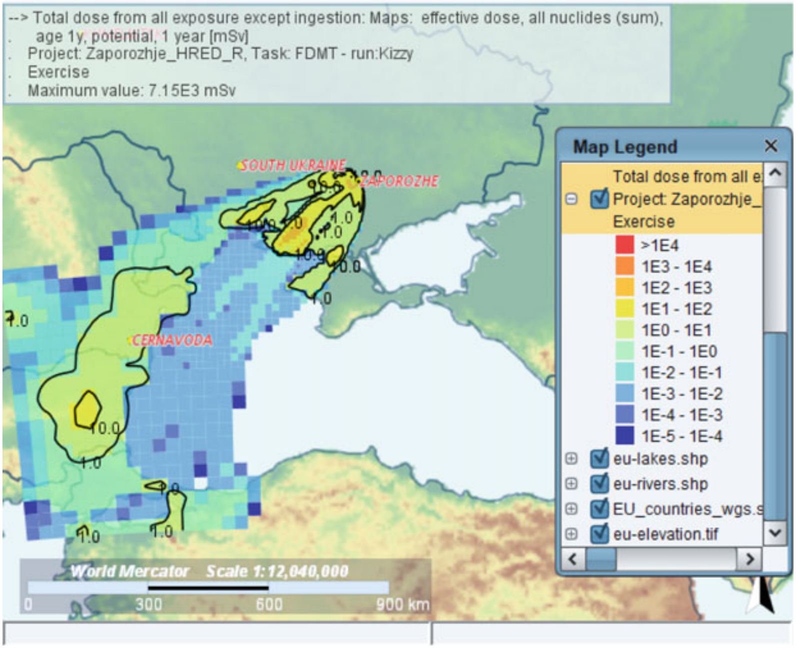


Fig. 6 Map of total effective dose received by 1-yr children from all terrestrial exposure pathways except ingestion and all radionuclides during 1 year after the accident simulated by JRODOS-FDMT model for the hypothetical accident scenario at ZNPP started on 27 June 2022, 01 h UTC. Isolines are plotted for the values 1 and 10 mSv

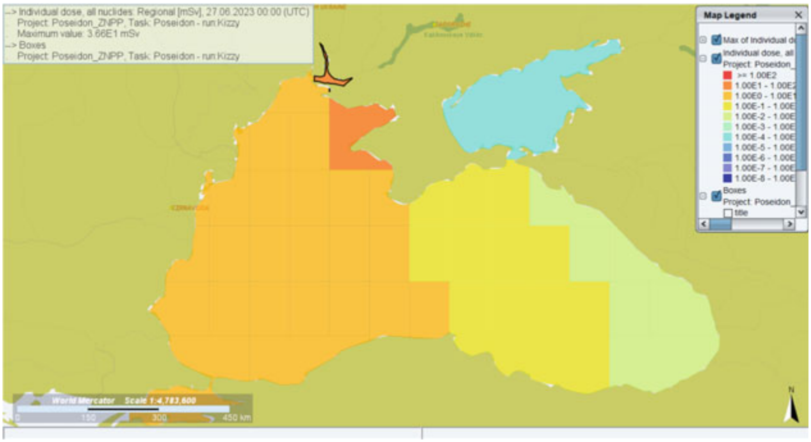


Fig. 7 Calculated individual doses to humans from all considered radionuclides for the first year after the considered accident due to the seafood consumption

5 Conclusions

In this study, the consequences of hypothetical accidental scenario at Zaporizhzhia NPP were evaluated by application of the atmospheric dispersion model, marine dispersion model and dose models, which are components of the EU nuclear emergency response system JRODOS. The source term scenario was taken from the available assessments of potential heavy accidents at ZNPP performed by official authorities in Ukraine (NAEC "Energoatom") with taking into account Fukushima lessons. The moderately conservative meteorological scenario leading to high depositions on the surface of the Black Sea was selected by processing available meteorological data for 3-months period (May to July 2022). The start date of release in the chosen scenario was 27 June 2022, 01 h UTC. The release duration was 25 h, while simulation of atmospheric transport was performed for 72 h. By the end of calculations of atmospheric dispersion model (30 June, 02 h UTC) the whole western part of Black Sea was contaminated by radionuclides. The deposition values of 400 kBq/m^2 were reached at distances up to 290 km from ZNPP. The doses from terrestrial pathways reached considerable levels at large territories. For example, the total effective dose received by 1-yr children from all terrestrial exposure pathways except ingestion and all radionuclides during 1 year after the accident, simulated by JRODOS-FDMT model reached the value of 14.6 mSv at distance of 840 km from ZNPP. The isoline of 1 mSv of the same kind of dose covered large parts of the territories of southern Ukraine, Romania and Bulgaria. Ingestion of food products would lead to an increase of the respective doses by several times.

For simulation the transfer of radionuclides in marine environment, the compartment model POSEIDON-R was applied. Within the JRODOS system, the atmospheric deposition of radionuclides on the sea surface calculated by the atmospheric dispersion model was automatically converted to the source term for the POSEIDON-R model. It was shown that the described heavy accident at the ZNPP could significantly and for a long time contaminate the Black See by radioactive materials. The maximum concentrations of radionuclides in the surface water immediately after the deposition could exceed 1000 Bq/m^3, while 20 years after the considered accident the concentration of ^{137}Cs in the Black Sea will be around 25 Bq/m^3, which is several times higher than the current concentration. According to model results, concentration of ^{137}Cs in fish could exceed 100 Bq/kg. While initially the concentration of ^{137}Cs in prey fish will be higher with maximum in regions where the highest deposition density took place, later the concentration of ^{137}Cs in predatory fish will become higher due to the ability of cesium for biomagnification in the food chain. In dose calculations the impact of all long-lived radionuclides were taken into account. Calculations were made for each compartment separately supposing that all seafood consumption was from this compartment. The calculated annual individual dose for adults due to seafood consumption in the half of the see will exceed 1 mSv, in the area near the north-western coast of the Crimea Peninsula, such dose could exceed 10 mSv.

References

1. MARIS (Marine Information System): Radioactivity and stable isotope data in the marine environment (2021). https://maris.iaea.org/explore. Accessed 16 Nov 2022
2. Charmasson, S., Bailly du Bois, P., Thebault, H., Boust, D., Fievet, B.: Radioactivity of anthropic origin in the marine environment. In: Monaco, A., Prouzet, P. (eds.) The Land-Sea Interactions, pp. 237–282. ISTE Ltd and Wiley (2014)
3. Bezhenar, R., Maderich, V., Schirone, A., Conte, F., Martazinova, V.: Transport and fate of ^{137}Cs in the Mediterranean and Black Seas system during 1945–2020 period: a modelling study. J. Environ. Radioact. **208–209**, 106023 (2019)
4. Landman, C., Päsler-Sauer, J., Raskob, W.: The decision support system RODOS. In: The Risks of Nuclear Energy Technology. Science Policy Reports. Springer, Heidelberg (2014). https://doi.org/10.1007/978-3-642-55116-1_21
5. Mikkelsen, T., Thykier-Nielsen, S., Hoe, S.: Chapter 2.16 Medium-range puff growth. In: Borrego, C., Renner, E. (eds.) Air Pollution Modeling and Its Application XVIII. Developments in Environmental Science, vol. 6, pp. 243–252. Elsevier, Amsterdam (2007). https://doi.org/10.1016/S1474-8177(07)06216-X
6. GFS homepage: National Oceanic and Atmospheric Administration. US Department of Commercem. https://www.emc.ncep.noaa.gov/emc/pages/numerical_forecast_systems/gfs.php. Accessed 06 Sept 2022
7. Müller, H., Gering, F., Pröhl, G.: Model description of the terrestrial food chain and dose module. FDMT in RODOS PV6.0. Report RODOS(RA3)-TN(03)06, Institut für Strahlenschutz, Neuherberg (2003). https://resy5.iket.kit.edu/RODOS/Documents/Public/HandbookV6/Volume3/FDM_Terra.pdf. Accessed 06 Sept 2022
8. Maderich, V., Bezhenar, R., Tateda, Y., Aoyama, M., Tsumune, D., Jung, K.T., de With, G.: The POSEIDON-R compartment model for the prediction of transport and fate of radionuclides in the marine environment. MethodsX **5**, 1251–1266 (2018)
9. CMEMS, Copernicus Marine Environment Monitoring Service. http://marine.copernicus.eu/. Accessed 06 Sept 2022
10. Additional targeted re-evaluation of safety of the reactors of Zaporizhzhia NPP with taking into account Fukushima Lessons. National Atomic Energy Generating Company 'Energoatom', Kyiv, Ukraine (2012)
11. Booklet to Provide Basic Information Regarding Health Effects of Radiation. Ministry of the Environment, Government of Japan (2012). https://www.env.go.jp/en/chemi/rhm/basic-info/index.html. Accessed 06 Sept 2022
12. Norms of Radiation Safety of Ukraine (NRBU-97). Ministry of Health of Ukraine, Kyiv (1997)
13. Gulin, S.B., Mirzoyeva, N.Yu., Egorov, V.N., Polikarpov, G.G., Sidorov, I.G., Proskurnin, V.Yu.: Secondary radioactive contamination of the Black Sea after Chernobyl accident: recent levels, pathways and trends. J. Environ. Radioact. **124**, 50–56 (2013)
14. Heling, R., Koziy, L., Bulgakov, V.: On the dynamical uptake model developed for the uptake of radionuclides in marine organisms for the POSEIDON-R model system. Radioprotection **37**(C1), 833–838 (2002)
15. Kasamatsu, F., Ishikawa, Y.: Natural variation of radionuclide ^{137}Cs concentration in marine organisms with special reference to the effect of food habits and trophic level. Mar. Ecol. Prog. Ser. **160**, 109–120 (1997)
16. IAEA (International Atomic Energy Agency): Generic models for use in assessing the impact of discharges of radioactive substances to the environment. Safety reports series no. 19. IAEA, Vienna, Austria (2001).
17. IAEA (International Atomic Energy Agency): Radiation protection and safety of radiation sources: international basic safety standards. General safety requirements. Interim edition. IAEA, Vienna, Austria (2011)

Flux Correction for Nonconservative Convection-Diffusion Equation

Sergii Kivva

Abstract Using the possibilities of the new concept of Flux-Corrected Transport (FCT), our goal is to construct the flux limiters of this method for a nonconservative convection-diffusion equation. The proposed approach treats the classical FCT method as an approximate solution to the corresponding optimization problem. As in the classical FCT method, we consider a hybrid difference scheme whose fluxes are a linear combination of low- and high-order fluxes. The flux limiter is computed as an approximate solution to an optimization problem with a linear objective function. The constraints for this optimization problem are derived from inequalities that are valid for the low-order monotone scheme and apply to the hybrid scheme. The proposed approach applies to both explicit and implicit schemes and can be extended to multidimensional differential equations. Numerical results for different approximations of the convective fluxes are given. It is shown that numerical results with the flux limiters, which are exact and approximate solutions to the optimization problem, are in good agreement.

Keywords flux-corrected transport · nonconservative convection-diffusion equation · difference scheme · linear programming

1 Introduction

The objective of this paper is to develop a flux limiter for the flux-corrected transport (FCT) method for a nonconservative convection-diffusion equation. The numerical solution of such equations arises in a variety of applications such as hydrodynamics, heat, and mass transfer. To the best of our knowledge, we are not aware of any formulas for computing the FCT flux limiter for a nonconservative convection-diffusion equation.

S. Kivva (✉)
Institute of Mathematical Machines and System Problems, Kyiv, Ukraine
e-mail: skivva@gmail.com

S. Shkarlet et al. (eds.), *Mathematical Modeling and Simulation of Systems*, Lecture Notes in Networks and Systems 667,
https://doi.org/10.1007/978-3-031-30251-0_2

On an interval $[a, b]$, we consider the initial boundary value problem (IBVP) for a nonconservative convection-diffusion equation

$$\frac{\partial \rho}{\partial t} + u(x,t)\frac{\partial \rho}{\partial x} + \lambda(x,t)\rho = \frac{\partial}{\partial x}\left(D(x,t)\frac{\partial \rho}{\partial x}\right) + f(x,t), \quad t > 0, \tag{1.1}$$

with initial condition

$$\rho(x,0) = \rho^0(x), \tag{1.2}$$

where $0 \le D(x,t) \le \mu = const$.

For simplicity and without loss of generality, we assume that the Dirichlet boundary conditions are specified at the ends of the interval $[a, b]$

$$\rho(a,t) = \rho_a(t), \qquad \rho(b,t) = \rho_b(t). \tag{1.3}$$

The two-step FCT algorithm was firstly developed by Boris and Book [1] for solving a transient continuity equation. Within this approach, the flux at the cell interface is computed as a convex combination of fluxes of a monotone low-order scheme and a high-order scheme. These two fluxes are combined by adding to one of them (basic flux) a limited flux that is the limited difference between the high-order and low-order fluxes at the cell interface. In the classical FCT method, the low-order flux is basic and the additional limited flux is antidiffusive. Kuzmin and his coworkers [5] consider the high-order flux as the basic with an additional dissipative flux. Such approach is now known as algebraic flux correction (AFC). The procedure of two-step flux correction consists of computing the time advanced low order solution in the first step and correcting the solution in the second step to produce accurate and monotone results. The basic idea is to switch between high-order scheme and positivity preserving low-order scheme to provide oscillation free good resolution in steep gradient areas, while at the same time preserve at least second-order accuracy in smooth regions. Later Zalesak [10, 11] extended FCT to multidimensional explicit difference schemes. Since the 1970s, FCT has been widely used in the modeling of various physical processes. Many variations and generalizations of FCT and their applications are given in [8].

In this paper, we derive the flux correction formulas for the nonconservative convection-diffusion equation using the approach proposed in [3]. As in the classical FCT method, we use a hybrid difference scheme consisting of a convex combination of low-order monotone and high-order schemes. According to [3], finding the flux limiters we consider as a corresponding optimization problem with a linear objective function. The constraints for the optimization problem derive from the inequalities which are valid for the monotone scheme and apply to the hybrid scheme. The flux limiters are obtained as an approximate solution to the optimization problem. Numerical results show that these flux limiters produce numerical solutions that are in good agreement with the numerical solutions, the flux limiters of which are calculated from optimization problem and correspond to maximal antidiffusive fluxes.

The advantage of such approach is that the two-step classical FCT method is reduced to one-step. For flux corrections in the classical FCT method, it is necessary to know the low-order numerical solution at the current time step. In the proposed approach [3], it is sufficient to know only the numerical solution at the previous time step.

The paper is organized as follows. In Sect. 2, we discretize the IVBP (1.1)–(1.3) by a hybrid scheme. An analog of the discrete local maximum principle for the monotone scheme is given in Sect. 3. The optimization problem for finding flux limiters and the algorithm of its solving are described in Sect. 4. An approximate solution of the optimization problem is derived in Sect. 5. The results of numerical experiments are presented in Sect. 6. Concluding remarks are drawn in Sect. 7.

2 Hybrid Difference Scheme

In this section, we discretize the IBVP (1.1)–(1.3) using a hybrid difference scheme, which is a linear combination of a monotone scheme and a high-order scheme.

On the interval $[a, b]$, we introduce a nonuniform grid Ω_h

$$\Omega_h = \left\{x_i : \; x_i = x_{i-1} + \Delta_{i-1/2}x, \; i = \overline{1, N}; \; x_0 = a, \; x_{N+1} = b\right\}. \tag{2.1}$$

Assuming that $u(x,t)$ and $\rho(x,t)$ are sufficiently smooth, we consider approximations of the convective term in (1.1). For this, we integrate it over an interval $[x_{i-1/2}, x_{i+1/2}]$ and, applying the left and right rectangular rules for numerical integration as well as central difference for the first-order derivative, we obtain the following upwind discretization

$$\begin{aligned}\int\limits_{x_{i-1/2}}^{x_{i+1/2}} u\frac{\partial\rho}{\partial x}dx &= \int\limits_{x_{i-1/2}}^{x_{i+1/2}} u^{+}\,\frac{\partial\rho}{\partial x}dx + \int\limits_{x_{i-1/2}}^{x_{i+1/2}} u^{-}\,\frac{\partial\rho}{\partial x}dx \\ &=\Delta x_i\left[u^{+}_{i-1/2}\frac{(\rho_i-\rho_{i-1})}{\Delta_{i-1/2}x} + u^{-}_{i+1/2}\frac{(\rho_{i+1}-\rho_i)}{\Delta_{i+1/2}x}\right] + O\left(\Delta x_i^2\right).\end{aligned} \tag{2.2}$$

where $\rho_i = \rho(x_i, t)$, $\Delta x_i = (x_{i+1} - x_{i-1})/2$ is the spatial size of the ith cell; $u^{\pm} = (u \pm |u|)/2$.

To obtain an approximation of a higher order, we apply the trapezoidal rule for numerical integration and central difference for the first-order derivative

$$\int\limits_{x_{i-1/2}}^{x_{i+1/2}} u\frac{\partial\rho}{\partial x}dx = \frac{\Delta x_i}{2}\left[u_{i-1/2}\frac{(\rho_i-\rho_{i-1})}{\Delta_{i-1/2}x} + u_{i+1/2}\frac{(\rho_{i+1}-\rho_i)}{\Delta_{i+1/2}x}\right] + O\left(\Delta x_i^3\right). \tag{2.3}$$

Besides, we rewrite the convective term in (1.1) as follows:

$$u\frac{\partial \rho}{\partial x} = \frac{\partial}{\partial x}(u\rho) - \rho\frac{\partial u}{\partial x}. \tag{2.4}$$

We discretize the terms on the right-hand side of (2.4) by the following difference relations

$$\int\limits_{x_{i-1/2}}^{x_{i+1/2}} \frac{\partial(u\rho)}{\partial x}dx = u^+_{i+1/2}\rho_i + u^-_{i+1/2}\rho_{i+1} - u^+_{i-1/2}\rho_{i-1} - u^-_{i-1/2}\rho_i + O(\Delta x_i), \tag{2.5}$$

$$\int\limits_{x_{i-1/2}}^{x_{i+1/2}} \frac{\partial(u\rho)}{\partial x}dx = \frac{1}{2}u_{i+1/2}(\rho_i + \rho_{i+1}) - u_{i-1/2}(\rho_{i-1} + \rho_i) + O\left(\Delta x_i^2\right), \tag{2.6}$$

$$\int\limits_{x_{i-1/2}}^{x_{i+1/2}} \rho\frac{\partial u}{\partial x}dx = \rho_i\left(u_{i+1/2} - u_{i-1/2}\right) + O\left(\Delta x_i^2\right). \tag{2.7}$$

Using a convex combination of (2.5) and (2.6) to approximate the divergent term in (2.4), we discretize the convective term as

$$\int\limits_{x_{i-1/2}}^{x_{i+1/2}} u\frac{\partial \rho}{\partial x}dx = \left[u^+_{i+1/2}\rho_i + u^-_{i+1/2}\rho_{i+1} + \beta_{i+1/2}\frac{|u_{i+1/2}|}{2}(\rho_{i+1} - \rho_i)\right.$$
$$\left. - u^+_{i-1/2}\rho_{i-1} - u^-_{i-1/2}\rho_i - \beta_{i-1/2}\frac{|u_{i-1/2}|}{2}(\rho_i - \rho_{i-1})\right] - \rho_i\left(u_{i+1/2} - u_{i-1/2}\right), \tag{2.8}$$

where $\beta_{i+1/2}$ is the flux limiter for the divergent part in square brackets of the convective flux. For a flux correction of the convective term in the divergent form, we refer to [3].

Below, to approximate the convective term in (1.1), we apply a convex combination of (2.2) and (2.3). Note that

$$\frac{1}{2}\left[u_{i+1/2}\frac{(\rho_{i+1} - \rho_i)}{\Delta_{i+1/2}x} + u_{i-1/2}\frac{(\rho_i - \rho_{i-1})}{\Delta_{i-1/2}x}\right] = \left[u^-_{i+1/2}\frac{(\rho_{i+1} - \rho_i)}{\Delta_{i+1/2}x}\right.$$
$$\left. + u^+_{i-1/2}\frac{(\rho_i - \rho_{i-1})}{\Delta_{i-1/2}x}\right] + \left[\frac{|u_{i+1/2}|}{2}\frac{(\rho_{i+1} - \rho_i)}{\Delta_{i+1/2}x} - \frac{|u_{i-1/2}|}{2}\frac{(\rho_i - \rho_{i-1})}{\Delta_{i-1/2}x}\right]. \tag{2.9}$$

The second term in square brackets on the right-hand side of (2.9) can be considered as an anti-diffusion.

We approximate (1.1)–(1.3) by the following weighted difference scheme

$$\frac{y_i^{n+1}-y_i^n}{\Delta t}+h_{i+1/2}^{-,(\sigma)}+h_{i-1/2}^{+,(\sigma)}+(\lambda y)_i^{(\sigma)}=f_i^{(\sigma)}, \tag{2.10}$$

where $y_i^n = y(x_i, t^n)$ is the grid function on Ω_h; Δt is the time step; $f_i^{(\sigma)} = \sigma f_i^{n+1} + (1-\sigma) f_i^n$, $\sigma \in [0, 1]$. The numerical flux $h_{i+1/2}^{\pm,n}$ is written in the form

$$h_{i\mp 1/2}^{\pm,n}=\left(u_{i\mp 1/2}^{\pm,n}+d_i^{\pm,n}-\alpha_i^{\pm,n}r_i^{\pm,n}\right)\frac{\Delta_{i\mp 1/2}y^n}{\Delta_{i\mp 1/2}x}, \tag{2.11}$$

where $\alpha_i^{\pm,n} \in [0, 1]$ is the flux limiter; $\Delta_{i+1/2}y^n = y_{i+1}^n - y_i^n$; the coefficients $d_i^{\pm,n}$ and $s_i^{\pm,n}$ are computed as

$$d_i^{\pm,n}=\pm\max\left(0,\frac{D_{i\mp 1/2}^n}{\Delta x_i}-\frac{\left|u_{i\mp 1/2}^n\right|}{2}\right), \tag{2.12}$$

$$r_i^{\pm,n}=\mp\min\left(0,\frac{D_{i\mp 1/2}^n}{\Delta x_i}-\frac{\left|u_{i\mp 1/2}^n\right|}{2}\right). \tag{2.13}$$

Note that for $\sigma = 0$ scheme (2.10) is explicit and implicit for $\sigma > 0$. Let us denote by y_0^n and y_{N+1}^n the values of $\rho(x,t)$ at the left and right ends of the interval $[a, b]$ at time t^n.

We rewrite the difference scheme (2.10) in matrix form as

$$\begin{aligned}\left[E+\Delta t\sigma\left(A^{n+1}+\Lambda^{n+1}\right)\right]\boldsymbol{y}^{n+1}-\Delta t\left[(B^-\boldsymbol{\alpha}^-)^{(\sigma)}+(B^+\boldsymbol{\alpha}^+)^{(\sigma)}\right]\\=\left[E-\Delta t(1-\sigma)\left(A^n+\Lambda^n\right)\right]\boldsymbol{y}^n+\Delta t\boldsymbol{g}^{(\sigma)},\end{aligned} \tag{2.14}$$

where $(B^{\pm}\boldsymbol{\alpha}^{\pm})^{(\sigma)} = \sigma B^{\pm,n+1}\boldsymbol{\alpha}^{\pm,n+1} + (1-\sigma)B^{\pm,n}\boldsymbol{\alpha}^{\pm,n}$; $B^{\pm} = diag\{b_i^{\pm}(\boldsymbol{y})\}_{i=1}^N$ is the diagonal matrix; E is the identity matrix of order N; $A = \{a_{ij}\}_i^j$ is tridiagonal square matrix of order N; $\Lambda = diag(\lambda_1, \ldots, \lambda_N)$ is the diagonal matrix; $\boldsymbol{\alpha}^{\pm} = (\alpha_1^{\pm}, \ldots, \alpha_N^{\pm})^T \in R^N$ are the numerical vectors of flux limiters; $\boldsymbol{g} = (g_1, \ldots, g_N)^T$ is the vector of boundary conditions and values of the function f at the points x_i. Components of the vector $\boldsymbol{g}$ are given by

$$g_1=\frac{(u_{1/2}^+ + d_1^+)y_0}{\Delta_{1/2}x}+f_1,\quad g_i=f_i,\quad g_N=\frac{-(u_{N+1}^- + d_N^-)y_{N+1}}{\Delta_{N+1/2}}+f_N. \tag{2.15}$$

Elements of the matrices A and $B^{\pm}$ are calculated as

$$a_{ii-1} = \frac{-u^{+}_{i-1/2} - d^{+}_{i}}{\Delta_{i-1/2}x}, \qquad b^{+}_{i} = r^{+}_{i}\,\frac{y_i - y_{i-1}}{\Delta_{i-1/2}x},$$
$$a_{ii+1} = \frac{u^{-}_{i+1/2} + d^{-}_{i}}{\Delta_{i+1/2}x}, \qquad b^{-}_{i} = r^{-}_{i}\,\frac{y_{i+1} - y_{i}}{\Delta_{i+1/2}x}, \qquad a_{ii} = -a_{ii-1} - a_{ii+1}. \tag{2.16}$$

3 Monotone Difference Scheme

We consider the system of equations (2.14) for $\boldsymbol{\alpha}^{\pm,n}, \boldsymbol{\alpha}^{\pm,n+1} = 0$

$$\begin{aligned}&\left[E + \Delta t\,\sigma A^{n+1} + \Delta t \sigma \Lambda^{n+1}\right]\boldsymbol{y}^{n+1} - \Delta t\,\sigma \boldsymbol{g}^{n+1}\\ =&\left[E - \Delta t\,(1-\sigma)A^{n} - \Delta t(1-\sigma)\Lambda^{n}\right]\boldsymbol{y}^{n} + \Delta t\,(1-\sigma)\boldsymbol{g}^{n}.\end{aligned} \tag{3.1}$$

In this section, we obtain the monotonicity condition for the difference scheme (3.1) and derive for it an analog of the discrete local maximum principle, which plays a key role in the flux correction design.

Definition 3.1 ([2]). *A difference scheme*

$$y^{n+1}_{i} = H(y^{n}_{i-k}, y^{n}_{i-k+1}, ..., y^{n}_{i}, ..., y^{n}_{i+l}) \tag{3.2}$$

is said to be monotone if H is a monotone increasing function of each of its arguments.

Theorem 3.1. *If Δt satisfies*

$$\Delta t\,\sigma \min_{1\le i\le N} \lambda^{n+1}_{i} < 1, \tag{3.3}$$

$$\Delta t\,(1-\sigma) \max_{1\le i\le N}\left[\frac{u^{+,n}_{i-1/2} + d^{+,n}_{i}}{\Delta_{i-1/2}x} - \frac{u^{-,n}_{i+1/2} + d^{-,n}_{i}}{\Delta_{i+1/2}x} + \lambda^{n}_{i}\right] \le 1, \tag{3.4}$$

then the difference scheme (3.1) *is monotone.*

Proof. If (3.3) holds, the matrix $\left[E + \Delta t\sigma(A^{n+1} + \Lambda^{n+1})\right]$ is a strictly row diagonally dominant M-matrix. Then the inverse matrix $\left[E + \Delta t\sigma(A^{n+1} + \Lambda^{n+1})\right]^{-1}$ is a matrix with nonnegative elements.

The nonnegativity of the elements of matrix $\left[E + \Delta t\sigma(A^{n+1} + \Lambda^{n+1})\right]^{-1}$ $\times [E - \Delta t(1-\sigma)(A^{n} + \Lambda^{n})]$ and, hence, the monotonicity of the scheme (3.1) follows from the nonnegativity of the elements $[E - \Delta t(1-\sigma)(A^{n} + \Lambda^{n}]$ for Δt satisfying (3.4).

Theorem 3.2. *If Δt satisfies*

$$\Delta t\,(1-\sigma) \max_{1\le i\le N}\left[\frac{u^{+,n}_{i-1/2} + d^{+,n}_{i}}{\Delta_{i-1/2}x} - \frac{u^{-,n}_{i+1/2} + d^{-,n}_{i}}{\Delta_{i+1/2}x}\right] \le 1, \tag{3.5}$$

then the numerical solution of the system of equations (3.1) *satisfies the following inequalities*

$$\begin{aligned}
&\min_{k\in S_i} y_k^n - \Delta t(1-\sigma)\lambda_i^n y_i^n + \Delta t(1-\sigma) f_i^n \\
\le & y_i^{n+1} + \Delta t\sigma \sum_j a_{ij}^{n+1} y_j^{n+1} + \Delta t\sigma\lambda_i^{n+1} y_i^{n+1} - \Delta t\sigma g_i^{n+1} \\
\le & \max_{k\in S_i} y_k^n - \Delta t(1-\sigma)\lambda_i^n y_i^n + \Delta t(1-\sigma) f_i^n,
\end{aligned} \tag{3.6}$$

where S_i is the stencil of the difference scheme (3.1) *for an ith grid node.*

Proof. Let us prove the right-hand side of inequality (3.6). We rewrite the ith row of the system of equations (3.1) in the form

$$\begin{aligned}
&y_i^{n+1} + \Delta t\sigma \sum_j a_{ij}^{n+1} y_j^{n+1} + \Delta t\sigma\lambda_i^{n+1} y_i^{n+1} - \Delta t\sigma g_i^{n+1} \\
=& \left[1 + \Delta t(1-\sigma)\left(\frac{u_{i+1/2}^{-,n} + d_i^{-,n}}{\Delta_{i+1/2}x} - \frac{u_{i-1/2}^{+,n} + d_i^{+,n}}{\Delta_{i-1/2}x}\right)\right] y_i^n - \Delta t(1-\sigma)\lambda_i^n y_i^n \\
&+\Delta t(1-\sigma)\left(\frac{u_{i-1/2}^{+,n} + d_i^{+,n}}{\Delta_{i-1/2}x} y_{i-1}^n - \frac{u_{i+1/2}^{-,n} + d_i^{-,n}}{\Delta_{i+1/2}x} y_{i+1}^n\right) + \Delta t(1-\sigma) f_i^n.
\end{aligned} \tag{3.7}$$

Under condition (3.5), the first and third terms on the right-hand side of (3.7) are a convex linear combination, therefore

$$\begin{aligned}
&y_i^{n+1} + \Delta t\sigma \sum_j a_{ij}^{n+1} y_j^{n+1} + \Delta t\sigma\lambda_i^{n+1} y_i^{n+1} - \Delta t\sigma g_i^{n+1} \\
\le & \max_{k\in S_i} y_k^n - \Delta t(1-\sigma)\lambda_i^n y_i^n + \Delta t(1-\sigma) f_i^n.
\end{aligned} \tag{3.8}$$

The lower bound (3.6) is obtained in a similar way, which proves the theorem.

Remark 3.1. Under condition (3.3), the matrix $G = \left[E + \Delta t\ \sigma(A^{n+1} + \Lambda^{n+1})\right]$ is a non-singular M-matrix, therefore G^{-1} is a nonnegative and isotone matrix [9, p.52, 2.4.3], i.e. if $\boldsymbol{x} \preceq \boldsymbol{y}$, then $G^{-1}\boldsymbol{x} \preceq G^{-1}\boldsymbol{y}$. Here $\preceq$ denotes the natural (componentwise) partial ordering on R^N, i.e. $\boldsymbol{x} \preceq \boldsymbol{y}$ if and only if $x_i \le y_i$ for all i. Thus, the change of the vector $\boldsymbol{y}^{n+1}$ can be controlled by changing the right-hand side of the equation (3.1).

Inequalities (3.6) hold for the right-hand side of (3.1) and will be used to obtain restrictions on flux limiters in the scheme (2.14). We can consider (3.6) as an analogue of discrete local maximum principle for the scheme (3.1). Note that to obtain restrictions (3.6), it is sufficient for us to know the numerical solution of (3.1) at a previous time step.

4 Finding Flux Limiters

To find fux limiters for scheme (2.14), we implement the approach proposed in [3]. Our goal is to find maximum values of the flux limiters for which the solution of the difference scheme (2.14) is similar to the solution of the monotone difference scheme (3.1). For this, we require that the difference scheme (2.14) satisfies inequalities (3.6). Then the finding flux limiters can be considered as the following optimization problem

$$\Im(\boldsymbol{\alpha}^{\pm,n},\boldsymbol{\alpha}^{\pm,n+1}) = \sum_{k=n}^{n+1}\sum_{i=1}^{N}\alpha_i^{+,k} + \sum_{k=n}^{n+1}\sum_{i=1}^{N}\alpha_i^{-,k} \to \max_{\boldsymbol{\alpha}^{\pm,n},\boldsymbol{\alpha}^{\pm,n+1}\in U_{ad}} \tag{4.1}$$

subject to (2.14) and

$$\begin{gathered}\underline{\boldsymbol{y}}^n + \Delta t(1-\sigma)\boldsymbol{f}^n \le \left[E - \Delta t(1-\sigma)A^n\right]\boldsymbol{y}^n \\ +\Delta t\left(B^+\boldsymbol{\alpha}^+ + B^-\boldsymbol{\alpha}^-\right)^{(\sigma)} + \Delta t(1-\sigma)\boldsymbol{g}^n \le \bar{\boldsymbol{y}}^n + \Delta t(1-\sigma)\boldsymbol{f}^n,\end{gathered} \tag{4.2}$$

where $\underline{\boldsymbol{y}}$ and $\bar{\boldsymbol{y}}$ are column vectors whose components are $\underline{y}_i = \min_{j\in S_i} y_j$ and $\bar{y}_i = \max_{j\in S_i} y_j$. U_{ad} is the set of vectors $\boldsymbol{\alpha}^{\pm,n}, \boldsymbol{\alpha}^{\pm,n+1}$, which is defined as the Cartesian product of N-vectors

$$U^{ad} = \left\{\left(\boldsymbol{\alpha}^{\pm,n},\boldsymbol{\alpha}^{\pm,n+1}\right) \in \left(R^N\right)^4 : \quad 0 \le \alpha_i^{\pm,k} \le 1, \quad k = n, n+1\right\}. \tag{4.3}$$

Note that for $\sigma = 0$ the optimization problem (4.1)–(4.3) and (2.14) is a linear programming problem, and for $\sigma > 0$ it is a nonlinear programming problem.

To solve the nonlinear optimization problem (4.1)–(4.3) and (2.14) in one time step, we use the following iterative process:

Step 1. Initialize positive numbers $\delta, \varepsilon_1, \varepsilon_2 > 0$. Set $p = 0$, $\boldsymbol{y}^{n+1,0} = \boldsymbol{y}^n$, $\boldsymbol{\alpha}^{\pm,n,0}$, $\boldsymbol{\alpha}^{\pm,n+1,0} = 0$.

Step 2. Find the solution $\boldsymbol{\alpha}^{\pm,n,p+1}, \boldsymbol{\alpha}^{\pm,n+1,p+1}$ of the following linear programming problem

$$\Im(\boldsymbol{\alpha}^{\pm,n,p+1},\boldsymbol{\alpha}^{\pm,n+1,p+1}) \to \max_{\boldsymbol{\alpha}^{\pm,n,p+1},\boldsymbol{\alpha}^{\pm,n+1,p+1}\in U_{ad}}, \tag{4.4}$$

$$\begin{aligned}&\min_{j\in S_i} y_j^n - y_i^n + \Delta t\,(1-\sigma)\sum_{j\ne i} a_{ij}^n\left(y_j^n - y_i^n\right) \\ &\le \Delta t\,(1-\sigma)\left(b_i^{+,n}\alpha_i^{+,n,p+1} + b_i^{-,n}\alpha_i^{-,n,p+1}\right) \\ &+ \Delta t\,\sigma\left(b_i^{+,n+1,p}\alpha_i^{+,n+1,p+1} + b_i^{-,n+1,p}\alpha_i^{-,n+1,p+1}\right) \\ &\le \max_{j\in S_i} y_j^n - y_i^n + \Delta t\,(1-\sigma)\sum_{j\ne i} a_{ij}^n\left(y_j^n - y_i^n\right),\end{aligned} \tag{4.5}$$

Step 3. For the $\boldsymbol{\alpha}^{\pm,n,p+1}$, $\boldsymbol{\alpha}^{\pm,n+1,p+1}$, find $y_i^{n+1,p+1}$ from the system of linear equations

$$\left[E + \Delta t\sigma \left(A^{n+1} + \Lambda^{n+1}\right)\right] \boldsymbol{y}^{n+1,p+1} = \left[E - \Delta t(1-\sigma)\left(A^n + \Lambda^n\right)\right] \boldsymbol{y}^n + \Delta t \left[\left(B^{+,p}\boldsymbol{\alpha}^{+,p+1}\right)^{(\sigma)} + \left(B^{-,p}\boldsymbol{\alpha}^{-,p+1}\right)^{(\sigma)}\right] + \Delta t\, \boldsymbol{g}^{(\sigma)}, \tag{4.6}$$

Step 4. Algorithm stop criterion

$$\max_i \frac{\left|y_i^{n+1,p+1} - y_i^{n+1,p}\right|}{\max\left(\delta, \left|y_i^{n+1,p+1}\right|\right)} < \varepsilon_1, \qquad \left|\Im\left(\boldsymbol{\alpha}^{\pm,n,p+1}, \boldsymbol{\alpha}^{\pm,n+1,p+1}\right) - \Im\left(\boldsymbol{\alpha}^{\pm,n,p}, \boldsymbol{\alpha}^{\pm,n+1,p}\right)\right| < \varepsilon_2. \tag{4.7}$$

If conditions (4.7) hold, then $\boldsymbol{y}^{n+1} = \boldsymbol{y}^{n+1,p+1}$. Otherwise, set $p = p + 1$ and go to **Step** 2.

The solvability of the linear programming problem (4.4)–(4.5) is considered in the theorem below.

Theorem 4.1. *Assume that Δt satisfies* (3.3)–(3.5)*, then the linear programming problem* (4.4)–(4.5) *is solvable.*

Proof. To prove that problem (4.4)–(4.5) is solvable, it is sufficient to show that the objective function $\Im(\boldsymbol{\alpha}^{\pm,n}, \boldsymbol{\alpha}^{\pm,n+1})$ is bounded and the feasible set is non-empty. The boundedness of the function (4.1) follows from the boundedness of the vectors $\boldsymbol{\alpha}^{\pm,n}$ and $\boldsymbol{\alpha}^{\pm,n+1}$ whose coordinates vary from zero to one. On the other hand, if the hypothesis of the theorem is true, then the zero vectors $\boldsymbol{\alpha}^{\pm,n}$ and $\boldsymbol{\alpha}^{\pm,n+1}$ satisfy the system of inequalities (4.5).

This completes the proof of the theorem.

5 Flux Limiter Design

In the iterative process described in the previous section, the flux limiters are found by solving the linear programming problem (4.4)–(4.5). Solving a linear programming problem requires additional computational cost. Therefore, in the iterative process at **Step** 2, instead of (4.4)–(4.5), it is reasonable to use its approximate solution.

The purpose of this section is to find a nontrivial approximate solution to the linear programming problem (4.4)–(4.5). Nonzero $\left(\boldsymbol{\alpha}^{\pm,n}, \boldsymbol{\alpha}^{\pm,n+1}\right) \in U_{ad}$ satisfy the system of inequalities (4.5), and, omitting the iteration number, we rewrite the latter in the form

$$
\begin{aligned}
&(1-\sigma)\left(b_i^{+,n}\alpha_i^{+,n}+b_i^{-,n}\alpha_i^{-,n}\right)+\sigma\left(b_i^{+,n+1}\alpha_i^{+,n+1}+b_i^{-,n+1}\alpha_i^{-,n+1}\right)\\
&\qquad\le\ \frac{1}{\Delta t}\left(\max_{j\in S_i} y_j^n-y_i^n\right)+(1-\sigma)\sum_{j\ne i}a_{ij}^n\left(y_j^n-y_i^n\right),
\end{aligned}
\tag{5.1}
$$

$$
\begin{aligned}
&(1-\sigma)\left(b_i^{+,n}\alpha_i^{+,n}+b_i^{-,n}\alpha_i^{-,n}\right)+\sigma\left(b_i^{+,n+1}\alpha_i^{+,n+1}+b_i^{-,n+1}\alpha_i^{-,n+1}\right)\\
&\qquad\ge\ \frac{1}{\Delta t}\left(\min_{j\in S_i} y_j^n-y_i^n\right)+(1-\sigma)\sum_{j\ne i}a_{ij}^n\left(y_j^n-y_i^n\right),
\end{aligned}
\tag{5.2}
$$

$$
0\le\alpha_i^{\pm,n}\le 1,\quad 0\le\alpha_i^{\pm,n+1}\le 1. \tag{5.3}
$$

For the left-hand sides of inequalities (5.1) and (5.2), the following estimates are valid

$$
\begin{aligned}
&(1-\sigma)\left(b_i^{+,n}\alpha_i^{+,n}+b_i^{-,n}\alpha_i^{-,n}\right)+\sigma\left(b_i^{+,n+1}\alpha_i^{+,n+1}+b_i^{-,n+1}\alpha_i^{-,n+1}\right)\\
&\qquad\le\alpha_i^{+,max}\Big[(1-\sigma)\left(\max(0,b_i^{+,n})+\max(0,b_i^{-,n})\right)\\
&\qquad\qquad+\sigma\left(\max(0,b_i^{+,n+1})+\max(0,b_i^{-,n+1})\right)\Big],
\end{aligned}
\tag{5.4}
$$

$$
\begin{aligned}
&(1-\sigma)\left(b_i^{+,n}\alpha_i^{+,n}+b_i^{-,n}\alpha_i^{-,n}\right)+\sigma\left(b_i^{+,n+1}\alpha_i^{+,n+1}+b_i^{-,n+1}\alpha_i^{-,n+1}\right)\\
&\qquad\ge\alpha_i^{-,max}\Big[(1-\sigma)\left(\min(0,b_i^{+,n})+\min(0,b_i^{-,n})\right)\\
&\qquad\qquad+\sigma\left(\min(0,b_i^{+,n+1})+\min(0,b_i^{-,n+1})\right)\Big],
\end{aligned}
\tag{5.5}
$$

where $\alpha_i^{+,max}$ and $\alpha_i^{-,max}$ are the maximums of the components $\alpha_i^{\pm,n}$ and $\alpha_i^{\pm,n+1}$ corresponding to the non-negative and non-positive coefficients $b_i^{\pm}$ on the left-hand sides of (5.4) and (5.5), respectively.

Substituting (5.4) into (5.1), and (5.5) into (5.2) yields

$$
\alpha_i^{\pm,k}=\begin{cases} R_i^+, & b_i^{\pm,k}>0,\\ R_i^-, & b_i^{\pm,k}<0,\end{cases}\qquad k=n,n+1, \tag{5.6}
$$

where

$$
R_i^{\pm}=\min\left(1,\alpha_i^{\pm,max}\right)=\min\left(1,Q_i^{\pm}/P_i^{\pm}\right), \tag{5.7}
$$

$$
Q_i^{+}=\frac{1}{\Delta t}\left(\max_{j\in S_i} y_j^n-y_i^n\right)+(1-\sigma)\sum_{j\ne i}a_{ij}^n\left(y_j^n-y_i^n\right), \tag{5.8}
$$

$$
Q_i^{-}=\frac{1}{\Delta t}\left(\min_{j\in S_i} y_j^n-y_i^n\right)+(1-\sigma)\sum_{j\ne i}a_{ij}^n\left(y_j^n-y_i^n\right), \tag{5.9}
$$

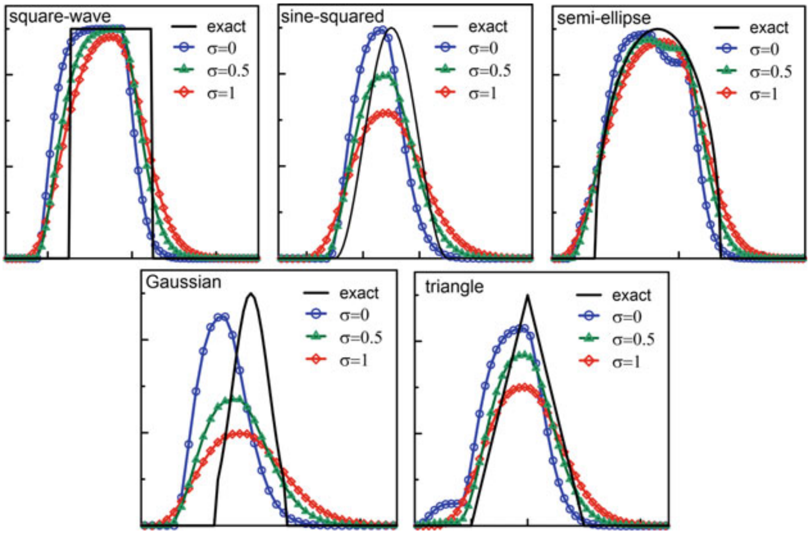


Fig. 1 Numerical results for the advection test (6.1) with NDVL scheme for various weights σ. Flux limiters are calculated using the linear programming problem (4.4)–(4.5)

$$
\begin{aligned}
P_i^+ &= (1-\sigma)\left(\max(0, b_i^{+,n}) + \max(0, b_i^{-,n})\right) \\
&\quad + \sigma\left(\max(0, b_i^{+,n+1}) + \max(0, b_i^{-,n+1})\right),
\end{aligned}
\tag{5.10}
$$

$$
\begin{aligned}
P_i^- &= (1-\sigma)\left(\min(0, b_i^{+,n}) + \min(0, b_i^{-,n})\right) \\
&\quad + \sigma\left(\min(0, b_i^{+,n+1}) + \min(0, b_i^{-,n+1})\right).
\end{aligned}
\tag{5.11}
$$

Remark 5.1. Note that this approach is also applicable for schemes with a high-order approximation of the convective-diffusive flux. Moreover, this method and formulas (5.6)–(5.11) can be easily generalized to the multidimensional case. For details we refer the reader to [3, 4]

6 Numerical Results

We conclude the paper with a number of numerical tests. The purpose of this section is to compare the results of the difference schemes considered in the paper. Below, we abbreviate by NDVL and NDVA the difference scheme (2.14), flux limiters of which are exact or approximate solutions of the linear programming problem (4.4)–(4.5). We also use DIV notation for the difference scheme, the flux correction of which is based on the divergent part of the convective flux (2.8).

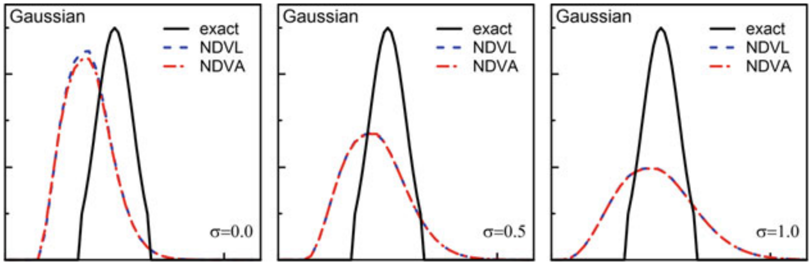


Fig. 2 Comparison of numerical results for the advection test (6.1) with NDVL and NDVA schemes for $\sigma = 0.5$

In our calculations, we apply the GLPK (GNU Linear Programming Kit) v.4.65 set of routines for solving linear programming problems. GLPK is available at https://www.gnu.org/software/glpk/.

6.1 One-Dimensional Advection

We consider the one-dimensional advection test of Leonard et al. [6] on the uniform grid with $\Delta x = 0.01$ and constant velocity. The initial scalar profile consists of five different shapes: square wave, sine-squared, semi-ellipse, Gaussian, and triangle. The initial profile is specified as

$$y(x_i) = \begin{cases} 1 & \text{if } \ 0.05 \le x_i \le 0.25 \ \text{(square wave)} \\ \sin^2\left[\dfrac{\pi}{0.2}(x_i - 0.85)\right] & \text{if } \ 0.85 \le x_i \le 1.05 \ \text{(sine} - \text{squared)} \\ \sqrt{1 - \left[\dfrac{1}{15\Delta x}(x_i - 1.75)\right]^2} & \text{if } \ 1.6 \le x_i \le 1.9 \quad \text{(semi} - \text{ellipse)} \\ \exp\left[-\dfrac{1}{2\gamma^2}(x_i - 2.65)^2\right] & \text{if } \ 2.6 \le x_i \le 2.7 \quad \text{(Gaussian)} \\ 10\,(x_i - 3.3) & \text{if } \ 3.3 \le x_i \le 3.4 \quad \text{(triangle)} \\ 1.0 - 10\,(x_i - 3.4) & \text{if } \ 3.4 \le x_i \le 3.5 \\ 0 & \text{otherwise} \end{cases} \tag{6.1}$$

The standard deviation for the Gaussian profile is specified as $\gamma = 2.5$.

Numerical results with the NDVL scheme after 400 time steps at a Courant number of 0.2 are shown in Fig. 1. The flux limiters are calculated using the linear programming problem (4.4)–(4.5). At the right edge of the semi-ellipse for $\sigma = 0$ and $\sigma = 0.5$, we observe the well-known "terracing" phenomenon, which is a nonlinear effect of residual phase errors. It is shown in [8, 11] that high-order FCT methods (above fourth-order) significantly reduce phase errors and that selective adding diffusion can also reduce terracing. In the numerical solution of the implicit scheme,

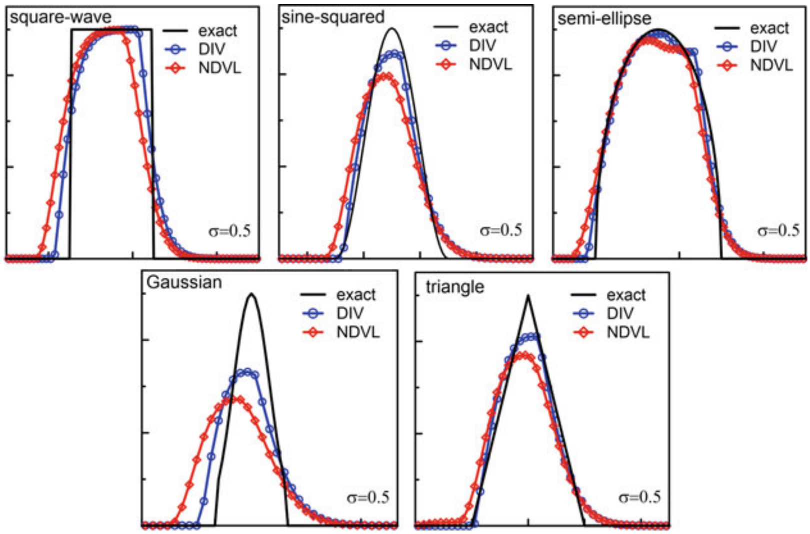


Fig. 3 Comparison of numerical results for the advection test (6.1) with DIV and NDVL schemes for $\sigma = 0.5$

there is no terracing. The implicit scheme is more diffusive than the previous two, and its numerical solution is also more diffusive.

The Gaussian test problem has a single moving maximum and shows the effects of "clipping" the solution. This is because the flux limiter cannot account for the true peak of the Gaussian as it passes between the grid points. The maximum is clipped less as the order of the algorithm increases. The key to good performance here is the application of a more flexible limiter and a more accurate estimate of the allowable upper and lower bounds on the solution [10, 11].

The numerical results for which the flux limiters are calculated using exact and approximate solutions of the linear programming problem (4.4)–(4.5) are slightly different. Their L^1-norm of errors and the maximum values are presented in Table 1. The comparison of the NDVL and NDVA results with $\sigma = 0.5$ is given in Fig. 2.

In Fig. 3 the solutions computed by the NDVL scheme are compared with the DIV scheme. Their L^1-norm of errors and the maximum values are presented in Table 1. Notice, that both the maximum values and the errors of the DIV scheme are better than the corresponding maximum values and errors of the NDVL scheme.

6.2 Solid Body Rotations

In this section, we consider the rotation of solid bodies [7, 10] under an incompressible flow that is described by the linear equation

Table 1 L^1-norm of errors and maximum values of numerical results for the advection test (6.1) with DIV, NDVL and NDVA schemes

	σ	DIV L^1 error	DIV y_{max}	NDVL L^1 error	NDVL y_{max}	NDVA L^1 error	NDVA y_{max}
wav	0.0	2.1811×10^{-2}	1.0000	8.1136×10^{-2}	1.0000	8.1182×10^{-2}	1.0000
	0.5	4.3933×10^{-2}	0.9997	6.5511×10^{-2}	0.9976	6.5527×10^{-2}	0.9973
	1.0	6.9477×10^{-2}	0.9843	7.6861×10^{-2}	0.9653	7.6774×10^{-2}	0.9650
sine	0.0	1.6883×10^{-2}	0.9938	4.6661×10^{-2}	0.9913	4.7052×10^{-2}	0.9766
	0.5	1.6423×10^{-2}	0.8895	3.2650×10^{-2}	0.7917	3.2759×10^{-2}	0.7899
	1.0	3.9029×10^{-2}	0.7043	4.5601×10^{-2}	0.6300	4.5694×10^{-2}	0.6286
elp	0.0	1.7926×10^{-2}	0.9973	4.9044×10^{-2}	0.9774	4.8959×10^{-2}	0.9775
	0.5	1.7913×10^{-2}	0.9810	2.8675×10^{-2}	0.9526	2.8660×10^{-2}	0.9524
	1.0	3.6078×10^{-2}	0.9601	3.9624×10^{-2}	0.9421	3.9603×10^{-2}	0.9422
gau	0.0	1.3639×10^{-2}	0.9764	6.9049×10^{-2}	0.8991	6.8116×10^{-2}	0.8661
	0.5	2.7592×10^{-2}	0.6629	4.5303×10^{-2}	0.5438	4.5337×10^{-2}	0.5417
	1.0	4.3681×10^{-2}	0.4828	4.8852×10^{-2}	0.3965	4.8882×10^{-2}	0.3949
tri	0.0	2.5205×10^{-2}	0.9389	4.8921×10^{-2}	0.8555	4.8870×10^{-2}	0.8517
	0.5	1.3843×10^{-2}	0.8216	2.6126×10^{-2}	0.7404	2.6180×10^{-2}	0.7391
	1.0	3.1245×10^{-2}	0.6655	3.7023×10^{-2}	0.6006	3.7123×10^{-2}	0.5991

wav = Square wave; sine = Sine-squared; elp = Semi-ellipse; gau = Gaussian; tri = Triangle.

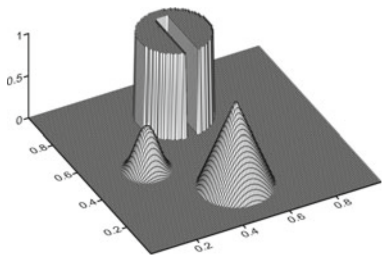


Fig. 4 Initial data and exact solution at the final time for the solid body rotation test

$$\frac{\partial \rho}{\partial t} + \boldsymbol{u} \cdot \nabla \rho = 0 \qquad \text{in} \quad \Omega = (0, 1) \times (0, 1) \tag{6.2}$$

with zero boundary conditions. The initial condition includes a slotted cylinder, a cone and a smooth hump (Fig. 4). The slotted cylinder of radius 0.15 and height 1 is centered at the point (0.5,0.75) and

$$\rho(x, y, 0) = \begin{cases} 1 & \text{if} \quad |x - 0.5| \geq 0.025 \quad \text{or} \quad y \geq 0.85 \\ 0 & \text{otherwise} \end{cases}$$

The cone of also radius $r_0 = 0.15$ and height 1 is centered at point $(x_0, y_0) = (0.25, 0.5)$ and

$$\rho(x, y, 0) = 1 - r(x, y)$$

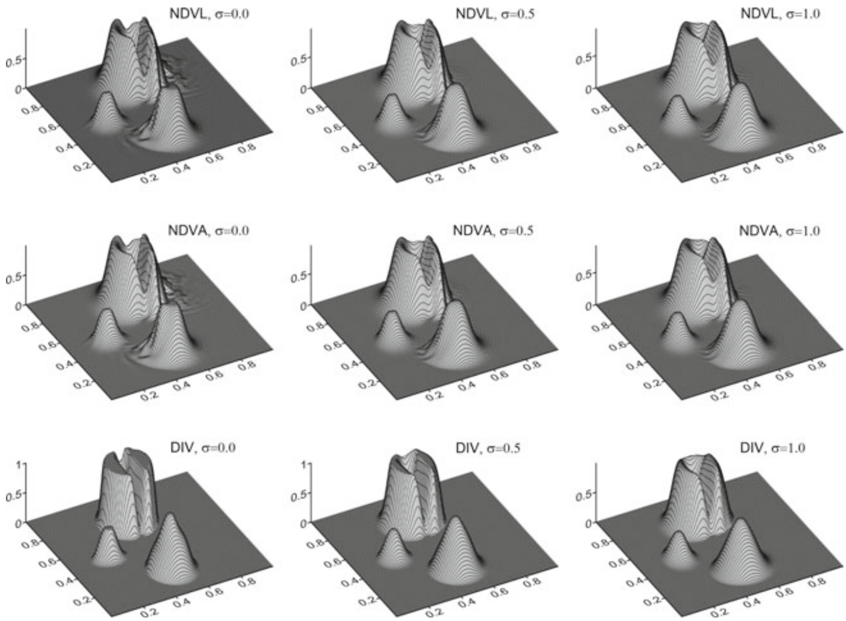


Fig. 5 Numerical results of the solid body rotation test after one revolution (5000 time steps) with NDVL, NDVA, and DIV schemes for various σ

Table 2 L^1-norm of errors and maximum values of numerical solutions for the solid body rotation test with DIV, NDVL, and NDVA schemes

		DIV		NDVL		NDVA	
	σ	L^1 error	y_{max}	L^1 error	y_{max}	L^1 error	y_{max}
Cyl	0.0	2.5900×10^{-2}	1.0000	4.4189×10^{-2}	0.9959	4.4337×10^{-2}	0.9946
	0.5	2.8022×10^{-2}	0.9912	4.0252×10^{-2}	0.9548	4.0256×10^{-2}	0.9547
	1.0	3.0557×10^{-2}	0.9681	3.9751×10^{-2}	0.9141	3.9749×10^{-2}	0.9139
Cn	0.0	2.9773×10^{-3}	0.8709	3.4419×10^{-3}	0.8144	3.4419×10^{-3}	0.8143
	0.5	2.1664×10^{-3}	0.8434	2.6798×10^{-3}	0.8094	2.6799×10^{-3}	0.8092
	1.0	2.4633×10^{-3}	0.8190	2.8654×10^{-3}	0.7905	2.8655×10^{-3}	0.7905
Hm	0.0	1.2495×10^{-3}	0.4947	2.1282×10^{-3}	0.4808	2.1283×10^{-3}	0.4804
	0.5	1.2132×10^{-3}	0.4645	1.7634×10^{-3}	0.4248	1.7636×10^{-3}	0.4247
	1.0	1.4077×10^{-3}	0.4247	1.7701×10^{-3}	0.3869	1.7703×10^{-3}	0.3868

Cyl = Slotted Cylinder; Cn = Cone; Hm = Hump.

where

$$r(x, y) = \frac{\min(\sqrt{(x - x_0)^2 + (y - y_0)^2}, r_0)}{r_0}$$

The hump is given by

$$\rho(x, y, 0) = \frac{1}{4}(1 + \cos(\pi r(x, y))$$

where $(x_0, y_0) = (0.5, 0.25)$ and $r_0 = 0.1$.

The flow velocity is calculated by $\boldsymbol{u}(x, y) = (-2\pi(y - 0.5), 2\pi(x - 0.5))$ and in result of which the counterclockwise rotation takes place about domain point (0.5, 0.5). The computational grid consists of uniform 128×128 cells. The exact solution of (6.2) reproduces by the initial state after each full revolution.

The numerical results produced with the NDVL, NDVA, and DIV schemes after one full revolution (5000 time steps) with different weights σ are presented in Fig. 5. The L^1-norm of errors and the maximum values of the numerical results are given in Table 2. As in the above advection test, we also note a good agreement between the numerical results obtained with the NDVL and NDVA schemes. Again, the solution obtained by the DIV scheme is more accurate than the solutions computed by the NDVL and NDVA schemes.

7 Conclusions

In this paper, we derive the formulas for calculating flux limiters for the FCT method for a nonconservative convection-diffusion equation. The flux limiter is computed as an approximate solution of the optimization problem that can be considered as a background of the FCT approach.

Following FCT, we consider a hybrid scheme which is a linear combination of monotone and high-order schemes. The difference between high-order flux and low-order flux is considered as an antidiffusive flux. The finding maximal flux limiters for the antidiffusive fluxes is treated as an optimization problem with a linear objective function. Constraints for the optimization problem are inequalities that are valid for the monotone scheme and applied to the hybrid scheme. This approach allows us to reduce classical two-step FCT to a one-step method for explicit difference schemes and design flux limiters with desired properties. It is also applicable to both explicit and implicit schemes and can be extended to the multidimensional case.

We have considered various approximations of convective fluxes. Numerical experiments show that the best results are obtained with a flux correction for the divergent part of the convective flux of a nonconservative convection-diffusion equation. We also note a good agreement between the numerical results for which the flux limiters are computed using exact and approximate solutions of optimization problem.

References

1. Boris, J.P., Book, D.L.: Flux-corrected transport. I. SHASTA, a fluid transport algorithm that works. J. Comput. Phys. **11**, 38–69(1973) https://doi.org/10.1016/0021-9991(73)90147-2
2. Harten, A., Hyman, J.M., Lax, P.D., Keyfitz, B.: On finite-difference approximations and entropy conditions for shocks. Comm. Pure Appl. Math. **29**, 297–322 (1976). https://doi.org/10.1002/cpa.3160290305
3. Kivva, S.: Flux-corrected transport for scalar hyperbolic conservation laws and convection-diffusion equations by using linear programming. J. Comput. Phys. **425**, 109874 (2021). https://doi.org/10.1016/j.jcp.2020.109874
4. Kivva, S.: Entropy stable flux correction for scalar hyperbolic conservation laws. J. Sci. Comput. **91**, 10 (2022). https://doi.org/10.1007/s10915-022-01792-0
5. Kuzmin, D., Möller, M.: Algebraic flux correction I. scalar conservation laws. In: Kuzmin, D., Lohner, R., Turek, S. (eds) Flux-Corrected Transport. Scientific Computation, pp. 155–206. Springer, Berlin (2006). https://doi.org/10.1007/3-540-27206-2_6
6. Leonard, B.P., Lock, A.P., Macvean, M.K.: The nirvana scheme applied to one-dimensional advection. Int. J. Numer. Methods Heat Fluid Flow **5**, 341–377 (1995). https://doi.org/10.1108/EUM0000000004120
7. Leveque, R.J.: High-resolution conservative algorithms for advection in incompressible flow. SIAM J. Numer. Anal. **33**, 627–665 (1996). https://doi.org/10.1137/0733033
8. Oran, E.S., Boris, J.P.: Numerical Simulation of Reactive Flow, 2nd edn. Cambridge University Press, Cambridge (2001). https://doi.org/10.1017/CBO9780511574474
9. Ortega, J.M., Rheinboldt, W.C.: Iterative Solution of Nonlinear Equations in Several Variables. Academic Press, New York (1970)
10. Zalesak, S.T.: Fully multidimensional flux-corrected transport algorithms for fluids. J. Comput. Phys. **31**, 335–362 (1979). https://doi.org/10.1016/0021-9991(79)90051-2
11. Zalesak, S.T.: The design of flux-corrected transport (FCT) algorithms for structured grids. In: Kuzmin, D., Loner, R., Turek, S. (eds) Flux-Corrected Transport. Scientific Computation, pp. 29-78. Springer, Berlin (2006). https://doi.org/10.1007/3-540-27206-2_2

Automated Parallelization of Software for Identifying Parameters of Intraparticle Diffusion and Adsorption in Heterogeneous Nanoporous Media

Mykhaylo Petryk, Anatoliy Doroshenko, Dmytro Mykhalyk, Pavlo Ivanenko, and Olena Yatsenko

Abstract The algorithm of the gradient procedure of identification of parameters of internal kinetics of systems of competitive diffusion for a heterogeneous catalytic nanoporous medium is presented. The results of designing and parallelizing a program implementing a Crank-Nicolson scheme using algebra-algorithmic specifications represented in a natural-linguistic form are given. The tools for automated design, synthesis, and auto-tuning of programs were applied that provided the translation of algebra-algorithmic schemes into source code in a target programming language and its tuning for the execution environment to increase the program performance. Numerical distributions of values of diffusion coefficients for intraparticle transfer along coordinate of medium thickness for various time snapshots were obtained. Based on the results of the identification, the models were checked for adequacy and numerical modeling and analysis of concentration and gradient fields of mass transfer were carried out. The experiment results of auto-tuning the software implementation demonstrated high multicore speedup on test data input.

Keywords Mathematical model · Mass transfer · Heterogeneous and nanoporous media · Automated program design · Software auto-tuning · Parallel computing

1 Introduction

Nowadays, controlling a scientific experiment and analysis of a state of complex multicomponent systems of mass transfer in heterogeneous nanoporous media is

---

A. Doroshenko · P. Ivanenko
Institute of Software Systems of National Academy of Sciences of Ukraine, Glushkov Prosp. 40, Kyiv 03187, Ukraine
e-mail: paiv@ukr.net

M. Petryk (✉) · D. Mykhalyk · O. Yatsenko
Ternopil Ivan Puluj National Technical University, Ruska Str. 56, Ternopil 46001, Ukraine
e-mail: mykhaylo_petryk@tntu.edu.ua

O. Yatsenko
e-mail: oayat@ukr.net

S. Shkarlet et al. (eds.), *Mathematical Modeling and Simulation of Systems*, Lecture Notes in Networks and Systems 667,
https://doi.org/10.1007/978-3-031-30251-0_3

closely associated with new developments in the field of system analysis and mathematical modeling of systems. Heterogeneous media consisting of thin layers of particles of forked porous structure with different physical–chemical properties are widely used in science-intensive technologies and priority sectors of industry, medicine, ecology, etc. Such layers are distributed systems of pores consisting of two main spaces: micro- and nanopores of particles and macropores and cavities between particles. Intraparticle space has a higher level of adsorptive capacity, and at the same time, has a lower velocity of diffusion intrusion in comparison with interparticle space [1–4].

In modeling concentration and gradient fields for various diffusible components, an important scientific problem is the identification of kinetic parameters of a transfer, predetermining mass transfer velocity on macro- and micro levels, and also equilibrium conditions. In [5], the Crank-Nicolson scheme was implemented for obtaining the numerical solution of a model of the distributed mass transfer system and identification of the distribution of diffusion coefficients based on the theory of optimal control of multicomponent systems state. As a result, a gradient procedure of identification of parameters of internal kinetics of a transfer was implemented and the distribution of diffusion coefficient values for intraparticle and interparticle transfer was obtained.

The purpose and novelty of this paper consist in applying algebra-algorithmic methodology and tools [6] for the automated design, generation, and optimization of a parallel program implementing the Crank-Nicolson scheme to be executed on a multicore processor. The methodology is based on Glushkov's systems of algorithmic algebras (SAA), term rewriting technique, and software auto-tuning. SAA is focused on solving the problems of formalized design of programs represented in the form of high-level specifications called schemes. The term rewriting technique is used to represent and automate transformations of program models in a high-level declarative style. The auto-tuning allows optimizing a program (tuning it to a specific computing environment) through the automated search for the best combination of program parameters' values out of a set of possible configurations, by executing each version and evaluating its performance on a specific multicore platform.

The main difference of our approach from existing automated software generation tools [7–9] consists in using high-level algebra-algorithmic schemes represented in a natural linguistic form that simplifies the comprehension of programs being designed. Recent research on software auto-tuning is outlined, for example, in papers [10–13]. Work [10] presents the tool for auto-tuning parallel programs without requiring information about applications and domain-specific knowledge and uses Bayesian optimization models to find the near-optimal parameter settings. In [11], an offline autotuning approach based on an analytic performance model is proposed. Paper [12] describes the auto-tuning approach focused on capturing the hidden structures of good configurations and using this knowledge to generate better ones. Work [13] proposes a generic approach for automatic program optimization by choosing the most suitable values of program parameters, such as the number of parallel threads. Our software tuning framework TuningGenie belongs to online autotuners and uses pragmas for describing program parameters. Its significant difference in comparison with other solutions consists in applying rewriting rules to transform input programs.

The paper has the following structure. Section 2 describes the problem of two-level transfer in heterogeneous nanoporous media. In Sect. 3, the algorithm implementing the gradient method of identification of intraparticle diffusion coefficients of the system is given. Section 4 is devoted to the automated design of the parallel program implementing the Crank-Nicolson method. In Sect. 5, the results of the numerical modeling and identification of kinetic parameters of the system and auto-tuning and execution of the parallel Crank-Nicolson program are given. The paper ends with conclusions in Sect. 6.

2 Statement of the Problem of Two-Level Transfer in a Heterogeneous System of Nanoporous Particles

The presented model is similar to the bipore model considered in [2–4, 14–18]. Mass transfer in the system of heterogeneous media consisting of small particles of nanoporous structure causes two types of mass transfer: diffusion in macropores, owing to interparticle space, and diffusion in the system of micro- and nanopores inside particles of the heterogeneous medium. For determining the contribution of each diffusion type in the system of overall mass transfer, it is necessary to know parameter values, define adsorption equilibriums, etc.

In this work, the heterogeneous nanoporous medium is considered, consisting of a large number $(n + 1)$ of thin layers, nanoporous spherical particles located perpendicular to the direction of input flow and linked via the system of conditions of n-interface interactions. This is the determining factor for heterogeneous thin nanoporous samples, especially in the case of gas diffusion before the adsorption equilibrium state taking into account the system of multi-interface interactions. Mass transfer occurs through the permeable surface of the bed in two directions: axial—in the macroporous space (z is a direction along bed height, perpendicular to the surfaces of layers) and radial—in the space of micro- and nanopores. The evolution of the system towards equilibrium is done by concentration gradients in macropores and micro- and nanopores of particles (from the surface to the center). Diffusion equilibrium in the model is described by Henry's condition [16].

The mathematical model of such transfer with the account of the above physical factors is described as a mixed boundary problem [1]: to build a solution of the system of partial differential equations, limited in the domain

$$D_n = \left\{ t > 0,\ r \in (0, R),\ z \in \bigcup_{k=1}^{n+1} (l_{k-1}, l_k);\ l_0 = 0;\ l_{n+1} \equiv l < \infty \right\}$$

and written in the matrix form:

$$\frac{\partial}{\partial t}\begin{bmatrix} c_1 \\ c_2 \\ \dots \\ c_{n+1} \end{bmatrix} = \frac{\partial}{\partial z}\left(\begin{bmatrix} D_{\text{inter}_1} & & \dots & 0 \\ & D_{\text{inter}_2} & \dots & \\ \dots & \dots & \dots & \dots \\ 0 & & \dots & D_{\text{inter}_{n+1}} \end{bmatrix}\frac{\partial}{\partial z}\begin{bmatrix} c_1 \\ c_2 \\ \dots \\ c_{n+1} \end{bmatrix}\right) -$$

$$-\left(\begin{bmatrix} \nu_1 D_{\text{intra}_1} & & \dots & 0 \\ & \nu_2 D_{\text{intra}_2} & \dots & \\ \dots & \dots & \dots & \dots \\ 0 & & \dots & \nu_{n+1} D_{\text{intra}_{n+1}} \end{bmatrix}\frac{\partial}{\partial r}\begin{bmatrix} q_1 \\ q_2 \\ \dots \\ q_{n+1} \end{bmatrix}\right)_{r=R}, \tag{1}$$

$$\frac{\partial}{\partial t}\begin{bmatrix} q_1 \\ q_2 \\ \dots \\ q_{n+1} \end{bmatrix} = \frac{1}{r^2}\frac{\partial}{\partial t}\left(\begin{bmatrix} D_{\text{intra}_1} & & \dots & 0 \\ & D_{\text{intra}_2} & \dots & \\ \dots & \dots & \dots & \dots \\ 0 & & \dots & D_{\text{intra}_{n+1}} \end{bmatrix} r^2\frac{\partial}{\partial r}\begin{bmatrix} q_1 \\ q_2 \\ \dots \\ q_{n+1} \end{bmatrix}\right) \tag{2}$$

with initial conditions:

$$c_k(t, z)_{t=0} = 0, q_k(t, z)_{t=0} = 0, \tag{3}$$

boundary conditions:

$$c_{n+1}(t, z = l) = c_{\infty_{n+1}}, \frac{\partial c_1}{\partial z}(t, z = 0) = 0, \tag{4}$$

$$\frac{\partial}{\partial r} q_k(t, r, z)_{r=0} = 0, q_k(t, r, z)_{r=R} = K_k c_k(t, z), \tag{5}$$

and the system of conditions of n-interface interactions along z coordinate:

$$\left[c_k(t, z) - c_{k+1}(t, z)\right]\Big|_{z=l_k} = 0; \left[\frac{\partial}{\partial z}c_k(t, z) - \upsilon_k\frac{\partial}{\partial z}c_{k+1}(t, z)\right]\Bigg|_{z=l_k} = 0, \tag{6}$$

where $K_k = \frac{q_{\infty_k}}{c_{\infty_k}}$; $\overline{q}_k(t, z) = \frac{1}{R^2}\int\limits_0^R q_k(t, x, z) r dr$; $\nu_k = \frac{3(1-\varepsilon_{\text{inter}_k})}{R\cdot\varepsilon_{\text{inter}_k}}$, $k = \overline{1, n+1}$.

The system of differential Eqs. (1) describes the transfer in the *interparticle space*, limited by the right parts of the system, taking into account the influence of mass transfer on external surfaces of particles and crystallites ($r = R$) for each k-th bed layer. The system of Eqs. (2) describes the transfer in micro- and nanopores of the *intraparticle space.* The connection between concentrations c_k in the *interparticle space* and concentrations q_k in the *intraparticle space* is determined by the system of right boundary conditions (5), which also defines the conditions of adsorption equilibrium on surfaces of spherical particles [16]; $\Delta l_k = l_{k+1} - l_k$ is the thickness of the k-th layer; R is particle radius. The problem is solved using the modified Crank-Nicolson finite-difference method [5].

3 The Algorithm of Implementation of the Gradient Method of Identification of Intraparticle Diffusion Coefficients of the Competitive Transfer System

The authors of the article paid attention to the issue of the adequacy of the mathematical model, namely by implementing the procedure for the model's parameter identification. Parameters that define limit mass transfer at the micro level—intraparticle diffusion coefficients are identified based on the results of NMR imaging (nuclear magnetic resonance) experimental studies, which were carried out in cooperation with the Ecole Supérieure de Physique et de Chimie Industrielles de Paris (ESPCI Paris) (within the SSHN/CampusFrance 2017, 2021 Grant). The concentration distributions of the absorbed masses of two components (benzene and hexane) were measured layer-by-layer with NMR scanning of a 10 mm nanoporous sample (10 layers of closely packed zeolite crystallites), into which the gas mixture of the adsorbate diffused. Since the system has a closed outlet, the last two layers of nanoporous crystallites in the system remained weakly saturated due to the reflection of gas flows from the boundary and contributed to a greater saturation of the middle and entrance zones of the layers (in the model, this is reflected by the impermeability condition (4)). As an identification criterion, the root-mean-square Lagrange functional has been used, which minimizes the discrepancy between the model and experimental values of adsorbed component masses distributions for each of the 10 layers. For the first 8 layers, essentially forming the main working adsorption medium, an almost full match was obtained between the model and experimental values of the adsorbed masses for each component due to the identified intraparticle diffusion coefficients. The relative error between experimental and model results is less than 1 percent. For the last two layers, the obtained results have slightly worse accuracy, due to the influence of the above-mentioned marginal effects of flow reflection, which has led to a sharp drop in concentrations to zero value. As result, the identified diffusion coefficients for these segments are also close to zero. A way out of this situation could be to scan the boundary initial segments with closer steps to avoid large concentration gradients, which will be taken into account in further studies.

The general procedure of implementation of the gradient method of identification of coefficients of intraparticle diffusion of the system for intraparticle space (D_{intra_m}, $m = \overline{1, n+1}$) is based on using a system state matrix $M_m\left(t_k, z_i, D^{\theta}_{\text{intra}_m}\right)$, which corresponds to the overall accumulation of a mass of a diffused component in layers of nanoporous particles in the interparticle and intraparticle spaces [15]. The matrix is defined by the formula:

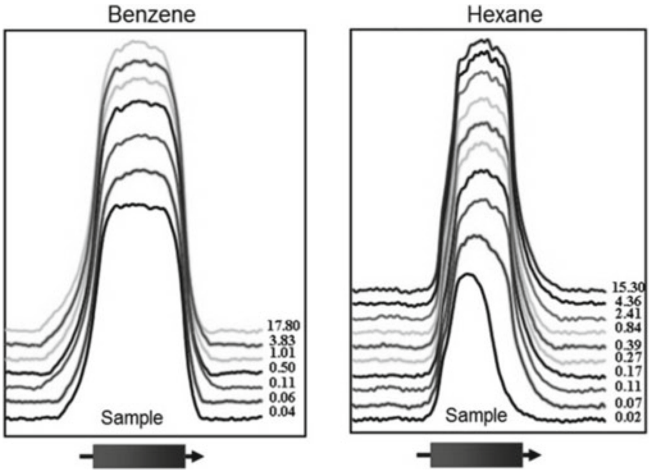


Fig. 1 Experimental data for competitive mass transfer in a heterogeneous nanoporous catalytic medium

$$M_m(t,z) = \left[U_m\left(t, z, D_{\text{intra}_m}\right) + \frac{1}{R}\int_0^R q_m\left(t, r, z, D_{\text{intra}_m}\right) r\,dr \right]_{L^2(\gamma_m),\, m=\overline{1,n+1}},$$

where $M_{\text{exp}} = \left[M_{\text{exp}_{km}}\right]_{\substack{m=\overline{1,n+1}\\ k=\overline{1,N}}}$ is the matrix of experimental data for i-th surface and k-th time layers (Fig. 1).

In the matrix $M_m\left(t_k, z_i, D^{\theta}_{\text{intra}_m}\right)$, time and space variables t and z define specific states of the competitive transfer system for heterogeneous (by direction z) catalytic medium of nanoporous particles, for which the kinetic parameters-coefficients of intraparticle diffusion D_{intra_m} $(m = \overline{1, n+1})$ for each of $n+1$ layers are identified.

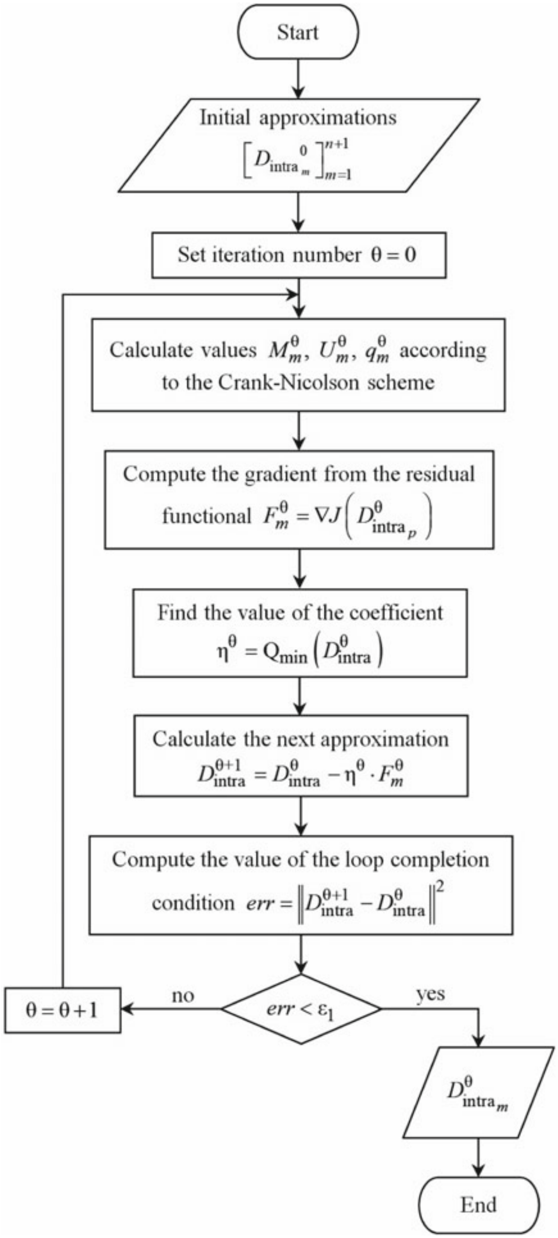
For identification of the distribution (vector) D_{intra_m}, one of the gradient methods is used. The mathematical substantiation of the use of such methods to parameterized identification of multicomponent distributed systems is given in [19]. Taking into account the specificity of the problem, the method of minimal errors is most applicable. For defining $(\theta + 1)$-th approximation of diffusion coefficient in the intraparticle space D_{intra}, the following gradient procedure of identification written in a matrix form is applied:

$$D^{\theta+1}_{\text{intra}_m} = D^{\theta}_{\text{intra}_m} - \eta^{\theta} \cdot \nabla J\left(D^{\theta}_{\text{intra}_1}, \ldots, D^{\theta}_{\text{intra}_{n+1}}\right);\ m = \overline{1, n+1},$$

where η^{θ} is the value of the coefficient for each θ-th step of the iteration.

The general scheme of the algorithm for the identification of coefficients of intraparticle diffusion D_{intra_m}, $m = \overline{1, n+1}$, is shown in Fig. 2.

Fig. 2 The overall scheme of the algorithm of identification of intraparticle diffusion coefficients



4 Automated Design of the Parallel Program Implementing the Crank-Nicolson Method

For designing the program implementing the Crank-Nicolson method, we apply the system of algorithmic algebra [6] focused on the high-level construction of algorithms represented in the form of schemes. SAA is the two-sorted algebra $GA = <\{Pr, Op\}; \Omega_{GA}>$, where Pr and Op are sets of conditions and operators; Ω_{GA} is the signature, which contains Boolean (disjunction, conjunction, negation) and operator constructs considered further.

In this paper, we use the natural linguistic form of operation representation. SAA is the basis for a language called SAA/1, which has the advantage to be human-friendly, uses a representation of algorithms close to natural language, and can be translated to target programming languages. The superpositions of SAA operations are called SAA schemes. Predicates and operators in SAA can be basic or compound. Basic elements are considered in SAA schemes as primary atomic abstractions. Identifiers of basic and compound predicates are enclosed in single quotes and identifiers of operators are written with double ones.

Compound operators are built from elementary ones by using the following operations:

- composition (sequential execution) of operators: "$operator$ 1"; "$operator$ 2";
- branching: IF '$condition$' THEN "$operator$ 1" ELSE "$operator$ 2" END IF;
- for loop: FOR ($counter$ FROM $start$ TO fin) "$operator$" END OF LOOP;
- asynchronous execution of p operators: PARALLEL($j = 1, ..., p$) ("$operator\ j$");
- control point: CP '$condition$', associated with the synchronization condition, which is false until the computation process reaches the point and true from the moment of reaching the point;
- synchronizer: WAIT '$condition$', which delays the computation until the value of the specified condition is true.

The developed Integrated toolkit for Designing and Synthesis of programs (IDS) [6] provides automated construction of algorithm schemes and generation of corresponding code in target programming languages (C, C++, Java). The user constructs an algorithm from top to bottom by selecting SAA operations from a list and adding them to an algorithm design tree. At each design step, the system provides a list of only those constructs, the substitution of which does not violate the syntactic correctness of the scheme being designed. The mapping of each SAA operation to a text in a programming language is specified as a code template in the IDS database.

Consider the process of the parallelization of one of the subroutines of the Crank-Nicolson scheme. The sequential SAA scheme of the subroutine designed using the IDS toolkit is given below. It contains a loop by the variable $k \in [1, ..., N]$, in which functions iterate_c(k) and iterate_q(k) compute the k-th layer for concentrations c_k and q_k in interparticle and intraparticle spaces, correspondingly.

```
SCHEME CRANK-NICOLSON SEQUENTIAL ====

"iterations"
==== FOR (k FROM 1 TO N)
         "iterate_c(k)";
         "iterate_q(k)"
        END OF LOOP

END OF SCHEME
```

The parallelization of the scheme consists in dividing the segment [1...*N*] into *NumThreads* sections to be processed in parallel. The SAA scheme of the parallelized algorithm is the following:

```
SCHEME CRANK-NICOLSON PARALLEL ====

"iterations"
==== PARALLEL(j = 1,..., NumThreads)
        (
           "IterateThread(j)"
        );
        WAIT 'Processing in all (NumThreads) threads is finished';

"IterateThread(j)"
==== "chunk := N / NumThreads";
        "start := (j – 1) * chunk + 1";
        "end := (j – 1) * chunk + chunk";
        IF (j = NumThreads) THEN
          "end := N"
        END IF;
        FOR (k FROM start TO end)
          "iterate_c(k )";
          "iterate_q(k)"
        END OF LOOP;
        CP 'Processing in thread (j) is finished';

END OF SCHEME
```

IDS toolkit automatically generated parallel Java program code based on the constructed SAA scheme.

The application of formal methods in IDS and term rewriting system TermWare [20] enables the automation of the manual work of programmers and a more advanced parallelization of algorithms. TermWare is a symbolic computation system that allows translating the source code of a program into a term and provides tools for its transformation with the help of rewriting rules. However, the performance of programs being designed can be further increased by using the TuningGenie framework [6]. TuningGenie is aimed at the automated generation of auto-tuning applications from a source code. The idea of an auto-tuner consists in the empirical evaluation of several versions of an input program and the selection of the best one with reduced execution time and higher result accuracy. The expert knowledge of a developer is stored in the source code of an input program in the form of special comments (pragmas). The pragmas contain information about parameter names and their value ranges. By exploiting such expert knowledge, the number of program versions to be evaluated is reduced, and therefore the auto-tuner performance is increased.

An example of one of TuningGenie pragmas (called *tunableParam*) is given below. The pragma sets the possible values for a thread count variable *NumThreads* in a range [1 . . . 8] with step 1:

```
//tunableParam name=NumThreads start=1 stop=8 step=1
int NumThreads = 1;
```

The *tunableParam* pragma applies to algorithms that use geometrical (data) parallelization: it allows finding the optimal decomposition of computation by estimating the size of a block that is executed on a single processor. It can also be applied when it is necessary to estimate the optimal number of some limited resources like the size of caches or the number of threads to be used in a program. The results of tuning the parallel program implementing the Crank-Nicolson scheme are given in Sect. 5.2.

5 Experimental Results

In this section, the results of the numerical modeling and identification of kinetic parameters of the heterogeneous system of nanoporous particles and auto-tuning and execution of the parallel Crank-Nicolson program on a multicore processor architecture are considered.

5.1 Numerical Modeling and Identification of Kinetic Parameters of the Heterogeneous System of Nanoporous Particles

The elements of the matrix of experimental data $\left[M_{\exp_{k_i}}\right]_{k=\overline{1,N}}^{i=\overline{1,M}}$ are the value of the distribution of the total absorbed mass along the coordinate z for various moments of a diffusion process. The results of identification carried out according to the considered methodology and using the above experimental data (Fig. 1) are shown in Fig. 3 and Fig. 4. Like the experimental results, they were obtained for various time snapshots for cases of a process of independent diffusion of benzene and hexane. Figure 3(a)

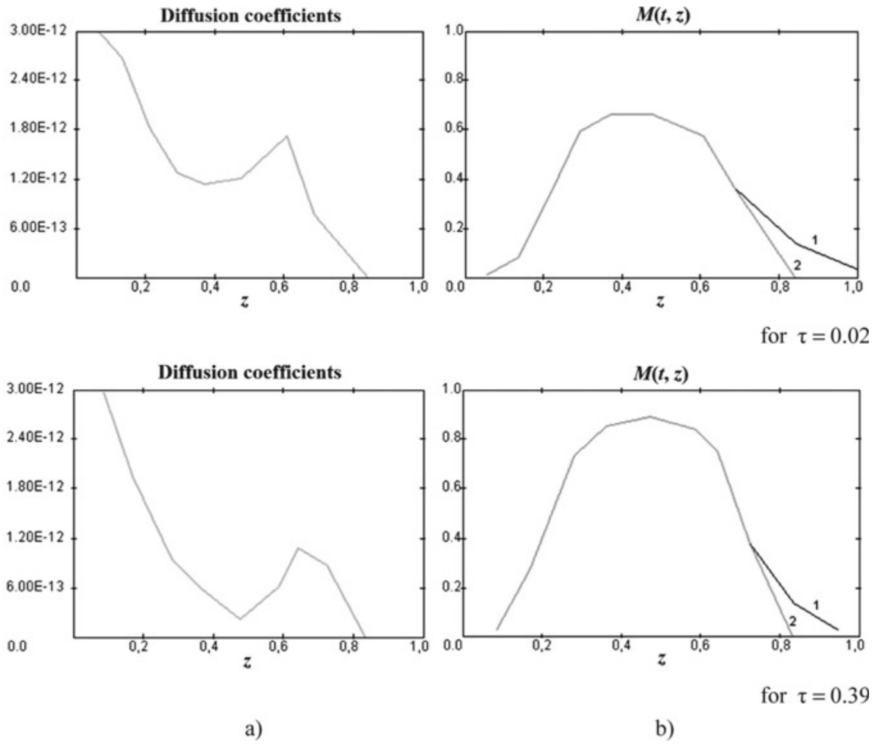


Fig. 3 Results of identification of diffusion coefficients for time $\tau = 0.02$ and $\tau = 0.39$ for benzene diffusion: a) distribution of diffusion coefficients D_{intra_m} in the intraparticle space; b) comparison of model (2) and experimental (1) curves

shows the graphical distribution of values of identified diffusion along coordinate z at benzene diffusion for time moments $\tau = 0.02$ h and $\tau = 0.39$ h.

The above graphical results of the process of identification of diffusion D_{intra_m} $(m = \overline{1, n+1})$ coefficients for the *intraparticle space* have general characteristic regularities consisting in some pseudo exponential descending of values of diffusion coefficients in the range $3 \cdot 10^{-12} \div 4 \cdot 10^{-14}$ m/s^2 (with an account of calculation errors).

A similar pattern can be observed for the hexane diffusion process as well, for which the procedure of identification of diffusion coefficients in the *intraparticle space* D_{intra_m} for time snapshots $\tau = 0.04$ h, $\tau = 0.50$ h, and $\tau = 3.83$ h was carried out. The results of the performed identification are shown in Fig. 4(a).

The results of identified distributions of diffusion coefficients in the intraparticle space along coordinate z (the main direction of the system's heterogeneity) allow modeling concentration fields and integral distributions of mass in heterogeneous catalytic nanoporous layer with reasonable accuracy. Figure 4(b) shows concentration profiles demonstrating comparative analysis of model curves (2) built as a result of the numerical solution of the problem (1)–(6) using the Crank-Nicolson scheme and

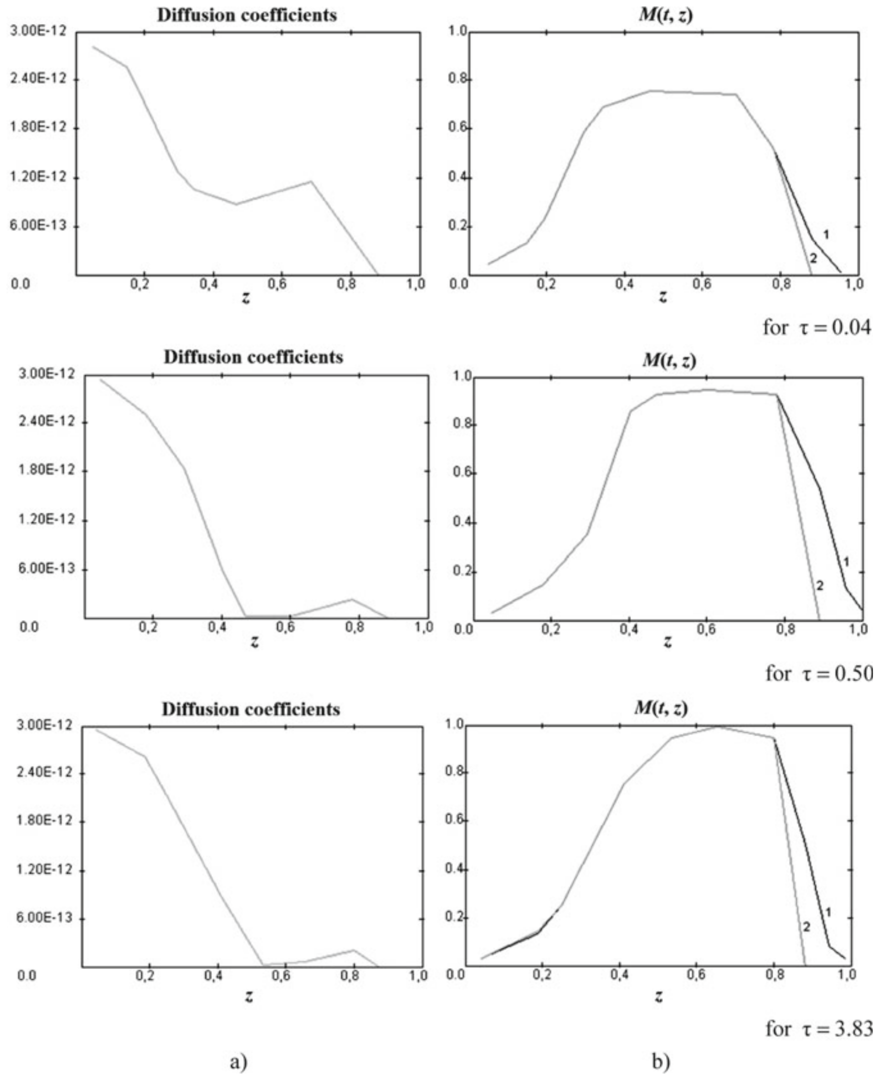


Fig. 4 Results of identification of diffusion coefficients for time $\tau = 0.04$, $\tau = 0.50$ and $\tau = 3.83$ for hexane diffusion: a) distribution of diffusion coefficients D_{intra_m} in the intraparticle space; b) comparison of model (2) and experimental (1) curves

results of identified coefficients of intraparticle diffusion with approximations of experimental curves of absorbed mass distribution in a nanoporous layer, used for the identification procedure.

As can be seen from the presented concentration distributions (Fig. 3 and 4, b), values of model and experimental profiles for all obtained graphs of integral mass $M(t, z)$ conform sufficiently owing to qualitative solutions of the reverse problem, i.e. distributions of diffusion D_{intra_m} coefficients. About 80% of length, model curves provide a complete match with experimental values. Such an approach provides a sufficiently high degree of adequacy of mathematical models and techniques of identification of parameters of researched heterogeneous distributed systems of multicomponent transfer.

5.2 Auto-tuning of the Parallel Program Implementing the Crank-Nicolson Scheme

TuningGenie framework was applied to try out various combinations of JVM performance-related compiler flags (i.e. -XX:-UseBiasedLocking/-XX:BiasedLockingStartupDelay) along with a different number of threads to find the optimal configuration for the parallel Crank-Nicolson program when executed on a multicore processor. Compiler flags didn't show a significant impact on computational performance, so the presented results will focus only on the *NumThreads* parameter.

The characteristics of the test environment were the following:

- CPU 2,7 GHz Quad-Core Intel Core i7-6820HQ (L1 Cache: 32k/32k ×4; L2/L3 Cache: 256k ×4, 8 MB);
- RAM 16 GB 2133 MHz LPDDR3;
- OpenJDK build 11.0.2+9;
- MacOS v11.6.

Intuition hints that the optimal number of spawned threads should be 8—it's a quad-core CPU with hyper-threading technology. This configuration is marked with a triangle on the chart (Fig. 5) and demonstrated a reasonable multicore speedup of 5.16 on test data input. However, a further increase in the number of threads resulted in an additional performance gain of about 140%. The best-performing programs' configuration with 62 threads (marked with a dot) achieved quite a decent multicore speedup of 6.5. This can be explained by the increased effectiveness of computer cache utilization. Fine-grained data decomposition is more likely to completely fit in L1–L3 caches and saves time on lookups to "slow" RAM. This effect is balanced by additional time costs for thread contention. Still, the overall effect appeared to be positive.

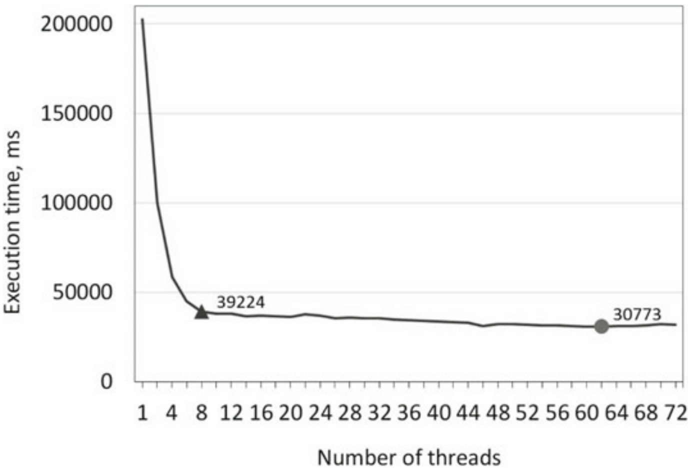


Fig. 5 The dependency of the execution time on the number of threads for the parallel Crank-Nicolson program

6 Conclusion

The algorithm of gradient procedure of identification of parameters of internal kinetics of systems of competitive diffusion for a heterogeneous catalytic nanoporous medium was presented. The software implementation of the Crank-Nicolson scheme was designed and parallelized using high-level algebra-algorithmic program specifications represented in a natural linguistic form. The developed software tools for constructing, synthesis, and auto-tuning of programs were applied for automated translation of the specifications into source code in a programming language and tuning of the code for the target computing environment to increase the performance.

Numerical distributions of values of diffusion coefficients for intraparticle transfer along coordinate of medium thickness for various time snapshots were obtained. Based on the results of the identification, the models were checked for adequacy and numerical modeling and analysis of concentration and gradient fields of mass transfer were carried out. The experiment results of auto-tuning the software implementation demonstrated high multicore speedup on test data input.

References

1. Kärger, J., Ruthven, D.M.: Diffusion and adsorption in porous solids. In: Schüth, F., Sing, K.S.W., Weitkamp, J. (eds.) Handbook of Porous Solids, pp. 2089–2173. Wiley-VCH, Weinheim (2002)
2. Kärger, J., Ruthven, D.M.: Diffusion in Zeolites and Other Microporous Solids. Wiley, New York (1992)

3. Chen, N.Y., Degnan, T.F., Smith, M.C.: Molecular Transport and Reaction in Zeolites: Design and Application of Shape Selective Catalysis. Wiley, New York (1994)
4. Ruthven, D.: Principles of Adsorption and Adsorption Processes. Wiley, New York (1984)
5. Petryk, M.R., Mykhalyk, D.M., Hoianiuk, I.V.: High-performance methods of identification of kinetic parameters for monodiffusion adsorption mass transfer. Bull. Natl. Univ. Water Environ. Eng. **4**(92), 91–104 (2020). (in Ukrainian)
6. Doroshenko, A., Yatsenko, O.: Formal and Adaptive Methods for Automation of Parallel Programs Construction: Emerging Research and Opportunities. IGI Global, Hershey (2021)
7. Jacindha, S., Abishek, G., Vasuki, P.: Program synthesis—a survey. In: Lecture Notes in Electrical Engineering, vol. 834, pp. 409–421 (2022)
8. Gulwani, S., Polozov, O., Singh, R.: Program synthesis. Found. Trends Program. Lang. **4**(1–2), 1–119 (2017)
9. David, C., Kroening, D.: Program synthesis: challenges and opportunities. Phil. Trans. R. Soc. A **375**(2104), 20150403 (2017)
10. Roy, R.B., Patel, T., Gadepally, V., Tiwari, D.: Bliss: auto-tuning complex applications using a pool of diverse lightweight learning models. In: 42nd ACM SIGPLAN International Conference on Programming Language Design and Implementation (PLDI 2021), pp. 1280–1295. ACM, New York (2021)
11. Seiferth, J., Korch, M., Rauber, T.: Offsite autotuning approach: performance model driven autotuning applied to parallel explicit ODE methods. In: High Performance Computing: 35th International Conference (ISC High Performance 2020), pp. 370–390. Springer, Heidelberg (2022)
12. Bao, L., Liu, X., Wang, F., Fang, B.: ACTGAN: automatic configuration tuning for software systems with generative adversarial networks. In: 34th IEEE/ACM International Conference on Automated Software Engineering (ASE 2019), New York, pp. 465–476. IEEE (2019)
13. Rasch, A., Haidl, M., Gorlatch, S.: ATF: a generic auto-tuning framework. In: 2017 IEEE 19th International Conference on High Performance Computing and Communications, Piscataway, NJ, pp. 64–71. IEEE Computer Society (2017)
14. N'Gokoli-Kekele, P., Springuel-Huet, M.-A., Fraissard, J.: An analytical study of molecular transport in a zeolite particle bed. Adsorption **8**(3), 35–44 (2002)
15. Kärger, J., Grinberg, F., Heitjans, P.: Diffusion Fundamentals. Leipziger Universitätsverlag, Leipzig (2005)
16. Cussler, E.L.: Diffusion: Mass Transfer in Fluid Systems. Cambridge University Press, Cambridge (2008)
17. Mehrer, H.: Diffusion in Solids. Springer, Berlin (2007)
18. Magalhães, F.D., Laurence, R.L., Conner, W.C., Springuel-Huet, M.-A., Nosov, F., Fraissard, J.: Study of molecular transport in beds of zeolite particles: semi-quantitative modeling of ^{129}Xe NMR experiments. J. Phys. Chem. **101**, 2277–2284 (1997)
19. Sergienko, I.V., Deineka, V.S.: System analysis of multicomponent distributed systems. Naukova dumka, Kyiv (2009). (in Russian)
20. Termware2. https://github.com/rssh/termware2. Accessed 09 Nov 2022

Adaptive Variants of the Extrapolation from the Past Method and the Operator Extrapolation Method

Serhii Denysov and Vladimir Semenov

Abstract In the article we study two novel algorithms for solving variational inequalities. We consider finite dimensional vector space and use Bregman projection instead of Euclidean one. The first algorithm is obtained by extending widely used two-stage Bregman method with a cost-effective step size adjustment procedure, which does not use operator's Lipschitz constant value. The second method, which we refer as adaptive operator extrapolation algorithm, is a result of Bregman divergence usage instead of Euclidean metric in the forward-reflected-backward algorithm. This method requires only a single Bregman projection onto a feasible set on each iterative step. For adaptive operator extrapolation method we also provide low-cost step adjustment rule, which does not use operator's Lipschitz constant and needs no extra operator calculations. We formulate and prove theorems about convergence of the methods in case of variational inequality with a pseudo-monotone and Lipschitz-continuous operator. We also use gap function to measure convergence speed, and prove corresponding theorems about complexity bounds (for a case of monotone operator and bounded feasible set). Behavior of the proposed algorithms also demonstrated with numerical experiments on a zero-sum game problem.

Keywords Variational inequality · Monotonicity · Pseudo-monotonicity · Bregman divergence · Kullback–Leibler divergence · convergence · Extrapolation from the past · Operator extrapolation

1 Introduction

Variational inequality (VI) is a powerful mathematical abstraction, which allows to formulate and study different problems of operations research and mathematical physics [1–5]. VIs are also well suited for solving saddle point problems, so many non-smooth convex minimization problems can be solved efficiently with cor-

S. Denysov · V. Semenov (✉)
Taras Shevchenko National University of Kyiv, Kyiv 01601, Ukraine
e-mail: semenov.volodya@gmail.com

S. Shkarlet et al. (eds.), *Mathematical Modeling and Simulation of Systems*, Lecture Notes in Networks and Systems 667,
https://doi.org/10.1007/978-3-031-30251-0_4

responding algorithms, as such problems are easily reformulated as saddle point problems [6].

Last years variational inequalities are increasingly used in machine learning area, mostly because of GANs learning problem is also formulated as saddle point problem [7, 8].

One of the first, widely known, studied and modified in numerous publications method for solving VIs is Korpelevich extragradient algorithm [9]. It's modifications with one projection onto a feasible set have been proposed in [10–12].

The noteworthy modern version of the extragradient method (often called mirror proximal method) was proposed by Nemirovski [6]. Roughly speaking this method can be derived from the Korpelevich extragradient algorithm by replacing Euclidean distance with Bregman divergence for projection on feasible set. For some sets, this approach can give substantial benefits, as the structure of the problem's feasible set can be used to simplify calculations. For example, using Kullback–Leibler divergence in case of a simplex as a feasible set, we obtain an explicitly computed projection operator (Kullback–Leibler divergence is a Bregman divergence constructed from negative entropy). Adaptive version of Nemirovski method were studied in [13, 14].

Another branch of research for VI related algorithms with application to equilibrium programming was started by work of L.D. Popov in 1980s, where he proposed a modification of Arrow–Hurwicz algorithm, which allows to find saddle point of convex-concave function [15]. Nowadays Popov's algorithm for monotone variational inequalities often appears in machine learning related papers and is called "Extrapolation from the Past" [7].

Later this algorithm was adapted to general Ky Fan inequalities, which resulted in two-stage proximal algorithm [16]. This algorithm also uses Bregman divergence instead of the Euclidean distance, and it was thoroughly studied in [17–20, 28]. Based on these ideas, so-called "forward-reflected-backward algorithm" [21] and related methods [22, 23] were developed and studied.

This work continues research of articles [18, 24] and studies two novel algorithms with Bregman projection for solving VIs in finite-dimensional vector space with norm. The first algorithm is a modification of two-stage Bregman method [18, 20] with an adaptive step size adjustment [24]. The adjustment procedure allows to use the algorithm without a priory knowledge of operator's Lipschitz constant. The second method is based on recent forward-reflected-backward algorithm [21], where Bregman projection is used instead of Euclidean one. This method can be called "adaptive operator extrapolation algorithm", and it also includes adaptive step size recalculation rule, which does not use Lipschitz constant and does not imply extra operator calculations. Big advantage of the second algorithm is only single projection and operator calculation on the iterative step.

In the case of Lipschitz-continuous pseudo-monotone operators we prove convergence theorems for both mentioned algorithms. We also prove theorems about $O\left(\frac{1}{\varepsilon}\right)$-complexity estimations in terms of the gap function. The latter theorems hold for the case of a monotone operator and bounded feasible set. We also numerically test Bregman and Euclidean versions of the algorithms on zero-sum matix games of different dimensions, which are formulated as saddle point problems.

Our paper is organized as follows. First, Sect. 2 provides important facts about VIs and Bregman divergence. Then, Sect. 3 contains the main results, including detailed description of the algorithms and main theoretical results. Finally, Sect. 4 demonstrates results of numerical experiments.

2 Preliminaries

We consider finite-dimensional real vector space E with it's dual space E^*. We denote the value of a linear function $a \in E^*$ at $b \in E$ by (a, b). Let $\|\cdot\|$ denote some norm on E (not necessary Euclidean) and $\|\cdot\|_*$ denote the norm on E^*, which is dual to $\|\cdot\|$.

Let function $\omega : E \to (-\infty, +\infty]$ be 1-strongly convex and uniformly continuous differentiable on $\operatorname{int}\operatorname{dom}\omega$. Corresponding Bregman divergence for ω is defined by the next formula [25]

$$W(a, b) = \omega(a) - \omega(b) - (\nabla\omega(b), a - b)\,, \quad \forall a \in \operatorname{dom}\omega, \quad \forall b \in \operatorname{int}\operatorname{dom}\omega.$$

The book [25] also gives practically important examples of a Bregman divergence. For the function

$$\omega(\cdot) = \frac{1}{2}\|\cdot\|_2^2$$

we have

$$W(a, b) = \frac{1}{2}\|a - b\|_2^2.$$

And for the negative entropy function (it is 1-strongly convex with respect to the ℓ_1-norm on probability simplex $\Delta^m = \left\{x \in R^m : \sum_{i=1}^m x_i = 1,\ x_i \geq 0\right\}$)

$$\omega(x) = \sum_{i=1}^m x_i \ln x_i, \quad x \in R_+^m = \left\{x \in R^m : x_i \geq 0\right\},$$

we get Kullback–Leibler divergence (KL-divergence)

$$W(x, y) = \sum_{i=1}^m x_i \ln \frac{x_i}{y_i} - \sum_{i=1}^m (x_i - y_i)\,, \quad x \in R_+^m, \quad y \in R_{++}^m = \operatorname{int} R_+^m.$$

The next useful 3-point identity holds [25]:

$$W(a, c) - W(a, b) - W(b, c) = (\nabla\omega(b) - \nabla\omega(c), a - b)\,.$$

Let $K \subseteq \operatorname{dom}\omega$ be a nonempty, closed, and convex set, $K \cap \operatorname{int}\operatorname{dom}\omega \neq \emptyset$. Let's consider strongly convex minimization problem

$$P_x^K(a) = \mathrm{argmin}_{y \in K} \left(-(a, y - x) + W(y, x) \right), \quad a \in E^*, \quad x \in \mathrm{int}\,\mathrm{dom}\,\omega. \quad (1)$$

The problem (1) has unique solution $z \in K$ [25], and

$$-(a, y - z) + W(y, x) - W(y, z) - W(z, x) \geq 0 \quad y \in K.$$

If $\omega(\cdot) = \frac{1}{2}\|\cdot\|_2^2$, the point $P_x^K(a)$ coincides with the Euclidean projection $P_K(x + a)$. For simplex Δ^m and KL-divergence we have [25]

$$P_x^{\Delta^m}(a) = \left(\frac{x_1 e^{a_1}}{\sum_{i=1}^m x_i e^{a_i}}, \frac{x_2 e^{a_2}}{\sum_{i=1}^m x_i e^{a_i}}, \ldots, \frac{x_m e^{a_m}}{\sum_{i=1}^m x_i e^{a_i}} \right), \quad a \in R^m, \quad x \in \mathrm{ri}\,\Delta^m.$$

Let C be a nonempty subset of space E, $V : E \to E^*$ be an operator, that acts from E to E^*. Consider the variational inequality problem:

$$\text{find } x \in C \text{ such that } (Vx, y - x) \geq 0 \quad \forall y \in C. \quad (2)$$

The set of solutions of the problem (2) is denoted S.

Throughout the paper we make the following standard assumptions:

- the set $C \subseteq E$ is convex and closed;
- operator $V : E \to E^*$ is pseudo-monotone and Lipschitz-continuous with a constant $L > 0$ on C;
- the set S is nonempty.

Recall, that operator V on the set C is called pseudo-monotone if for all $x, y \in C$ from $(Vx, y - x) \geq 0$ follows $(Vy, y - x) \geq 0$. Operator V is monotone on C if $(Vx - Vy, x - y) \geq 0$ for all $x, y \in C$.

In what follows, we will assume that the following condition is satisfied:

$$C \subseteq \mathrm{int}\,\mathrm{dom}\,\omega.$$

The quality of approximate solution of variational inequality (2) will be measured using the non-negative gap function (duality gap) [6]

$$\mathrm{gap}(x) = \sup_{y \in C} (Vy, x - y). \quad (3)$$

Obviously, for definition (3) to be correct, feasible set should be bounded. The next lemma holds.

Lemma 1 **([6]).** *Let operator V be monotone. If $x \in C$ is a solution of (2), then*

$$\mathrm{gap}(x) = 0.$$

If for some $x \in C$ we have $\mathrm{gap}(x) = 0$, *then x is a solution of (2).*

3 Methods

Let's consider the following algorithms for approximate solving the variational inequality problem (2).

Algorithm 1. Extrapolation from the Past with adaptive step size.

Initialization. Select $y_0 \in E$, $x_0 \in \operatorname{int} \operatorname{dom} \omega$, $\tau \in \left(0, \frac{1}{3}\right)$ and a real number $\lambda_1 > 0$. Set $n \leftarrow 1$.

1) Compute
$$y_n = P_{x_n}^C \left(-\lambda_n V y_{n-1}\right).$$

2) Compute
$$x_{n+1} = P_{x_n}^C \left(-\lambda_n V y_n\right).$$

3) If $y_n = x_n = x_{n+1}$, then STOP and $x_{n+1} \in S$, otherwise compute
$$\lambda_{n+1} = \begin{cases} \min\left\{\lambda_n, \tau \frac{\|y_n - y_{n-1}\|}{\|V y_n - V y_{n-1}\|_*}\right\}, & \text{if } V y_n \neq V y_{n-1}, \\ \lambda_n, & \text{otherwise.} \end{cases}$$

Set $n \leftarrow n + 1$ and go to 1.

Remark 1. Algorithm 1 is an adaptive modification of two-stage non-Euclidean proximal algorithm [17, 20].

Algorithm 2. Operator extrapolation with adaptive step size.

Initialization. Select $x_0 \in E$, $x_1 \in \operatorname{int} \operatorname{dom} \omega$, $\tau \in \left(0, \frac{1}{2}\right)$ and a real numbers λ_0, $\lambda_1 > 0$. Set $n \leftarrow 1$.

1) Compute
$$x_{n+1} = P_{x_n}^C \left(-\lambda_n V x_n - \lambda_{n-1} \left(V x_n - V x_{n-1}\right)\right).$$

2) If $x_{n-1} = x_n = x_{n+1}$, then STOP and $x_{n+1} \in S$, otherwise compute
$$\lambda_{n+1} = \begin{cases} \min\left\{\lambda_n, \tau \frac{\|x_{n+1} - x_n\|}{\|V x_{n+1} - V x_n\|_*}\right\}, & \text{if } V x_{n+1} \neq V x_n, \\ \lambda_n, & \text{otherwise.} \end{cases}$$

Set $n \leftarrow n + 1$ and go to 1.

Remark 2. Algorithm 2 is an adaptive modification of forward-reflected-backward algorithm [21], where Bregman divergence based projection is used instead of the Euclidean one. Only a single operator value calculation and Bregman projection needed at the iteration step, which considerably reduces calculation for problems with complex feasible sets or operators.

Now let us turn to the analysis of the algorithms. For Algorithm 1 we have the next convergence results.

Lemma 2 **([24]).** *For the sequences (x_n) and (y_n), generated by Algorithm 1, the following inequality holds*

$$W(z, x_{n+1}) \le W(z, x_n) - \left(1 - 2\tau \frac{\lambda_n}{\lambda_{n+1}}\right) W(x_{n+1}, y_n) -$$

$$- \left(1 - \tau \frac{\lambda_n}{\lambda_{n+1}}\right) W(y_n, x_n) + \tau \frac{\lambda_n}{\lambda_{n+1}} W(x_n, y_{n-1}),$$

where $z \in S$.

Theorem 1 **([24]).** *Let operator $V : E \to E^*$ is pseudo-monotone and Lipschitz-continuous with constant $L > 0$, $C \subseteq E$ be a nonempty convex and closed set. Let $S \neq \emptyset$. Then the sequences (x_n) and (y_n), generated by Extrapolation from the Past Algorithm with adaptive step size (Algorithm 1), converge to some $z \in S$.*

For Algorithm 2 we have the next results.

Lemma 3. *For the sequence (x_n), generated by Algorithm 2, the following inequality holds*

$$W(z, x_{n+1}) + \lambda_n (V x_n - V x_{n+1}, x_{n+1} - z) + \tau \frac{\lambda_n}{\lambda_{n+1}} W(x_{n+1}, x_n) \le$$

$$\le W(z, x_n) + \lambda_{n-1} (V x_{n-1} - V x_n, x_n - z) +$$

$$+\tau \frac{\lambda_{n-1}}{\lambda_n} W(x_n, x_{n-1}) - \left(1 - \tau \frac{\lambda_{n-1}}{\lambda_n} - \tau \frac{\lambda_n}{\lambda_{n+1}}\right) W(x_{n+1}, x_n),$$

where $z \in S$.

Theorem 2. *Let $C \subseteq E$ be a nonempty convex and closed set and operator $V : E \to E^*$ is pseudo-monotone and Lipschitz-continuous with constant $L > 0$. Let $S \neq \emptyset$. Then sequence generated by Algorithm 1 (x_n) converge to $z \in S$.*

Remark 3. Results like Theorems 1 and 2 are valid for versions of Algorithms 1 and 2 with such nonmonotonic λ updating rules:

$$\lambda_{n+1} = \begin{cases} \min\left\{\lambda_n + \eta_n, \tau \frac{\|y_n - y_{n-1}\|}{\|V y_n - V y_{n-1}\|_*}\right\}, & \text{if } V y_n \neq V y_{n-1}, \\ \lambda_n + \eta_n, & \text{otherwise;} \end{cases}$$

$$\lambda_{n+1} = \begin{cases} \min\left\{\lambda_n + \eta_n, \tau \frac{\|x_{n+1} - x_n\|}{\|V x_{n+1} - V x_n\|_*}\right\}, & \text{if } V x_{n+1} \neq V x_n, \\ \lambda_n + \eta_n, & \text{otherwise;} \end{cases}$$

where (η_n) is a summable nonnegative sequence.

Let operator V be monotone and Lipschitz-continuous with constant $L > 0$. For algorithms with a stationary (non-adaptive, $\lambda_n = \lambda$) choice of the step size, such complexity estimates are obtained.

Theorem 3. *Let (x_n), (y_n) are sequences, generated by Algorithm 1 with $\lambda = \frac{1}{3L}$. Then for the means sequence $z_N = \frac{1}{N}\sum_{n=1}^{N} y_n$ the next inequality holds:*

$$\operatorname{gap}(z_N) \le L\frac{3\sup_{y\in C} W(y, x_1) + W(x_1, y_0)}{N}.$$

Theorem 4. *Let (x_n) be a sequence, generated by Algorithm 2 with $\lambda = \frac{1}{2L}$. Then for the means sequence $z_{N+1} = \frac{1}{N}\sum_{n=1}^{N} x_{n+1}$ the next inequality holds*

$$\operatorname{gap}(z_{N+1}) \le \frac{2L\sup_{y\in C} W(y, x_1)}{N}.$$

Remark 4. The feasible set C in Theorems 3 and 4 is naturally bounded.

Therefore, Algorithms 1 and 2 with a stationary step size need to do $O\left(\frac{LD}{\varepsilon}\right)$ iterations to obtain a feasible point $z \in C$ with $\operatorname{gap}(z) \le \varepsilon$, $\varepsilon > 0$, where

$$D = \sup_{a,b\in C} W(a, b) < +\infty.$$

4 Numerical Experiments

For the numerical experiments, algorithms were implemented in Python 3.9 using NumPy 1.21. Tests were run on a 64-bit Intel Core i7-1065G7 (1.3 - 3.9GHz) PC with 16GB RAM (Ubuntu Linux).

We consider a bilinear saddle point problem

$$\min_{x\in X}\max_{y\in Y}(Px, y),$$

where P is an $m \times n$ real matrix, $X = \Delta^n = \left\{x \in R^n : \sum_{i=1}^{n} x_i = 1,\ x_i \ge 0\right\}$ and $Y = \Delta^m$. The problem is reformulated as variational inequality as below:

$$Vz = V(x, y) = \begin{pmatrix} P^* y \\ -Px \end{pmatrix}, \quad C = X \times Y.$$

Table 1 Time to reach gap 0.01, seconds

Matrix size	Algorithm 1 - E	Algorithm 1 - KL	Algorithm 2 - E	Algorithm 2 - KL
150×100	2.03	1.11	1.13	0.74
750×500	5.47	3.09	3.01	1.53
1500×1000	22.01	13.35	13.53	8.07

We measure performance with a function $\text{gap}(z) = \sup_{v \in C}(Vv, z - v)$, which is called duality gap and can be simply computed as $\max_i (Px)_i - \min_j (P^* y)_j$ due to simplex constraints.

We compare adaptive variants of studied algorithms, using Euclidean metrics (Algorithm 1 - E, Algorithm 2 - E) and Kullback-Leibler divergence (Algorithm 1 - KL, Algorithm 2 - KL) for the projection. The gap is calculated for averaged approximations on each iteration.

For all tests starting point is $z_0 = \left(\frac{1}{n}, \ldots, \frac{1}{n}, \frac{1}{m}, \ldots, \frac{1}{m}\right)$, which belongs to the feasible set by construction. As for adaptivity parameter τ, we take 0.3 for both flavors of Algorithm 1, and 0.45 for Algorithm 2, which are close to the maximal values from convergence theorems. For the initial step sizes λ, we take 0.05 for Euclidean versions, and 0.5 for Kullback–Leibler ones (Table 1).

As expected, Algorithm 2 behaves better with relation to calculation time, as it needs twice less projections on each iteration—and for the matrix game problem, we need to project on simplices of corresponding dimensions, which require noticeable amount of computational time. Also both KL versions looks to outperform their Euclidean counterpart—but it's only because of gap behavior on averaged approximation. If we consider gap for x_n on the last iteration, it will be less for Euclidean version of algorithms. For example, after 2000 iterations for 750×500 matrix, the minimal gap is reached with Euclidean flavor of Algorithm 2 (approx. 0.001), while KL version has gap on averaged value 0.004, and gap on the last iteration 0.16.

Behavior on figures is also mostly expected—all algorithms converge with the same asymptotic speed. Somewhat surprisingly, Algorithm 2 slightly outperforms Algorithm 1 in terms of iteration count, which is probably related to greater adaptivity parameter τ, which leads to a bit bigger step sizes.

We should note, that relative performance depends on the problem matrix and initial values selection. In our experiments we have had scenarios, when Algorithm 2 performed the same or even slightly better in terms of iterations on the same kind of problem (zero sum matrix game). But in most tests the behavior was close to the shown on Figs. 1 and 2.

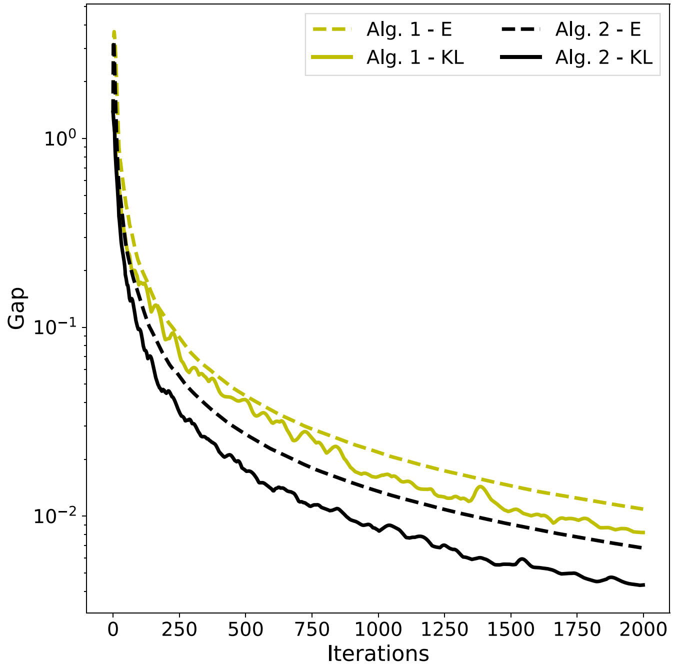


Fig. 1 Gap from averaged approximation, 750 × 500 matrix

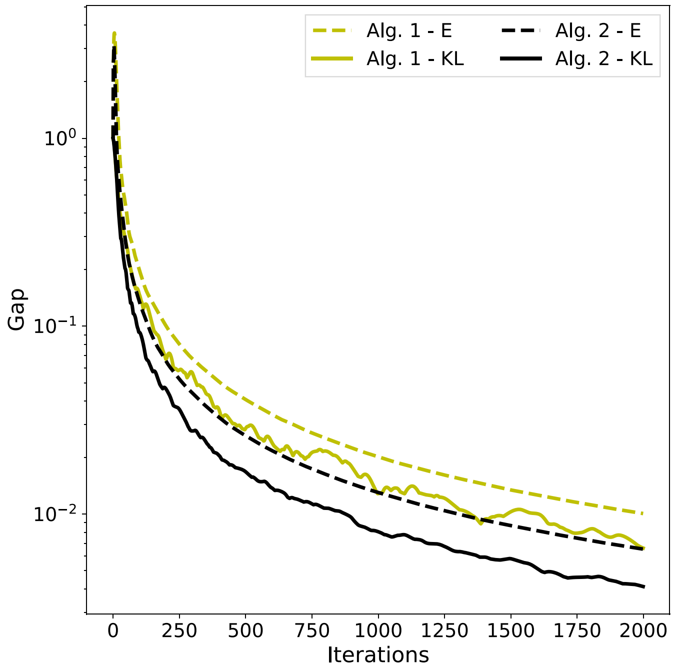


Fig. 2 Gap from averaged approximation, 1500 × 1000 matrix

5 Conclusions

The paper studies two new iterative methods with Bregman projection for the approximate solution of VIs. The first method was obtained by introducing a new adaptive step size regulation into the two-stage Bregman method [17]. The second method is obtained by replacing the Euclidean metric with Bregman divergence in the method from [21] (forward-reflection-backward algorithm). The optimistic gradient descent algorithm, which is popular in the field of machine learning, is a special example of this method [7, 8, 26, 27]. Note that methods of this type have long become workhorses in the calculation of equilibria in problems of economic content [31].

For variational inequalities with pseudo-monotone, Lipschitz-continuous operators, convergence theorems are presented. The $O(\frac{1}{\varepsilon})$-complexity bounds in terms of the gap function are proved for algorithms with stationary (non-adaptive) step size and for variational inequalities with Lipschitz-continuous monotone operators and convex closed bounded feasible set.

In the theory of algorithms for saddle point problems and variational inequalities a fundamental problem arose about the development of an iterative algorithm with single computation of the operator's value and projection onto the feasible set at the iteration step and $O(\frac{1}{\varepsilon})$-complexity bound. Algorithm 2 (operator extrapolation) answers this question.

The results of numerical experiments are also presented to confirm the effectiveness of the proposed algorithms.

In upcoming works, we plan to consider a multi-block version of the adaptive method of operator extrapolation. Another interesting problem is the analysis of the convergence of distributed and/or stochastic versions of algorithms of versions with variance reduction [28–30].

The Ministry of Education and Science of Ukraine is funding this work (research project "Computational algorithms and optimization for artificial intelligence, medicine and defense", 0122U002026, 2022–2024).

References

1. Kinderlehrer, D., Stampacchia, G.: An Introduction to Variational Inequalities and Their Applications. Society for Industrial and Applied Mathematics, Philadelphia (2000). https://doi.org/10.1137/1.9780898719451
2. Royset, J.O., Wets, R.J.-B.: An Optimization Primer. SSORFE, Springer, Cham (2021). https://doi.org/10.1007/978-3-030-76275-9
3. Bauschke, H.H., Combettes, P.L.: Convex Analysis and Monotone Operator Theory in Hilbert Spaces. CBM, Springer, Cham (2017). https://doi.org/10.1007/978-3-319-48311-5
4. Alber, Y., Ryazantseva, I.: Nonlinear ill-Posed Problems of Monotone Type. Springer, Dordrecht (2006). https://doi.org/10.1007/1-4020-4396-1
5. Konnov, I.V.: Combined Relaxation Methods for Variational Inequalities. Springer-Verlag, Heidelberg (2001). https://doi.org/10.1007/978-3-642-56886-2

6. Nemirovski, A.: Prox-method with rate of convergence O(1/T) for variational inequalities with Lipschitz continuous monotone operators and smooth convex-concave saddle point problems. SIAM J. Optim. **15**, 229–251 (2014). https://doi.org/10.1137/S1052623403425629
7. Gidel, G., Berard, H., Vincent, P., Lacoste-Julien, S.: A Variational Inequality Perspective on Generative Adversarial Networks. arXiv:1802.10551 (2018)
8. Goodfellow, I.: NIPS 2016 tutorial: Generative adversarial networks. arXiv:1701.00160 (2016)
9. Korpelevich, G.M.: An extragradient method for finding saddle points and for other problems. Matecon. **12**(4), 747–756 (1976)
10. Tseng, P.: A modified forward-backward splitting method for maximal monotone mappings. SIAM J. Control. Optim. **38**, 431–446 (2000). https://doi.org/10.1137/S0363012998338806
11. Censor, Y., Gibali, A., Reich, S.: The subgradient extragradient method for solving variational inequalities in Hilbert space. J. Optim. Theory Appl. **148**, 318–335 (2011). https://doi.org/10.1007/s10957-010-9757-3
12. Yang, J., Cholamjiak, P., Sunthrayuth, P.: Modified Tseng's splitting algorithms for the sum of two monotone operators in Banach spaces. AIMS Math. **6**(5), 4873–4900 (2021). https://doi.org/10.3934/math.2021286
13. Bach, F., Levy, K.Y.: A Universal Algorithm for Variational Inequalities Adaptive to Smoothness and Noise. arXiv:1902.01637 (2019)
14. Antonakopoulos, K., Belmega V., Mertikopoulos, P.: An adaptive mirror-prox method for variational inequalities with singular operators. In: Advances in Neural Information Processing Systems, vol. 32 (NeurIPS), pp. 8455-8465. Curran Associates, Inc. (2019)
15. Popov, L.D.: A modification of the Arrow-Hurwicz method for search of saddle points. Math. Notes Acad. Sci. Ussr. **28**, 845–848 (1980). https://doi.org/10.1007/BF01141092
16. Hieu, D.V.: Convergence analysis of a new algorithm for strongly pseudomontone equilibrium problems. Numer. Algor. **77**, 983–1001 (2018). https://doi.org/10.1007/s11075-017-0350-9
17. Gibali, A., Thong, D.V.: A new low-cost double projection method for solving variational inequalities. Optim. Eng. **21**, 1613–1634 (2020). https://doi.org/10.1007/s11081-020-09490-2
18. Semenov, V.V.: Modified Extragradient Method with Bregman Divergence for Variational Inequalities. J. Autom. Inf. Sci. **50**(8), 26–37 (2018). https://doi.org/10.1615/JAutomatInfScien.v50.i8.30
19. Gibali, A.: A new Bregman projection method for solving variational inequalities in Hilbert spaces. Pure Appl. Funct. Anal. **3**(3), 403–415 (2018)
20. Chabak, L., Semenov, V., Vedel, Y.: A new non-Euclidean proximal method for equilibrium problems. In: Chertov, O., Mylovanov, T., Kondratenko, Y., Kacprzyk, J., Kreinovich, V., Stefanuk, V. (eds.) ICDSIAI 2018. AISC, vol. 836, pp. 50–58. Springer, Cham (2019). https://doi.org/10.1007/978-3-319-97885-7_6
21. Malitsky, Y., Tam, M.K.: A forward-backward splitting method for monotone inclusions without cocoercivity. SIAM J. Optim. **30**(2), 1451–1472 (2020). https://doi.org/10.1137/18M1207260
22. Csetnek, E.R., Malitsky, Y., Tam, M.K.: Shadow Douglas-Rachford splitting for monotone inclusions. Appl. Math. Optim. **80**, 665–678 (2019). https://doi.org/10.1007/s00245-019-09597-8
23. Cevher, V., Vu, B.C.: A reflected forward-backward splitting method for monotone inclusions involving Lipschitzian operators. Set-Valued Var. Anal. **29**, 163–174 (2021). https://doi.org/10.1007/s11228-020-00542-4
24. Semenov, V.V., Denisov, S.V., Kravets, A.V.: Adaptive two-stage Bregman method for variational inequalities. Cybern. Syst. Anal. **57**(6), 959–967 (2021). https://doi.org/10.1007/s10559-021-00421-2
25. Beck, A.: First-Order Methods in Optimization. Society for Industrial and Applied Mathematics, Philadelphia (2017)
26. Daskalakis, C., Ilyas, A., Syrgkanis, V., Zeng, H.: Training GANs with optimism. arXiv:1711.00141 (2018)

27. Liu, M., et al.: A decentralized parallel algorithm for training generative adversarial nets. In: Advances in Neural Information Processing Systems (NeurIPS) (2020)
28. Franci, B., Grammatico, S.: Distributed projected-reflected-gradient algorithms for stochastic generalized Nash equilibrium problems. arXiv:2003.10261 (2021)
29. Alacaoglu, A., Malitsky, Y.: Stochastic Variance Reduction for Variational Inequality Methods. arXiv:2102.08352 (2021)
30. Dung, N.V., Vu, B.C.: Convergence analysis of the stochastic reflected forward-backward splitting algorithm. arXiv:2102.08906 (2021)
31. Polyak, R.: Finding Nonlinear Production-Consumption Equilibrium. arXiv:2204.04496 (2022)

Gender Impact Assessment of the COVID-19 Pandemic in Ukraine

Ihor Ivanov and Igor Brovchenko

Abstract Sex differences in the epidemic and demographic statistics of Ukraine are considered. Unlike the majority of countries in the world, there were more absolute excess deaths in Ukraine among women than among men, which is primarily associated with the demographics of the population, which in closely related with age-sex data of COVID-19 cases and life expectancy, which for decades was around ten years higher for women. We calculated monthly life expectancy and estimated that compared to 2019, it decreased by 0.5 and 0.3 years, for women and men, respectively in 2020. While for July 2020 it was higher by 0.4 and 0.6 years, respectively, since the peak of the first epidemic wave in Ukraine was in autumn. This suggests nearly equal drop of life expectancy for each sex in 2020 compared to July 2020 that can be considered as impact of first COVID-19 wave. At the same time, the decreasing of life expectancy was significantly deeper for women than for men, both according to State Statistics Service of Ukraine (SSSU) report and our estimates based only on the number of deaths by sex distribution. Additionally, it was found that during the second and third waves with dominant variants Alpha and Delta, respectively, the proportion of dying women among all COVID-19 deaths was increased, which was not observed during the first and fourth waves with the dominant wild type and Omicron variant, respectively. Moreover, according to the data available for the first three waves, excess deaths repeated the differences in the gender structure of COVID-19 deaths with significantly increased amplitude. Analysis of case fatality rate by sex, age, and epidemic wave showed consistent results: although the case fatality rate for all waves remained higher for men, during the second and third epidemic waves the difference was significantly reduced. The male to female ratio of age-standardized case fatality rates were estimated as 1.73 (95% CI: 1.69 to 1.78) for the first wave, 1.42 (95% CI: 1.39 to 1.46) for the second wave, 1.30 (95% CI: 1.28 to 1.33) for the third wave and 1.58 (95% CI: 1.51 to 1.66) for the fourth wave.

I. Ivanov (✉) · I. Brovchenko
Institute of Mathematical Machines and System Problems NAS of Ukraine, Glushkov Prosp., 42, Kyiv 03187, Ukraine
e-mail: ihorivanov@ukr.net

I. Ivanov
S. P. Timoshenko Institute of Mechanics NAS of Ukraine, Nesterov Str., 3, Kyiv 03057, Ukraine

S. Shkarlet et al. (eds.), *Mathematical Modeling and Simulation of Systems*, Lecture Notes in Networks and Systems 667,
https://doi.org/10.1007/978-3-031-30251-0_5

The obtained results can be used be age and sex structured epidemic mathematical models.

Keywords COVID-19 · Sex differences · Excess mortality · Demography · Life expectancy · Epidemic mathematical models

1 Introduction

It is widely established that infected with SARS-COV-2 men have a higher chance of death than women of the same age [1–5]. Moreover, standardized by age categories male/female ratio of the fatality risk remains higher than male/female ratio of all-causes mortality risk [8] taking into account the fact that women outlive men in almost all countries of the world [6]. As a result, in the vast majority of countries with available information on the sex composition of COVID-19 deaths, there were more fatalities among men than among women (see [7]).

Although hundreds of studies have been devoted to the male/female ratio of the mortality risk (see review [3] and references therein), its change depending on SARS-CoV-2 variants of concern remains unclear.

Analysis of sex-specific demographic impact of COVID-19 is widely discussed in terms of excess mortality [9, 10] and life expectancy reduction [11–14]. From cross-country comparisons [11, 12] it follows that the decrease in life expectancy was typically greater for men than for women in countries that were significantly affected by the pandemic in 2020. In particular, in Russia, where the age and sex composition of the population is close to Ukrainian, and where, as in Ukraine, women live on average 10 years longer than men, the reduction in life expectancy for each sex was quite close: 2.25 years for women and 2.50 years for men [12].

At the same time, except of large cross-countries studies with the most general conclusions, only few studies have been devoted to relationship between the COVID-19 pandemic in Ukraine and all-causes mortality [15–19]. In [15] the excess mortality in 2020 by age category was discussed, concluding that, similarly to other countries of the world, excess mortality in Ukraine was observed primarily among the older people. The paper [16] for Ukraine and 15 other countries compares the risks of mortality from COVID-19 among the entire population with the risks of dying from all causes in accordance with live tables. The work [17] is devoted to excess deaths in Ukraine at the national and subnational levels, where the number of excess deaths in Ukraine for the period from March 2020 to January 2022 is estimated by various methods from 183,000 to 209,000.

2 Demographic and Epidemic Sex Overview

2.1 All-Causes Deaths

In addition to public demographic data from the State Statistics Service of Ukraine (SSSU) [20], there is available by request the raw data from the Ministry of Justice of Ukraine [21], which is then sent to the SSSU, where they are processed and corrected. Since the accumulated difference between the datasets from these two sources for each year does not exceed 1%, the Ministry of Justice of Ukraine data for some tasks can be considered as a good approximation of the SSSU data.

This data provides us information about sex specified daily deaths in Ukraine (see Fig. 1). Before the pandemic, in 2015–2019, there was an approximate sex parity in the number of deaths (women accounted for 50.7% of all deaths). At the same time, in the cold season, and especially during epidemics of influenza-like diseases, the proportion of women among all the dead increased up to 52%, so that in some months the number of deaths among women was almost 10% higher than that among men. Although there was sex parity in the number of deaths during the warm season, which was not violated during periods of excess mortality due to extreme heat as well.

With the onset of the pandemic, the proportion of women who died has risen significantly above 50%, although not immediately. In 2021, women accounted for

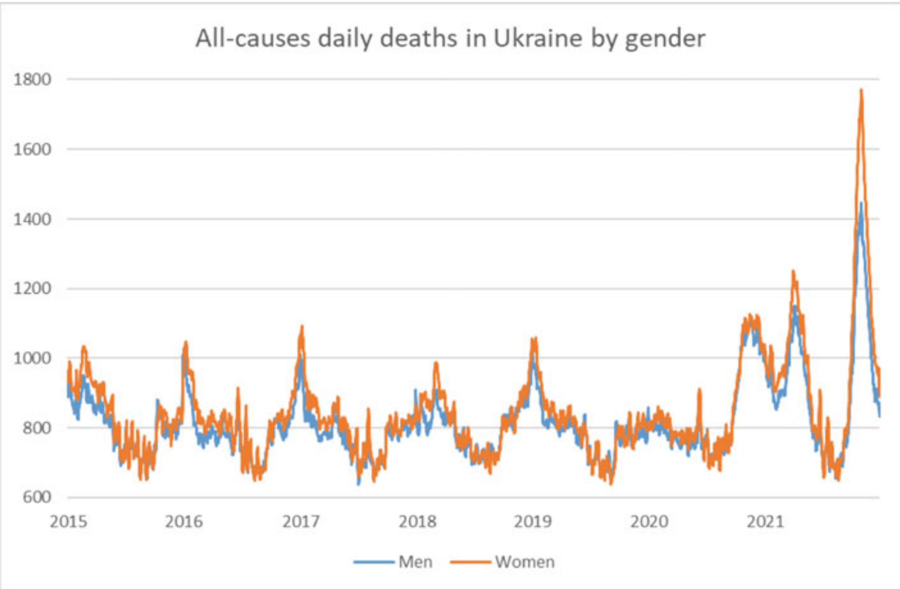


Fig. 1 All-causes deaths in 2015–2021, symmetrical weighted 7-day moving average with weight: 0.05, 0.1, 0.2, 0.3, 0.2, 0.1, 0.05. Data from the Ministry of Justice of Ukraine

more than 51.9% of all deaths, that may seem counterintuitive given that covid affects men more than women. However, it should be taken into account that as of December 2020, women accounted for 53% of the all population of Ukraine and 66.8% of the population of Ukraine over 65 years old, according to the SSSU. In addition, women accounted for 59.3% of all infected.

2.2 Life Expectancy (LE)

According to the SSSU, women's LE for decades was approximately 10 years higher than that of men (triangles on Fig. 2).

On Fig. 2 solid lines show our monthly calculations for 12-months' time spans based on data on the sex and age composition of the deceased from the Ministry of Justice of Ukraine for 2015–2020.

Calculations for 2021 were performed based on information on the sex composition of the deceased only, assuming the constancy of their age composition and scaling it by the number of deaths of each sex (taking into account that the age distributions of those who died from covid and those who died from all other causes are quite close).

Life expectancy at birth e_0 has been found according to Farr's method [22]:

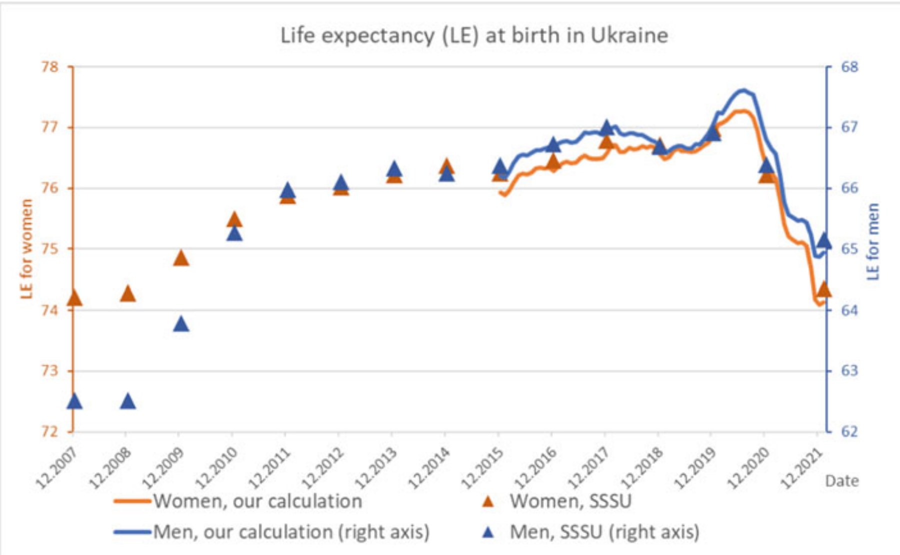


Fig. 2 Life expectancy at birth. Calculations for 2021 do not take into account the age structure of the deceased, 12-months moving average

$$e_0 = \sum\nolimits_{x \geq 1} l_x, \tag{1}$$

$$l_x = \prod\nolimits_{i=0}^{x-1} p_i, \tag{2}$$

$$p_x = 1 - q_x, \tag{3}$$

$$q_x = \frac{M_x}{B_x + \frac{M_x}{2}} \tag{4}$$

where l_x—the probability of living to age x, p_x– the probability of surviving at the age of x full years for the considered period (with a duration of 12 months for our calculations), q_x—is the probability of dying at the age of x full years for the same period, M_x—is the number of deaths at the age of x full years, B_x—is the average number of people at the age of x full years.

Our calculations are consistent with SSSU report that in 2020 LE decreased more for women than for men (Table 1), although the deviation from the SSSU data was relatively high. This phenomenon seems not to be directly related to mortality from COVID-19, since as of July 2020 (i.e., for the period from August 2019 to July 2020), when life expectancy reached a local peak in 2020, its growth, compared to 2019, was 0.6 years for men and 0.4 years for women as the peak of the first epidemic wave in Ukraine fell on autumn. Consequently, compared to July 2020 the fall of LE was almost equal for women and men: 0.9 and 0.8 years, respectively.

Much greater sex disparity in the decrease of LE was observed in 2021. It fell by 2.6 years for women and 1.8 for men relative to 2019 according SSSU, and, respectively, by 2.8 and 2.1 years according our estimations. The fall was sharper with still essential sex difference when comparing to July 2020: by 3.2 years for women and 2.7 for men.

Table 1 LE at birth in Ukraine

	Dec 2015	Dec 2016	Dec 2017	Dec 2018	Dec 2019	Jul 2020	Dec 2020	Dec 2021
SSSU								
Women	76.25	76.46	76.78	76.72	76.98	-	76.22	74.36
Men	66.37	66.73	67.02	66.69	66.92	-	66.39	65.16
Our estimations								
Women	75.94	76.28	76.59	76.56	76.89	77.28	76.39	74.12*
Men	66.20	66.64	66.92	66.69	67.06	67.62	66.81	64.94*

* based on the number of deaths by sex only

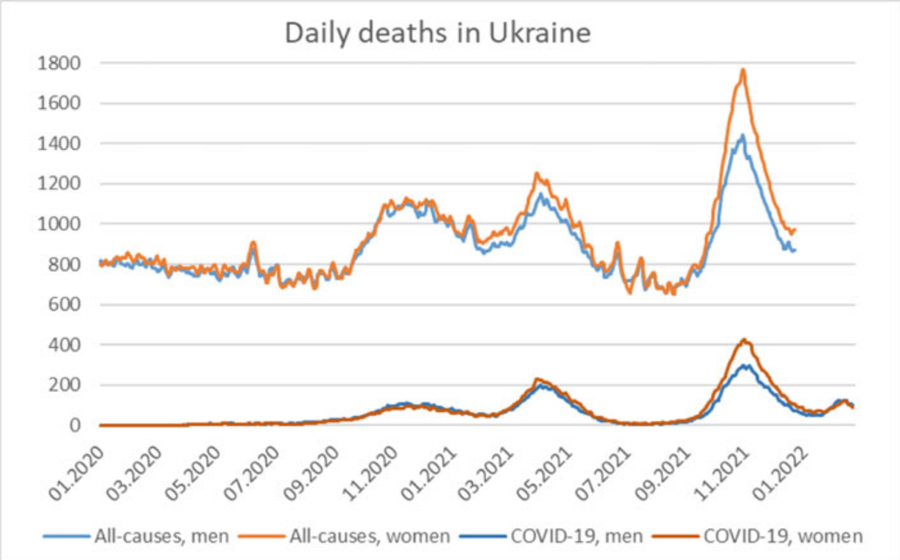


Fig. 3 All-causes deaths (Ministry of Justice of Ukraine) and COVID-19 fatalities since 2020 (Ministry of Health). Symmetrical weighted 7-day moving average with weight: 0.05, 0.1, 0.2, 0.3, 0.2, 0.1, 0.05

2.3 Confirmed COVID-19 Fatalities

Information about the sex composition of the daily number of COVID-19 deaths was public until the start of the war in Ukraine. As follows from Fig. 3, this data provides evidence that pandemic-related excess mortality in Ukraine, in addition to confirmed COVID-19 fatal cases, is predominantly undiagnosed deaths directly related to SARS-CoV-2 infection. At the same time, during the second and third waves with peaks in spring and autumn of 2021, respectively, the sex disparity observed in the statistics of COVID-19 fatalities corresponded to a significantly increased sex disparity in the statistics of all-causes deaths.

3 COVID-19 Case Fatality Rate (CFR) Analysis

3.1 Methods

All the available timeline of COVID-19 epidemic data was divided on 4 waves:

- 1-st wave with the beginning at the beginning of the pandemic and until 15.02.2021 (“wild” type);
- 2-nd wave starting on 15.02.2021 and ending on 15.07.2021 (variant Alpha)

Table 2 Size of epidemic data

	Wave 1 Wild	Wave 2 Alpha	Wave 3 Delta	Wave 4 Omicron
COVID-19 cases				
Women	765,202	576,697	851,593	592,556
Men	516,687	396,306	589,208	409,258
COVID-19 deaths				
Women	12,268	14,600	26,090	3566
Men	13,732	12,499	18,960	3467

- 3-rd wave starting on 15.07.2021 and ending on 01.01.2022 (variant Delta);
- 4-th wave starting on 01.01.2022 and ending on 15.02.2022 (variant Omicron, subvariants BA.1, BA.2).

The dates of wave demarcation are chosen so that they roughly correspond to the local minima between waves. According to GISAID database [23] each wave is associated with an outbreak of a single virus variant (specified in brackets in the list above).

The available database of COVID-19 cases in Ukraine contains information on the sex and age composition of cases, as well as information on whether the case was fatal. Its size, broken down into epidemic waves, is shown in Table 2.

For each wave and sex, the CFR is calculated as the ratio of deaths to all cases for a given day (the database allows to associate each fatal case with the date of registration of corresponding case of illness) for every age category.

For a given age-sex group and epidemic wave, the CFR is calculated by the formula:

$$\mathrm{CFR} = \frac{\begin{array}{c}\text{Number of COVID}-19\text{ deaths caused by}\\ \text{illness cases registered during a given epidemic wave}\end{array}}{\text{Number of all COVID}-19\text{ cases registered during the same epidemic wave}}. \quad (5)$$

To obtain confidence intervals, each sample was presented by zeros and ones, where one means a COVID-19 case that had a fatal outcome, and zero means any other COVID-19 case. Thus, the sample data can be considered as realizations of Bernoulli random variables with a binomial sample mean, which admits a normal approximation. Figure 4 shows CFRs with based on this approximation standard error and confidence interval.

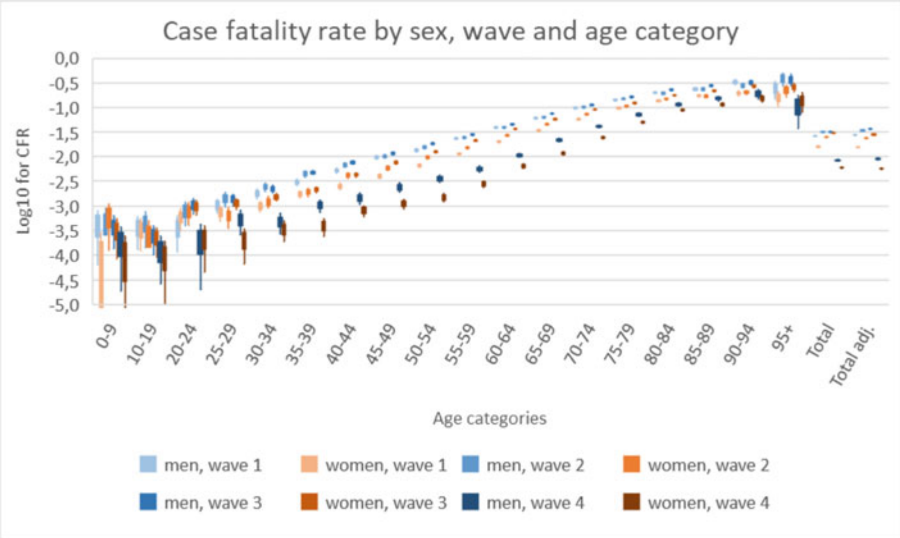


Fig. 4 Box and whisker diagram for case fatality rate by wave and sex-age category. Box shows standard error, whisker shows 95% confidence interval. "Total" means CFR for overall population of a given sex. "Total adjusted" means CFR for age standardized COVID-19 cases

Since the ratio of the CFR for men and women also demonstrates asymptotically normal behavior, we construct for it a confidence interval and a standard error based on the assumption of normal distribution. The variance estimate for this ratio is found using the delta method [24].

Consider random variables ξ and η with finite means and variances, where η either has no mass at 0 (discrete) or has support $[0, \infty)$. Then we can estimate the variance of the ratio as

$$\mathrm{Var}\left(\frac{\xi}{\eta}\right) \approx \frac{\mu_\xi^2}{\mu_\eta^2}\left(\frac{\sigma_\xi^2}{\mu_\eta^2} - 2\frac{\mathrm{Cov}(\xi, \eta)}{\mu_\xi \mu_\eta} + \frac{\sigma_\xi^2}{\mu_\eta^2}\right), \tag{6}$$

where μ_ξ, μ_η—means for ξ and η respectively, σ_ξ, σ_η—standard deviations for ξ and η respectively, $\mathrm{Cov}(\xi, \eta) = \mathrm{E}\big[\big(X - \mu_\xi\big)\big(Y - \mu_\eta\big)\big]$.

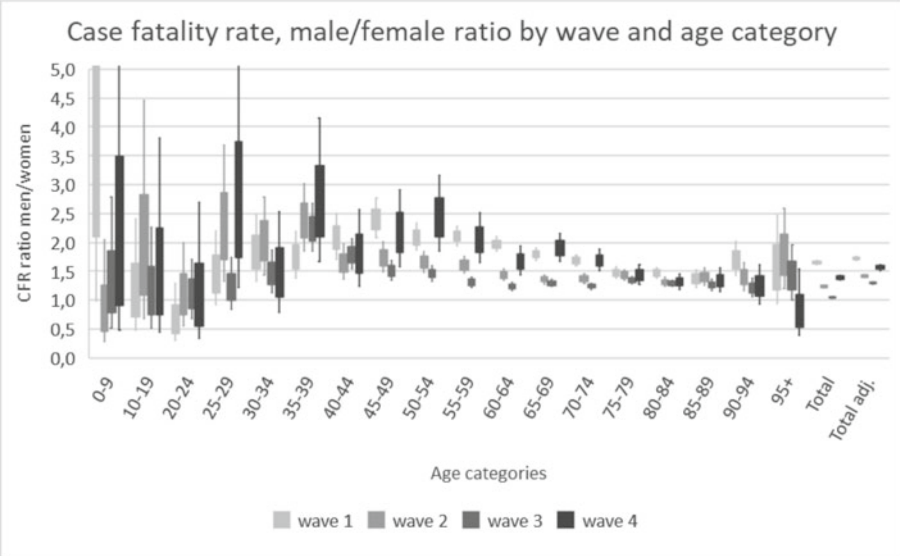


Fig. 5 Box and whisker diagram for a sex ratio of case fatality rates. Box shows standard error, whisker shows 95% confidence interval. “Total” means CFR for overall population of a given sex. “Total adjusted” means CFR for age standardized COVID-19 cases

Since ξ and η independent, the (6) takes the form

$$\mathrm{Var}\left(\frac{\xi}{\eta}\right) \approx \frac{\mu_{\xi}^{2}}{\mu_{\eta}^{2}}\left(\frac{\sigma_{\xi}^{2}}{\mu_{\eta}^{2}} + \frac{\sigma_{\xi}^{2}}{\mu_{\eta}^{2}}\right). \tag{7}$$

Table 2 and Fig. 5 shows the ratio of CFR for men and women with based on (7) confidence interval and standard error. To build interval estimates the ratios were taken logarithmically.

Finally, aggregated male–female ratio of age standardized CFR was computed for a unified age structure of COVID-19 cases among each sex (Table 3).

Table 3 Male to female CFR ratio with 95% CI. “Total” means CFR for overall population of a given sex. “Total adjusted” means CFR for age standardized COVID-19 cases

Age	Wave 1 Wild	Wave 2 Alpha	Wave 3 Delta	Wave 4 Omicron
0–9	4.59 (0.99 to 21.4)	0.76 (0.28 to 2.04)	1.20 (0.52 to 2.79)	1.78 (0.48 to 6.64)
10–19	1.09 (0.49 to 2.41)	1.75 (0.69 to 4.46)	1.09 (0.52 to 2.26)	1.30 (0.44 to 3.81)
20–24	0.62 (0.39 to 1.29)	1.04 (0.54 to 2.00)	1.08 (0.68 to 1.71)	0.95 (0.34 to 2.70)
25–29	1.41 (0.91 to 2.20)	2.20 (1.32 to 3.68)	1.21 (0.85 to 1.74)	2.55 (1.21 to 5.38)
30–34	1.81 (1.33 to 2.48)	2.00 (1.44 to 2.78)	1.46 (1.13 to 1.87)	1.42 (0.79 to 2.53)
35–39	1.73 (1.38 to 2.18)	2.36 (1.85 to 3.02)	2.22 (1.84 to 2.68)	2.64 (1.67 to 4.16)
40–44	2.07 (1.71 to 2.49)	1.64 (1.36 to 1.97)	1.78 (1.53 to 2.06)	1.78 (1.23 to 2.56)
45–49	2.39 (2.07 to 2.76)	1.73 (1.49 to 2.01)	1.51 (1.34 to 1.69)	2.14 (1.57 to 2.92)
50–54	2.08 (1.86 to 2.33)	1.65 (1.46 to 1.85)	1.46 (1.33 to 1.60)	2.41 (1.84 to 3.17)
55–59	2.11 (1.94 to 2.29)	1.60 (1.47 to 1.75)	1.31 (1.22 to 1.40)	2.04 (1.65 to 2.51)
60–64	1.97 (1.84 to 2.10)	1.44 (1.34 to 1.54)	1.24 (1.17 to 1.30)	1.67 (1.44 to 1.94)
65–69	1.80 (1.69 to 1.91)	1.36 (1.28 to 1.44)	1.29 (1.23 to 1.35)	1.90 (1.67 to 2.16)
70–74	1.68 (1.59 to 1.78)	1.37 (1.30 to 1.45)	1.24 (1.19 to 1.29)	1.69 (1.51 to 1.88)
75–79	1.47 (1.37 to 1.57)	1.43 (1.34 to 1.53)	1.34 (1.27 to 1.41)	1.43 (1.27 to 1.61)
80–84	1.47 (1.38 to 1.56)	1.31 (1.23 to 1.39)	1.29 (1.23 to 1.35)	1.32 (1.19 to 1.46)
85–89	1.36 (1.22 to 1.53)	1.40 (1.25 to 1.55)	1.26 (1.16 to 1.36)	1.33 (1.14 to 1.56)
90–94	1.69 (1.41 to 2.02)	1.39 (1.16 to 1.66)	1.21 (1.05 to 1.38)	1.23 (0.94 to 1.62)
95 +	1.52 (0.94 to 2.48)	1.76 (1.19 to 2.59)	1.41 (1.01 to 1.96)	0.77 (0.39 to 1.54)
Total				
Unadjusted	1.66 (1.62 to 1.70)	1.25 (1.22 to 1.28)	1.05 (1.03 to 1.07)	1.41 (1.34 to 1.47)
Adjusted	1.73 (1.69 to 1.78)	1.42 (1.39 to 1.46)	1.30 (1.28 to 1.33)	1.58 (1.51 to 1.66)

3.2 Results

As expected, the results (see Fig. 4) show that the transition from the first to the second, and then to the third wave was accompanied by an increase in CFR, after which this indicator fell sharply to the fourth wave.

In addition, sex differences in CFR indicate a persistence of higher COVID-19 mortality for males. At the same time, during the first and fourth waves, this sex gap in mortality was higher (see Fig. 4, 5 and Table 3), especially at the middle of the age distribution.

Findings on male/female ratio of CFRs for the first wave are in a good agreement with the results [2, 4, 5].

4 Conclusion

Ukraine has untypical demographic structure with a significant proportion of older women and life expectancy 10 years higher for women than for men. Thus, despite the higher risk of death from COVID-19 for men than for women, the pandemic has led to more deaths among women than among men in absolute terms. As a result, the sex difference in life expectancy has been narrowed by 0.8 years by the end of 2021 compared with 2019 although its decline in 2020 were mixed.

Notably, that during the first wave, sex parity was maintained both in the number of deaths from all causes and in the number of COVID-19 fatalities, while the subsequent two waves led to a larger number of COVID-19 fatalities among women than among men, which was accompanied by excess deaths with an even greater gap between females and males. This indicates a similar nature of unaccounted excess mortality and confirmed COVID-19 mortality.

The obtained results show that for the second and third epidemic waves that the difference in CFR between men and women were the highest, while during the first and fourth waves there was an approximate parity with a slight pre-dominance of deaths from COVID-19 among men. At the same time, it follows from the analysis of age, sex and wave specific CFR that changes in the sex structure of excess mortality should primarily be associated with changes in the pathological properties of the virus.

Although there are no daily demographic data on deaths from all causes for 2022 due to the outbreak of hostilities in Ukraine, it can be predicted that the number of excess deaths during the fourth wave will probably be the same among men and women. The obtained results can be used be age and sex structured epidemic mathematical models.

Acknowledgements This research was supported by the National Research Foundation of Ukraine, Grant 2021.01/0311, NAS of Ukraine 15.04.2021, №6.2021.

References

1. Huang, B., Cai, Y., Li, N., et al.: Sex-based clinical and immunological differences in COVID-19. BMC Infect. Dis. **21**, 647 (2021). https://doi.org/10.1186/s12879-021-06313-2
2. Gebhard, C., Regitz-Zagrosek, V., Neuhauser, H.K., et al.: Impact of sex and gender on COVID-19 outcomes in Europe. Biol. Sex Differ. **11**, 29 (2020). https://doi.org/10.1186/s13293-020-00304-9
3. Pijls, B.G., Jolani, S., Atherley, A., Dijkstra, J.I.R., Franssen, G.H.L., Hendriks, S., Yu, E.Y.-W., Zalpuri, S., Richters, A., Zeegers, M.P.: Temporal trends of sex differences for COVID-19 infection, hospitalisation, severe disease, intensive care unit (ICU) admission and death: a meta-analysis of 229 studies covering over 10M patients. F1000Res. **11**, 5 (2022). https://doi.org/10.12688/f1000research.74645.1
4. Ahrenfeldt, L.J., Otavova, M., Christensen, K., Lindahl-Jacobsen, R.: Sex and age differences in COVID-19 mortality in Europe. Wien. Klin. Wochenschr. **133**, 393–398 (2021). https://doi.org/10.1007/s00508-020-01793-9

5. Green, M.S., Nitzan, D., Schwartz, N., Niv, Y., Peer, V.: Sex differences in the case-fatality rates for COVID-19—a comparison of the age-related differences and consistency over seven countries. PLoS ONE **16**(4), e0250523 (2021)
6. Barford, A., Dorling, D., Davey, S.G., Shaw, M.: Life expectancy: women now on top everywhere. BMJ **332**(7545), 808 (2006)
7. Ramírez-Soto, M.C., Ortega-Cáceres, G., Arroyo-Hernández, H.: Sex differences in COVID-19 fatality rate and risk of death: an analysis in 73 countries, 2020–2021. Infez. Med. **29**(3), 402–407 (2021). PMID: 35146345. PMCID: PMC8805484. https://doi.org/10.53854/liim-2903-11
8. Guilmoto, C.Z.: COVID-19 death rates by age and sex and the resulting mortality vulnerability of countries and regions in the world. MedRxiv (2020)
9. Nielsen, J., Nørgaard, S.K., Lanzieri, G., et al.: Sex-differences in COVID-19 associated excess mortality is not exceptional for the COVID-19 pandemic. Sci. Rep. **11**, 20815 (2021). https://doi.org/10.1038/s41598-021-00213-w
10. Dahal, S., Banda, J.M., Bento, A.I., et al.: Characterizing all-cause excess mortality patterns during COVID-19 pandemic in Mexico. BMC Infect. Dis. **21**, 432 (2021). https://doi.org/10.1186/s12879-021-06122-7
11. Aburto, J.M., Schöley, J., Kashnitsky, I., Zhang, L., Rahal, C., Missov, T.I., Mills, M.C., Dowd, J.B., Kashyap, R.: Quantifying impacts of the COVID-19 pandemic through life-expectancy losses: a population-level study of 29 countries. Int. J. Epidemiol.**51**(1), 63–74 (2022). https://doi.org/10.1093/ije/dyab207
12. Islam, N., Jdanov, D.A., Shkolnikov, V.M., Khunti, K., Kawachi, I., White, M., Lacey, B.: Effects of COVID-19 pandemic on life expectancy and premature mortality in 2020: time series analysis in 37 countries. BMJ **375**, e066768 (2021)
13. Chan, E.Y.S., Cheng, D., Martin, J.: Impact of COVID-19 on excess mortality, life expectancy, and years of life lost in the United States. PloS One **16**(9), e0256835 (2021)
14. García-Guerrero, V.M., Beltrán-Sánchez, H.: Heterogeneity in excess mortality and its impact on loss of life expectancy due to COVID-19: evidence from Mexico. Can. Stud. Popul. **48**, 165–200 (2021). https://doi.org/10.1007/s42650-021-00051-1
15. Mehta, N., Honchar, I., Doroshenko, O., Brovchenko, I., Pak, K., Danyuk, M., Polikarchuk, P.: Excess deaths associated with the COVID-19 pandemic in Ukraine in 2020. medRxiv (2021)
16. Bauer, P., Brugger, J., König, F., et al.: An international comparison of age and sex dependency of COVID-19 deaths in 2020: a descriptive analysis. Sci. Rep. **11**, 19143 (2021). https://doi.org/10.1038/s41598-021-97711-8
17. Ivanov, I., Brovchenko, I.: Estimation of associated with COVID-19 excess mortality in Ukraine. Math. Mach. Syst. (3), 48–57 (2022). (in Ukrainian)
18. Ugarte, M.P., Achilleos, S., Quattrocchi, A., et al.: Premature mortality attributable to COVID-19: potential years of life lost in 17 countries around the world, January–August 2020. BMC Public Health **22**, 54 (2022). https://doi.org/10.1186/s12889-021-12377-1
19. Kurylo, I.: Tendencies of mortality and life expectancy in Ukraine before the Russian full scale military invasion. Econ. Sociol. (1), 58–71 (2022)
20. State Statistic Service of Ukraine. https://ukrstat.gov.ua/. Accessed 12 Sept 2022
21. The Quarantine Economy. https://q.rating.zone/. Accessed 13 Sept 2022
22. Farr, W.: Report upon the mortality of lunatics. J. Stat. Soc. Lond. **4**(1), 17–33 (1841)
23. GISAID. https://gisaid.org/hcov19-variants/. Accessed 12 Sept 2022
24. Stuart, A., Ord, K.: Kendall's Advanced Theory of Statistics, 6th edn., vol. 1, p. 351. Arnold, London (1994)

Mathematical Modeling and Simulation of Systems in Manufacturing

Modelling Shrinkage of a Polymer Binder During Curing

Andrii Kondratiev, Maryna Shevtsova, Oleksii Vambol, Anton Tsaritsynskyi, and Tetyana Nabokina

Abstract The quality of modern polymer composite materials and composite-based structures is largely determined by technological conditions of their production. The paper deals with the optimization of technological conditions of heat treatment of cured composite products, carried out based on the results of mathematical and computer modelling of thermochemical and thermophysical properties of materials, as well as heat transfer processes accompanied by corresponding physical and chemical transformations in the product. The results of the studies and subsequent mathematical modelling of shrinkage of the epoxy binder EA9396 during curing are presented. The experimental data are obtained using the differential scanning calorimetry. An analytical dependence describing the change in the degree of curing of a binder with a change in the conditions of product moulding has been proposed. The dependence makes it possible to determine the state of a polymer binder (viscous, viscoelastic, elastic) at any time of the curing process, to establish the point at which stresses occur in the cured structure as well as to detect the physical and mechanical properties of the composite corresponding to the degree of curing and viscosity. A model of polymer binder shrinkage during curing has been developed. The shrinkage model satisfactorily describes the experimental results. The curves constructed using the experimentally received reaction rate data and the analytical model differ by no more than 5%. The dependence of the shrinkage rate of the polymer binder on the reaction rate is presented. A software that allows obtaining the dependence of the degree of curing and shrinkage on time for a given temperature range has been developed. Based on the results received, the selection of the optimal curing mode can be carried out, which makes it possible to compensate shrinkage deformations with temperature ones and reduce technological stresses in the composite product.

Keywords Temperature-time regime · Heating rate · Reaction rate · Viscosity

A. Kondratiev (✉)
O. M. Beketov National University of Urban Economy in Kharkiv, 17 Marshal Bazhanov St., Kharkiv 61002, Ukraine
e-mail: andrii.kondratiev@kname.edu.ua

M. Shevtsova · O. Vambol · A. Tsaritsynskyi · T. Nabokina
National Aerospace University "Kharkiv Aviation Institute", 17 Chkalov St., Kharkiv 61070, Ukraine

S. Shkarlet et al. (eds.), *Mathematical Modeling and Simulation of Systems*, Lecture Notes in Networks and Systems 667,
https://doi.org/10.1007/978-3-031-30251-0_6

1 Introduction

The trend towards introducing composites and replacing classical (metal) structural materials with composite materials has become prevailing in recent years [1, 2], structural and functional polymer composite materials (PCMs) being used most extensively [3, 4]. PCMs are widely applied in critical engineering structures, aerospace and energy industries, civil and industrial construction, etc. [5, 6]. The main physical and chemical process that occurs at manufacturing PCMs and products from them is the curing process [7, 8]. This process serves as a basis for creating PCMs with required technological, strength and thermal properties [9, 10]. At the same time, the kinetics of the curing reaction is the determining factor in choosing the temperature-time regime for obtaining a PCM [11]. A proper selection of curing conditions allows compensating shrinkage deformations with temperature ones and reducing technological stresses [12, 13]. In this regard, the study of shrinkage deformations during PCM curing and their subsequent mathematical modelling is an important scientific and practical task.

2 Literature Review

Viscosity of a material, as shown in the works [14, 15], is functionally dependent on the temperature and time of the process: with an increase in the temperature and time, the viscosity also increases. An increase in the viscosity of a material leads to the occurrence of shrinkage and temperature stresses in the moulded material-product. Shrinkage stresses are associated primarily with transformations in the PCM and the degassing process. However, these works do not consider the effect of temperature on the curing process.

The results obtained in the works [16–18] have allowed simulating the curing process. However, these works do not analyse the effect of shrinkage deformations.

The occurrence of stress relaxation and creep phenomena in the composite at the stage of temperature holding are considered in the works [19, 20]. However, the study of the distribution of the temperature field during the curing process is not carried out in these works.

The research [21] demonstrates that at the stage of heating, along with shrinkage, temperature phenomena also contribute to the occurrence of stresses and deformations. It is shown that temperature stresses become comparable with shrinkage stresses at reaching the viscosity of the material corresponding to 60…70% of the conversion in the binder.

The paper [22], presents a description of the occurrence of temperature stresses at the stage of PCM heating to the polymerization temperature, which is based on experimental studies. The authors neglect the occurrence of deformations due to uneven distribution of temperature through the thickness, believing that at this stage the binder is in a viscous-flow state and is unable to fully take and transfer the load.

In the work [23] a method for determining technological stresses that arise in PCMs during curing is developed. The method has made it possible to take into account shrinkage, changes in physical and mechanical properties, and rheological processes occurring in the binder during moulding. It is shown that, at the initial stage of the heating process, the stresses are equal to zero, and they appear after the formation of a new structure begins (the degree of curing increases). Further, it is noted that at a high heating rate, shrinkage deformations are compensated by heating at the initial stage and, therefore, the stress values do not fall in the negative zone. At low heating rates, the temperature deformations are not as large as the shrinkage ones, so the stress values have a positive sign.

The study of a particular case of composite curing by the autoclave method is described in the works [24, 25].

To determine the degree of curing, a number of methods that describe the process are employed. The commonly used ones include the Carothers equation, the Arrhenius equation, as well as equations obtained by approximating experimental data [26, 27]. Unfortunately, all these methods consider only individual factors. The Carothers equation deals only with the molecular structure, without regard to the kinetics and physics of the process. The results obtained with its help do not directly reflect the change in the degree of curing depending on the technological parameters of the moulding process (temperature and time). Often, to determine the degree of curing of a material, researchers use the Arrhenius equation, which takes into account the kinetics of the process and gives a complete picture of the change in the curing degree depending on temperature. However, the Arrhenius equation allows obtaining a result only with a certain degree of accuracy. Moreover, the dependences for determining the degree of polymerization obtained with the help of the mentioned equations are quite difficult to implement in the production process. First of all, this is due to a fairly large range of resins, hardeners, modifiers and other components that make up the binder. This, in turn, leads to a large number of experiments to determine the required design parameters.

In most cases, to measure the degree of polymerization and describe the kinetics of the curing and shrinkage processes, empirical dependences are used [26, 27]. Such dependences usually include the main technological parameters of the curing process. However, the dependences obtained by approximating the experimental data do not fully reflect the curing mechanism. For practical calculations, it is more convenient to employ analytical expressions. This makes it possible to use a smaller amount of experimental data when constructing a model of shrinkage deformations during the curing of a polymer binder.

3 Research Methodology

Shrinkage of polymer binders depends on the rate of heating during curing, and, consequently, on the rate of the polymerization reaction [23, 26]. Based on the analysis of possible equations describing the kinetics of the curing reaction of a polymer binder, in general terms, the analytical model can be represented as follows:

$$\frac{d\eta}{dt} = f(\eta, T) \tag{1}$$

where $f(\eta, T)$ is the function that determines the reaction rate depending on the curing degree and temperature.

This equation should be integrated under the initial condition $\eta(0) = \eta_0$ where η_0 is the initial degree of curing (as a rule, $\eta_0 = 0$ [23, 26]).

If the function f is known, using the Eq. (1) it is possible to find the dependence of the degree of curing on time $\eta(t)$ for the set temperature-time regime $T(t)$. If there is a sufficient amount of experimental data for a particular binder, the function f can be defined in tabular form. Table 1 shows the data on the reaction rate for the EA9396 epoxy binder.

The data are obtained using the differential scanning calorimetry. This method implies determining the degree of curing of the binder based on the amount of heat released due to the exothermic effect of the polymerization reaction [28].

At performing numerical integration of the Eq. (1), defining the function f for determining the reaction rate at intermediate points with the help of the tabular data, interpolation should be applied.

Having analysed the existing models and experimental data, we will take the following analytical expression to calculate the reaction rate:

$$f(\eta, T) = \begin{cases} K(T)\eta^{m(T)}\left(\frac{\eta^*(T)-\eta}{\eta^*(T)}\right)^{n(T)} if\ \eta \le \eta^*(T); \\ 0\ \ if\ \eta > \eta^*(T), \end{cases} \tag{2}$$

where $\eta^*(T)$ is the maximum degree of curing that can be achieved at the temperature T; $K(T)$, $m(T)$, $n(T)$ are certain temperature functions.

For a number of temperature values for the EA9396 binder (Table 1), the values of the functions η^*, K, m, n, best approximating the dependence of the reaction rate on the curing degree at a fixed temperature were selected. The obtained values of the functions m, n were approximated by the linear dependences:

$$\begin{aligned} m(T) &= 0.062 + 0.0038T; \\ n(T) &= 0.064 + 0.0074T. \end{aligned} \tag{3}$$

Table 1 Curing reaction rate of the EA9396 epoxy binder depending on the curing degree and temperature

Curing degree, %	Temperature, °C												
	24	36	45	60	75	90	102	114	126	141	156	165	174
1	0.00119	0.00370	0.0068	0.0110									
10	0.00146	0.00364	0.00732	0.02199	0.05842	0.13012							
20	0.00148	0.00422	0.00728	0.02257	0.05167	0.15229	0.23694						
30	0.00138	0.00406	0.00840	0.02192	0.05699	0.15020	0.28083	0.37513					
40	0.00107	0.00365	0.0075	0.02020	0.05356	0.13982	0.24432	0.38034					
50	0	0.00299	0.00629	0.01776	0.04705	0.12356	0.22532	0.34606					
60	0	0.00101	0.00346	0.01454	0.0348	0.10201	0.18144	0.2824	0.46946				
70	0	0	0	0.00684	0.02421	0.07729	0.1281	0.21709	0.35119				
80	0	0	0	0	0.00128	0.04097	0.05786	0.13553	0.22129				
90	0	0	0	0	0	0	0	0.00253	0.07166	0.13430			
96	0	0	0	0	0	0	0	0	0	0.01890	0.0397		
97	0	0	0	0	0	0	0	0	0	0.00520	0.01775		
98	0	0	0	0	0	0	0	0	0	0	0.00856	0.01336	
99	0	0	0	0	0	0	0	0	0	0	0.00207	0.00924	0.00839
100	0	0	0	0	0	0	0	0	0	0	0	0	0

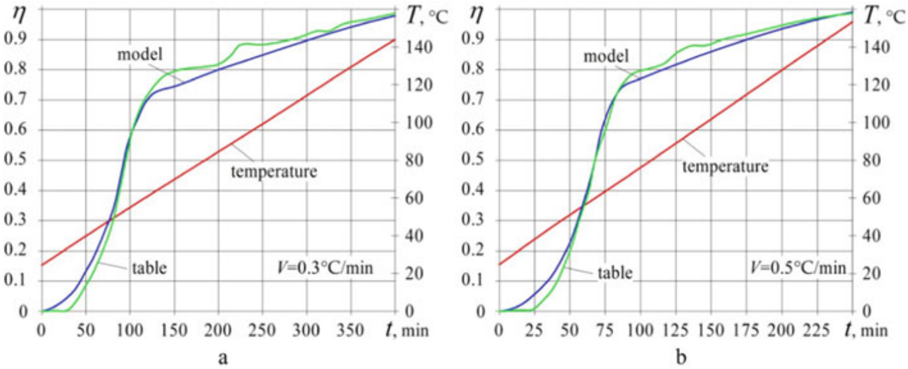


Fig. 1 Results of calculating the degree of curing of the EA9396 binder, obtained using the tabular and analytical representation of the function $f(\eta, T)$, for different heating rates: a – $V = 0.3$ °C/min; b – $V = 0.5$ °C/min

Further, we selected the optimal values of the functions η^*, K, at m, n, obtained from the dependences (3), which were quite accurately approximated by the expressions:

$$\begin{aligned} K(T) &= 0.000472e^{0.0948(T-5.21)+[0.0147(T-5.21)]^2}; \\ \eta^*(T) &= 0.298lnT - 0.491. \end{aligned} \tag{4}$$

Figure 1 shows the example results of calculating the degree of curing of the EA9396 binder, obtained using the tabular and analytical representation of the function $f(\eta, T)$. To solve the kinetic Eq. (1), the Runge-Kutta method of the second order was used. When setting the function $f(\eta, T)$ with the help of the tabular data, values at the intermediate points were calculated using linear interpolation.

To calculate the shrinkage depending on the curing degree, we used the following model:

$$\frac{d\xi}{d\eta} = c_1 + c_2 exp\left(-c_3\frac{d\eta}{dt}\right), \tag{5}$$

where c_1, c_2, c_3 are experimentally determined coefficients.

4 Results

Figure 2 shows the results of measuring the shrinkage of the EA9396 binder over time as well as the curves obtained using the dependence (5). The Eq. (5) was solved by the method of trapezoids, based on the data obtained by solving the kinetic equation using the reaction rates presented in Table 1 and the proposed analytical model (1). Thus, we obtained the following coefficient values: $c_1 = 0$; $c_2 = 0.04$; $c_3 = 7$.

The proposed shrinkage model describes the experimental results quite satisfactorily, and the curves, constructed using the reaction rate data obtained from Table 1 and the analytical model (1) differ insignificantly (no more than 5%).

Figure 3 presents the dependence of the shrinkage rate on the reaction rate for the binder under study. For a numerical implementation of the proposed models, we developed a software that allows, for a given temperature regime (Fig. 4, a), obtaining the dependence of the curing degree and shrinkage on time (Fig. 4, b).

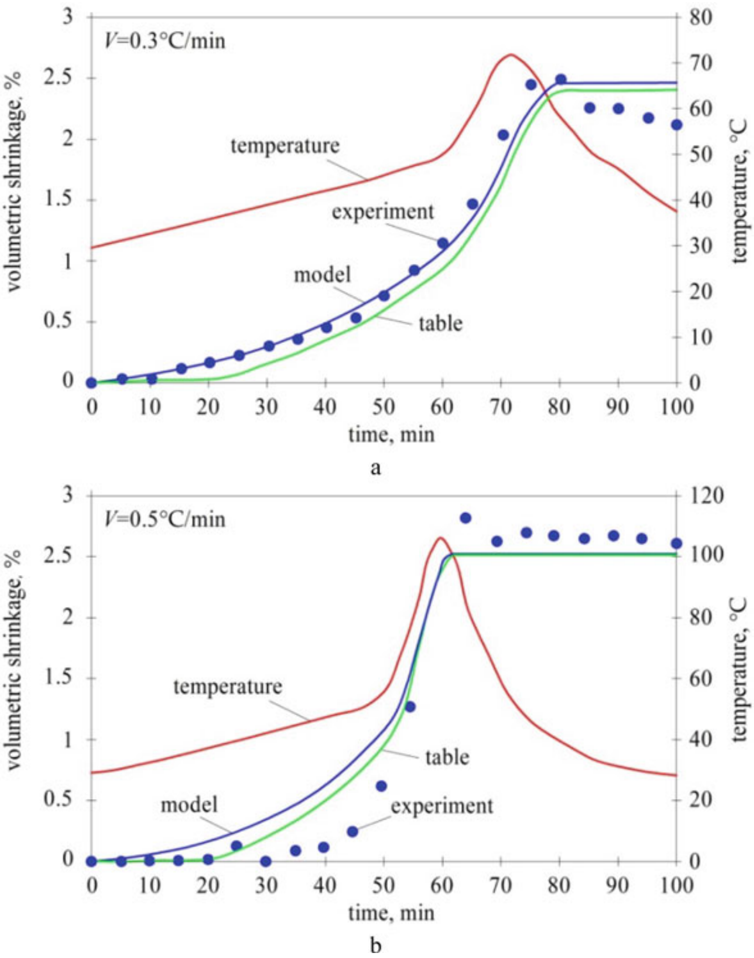


Fig. 2 Results of the EA9396 binder shrinkage modelling for different heating rates: a – $V = 0.3$ °C/min; b – $V = 0.5$ °C/min

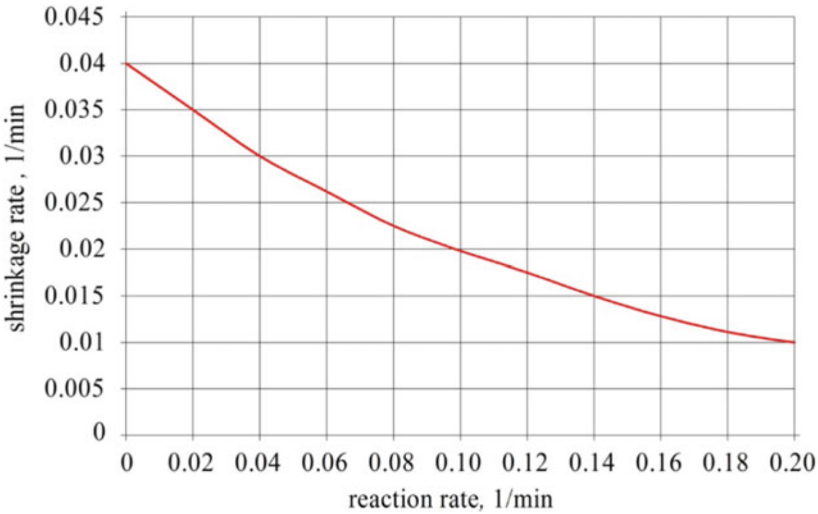


Fig. 3 Dependence of the shrinkage rate on the reaction rate

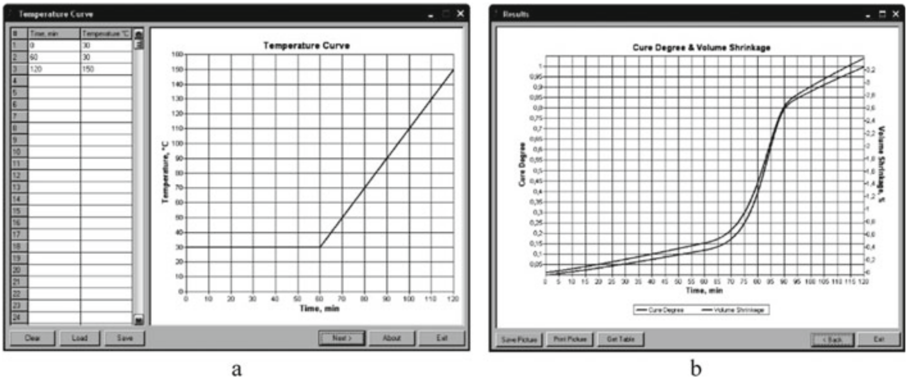


Fig. 4 Developed software that allows, for a given temperature regime (a), obtaining the dependence of the curing degree and shrinkage on time (b)

5 Conclusions

A general analytical dependence describing the change in the degree of curing of any binder with a change in the moulding conditions has been proposed. The dependence makes it possible to determine the PCM state (viscous, viscoelastic, elastic) at any time of the curing process, to establish the point at which stresses occur in the cured structure, as well as to detect the physical and mechanical properties of the composite that correspond to the degree of curing and viscosity. A model of shrinkage deformations that satisfactorily describes the experimental results has been developed. The dependence of the shrinkage rate on the reaction rate is presented. A software that allows obtaining the dependence of the degree of curing and shrinkage on time for a given temperature regime has been developed.

References

1. Karpus, V., Ivanov, V., Dehtiarov, I., Zajac, J., Kurochkina, V.: Technological assurance of complex parts manufacturing. In: Advances in Design, Simulation and Manufacturing. DSMIE 2018. Lecture Notes in Mechanical Engineering, pp. 51–61. Springer, Cham (2019). https://doi.org/10.1007/978-3-319-93587-4_6
2. Otrosh, Y., Kovalov, A., Semkiv, O., Rudeshko, I., Diven, V.: Methodology remaining lifetime determination of the building structures. In: MATEC Web Conferences 2018, vol. 230, p. 02023 (2018). https://doi.org/10.1051/matecconf/201823002023
3. Fomin, O., Lovskaya, A., Plakhtiy, A., Nerubatsky, V.: The influence of implementation of circular pipes in load-bearing structures of bodies of freight cars on their physico-mechanical properties **6**(162), 89–96 (2017)
4. Galych, I., Antoshchenkov, R., Antoshchenkov, V., Lukjanov, I., Diundik, S., Kis, O.: Estimating the dynamics of a machine-tractor assembly considering the effect of the supporting surface profile. Eastern-Eur. J. Enterp. Technol. **1**(7(109), 51–62 (2021). https://doi.org/10.15587/1729-4061.2021.225117
5. Smetankina, N., Merkulova, A., Merkulov, D., Postnyi, O.: Dynamic response of laminate composite shells with complex shape under low-velocity impact. In: International Scientific and Technical Conference on Integrated Computer Technologies in Mechanical Engineering - Synergetic Engineering, ICTM 2020, vol. 188, pp. 267–276 (2021). https://doi.org/10.1007/978-3-030-66717-7_22
6. Slyvynskyi, V.I., Sanin, A.F., Kharchenko, M.E., Kondratyev, A.V.: Thermally and dimensionally stable structures of carbon-carbon laminated composites for space applications. In: 65th International Astronautical Congress 2014, IAC 2014, vol. 8, pp. 5739–5751 (2014)
7. Rajak, D.K., Pagar, D.D., Menezes, P.L., Linul, E.: Fiber-reinforced polymer composites: manufacturing, properties, and applications. Polymers **11**(10), 1667 (2019). https://doi.org/10.3390/polym11101667
8. Rodionov, V.V.: Optimization of molding the polymeric composite material with improved characteristics. Plast. Massy **3–4**, 55–58 (2019). https://doi.org/10.35164/0554-2901-2019-3-4-55-58
9. Saba, N., Jawaid, M.: A review on thermomechanical properties of polymers and fibers reinforced polymer composites. J. Ind. Eng. Chem. **67**, 1–11 (2018). https://doi.org/10.1016/j.jiec.2018.06.018
10. Tiwary, A., Kumar, R., Chohan, J.S.: A review on characteristics of composite and advanced materials used for aerospace applications. Mater. Today Proc. **51**(1), 865–870 (2022). https://doi.org/10.1016/j.matpr.2021.06.276
11. Hsissou, R., Seghiri, R., Benzekri, Z., Hilali, M., Rafik, M., Elharfi, A.: Polymer composite materials: a comprehensive review. Compos. Struct. **262**, 113640 (2021). https://doi.org/10.1016/j.compstruct.2021.113640
12. Gaidachuk, V.E., Kondratiev, A.V., Chesnokov, A.V.: Changes in the thermal and dimensional stability of the structure of a polymer composite after carbonization. Mech. Compos. Mater. **52**(6), 799–806 (2017). https://doi.org/10.1007/s11029-017-9631-6
13. Kombarov, V., Kryzhyvets, Y., Biletskyi, I., Tsegelnyk, Y., Aksonov, Y., Piddubna, L.: Numerical control of fiberglass pipe bends manufacturing. In: IEEE 2nd KhPI Week on Advanced Technology 2021, pp. 357–362 (2021). https://doi.org/10.1109/KhPIWeek53812.2021.9570068

14. Li, D.N., Li, X.D., Dai, J.F., Xi, S.B.: A comparison of curing process-induced residual stresses and cure shrinkage in micro-scale composite structures with different constitutive laws. Appl. Compos. Mater. **25**, 67–84 (2018). https://doi.org/10.1007/s10443-017-9608-6
15. Yuan, Z.Y., Wang, Y.J., Yang, G.G., Tang, A.F., Yang, Z.C., Li, S.J., Li, Y., Song, D.L.: Evolution of curing residual stresses in composite using multi-scale method. Compos. Part B-Eng. **155**, 49–61 (2018). https://doi.org/10.1016/j.compositesb.2018.08.012
16. Carlone, P., Rubino, F., Paradiso, V., Tucci, F.: Multi-scale modeling and online monitoring of resin flow through dual-scale textiles in liquid composite molding processes. Int. J. Adv. Manuf. Technol. **96**, 2215–2230 (2018). https://doi.org/10.1007/s00170-018-1703-9
17. Lionetto, F., Moscatello, A., Totaro, G., Raffone, M., Maffezzoli, A.: Experimental and numerical study of vacuum resin infusion of stiffened carbon fiber reinforced panels. Materials **13**(21) (2020). https://doi.org/10.3390/ma13214800
18. Rocha, H., Semprimoschnig, C., Nunes, J.P.: Sensors for process and structural health monitoring of aerospace composites: a review. Eng. Struct. **237**, 112231 (2021). https://doi.org/10.1016/j.engstruct.2021.112231
19. Brauner, C., Frerich, T., Herrmann, A.S.: Cure-dependent thermomechanical modelling of the stress relaxation behaviour of composite materials during manufacturing. J. Compos. Mater. **51**, 877–898 (2017). https://doi.org/10.1177/0021998316656924
20. Cameron, C.J., Saseendran, S., Stig, F., Rouhi, M.: A rapid method for simulating residual stress to enable optimization against cure induced distortion. J. Compos. Mater. **55**(26), 3799–3812 (2021). https://doi.org/10.1177/00219983211024341
21. Muliana, A.H.: Spatial and temporal changes in physical properties of epoxy during curing and their effects on the residual stresses and properties of cured epoxy and composites. Appl. Eng. Sci. **7**, 100061 (2021). https://doi.org/10.1016/j.apples.2021.100061
22. Kondratiev, A., Píštěk, V., Smovziuk, L., Shevtsova, M., Fomina, A., Kučera, P.: Stress–strain behaviour of reparable composite panel with step–variable thickness. Polymers **13**(21), 3830 (2021). https://doi.org/10.3390/polym13213830
23. Startsev, O.V., Vapirov, Y.M., Lebedev, M.P., Kychkin, A.K.: Comparison of glass-transition temperatures for epoxy polymers obtained by methods of thermal analysis. Mech. Compos. Mater. **56**(2), 227–240 (2020). https://doi.org/10.1007/s11029-020-09875-5
24. Deng, B., Shi, Y.Y., Yu, T., Zhao, P.: Influence mechanism and optimization analysis of technological parameters for the composite prepreg tape winding process. Polymers **12**(8), 1843 (2020). https://doi.org/10.3390/polym12081843
25. Budelmann, D., Schmidt, C., Meiners, D.: Prepreg tack: a review of mechanisms, measurement, and manufacturing implication. Polym. Compos. **41**(9), 3440–3458 (2020). https://doi.org/10.1002/pc.25642
26. Brinson, H.F., Brinson, L.C.: Polymer Engineering Science and Viscoelasticity: An Introduction, 2nd edn., p. 482. Springer (2015). https://doi.org/10.1007/978-1-4899-7485-3
27. Baran, I., Cinar, K., Ersoy, N., Akkerman, R., Hattel, J.H.: A review on the mechanical modeling of composite manufacturing processes. Arch. Comput. Methods Eng. **24**, 365–395 (2017). https://doi.org/10.1007/s11831-016-9167-2
28. Shah, V.: Handbook of Plastics Testing and Failure Analysis, 4th edn. Wiley, Hoboken (2020)

Development of a Digital Twin of Reservoir Filling by Gas Mixture Component

Olga Shypul, Vadim Garin, Denys Tkachenko, Sergiy Zaklinskyy, Oleg Tryfonov, and Sergiy Plankovskyy

Abstract The subject of research is a development of a digital twin of the process of filling the reservoir with gas mixture components. The problem is motivated by the concept of digitalization of modern production and is considered in relation to the researched technology of thermal pulse processing. The features of the operation of the fuel mixture generator according to the method of critical holes are determined and the corresponding determining equations for the dosing of gas mixture components and the reservoir filling time is given. It analyzed the modes of operation of the digital twin and selected the settings of the model of operation of the digital twin, which ensure a rational reproduction of the gas-dynamic non-stationary process of filling the tank with a component of the gas mixture. A gas-dynamic unsteady flow model was built and a numerical study was carried out using the ANSYS Fluent software. A reduced-order model (ROM mod-el of ANSYS Fluent) was developed and used in ANSYS Twin Builder for the construction of a digital twin. Developed examples of digital twins of the system of filling the tank with a gas mixture using standard elements of the Twin Builder and Modelica libraries.

Keywords Digital Twin · Reduced Order Model · Numerical Simulation · Filling the Reservoir

1 Introduction

Important quality indicators of the precision parts and mechanisms, such as service live and reliability, are mostly determined by the quality of edge processing

O. Shypul (✉) · V. Garin · D. Tkachenko · S. Zaklinskyy · O. Tryfonov
National Aerospace University "Kharkiv Aviation Institute", 17 Chkalova Str, Kharkiv 61070, Ukraine
e-mail: o.shipul@khai.edu

S. Plankovskyy
O.M.Beketov National University of Urban Economy, 17 Marshala Bazhanova Str, Kharkiv 61002, Ukraine

S. Shkarlet et al. (eds.), *Mathematical Modeling and Simulation of Systems*, Lecture Notes in Networks and Systems 667,
https://doi.org/10.1007/978-3-031-30251-0_7

and surface cleaning. Despite the almost perfect mechanical processing technologies, the formation of various contaminations during the parts production is inevitable. Finishing technologies, for example, deburring, precision machining, surface cleaning, production of typical reliefs, etc., take a significant share of the overall labor intensity of the production of parts [1–4]. Finishing operations by fuel mixture combustion products, which are carried out by various thermal methods, including thermal pulse, have unique advantages in terms of efficiency and technological possibilities [5, 6]. However, an urgent issue regarding the implementation of these technologies is the difficulty of determining, configuring and ensuring the optimal equipment operating regimes in order to achieve the expected accuracy of processing [7].

The technology of thermal pulse finishing takes place due to a set of complex physical and chemical processes. It can be carried out by various fuel mixtures with different mixture composition and homogeneity, different initial pressure conditions and combustion modes, etc. The multifactorial nature of these processes is the main difficulty not only for determining the necessary technological regimes, but also for predicting the of both the processing progress and the processing result.

In accordance with the strategy of the industrial revolution Industry 4.0 [8, 9], as well as with flexibility, productivity, automation, requirements are placed to the digitalization of production processes and the construction of processes digital twins (Digital Twin).

A digital twin is created as a virtual copy of a physical object for use as a simulator, predictive model and/or tool for diagnosing and adjusting equipment through real-time monitoring. At controlling of the technological process, the process digital twin allows you to test and verify different flow strategies, while the performance of the control system depends on the available sensors and executive mechanisms of the digital twin [10].

Regarding the technology of thermal pulse processing, the construction of a process digital twin requires usage of a complex model of all equipment operation processes, including gas-dynamic processes at generation and working chamber filling by gas mixture [11], fuel ignition [12] and combustion [13], heat exchange at objects processing [14], release of combustion products [15], etc. Obviously, the specified complex model creation, moreover, the implementation of calculations based on it, is an unnecessarily difficult task. Therefore, it is preferable to build digital twins of individual physical and chemical processes inherent to the thermal pulse processing [16], and in the future the total digitalization of the whole technology.

A typical cycle of the development digital twin development and its subsequent use for simulation the technological process is considered in works [17–24]. The main activities during the development of the model are the problem definition and the goal simulation, collection and analysis of data from the real system, development and verification of the digital twin model concept that represents the real system, the application of the digital twin into the technological process and comparison with the real equipment operation.

In this study, the methodology of digital twins is applied to the fuel mixture generation system of the thermal pulse processing set-up. In particular, within the framework of this work, the subsystem of reservoir filling by a gas mixture was considered in order to determine and justify the choice of the digital twin model for the specified system.

2 Fuel Mixture Generator for Thermal Pulse Processing

To ensure a stable value of the specific heat flow during cyclic thermal pulse processing, simultaneous control of the fuel mixture mass and pressure charge in the combustion chamber is required directly during the mixture generation and filling process, with subsequent correction of the processing time according to the determined initial temperature of the fuel mixture in the chamber. According to the specified requirement, the fuel mixture generation system of the thermal pulse processing set-up was developed, by the authors of this work, and is detailed described in [7]. Thus, the fuel mixture generator (Fig. 1) includes fuel, oxidizer, and neutral gas supply units, which consist of the following main components: gas cylinders (20, 30, 40), reducers (21, 31, 41) and valves (25, 35, 45) at the entrance to the installation. Pressure sensors BP81, BP91, BP109 designed for pressure up to 24 MPa are installed in the cylinders. The block of receivers (intermediate vessels) consists of the valves YA81, YA91, YA110 for gas filling of the receivers (oxidizer TANK82, fuel TANK92, neutral gas TANK110), the pressure sensors BP82, BP92(1), BP92(2), BP110, the temperature sensors BK82, BK92(1), BK92(2), BK110. The mixer unit consists of the two-way electromagnetic valves YA83, YA93, YA111, the MIXER100 mixer with replaceable nozzles. The filling chamber CHAMBER60 (the reservoir) is equipped with the pressure sensors BP61(1), BP61(2) and the temperature sensors BK61(1), BK61(2).

To ensure the specified accuracy of the fuel mixture in terms of its component ratio and homogeneity, the operation of this system is carried out according to the dynamic method, which is based on the effect of supercritical outflow of gases from receivers through calibrated holes in the mixer [25–27].

Determining equations for dosing the components of the gas mixture and the time to fill the reservoir by the mixture. According to the method given in [28], the instantaneous value of the mass flow rate through the critical opening at a supercritical pressure drop is determined by the equation:

$$G = \frac{\mu F P}{\sqrt{RT}}\psi, \tag{1}$$

where $\psi = \sqrt{k\left(\frac{2}{(k+1)}\right)^{\frac{k+1}{k-1}}}$; μ is the nozzle discharge coefficient; F is the cross-section area of the nozzle critical hole; k, R is the adiabatic exponent and the gas constant of outflowing gas.

The current values of gas temperature and pressure in the vessel under the condition of adiabatic flow are determined by the equations

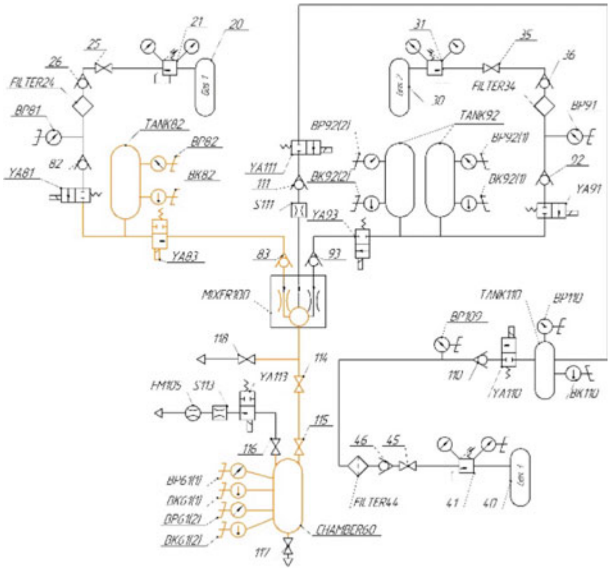


Fig. 1 The schematic of the fuel mixture generator

$$T = T_0 \left(\frac{P}{P_0} \right)^{\frac{k-1}{k}}, \tag{2}$$

$$P = P_0 (1 + Bt)^{\frac{-2k}{k-1}}, \tag{3}$$

where T_0, P_0 is the initial gas temperature and pressure; V is the volume of the vessel and

$$B = \frac{(k-1) F \sqrt{R T_0}}{2V} \psi.$$

The critical holes areas ratio is established based on the conditions at the initial time instance: $\frac{G_1}{G_2} = \beta$, $P_1 = P_2 = P_0$, $T_1 = T_2 = T_0$. Then, using (1) we obtain:

$$F_1 = \beta \cdot F_2 \frac{\mu_2}{\mu_1} \frac{\sqrt{M_2}}{\sqrt{M_1}} \frac{\psi_2}{\psi_1}, \tag{4}$$

where $M_1,\ M_2$—molar masses of the gases which form the mixture.

We will consider equal temperature in the both fuel and oxidizer receivers. Substituting expressions (3) and (4) for both components into formula (2) for the current temperature, we obtain that the ratio $T_1 = T_2$ holds identically at the equality of the initial temperatures $T_{10} = T_{20} = T_0$ and the ratio of the volumes of the receivers determined by the expression:

$$V_1 = \beta \cdot V_2 \frac{\mu_2}{\mu_1} \frac{M_2}{M_1} \frac{k_1 - 1}{k_2 - 1}. \tag{5}$$

Taking into account the fact that the pressure in the gases receivers and, respectively, the mass flow rates of the components during filling will change differently, to provide the specified accuracy of the mass concentration of gases in the mixture it is necessary to set the initial pressures in the receivers based on the expression:

$$\frac{\int_0^\tau G_1 dt}{\int_0^\tau G_2 dt} = \beta, \tag{6}$$

where τ – time of filling the chamber by the mixture.

After substituting the expressions (1–3) in the (6), it is transformed to the form as following:

$$\frac{P_{01}}{P_{02}} = \frac{\int_0^\tau (1 + B_2 t)^{-\frac{k_2+1}{k_2-1}} dt}{\int_0^\tau (1 + B_1 t)^{-\frac{k_1+1}{k_1-1}} dt}. \tag{7}$$

The time of reservoir filling by the mixture is defined as the time during which, for one of the components, the mass fed into the chamber becomes equal to the set value. Based on the accepted assumptions, this leads to the following dependence for one of the components:

$$m_2 = \int_0^\tau G_2 dt, \tag{8}$$

after integration taking into account (1), (2), (3), it determines the time of filling:

$$\tau = \frac{1}{B_2}\left(\left(\frac{A_2}{A_2 - C_2}\right)^{\frac{k_2-1}{2}} - 1\right), \tag{9}$$

where $A_2 = \mu_2 F_2 \psi_2 P_{02}(k_2 - 1)$ and $C_2 = 2m_2 B_2 \sqrt{R_2 T_0}$.

Accepted Assumptions and Deviations from the Real Filling Process.
Firstly, the time calculated according to dependence (7) refers to the time of gas leakage from the receivers, and differs from the time of filling the reservoir by the amount needed for components flow through the pipe line, valves operation, mixing in the mixer, etc.

Secondly, the volume of the components of the mixture remaining in the pipeline can be determined based on the geometry of the intermediate tract, however, due to the significant influence of heat exchange with the walls of the pipeline, the error in the mixture components mass calculation with the assumption of adiabaticity of the process significantly affects the effective dosing accuracy.

Thirdly, the given technique for calculating the filling time assumes a defined ratio of the receiver volumes in accordance with the ratio of the gas mixture components, i.e. when the fuel mixture is changed, the volume of one of the receivers needs to be adjusted.

And most importantly, monitoring the fuel mixture quality according to its component composition in the generator control system should be based on the mass of entered the reservoir (combustion chamber) components, calculated according to the equation of state for an ideal gas (Clapeyron-Mendeleev law):

$$PV = \frac{m}{M}RT. \tag{10}$$

However, it should be noted that the problem of correct mass calculation in such a case is the determination of the gas temperature averaged over the volume of the reservoir, which, due to slow mixing, differs significantly from the temperature at the point of measurement by the corresponding sensor. Moreover, it is impossible to place any sensors in the working combustion chamber of the thermal pulse machine at all due to high operating temperatures and pressure. Therefore, it makes sense to install pressure and temperature sensors at the junction of the main tract with the reservoir (combustion chamber). In this case, the mixture components in the reservoir can be determined as the difference between the masses of the component that left the receiver and remained in the main tract.

In addition, control of the mixture generation process using available pressure sensors requires the development of the pressure change law in the reservoir taking into account the time of filling, the heat exchange of gases with the walls of the combustion chamber during its cyclic operation.

3 Numerical Study of the Reservoir Filling Subsystem with a Gas Mixture Component

The fuel, oxidizer, and neutral gas supply units of the fuel mixture generator are built according to the same schemes. Therefore, for further research without loss of commonality, it was considered one subsystem, in particular, the oxidizer unit (Fig. 2), for which the task of the digital twin building of the reservoir filling process by the gas mixture component was solved.

Using the finite element method, the reservoir (combustion chamber of the thermal pulse unit) filling process by the mixture gas component, in particular nitrogen, was investigated. The simulation was carried out using the ANSYS Fluent software [29]. A calculation 3D model is shown on Fig. 3.

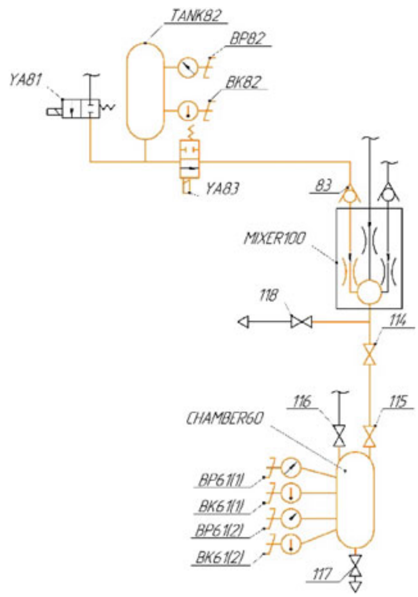


Fig. 2 The schematic of the gas mixture component filling block

Fig. 3 The 3D model of the oxidizer supply unit

A detailed description of the similar model creation, defining equations and settings of the solver are given in [28]. The following values for N2 are accepted as initial data for calculations: $\rho = 1{,}251$ kg/m^3, $M = 28{,}0134 \times 10^{-3}$ kg/Mole, $k = 1{,}34$, $R = 297$ J/(kg·K). The initial values of the pressure in the receiver and the reservoir are taken as equal to 0.5 MPa and 0.1 MPa, respectively. The initial temperature in the subsystem is 293 K. The volume of the reservoir is 15×10^{-3} m^3. The diameter of the critical hole in the mixer is 1 mm. The gas flow time is 10 s.

The following simulation results were obtained for further creation, adjustment and verification of the digital twin operation.

The main thermodynamic parameters of the non-stationary gas-dynamic flow during the filling period were determined, particularly, pressure change in the control point of the reservoir volume during the filling time (Fig. 4, a); temperature change at the control point of the reservoir volume during the filling time (Fig. 4, b); heterogeneity of gas temperature distribution in the volume (Fig. 5) and deviation of the current gas temperature at the control point from the averaged one in the volume, determined by dependence (11) (Fig. 6); change of the mixture component mass in the reservoir during the filling time (Fig. 7), calculated using to dependence (12); etc.

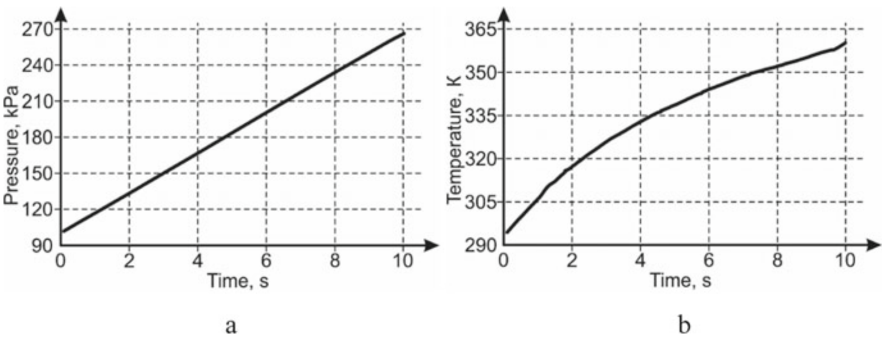
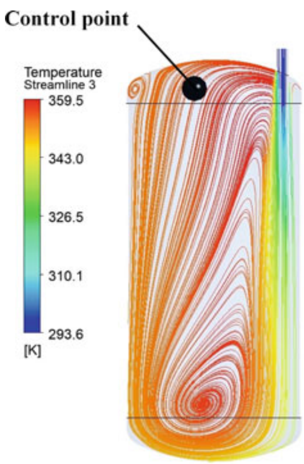


Fig. 4 Pressure (a) and temperature (b) change in the reservoir vs the filling time

Fig. 5 Lines of the gas temperature level at the instance of the reservoir filling



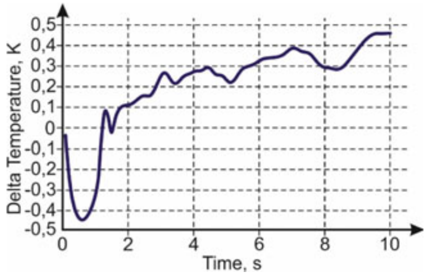


Fig. 6 Deviation of the current gas temperature in the control point from the volume-averaged temperature

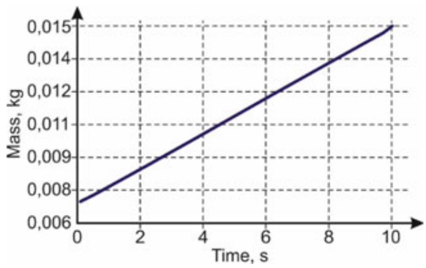


Fig. 7 Change of the gas mass in the reservoir vs the filling time

The CEL language was used to determine the time dependence of the calculated parameters of the process. Deviation of the gas temperature in the control point from the volume-averaged temperature is determined by the dependence:

$$\Delta(t) = \left(\frac{T(t)@Point - volumeAve(T(t))@Reservoir}{volumeAve(T(t))@Reservoir}\right) \cdot 100\%. \qquad (11)$$

The gas mass in the calculated volume during the chamber filling is determined by integrating the gas density value:

$$m(t) = volumeInt(\rho(t))@Reservoir. \qquad (12)$$

The presence of a numerical model of the process allows to obtain a qualitative picture of the functioning of the filling subsystem, and further research shows its use in the structure of a digital twin.

4 Development of a Digital Twin of Reservoir Filling by Gas Mixture Component

There are several ways to build digital twins of processes, but they have common principle of processing and comparing information which come from the real sensors at measuring of the process parameters with information from the virtual sensors of a digital twin. In addition to general process management, the digital twin allows to detect and prevent disruptions to the normal operation of the equipment or the process going and establishing the causes of their occurrence. The work of a digital twin can be organized either online or offline. Additionally, digital twins can be used to evaluate the results of different management strategies. Exactly the digital twin's construction of the processes guarantees quick reconfiguration of the system for specific tasks with automatic determination of the optimal regime for achieving the predicted processing accuracy [9, 10].

Ansys Inc. proposes the concept of a digital twin using ANSYS Twin Builder numerical simulation technologies [30]. The results of modeling using digital twins can be calibrated based on the operating parameters of the physical process. The direct connection between the model of finite elements in its common understanding (for example, Fig. 3) and the control scheme leads to a slow speed of calculation and predicting of the initial parameters. Therefore, at the development of the digital twin, reduced order models (ROM) are used. Creating a ROM model is possible by studying the physics of the process and extracting its global behavior from standalone simulation. It can be directly obtained based on the preliminary finite-element modeling results, taking into account only the main features of the system and process. Since the calculation of the result in a separate digital ROM-object requires only simple algebraic operations (for example, summation of vectors and evaluation of response surfaces), the calculation process does not require a lot of technical time resources, and the result can be obtained almost in real time times with parametric ROM simulation, you can evaluate the model and quickly explore the variation of the results depending on the input parameters values. Its underlines the importance of a qualitatively created ROM model at simulating of the complex multifunctional system operation.

The same physical system can be reproduced by a different set of tools from the Twin Builder and Modelica libraries (which can also be accessed in ANSYS Twin Builder), depending on the particularities of the study and the availability of input and desired output data. During the next step we will list the main components that were used to create a digital twin of the considered thermal pulse unit line (Fig. 2).

Calculation Schemes of the Numerical Twin.
The main element of the digital twin is a calculation scheme that provides integration of the ANSYS FLUENT solver [29] (Fig. 8), determination of equipment operation regimes and equipment operation simulation for each of them.

The left side of the "Static ROM" scheme (Fig. 8) shows the input parameters of the process, which are received either in the tabular data from the program code, or can be set in an analytical form, in particular, to study pressure and temperature changes at the entrance to the reservoir. The right side of this scheme presents a list of possible operating regimes of the object, particularly, in our case, the expected mass of the mixture component, etc.

More advanced way of the digital twin creating with ability to monitor and change the input data for the estimated ROM model requires writing special software applications, user extensions, or macros. For this, in ANSYS Twin Builder there are

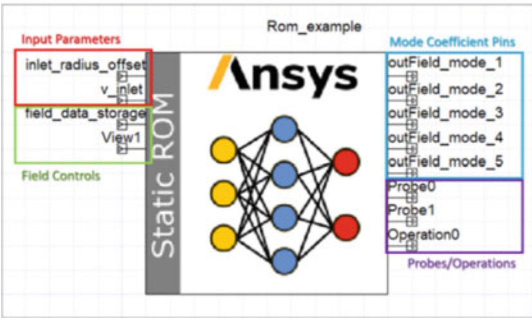


Fig. 8 An example of a reduced-order ANSYS model control element

special libraries of standard component parts in (Fig. 9), and the component-oriented Modelica language integrated into it, according to which the digital twin of the researched process acquires a schematic form (Fig. 10).

At creation of the digital twins of the filling system elements, the standard elements from the TwinBuilder and Modelica libraries [30] were used: table of initial conditions, source tank, pipes, jet, receiver tank, sensors and control elements of output parameters (pressure, temperature), etc.

In further research, the development of ROM models for the digital twin of the two-component gas mixture mixer is planned. Comparison of the digital twin and equipment operations, assessment of the accuracy of the numerical experiment's initial parameters of with natural ones. At the final stage, the digital twin will be integrated into the automated control system of the mixture formation process.

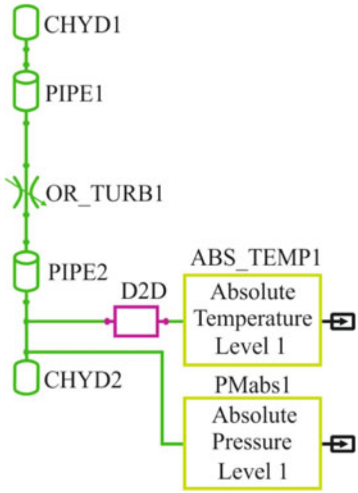


Fig. 9 A digital twin of the reservoir filling subsystem by the gas mixture component, created using the TwinBuilder standard library

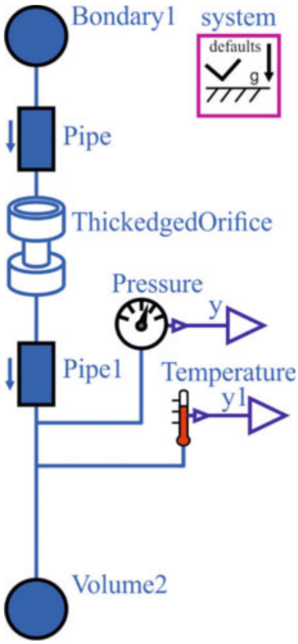


Fig. 10 A digital twin of the reservoir filling subsystem by the gas mixture component, created using the component-oriented Modelica language

5 Conclusion

In accordance with the concept of digitalization of modern production in relation to the researched technology of thermal pulse processing, the need of digital twin development for the inherent separate physical and chemical processes is substantiated. The features of the fuel mixture generator operation by the method of critical holes are determined and the corresponding determining equations for the components of the gas mixture dosing as well as the time of filling the tank are given.

For further use in the structure of a digital twin of a separate reservoir filling subsystem by one of the gas mixture components, a gas-dynamic unsteady flow model was built and a numerical study was carried out using the ANSYS Fluent software. A reduced-order model (ROM model, ANSYS Fluent) was developed and used in the ANSYS Twin Builder for the construction of the digital twin. Developed examples of the digital twins of the reservoir filling system by the gas mixture using standard elements of the Twin Builder and Modelica libraries.

References

1. Ramachandran, N., Pande, S.S., Ramakrishnan, N.: The role of deburring in manufacturing: a state-of-the-art survey. J. Mater. Process. Technol. **44**(1–2), 1–13 (1994). https://doi.org/10.1016/0924-0136(94)90033-7

2. Biermann, D., Heilmann, M.: Burr minimization strategies in machining operations. In: Aurich, J., Dornfeld, D. (Eds.), Burrs – Analysis, Control and Removal, pp. 13–20. Springer, Heidelberg (2010).https://doi.org/10.1007/978-3-642-00568-8
3. Jin, S.Y., Pramanik, A., Basak, A.K., et al.: Burr formation and its treatments – A review. Int. J. Adv. Manuf. Technol. **107**(5–6), 2189–2210 (2020). https://doi.org/10.1007/s00170-020-05203-2
4. Obeidi, M.A., McCarthy, E., Ul Ahad, I., et al.: Improving the surface finish and other properties of engineering metal parts. In: Key Engineering Materials, vol. 813, pp. 197–202 (2019). https://doi.org/10.4028/www.scientific.net/KEM.813.197
5. Benedict, G.F.: Thermal energy method: deburring (TEM). Non-traditional Manufacturing Processes, pp. 349–361. CRC Press, Boca Raton (2017)
6. Fritz, A., et al.: Experimental analysis of thermal energy deburring process by design of experiment. In: Proceedings of the ASME 2012 International Mechanical Engineering Congress and Exposition, vol. 3, pp. 2035–2041. ASME (2012)
7. Plankovskyy, S., Popov, V., Shypul, O., et al.: Advanced thermal energy method for finishing precision parts. In: Gupta, K., Pramanik, A. (eds.) Advanced Machining and Finishing, pp. 527–575. Elsevier, Amsterdam (2021). https://doi.org/10.1016/B978-0-12-817452-4.00014-2
8. Xu, L.D., Xu, E.L., Li, L.: Industry 4.0: state of the art and future trends. Int. J. Prod. Res. **56**(8), 2941–2962 (2018). https://doi.org/10.1080/00207543.2018.1444806
9. Mubarak, M.F., Petraite, M.: Industry 4.0 technologies, digital trust and technological orientation: what matters in open innovation? Technol. Forecast. Soc. Change **161**(120332) (2020). https://doi.org/10.1016/j.techfore.2020.120332
10. Reed, S., Löfstrand, M., Andrews, J.: Modelling cycle for simulation digital twins. Manuf. Lett. **28**, 54–58 (2021). https://doi.org/10.1016/j.mfglet.2021.04.004
11. Liu, J., Zheng, S., Zhang, Z., et al.: Numerical study on the fast filling of on-bus gaseous hydrogen storage cylinder. Int. J. Hydrogen Energy **45**(15), 9241–9251 (2020). https://doi.org/10.1016/j.ijhydene.2020.01.033
12. Raizer, Y.P.: Gas Discharge Physics. Springer, Berlin (1991)
13. Assovskii, I.G., Merzhanov, A.G.: Validity of experimental and theoretical modeling of combustion of high-energy materials. Combust. Explosion Shock Waves **49**(3), 264–272 (2013). https://doi.org/10.1134/S0010508213030027
14. Quintens, H., et al.: Experimental wall heat transfer measurements for various combustion regimes: deflagration, autoignition and detonation. In; AIAA Propulsion and Energy 2019 Forum, p. 4381 (2019).https://doi.org/10.2514/6.2019-4381
15. Sobieski, W., Grygo, D.: Fluid flow in the impulse valve of a hydraulic ram. Tech. Sci. **22**(3), 205–220 (2019). https://doi.org/10.31648/ts.4352
16. Gehrmann, C., Gunnarsson, M.: A digital twin based industrial automation and control system security architecture. IEEE Trans. Industr. Inf. **16**(1), 669–680 (2019). https://doi.org/10.1109/TII.2019.2938885
17. Cimino, C., Negri, E., Fumagalli, L.: Review of digital twin applications in manufacturing. Comput. Ind. **113**, 1–15 (2019). https://doi.org/10.1016/j.compind.2019.103130
18. Qinglin, Q., Fei, T., Tianliang, H., et al.: Enabling technologies and tools for digital twin. J. Manuf. Syst. **58**(B), 3–21 (2021). https://doi.org/10.1016/j.jmsy.2019.10.001
19. Söderberg, R., Wärmefjord, K., Carlson, J.S., et al.: Toward a digital twin for real-time geometry assurance in individualized production. Manuf. Technol. **66**(1), 137–140 (2017). https://doi.org/10.1016/j.cirp.2017.04.038
20. Kritzinger, W., Karner, M., Traar, G., et al.: Digital twin in manufacturing: a categorical literature review and classification. IFAC-PapersOnLine **51**(11), 1016–1022 (2018). https://doi.org/10.1016/J.IFACOL.2018.08.474
21. Sinner, P., Daume, S., Herwig, C., Kager, J.: Usage of digital twins along a typical process development cycle. Adv. Biochem. Eng. Biotechnol. **176**, 71–96 (2021). https://doi.org/10.1007/10_2020_149
22. Brovkova, M., Molodtsov, V., Bushuev, V.: Implementation specifics and application potential of digital twins of technological systems. Int. J. Adv. Manuf. Technol. **117**, 2279–2286 (2021). https://doi.org/10.1007/s00170-021-07141-z

23. Robles, J., Baca, G., Chong, J., et al.: Nonsingular terminal sliding mode control for a variable speed wind turbine system using face mock-up interface co-simulation. In: 11th International Conference on Power, Energy and Electrical Engineering (CPEEE), pp. 158–164. IEEE (2021), Rutgers University.https://doi.org/10.1109/CPEEE51686.2021.9383360
24. Nauri, I.M., Ihwanudin, M., et al.: Development of integrated hardware-in-the-loop (hil) test bench anti-lock brake system (abs) instrument. J. Phys.: Conf. Ser. **1700**, 012097 (2020). https://doi.org/10.1088/1742-6596/1700/1/012097
25. Plankovskyy, S., Shypul, O., Zaklinskyy, S., Tsegelnyk, Y., Kombarov, V.: A method of rapid measurement of vessels volume with complex shape by critical nozzles. In: Nechyporuk, M., Pavlikov, V., Kritskiy, D. (eds.) Integrated Computer Technologies in Mechanical Engineering – 2020, ICTM 2020. Synergetic Engineering Kharkov Ukraine 29 10 2020–30 10 2020. Lecture Notes in Networks and Systems LNNS, vol. 188, pp. 247–255. Springer Cham (2021). https://doi.org/10.1007/978-3-030-66717-7_20
26. Sysoiev, Y.O.: The creation of multicomponent gas mixtures for ion-plasma technologies. Prob. Atomic Sci. Technol. **2**, 137–142 (2014)
27. Song, E., Liu, Z., Yang, L., Yao, C., Sun, J., Dong, Q.: Effects of nozzle structure on the gas mixture uniformity of marine gas engine. Ocean Eng. **142**, 507–520 (2017). https://doi.org/10.1016/j.oceaneng.2017.07.011
28. Shypul, O., Plankovskyy, S., Zaklinskyy, S., Pavlenko, O., Garin, V. (2022). Determination of the Mass of Gas in a Reservoir at Filling with a Mixture Component Under the Pressure. In: Nechyporuk, M., Pavlikov, V., Kritskiy, D. (eds) Integrated Computer Technologies in Mechanical Engineering - 2021. ICTM 2021. Lecture Notes in Networks and Systems, vol. 367, pp. 166–177. Springer, Cham (2021). https://doi.org/10.1007/978-3-030-94259-5_16
29. ANSYS Fluent Reference Guide (2022). https://ansyshelp.ansys.com/account/secured?returnurl=/Views/Secured/prod_page.html?pn=Fluent&pid=Fluent&lang=en. Accessed 14 Feb 2022
30. ANSYS Twin Builder Reference Guide (2022). https://ansyshelp.ansys.com/account/secured?returnurl=/Views/Secured/prod_page.html?pn=Twin%20Builder&pid=TwinBuilder&lang=en. Accessed 14 Feb 2022

Robust and Adaptive Control Systems for the Management of Technological Facilities of the Food Industry

Nataliia Lutska, Lidiia Vlasenko, Tetyana Herasymenko, and Serhii Hrybkov

Abstract The article is devoted to a topical problem related to the stage of creating prototypes of systems. Prototyping is one of the stages typical for the design and creation of digital duplicates and cyber-physical systems of manufacturing enterprises within the Industry 4.0 concept. The paper considers the operation of a robust and adaptive control system based on the switching between the robust and optimal controllers, while the change in the system structure depends on the change in the specified acceptable range of values of the defined factors. For this purpose, there was carried out an analysis of aspects on which depend the structure and parameters of the control system being designed for technological facilities of the food industry. The results are provided in the form of a cause and effect diagram.

The paper offers a structure of an adaptive control system with an identifier based on robust and optimal controllers and a fuzzy inference unit. To reveal the contents of the latter, there is given a fragment of the decision tree on the choice of the controller structure. In addition, the work provides schemes and results of the simulation modeling of the robust and adaptive control system with switching for a heat exchanger at nominal values of object uncertainties. Control system design is carried out in Matlab/Simulink. In the course of the research, there was conducted a series of experiments under different specified conditions and requirements. The results of the design and their subsequent analysis are presented in a convenient format which proves the efficiency of the proposed method.

In the future, it is planned to improve the system by adding new model structures and intelligent controllers of various types. Moreover, it is planned to conduct independently a series of studies to improve the logical inference unit by including additional fuzzy rules and ontology-based solution schemes.

N. Lutska (✉) · T. Herasymenko · S. Hrybkov
National University of Food Technologies, Kiev 01601, Ukraine
e-mail: lutskanm2017@gmail.com

S. Hrybkov
e-mail: sergio_nuft@nuft.edu.ua

L. Vlasenko
State University of Trade and Economics, Kiev 02156, Ukraine

S. Shkarlet et al. (eds.), *Mathematical Modeling and Simulation of Systems*, Lecture Notes in Networks and Systems 667,
https://doi.org/10.1007/978-3-031-30251-0_8

Keywords Controller · Switching · Control system · Technological process

1 Introduction

The creation of digital duplicates and the construction of cyber-physical systems are nowadays one of the actively developing directions of research [1–3]. These principles are fundamental for the construction of a modern industrial manufacturing enterprise. Technological facilities of the food industry (TFFI) have always been and remain very attractive and promising for the introduction of new technologies. In particular, it is due to the fact that nutrition is always relevant to humanity and technologies which produce food always improve [4].

One of the key requirements for the construction of an adequate digital duplicate of the production and the creation of a diverting cyber-physical system is the use of mathematical models with the specified accuracy [5]. TFFI refers to the objects which have uncertainties, and which are non-linear, non-stationary, multidimensional, etc. It is often the case that traditional controllers do not provide accuracy, speed, and quality of regulation that would be sufficient for the achievement of the specified energy efficiency and energy saving. Therefore, not only the use of robust and optimal controllers but also their integration into the control system becomes relevant. The use of systems with switching is considered to be particularly effective [6].

Characteristic features of technological facilities (technological processes, machines, complexes) are significant nonlinearities, nonstationarity, distribution of coordinates, which are associated with the simultaneous processes of heat and mass transfer, hydrodynamics, and physicochemical transformations of the substance to ensure the production output of the specified volume and quality. In the tasks of analysis and synthesis of the automation system of such objects, there arise significant difficulties and additional requirements to ensure stable states of operation in different modes [7–9].

Analysis of the current state of automated control systems of TFFI aimed at the improvement of the automation system quality provides for the feasibility of the use of some special methods (adaptation, invariance to disturbances, delay compensation, sensitivity theory, etc.), and some approaches including combining adaptive, robust, optimal and switching systems [10–13].

The efficiency of classical PID controllers has been proven in many studies. The literature describes various methods for adjusting the controller, including the Ziegler-Nichols method, the Tyreus-Leuben method, damped oscillations, the Cohen and Kuhn method, and CHR [14–21]. However, the use of classical controllers in complex systems implies several problems associated with restrictions of various types. To compensate for them, it is proposed to use adaptive and robust control systems, which, among others, allow distinguishing disturbance signals and compensating for them.

2 Methods

Control systems with adaptation, which are based on the identification of the object parameters, operate in two stages: the identification of the control object parameters and the calculation of the controller's optimal settings. Adaptive control systems with an identifier have several disadvantages, the main of which is in the essence of the identification problem. Identification of TFFI in real time is a rather complex task that is impossible to solve adequately without test input signals (an active identification), since TFFI is constantly influenced by broad-spectrum disturbances, including low-frequency ones. TFFI cannot be constantly under the influence of test signals because in this case the overall quality of system management will de-crease. Unknown external disturbances also signficantly complicate the adaptation algorithm or do not take it into account at all, thus optimizing only the feedback on the change of the task. Therefore, there is a possibility that the system may lose its stability. Moreover, the more complex the model of the control object, the more time it takes to identify it, and at the same time the simpler the model, the lower the quality of the transition control processes built on it. For this reason, as a rule, the adaptation circuit lists the values of the control device parameters at appropriate intervals (quite considerable) or on demand, when the control quality is unsatisfactory. Consequently, adaptive systems of TFFI control based on the identification are only suitable for quasi-stationary facilities.

On the other hand, the robust controller guarantees the system stability and quality in the whole range of changes in the specified uncertainty, but if the area of uncertainty is too large, the control system becomes too rough, and the control system in the nominal mode has poor quality.

There were analyzed the factors on which depend the structure and parameters of the control system being designed for TFFI. The results are provided in the form of a cause and effect diagram (see Fig. 1). First, there were identified the main factors that determine the type of controller to be selected: robust, classic optimal, or robust and adaptive. In the course of the analysis, the factors were ranked, insignificant ones were rejected and the most important and influential ones were kept. The reasons for the first order included system requirements, the degree of object identification, the type of technological facility, the presence of uncertainties, and the mathematical model typical for the selected object. After that, they were detailed with the reasons of the second order.

Evaluating the current information, which includes the characteristics of input and output signals, and the identified model with uncertainties, it is possible to construct a robust and adaptive control system with switching. In this case, if there is enough information, an adaptive system is chosen, if not – then the robust one. The control device in such a system can change the structure when making a transition from one system to another, or not change it, and thus set only the controller parameters, such as PID. The switching system itself can be created on the basis of a fuzzy logic conclusion, which will help get rid of the rigid boundaries of the transition between the types of control.

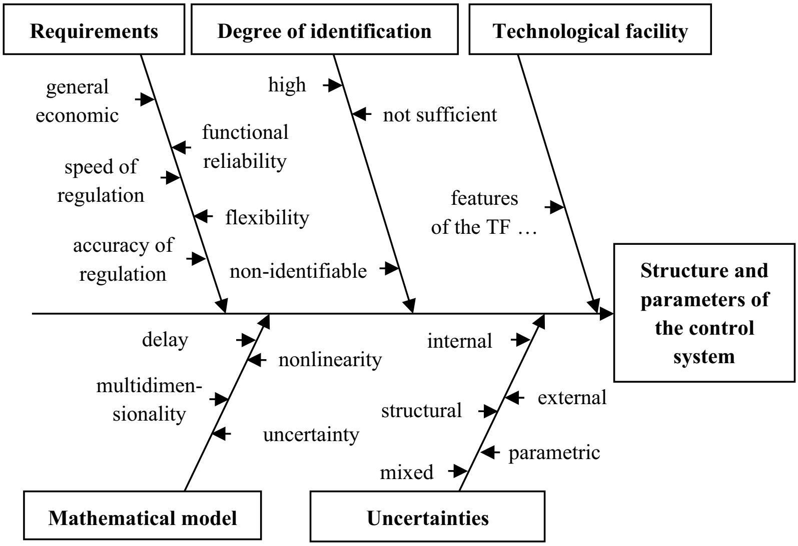


Fig. 1 Cause and effect Ishikawa diagram

2.1 The Structure of the Control System with Switching

Figure 2 shows a block diagram of an adaptive controller with an identifier that uses a fuzzy conclusion regarding the application of the best control law from a number of the specified ones, particularly in robust or optimal.

The control system operates in an iterative mode as follows. During the system initialization, a mathematical model of the object is identified. The identifier includes a unit for generating and implementing test input signals, according to which there is identified the mathematical model of the automation object and used model structures from the set of such structures. During the identification, the control object is open, and the change of the control signal should not result in exceeding the limits of the technological regulations. For this purpose, finite-frequency method of identification [22] or other known parametric methods of identification can be applied. As a result of the identification, we obtain a mathematical model of the object, as well as the dispersion of its parameters. The first step includes the calculation of the structure and parameters of the robust controller which is implemented in a closed circuit.

In the second step, which is activated after a specified period of time, after the identification, the optimal controller is implemented, and the criteria of control quality and dispersion of model parameters are compared.

The first step can also be implemented by the optimal controller, provided that the standard deviation of the parameters fits 20% of the parameter value.

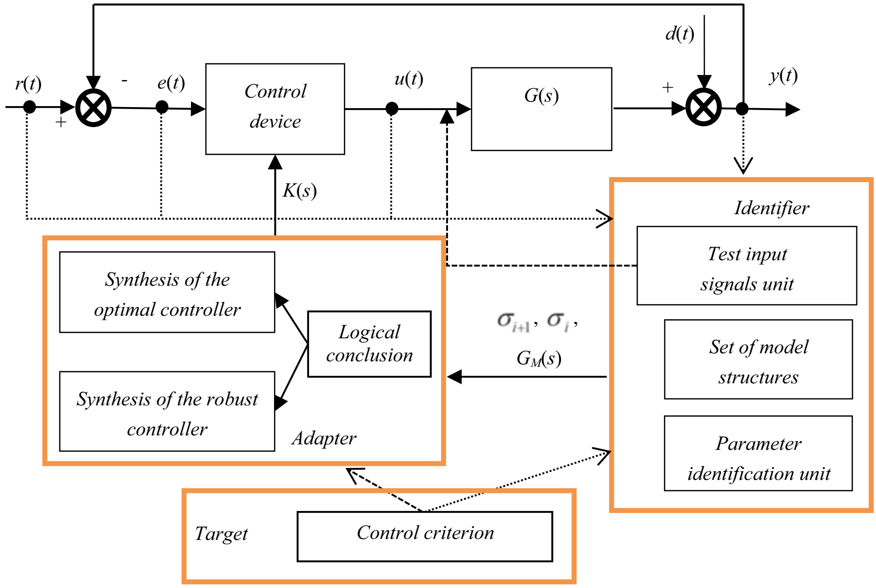


Fig. 2 The structure of the adaptive control system with an identifier based on robust and optimal controllers, as well as a fuzzy inference unit

Subsequent steps of identification are activated in one of the following ways:

– after the corresponding equal periods of time set by the project designer;
– in case of gross violation of technological regulations;
– in case of a significant increase in the control criterion.

After the identification, the following ratio is obtained:

$$\Delta_\sigma = \left[\frac{\sigma^i_{\theta_j} - \sigma^{i-1}_{\theta_j}}{\sigma^{i+1}_{\theta_j}}\right], \tag{1}$$

where Δ_σ is a vector of relative error of identification quality improvement; $\sigma^i_{\theta_j}$ is a standard deviation of the j-th parameter in the i-th step. It is important to note that vector Δ_σ can also be interpreted as the amount of received (when $\Delta_\sigma > 0$) or lost (when $\Delta_\sigma < 0$) information for the given period.

Vector Δ_σ is an input vector of the logical conclusion unit, whose result is a binary signal that switches the controller from the optimal to the robust algorithm.

In Fig. 3, there is provided a fragment of the decision tree for the unit "Logical conclusion" (Fig. 2). In addition, it shows how the choice of the controller (robust, classical optimal, or robust and adaptive) is influenced by the reasons of another kind (Fig. 1) – accuracy of regulation, speed of regulation, and uncertainty.

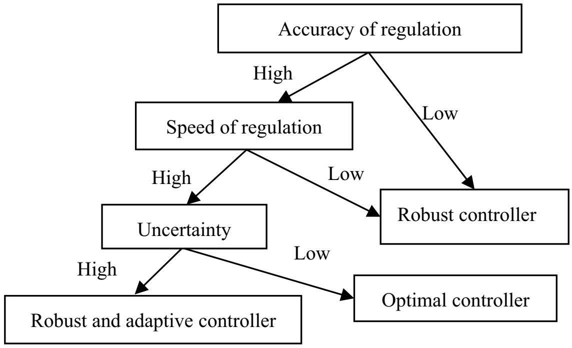


Fig. 3 Fragment of the decision tree regarding the choice of the controller

The proposed control system can be simplified or complemented by additional functions. In particular, the simplest such system can be a control system with different settings of the controller with one structure according to two criteria – optimal and robust. The switching algorithm functions as a logical conclusion on the value of the error of operational identification of the TFFI model.

2.2 *Modeling a Robust and Adaptive Control System*

Let's consider an example of a control system with switching for an industrial shell-and-tube heat exchanger operating under the influence of intense external disturbances and internal changes. Such heat exchangers are used at most Ukrainian food enterprises and especially at sugar factories.

Figure 4 demonstrates a fragment of the scheme for modeling a robust and adaptive control system of a technological facility with switching. The scheme consists of an object whose parameters change *Object*, two PID controllers *PID* and *Ro*), an identification unit *Identification* that evaluates the current values of the parameters of the mathematical model of the object, and a Switching Controller unit, which is a logic output unit. Input disturbance signals were simulated as the sum of random signals with different statistical characteristics.

The control criteria and the obtained system settings are provided in Table 1 (all settings are given in absolute units). The following notation keys are used in the table: $H(s)$, $H_{\text{ref}}(s)$ are a transfer function of the system and its specified form according to the channel of disturbance (an output regulated variable); $S(s)$, $T(s)$ are a transfer function of sensitivity and additional sensitivity; $W_S(s)$, $W_T(s)$, $W_L(s)$, $W_R(s)$ are weight transfer functions; *tau* is the set value of the time constant of the desired transfer function (an aperiodic link of the first order); *gap* – is a maximum relative error between the target and actual response ($gap = \frac{\|y(t)-y_{ref}(t)\|_2}{\|1-y_{ref}(t)\|_2}$); $e(k)$, e_{ref} are an identification error at the control step k and its set value.

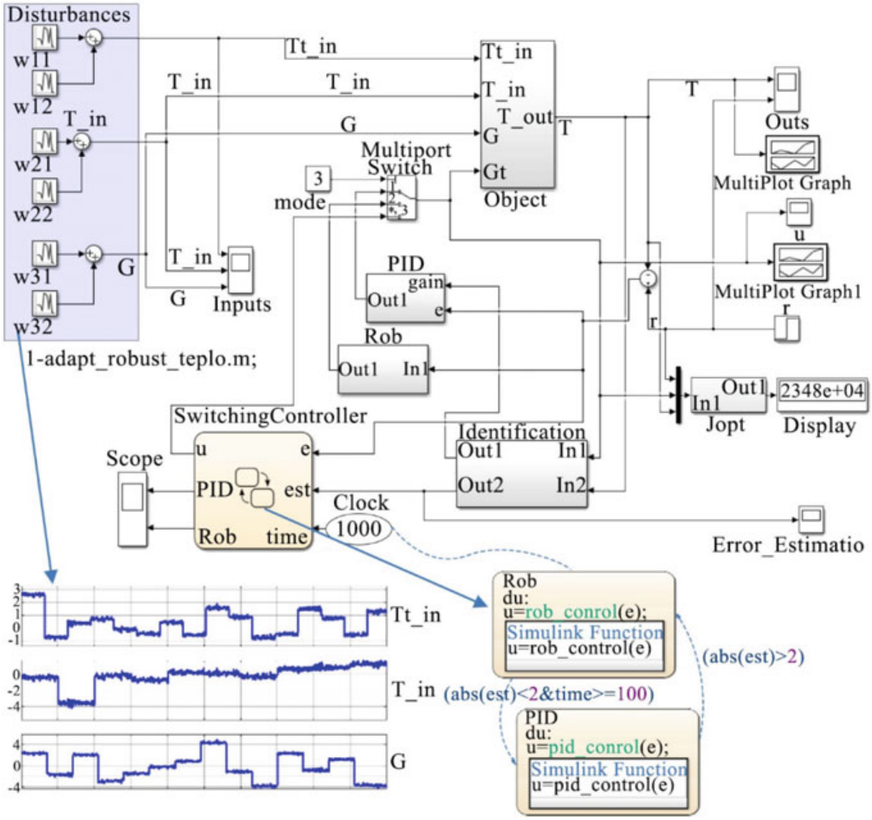


Fig. 4 Simulation scheme (Matlab/Simulink)

The simulation was performed as a series of 11 experiments during which the parameters of the object and the characteristics of the disturbances in a specified wide range were changed. In each experiment, the output value (temperature), the control signal (coolant flow), the status of the identification estimation, and the status of the switching system were observed.

3 Result and Discussion

Simulation of control systems with different settings in the case of combined actions of input disturbances and for nominal and random values of uncertainties showed the expected results. Taking into account the robust properties of the optimal controller (the criterion also uses H_2 and H_∞ norms), there can be used all three control systems, but the third (switching) system has the best quality. As can be seen from the graph of switching states, at the initial stage the robust controller functions, then the

Table 1 Criteria of system management and configuration

Type of controller	Control criterion	System settings
Optimal PID	$f_1 = \begin{cases} \left\| \frac{H(s)}{H_{ref}(s)} \right\|_\infty \\ \frac{\left\| T(s) - \frac{1/tau}{s(s+1/tau)} \right\|_2}{gap \left\| \frac{1}{s} \left(\frac{1/tau}{s+1/tau} - 1 \right) \right\|_2} \end{cases}$	$kp = 2.56$ $ki = 0.19$ $kd = 22.7$ $Tf = 1.58$ s $Ts = 1$ s tau $= 20$ s $H_{ref} : \{y_{max} = 2.5; t = 20\,sec\}$
Robust PID	$f_2 = \left\| \begin{matrix} W_S(s)S(s) \\ W_T(s)T(s) \\ W_L(s)H(s)W_R(s) \end{matrix} \right\|_\infty$	$kp = 9.05$ $ki = 0.007$ $kd = 28.2$ $Tf = 1.58$ s $Ts = 1$ s $W_L = 1$ $W_R = 50$
Switching PID (robust-adaptive PID)	$f_3 = \begin{cases} f_1, e(k) > e_{ref} \\ andk > 1, \\ f_2, e(k) < e_{ref} \end{cases}$	$k = 50$ $e_{ref} = 2$ $Ts = 1$ s

identification of the object model becomes acceptable and the optimal controller is switched on. After that, the robust and optimal controllers are switched on alternately. Figure 5. shows the transitional processes obtained for the nominal control system. The switching system has smaller dynamic errors at each segment of disturbances.

During the analysis of systems, the following evaluation criteria were chosen: integral quadratic criterion for nominal (Nom) and internally perturbed system (UC),

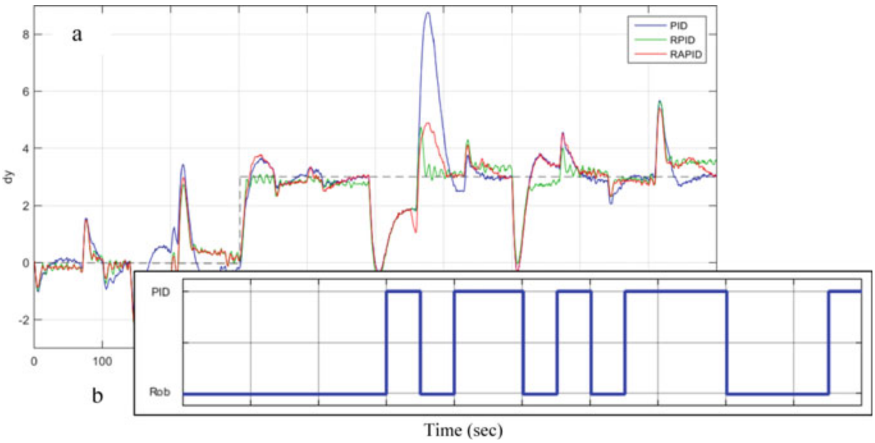


Fig. 5 Simulation results: a – transitional processes in systems with optimal PID controller (PID), robust PID controller (RPID), a robust and adaptive PID controller (RAPID), and a dotted line as a set point; b – states of the switching system

Table 2 Comparative characteristics of control systems

Type of controller	Comparison criteria			
	Integral-quadratic		$\|H^{w-y}(s)\|_\infty$	$\|H^{w-y}(s)\|_2$
	Nom	UC		
Optimal PID	$2.2 \bullet 10^4$	$[2.1 - 4.6] \bullet 10^4$	0.282	0.073
Robust PID	$2.5 \bullet 10^4$	$[1.6 - 2.8] \bullet 10^4$	0.156	0.059
Switching PID (robust-adaptive PID)	$1.8 \bullet 10^4$	$[1.5 - 2.5] \bullet 10^4$	–	–

H_∞- and H_2-norms of closed systems. These estimates are provided in Table 2. On their basis, there can be made a conclusion that the integral quadratic criterion is the smallest one in the switching system.

4 Conclusion

The paper provides schemes and the results of simulation modeling of a robust and adaptive control system with switching for a plate heat exchanger at nominal values of object uncertainties. According to the conducted modeling and the results of the comparison in the proposed system in the area of uncertainty, the quality of transitional processes is improved and the resource for control is reduced in comparison with individual systems.

The offered concept will be complemented by additional model structures, control-ler structures of different types (in particular, certain types of intelligent controllers), as well as improving the logical conclusion unit (introduction of fuzzy rules and addi-tional ontology-based solution schemes).

Due to the rapid development of Industry 4.0 and Digital Twin Technology trends, the solved task is the stage of developing a prototype system based on object models, which will expedite its development and simplify its implementation.

References

1. Oks, S.J., Jalowski, M., Fritzsche, A., Möslein, K.M.: Cyber-physical modeling and simulation: a reference architecture for designing demonstrators for industrial cyber-physical systems. Procedia CIRP **84**, 257–264 (2019). https://doi.org/10.1016/j.procir.2019.04.239
2. Enterprise-control system integration. Part 1 – Part 5.: ANSI/ISA-95.00.(01-05)-(2010-2013) [ISO/IEC 62264-1-2 Mod]
3. Digital Twins for Industrial Applications, An Industrial Internet Consortium: White Paper (2018)
4. Ladanyuk, A.P., Lutskaya, N.N.: Problems features of the robust control of process plants. Part I. Process Plants Math. Models. J. Autom. Inf. Sci. **48**(9), 75–83 (2016). https://doi.org/10.1615/JAutomatInfScien.v48.i9.60.

5. Derler, P., Lee, E.A., Vincentelli, A.S.: Modeling cyber–physical systems. Proc. IEEE **100**(1), 13–28 (2011). https://doi.org/10.1109/JPROC.2011.2160929
6. Yu, L., Fei, S., Sun, L., et al.: Design of robust adaptive neural switching controller for robotic manipulators with uncertainty and disturbances. J. Intell. Robot. Syst. **77**, 571–581 (2015). https://doi.org/10.1007/s10846-013-0008-3
7. Reynoso-Meza, G., Blasco, X., Sanchis, J., Martinez, M.: Controller tuning using evolutionary multi-objective optimisation: current trends and applications. Control. Eng. Pract. **28**, 58–73 (2014). https://doi.org/10.1016/j.conengprac.2014.03.003
8. Divya, N., Manoharan, S., Arulvadivu, J., Palpandian, P.: An efficient tuning of fractional order PID controller for an industrial control process. In: Arya, R.K., Tiwari, A.K.: International Chemical Engineering Conference 2021 (100 Glorious Years of Chemical Engineering & Technology), vol. 57, pp. 1654–1659 (2021). https://doi.org/10.1016/j.matpr.2021.12.255
9. Zhang, Z., Shen, H., Li, J.: Adaptive stabilization of uncertain unified chaotic systems with nonlinear input. Appl. Math. Comput. 4260–4267 (2011). https://doi.org/10.1016/j.amc.2011.09.056
10. Korobiichuk, I, et al.: Synthesis of optimal robust regulator for food processing facilities. In: Szewczyk, R., Zieliński, C., Kaliczyńska, M. (eds.) Automation 2017: Innovations in Automation, Robotics and Measurement Techniques, Advances in Intelligent Systems and Computing, vol. 550, pp. 58–66, Springer, Cham (2017). https://doi.org/10.1007/978-3-319-54042-9_5
11. Sebastian, A., Salapaka, S.: Design methodologies for robust nano-positioning. IEEE Trans. Control Syst. Technol. **13**(6), 868–876 (2005). https://doi.org/10.1109/TCST.2005.854336
12. Apkarian, P., Noll, D.: Nonsmooth optimization for multiband frequency domain control design. Automatica **43**(7), 724–731 (2007). https://doi.org/10.1016/j.automatica.2006.08.031
13. Yan, S., Sun, Z.: Study on separation principles for T-S fuzzy system with switching controller and switching observer. Neurocomputing **73**(13–15), 2431–2438 (2010). https://doi.org/10.1016/j.neucom.2009.12.037
14. Chang, W., Yan, J.: Adaptive robust PID controller design based on a sliding mode for uncertain chaotic systems. Chaos Solitons Fractals **26**(1), 167–175 (2005). https://doi.org/10.1016/j.chaos.2004.12.013
15. Tavakoli, S., Tavakoli, M.: Optimal tuning of PID controllers for first order plus time delay models using dimensional analysis. In: 4th International Conference on Control and Automation Proceedings, pp. 942–946. IEEE (2003). https://doi.org/10.1109/ICCA.2003.1595161
16. Li, Y., Xu, Q.: Adaptive sliding mode control with perturbation estimation and PID sliding surface for motion tracking of a piezo-driven micromanipulator. IEEE Trans. Control Syst. Technol. **18**, 798–810 (2010). https://doi.org/10.1109/TCST.2009.2028878
17. Toscano, R.: A simple robust PI/PID controller design via numerical optimization approach. J. Process Control **15**, 81–88 (2005). https://doi.org/10.1016/j.jprocont.2004.03.005
18. Wang, Q.-G., Fung, H.-W., Zhang, Y.: PID tuning with exact gain and phase margins. ISA Trans. **38**, 243–249 (1999). https://doi.org/10.1016/S0019-0578(99)00020-8
19. Arora, A., Hote, Y.V., Rastogi, M.: Design of PID controller for unstable system. In: Balasubramaniam, P. (ed.) Control, Computation and Information Systems. ICLICC 2011. Communications in Computer and Information Science, vol. 140, pp. 19–26. Springer, Berlin (2011). https://doi.org/10.1007/978-3-642-19263-0_3
20. Yaniv, O., Nagurka, M.: Design of PID controllers satisfying gain margin and sensitivity constraints on a set of plants. Automatica **40**(1), 111–116 (2004). https://doi.org/10.1016/j.automatica.2003.08.005.
21. Kadu, C., Patil, C.: Design and implementation of stable PID controller for interacting level control system. Procedia Comput. Sci. **79**, 737–746 (2016)
22. Alexandrov, A.G.: Finite-frequency method of identification. In: 10-th IFAC Symposium on System Identification, vol. 2, pp. 523–527 (1994). Preprints (1994). https://doi.org/10.1016/S1474-6670(17)47809-8.

Substantiation of Time Distribution Law for Modeling the Activity of Automated Control System Operator

Sergiy Tymchenko, Yevgen Kaplun, Evgeniy Roshchupkin, and Sergiy Kukobko

Abstract The existing models for ensuring the activities of the operator of the automated control system (OACS) or its individual elements use simplified data on the time spent on performing elementary actions. Various laws of time distribution for performing elementary actions are investigated to select an adequate one in modeling the activities of automated control system operators. The step-by-step modeling method is the basis for the formation of a model of the operator's activity. The analysis of the general structure of its activities reflects the main elements, namely, the receipt of information and its processing by the operator, decision-making, interaction with automation tools. The operator performs the basic operations, which are a set of successively performed elementary actions by the operator, such as search movements of the eyes, fixation of the gaze, perception and evaluation of forms or signs. To estimate the average execution time of several elementary actions, the sum of the values of each elementary operation was used. The random nature of these quantities requires the choice of the distribution laws of the time for the operator to solve various control problems during the simulation. Hypotheses about the laws of distribution of time for solving problems by the operator are put forward and tested: truncated normal distribution, gamma distribution, beta distribution, logarithmically normal distribution. The agreement of the empirical and theoretical distributions is checked, histograms and distribution densities of the decision time by the operator are shown.

Keywords Operator activity modeling · Simulation · Elementary action

S. Tymchenko · Y. Kaplun
National Academy of National Guard of Ukraine, Kharkiv, Ukraine

E. Roshchupkin
Ivan Kozhedub Kharkiv National Air Force University, Kharkiv, Ukraine

S. Kukobko (✉)
State Scientific Research Institute of Armament and Military Equipment Testing and Certification, Chernigiv, Ukraine
e-mail: sergiy.kukobko@gmail.com

S. Shkarlet et al. (eds.), *Mathematical Modeling and Simulation of Systems*, Lecture Notes in Networks and Systems 667,
https://doi.org/10.1007/978-3-031-30251-0_9

1 Introduction

Development and implementation of new information technologies in modern automated control systems (ACS) entails a change in the structure of the information support system for the operator's activities or its individual elements [1–3]. The noted transformation will lead to a change in the structure of the operator's activities. The use of new approaches to the formation and management of information models (IM) [4–6] in promising automated control systems determines the need for additional studies of the operator's activities at the design stage of such systems in order to take into account the changed operating conditions of the operator.

Expanding the range of factors taken into account through the use of appropriate models of the operator's activity makes it possible to assess the time of solving of the decision-making problems, as well as to estimate the time spent on performing individual operations in the structure of the operator's activity and identifying problem areas in the design of operator activities. A significant limitation when using these models is that they are developed using simplified data on the time spent on performing elementary actions. This requires additional research to substantiate the laws of time distribution used in modeling the activities of ACS operators.

A fairly large number of works are devoted to the development and description of models of human activity in ergatic systems, for example [7–9]. A number of works [10–12] are devoted to the principles and theoretical foundations of the formation of models of human activity, in others of them models of simple actions or processes that are elements of complex human activity are considered. However, issues related to the substantiation and choice of the laws of time distribution remain uncovered.

Substantiation of the choice of the law of distribution of time when modeling the activity of an ACS operator.

2 Main Material

It is proposed to take the method of step-by-step modeling as the basis for the method of forming a model of operator's activity [9, 13]. It is proposed to start developing a model of the operator's activity with an analysis of the general structure of his activity [10, 14]. When modeling the operator's activities, we will strive to carry out not external copying, but the psychological correspondence of the model and real decision-making processes [11, 15].

The formed structure of the operator's activity reflects the main elements of the activity associated with the receipt and processing of information, decision-making and interaction with automation tools [2] (Fig. 1).

The main operations performed by the operator are a set of sequentially performed elementary actions (EA) (time of their execution – t_{ea}). When working with information models (IM), the most frequently performed EA include: search movements of the operator's eyes, gaze fixation, perception and assessment of critical objects

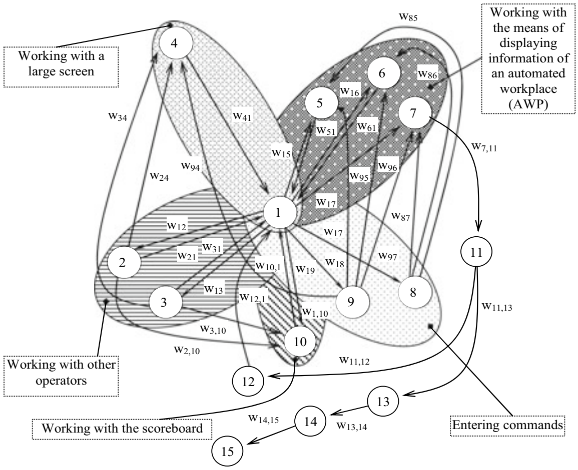


Fig. 1 A structure of the operator's activity model

(forms, signs, etc.). Therefore, to estimate the average execution time of several EA (t_e), the sum of ted values tea can be used.

It is important to note that, due to the influence of a large number of objective and subjective factors, t_{ea} and t_e are random variables. Therefore, in the interests of increasing the reliability of estimates of the performance indicators of individual operations, we use the probability distributions of estimates $t_{ea} - f(\tau)$. Knowledge of the laws of distribution of time for the operator to solve various control problems is necessary both for the distribution of functions between a person and a machine in an automated control system and a rational choice of a set of display devices for a control system, and for assessing the effectiveness of an automated control system and computer modeling of the operator's activity.

The results of the study [1, 2, 7, 16, 17] confirm the assumption about the asymmetry of the boundedness (on the left) of the laws of time distribution for the operator to solve various problems. The authors of these works used various prior distributions, which are in fairly good agreement with the empirical ones (the assessment was made using the Pearson criterion χ^2).

So, in the studies [1] it is shown that the time of solving problems by the operator is subject to the truncated normal distribution law:

$$f(x) = \begin{cases} \dfrac{C}{\delta\sqrt{2\pi}} \exp\left\{-(x - t_1 - m_x)^2 \dfrac{1}{2\delta^2}\right\}, x > t_1; \\ 0, x \le t_1, \end{cases} \tag{1}$$

where t_1 is the minimum time for solving the problem; m and δ are the parameters of the non-truncated normal law; C is the normalizing factor.

It was shown in [1, 12] that the time for solving problems is subject to Pearson's law, and in [2, 11] – to Erlang's law. Both of these laws are partial cases of the gamma distribution [16]:

$$f(x) = \begin{cases} \frac{\alpha^{\nu}}{G(\nu)}(x - t_1)^{\nu - 1}\exp\{-\alpha(x - t_1)\}, x > t_1; \\ 0, x \leq t_1, \end{cases} \tag{2}$$

where t_1 is the minimum time for solving the problem; α is the scale parameter $\alpha > 0$; $\frac{\nu}{\alpha}$ is the expectation; G(ν) is the gamma-function.

Considering the process of solving the problem by the operator, it is quite natural to assume that the solution of the problem in a time arbitrarily close to t_1 is unlikely. From this point of view, one should give preference to distribution (2).

In the course of the study, in addition to distributions (1) and (2), two more hypotheses were put forward and tested about the distribution laws of the time for solving problems by the operator.

The first of them includes the beta distribution, the density of which is described by the expression:

$$f(x) = \begin{cases} (x - t_1)^m (t_2 - x)^n, x > t_1, x < t_2; \\ 0, x \leq t_1, x \geq t_2, \end{cases} \tag{3}$$

where t_1 and t_2 are the limits of the distribution area of the random variable; m and n are the power exponents (m > -1, n > -1); C is the normalizing factor.

Distribution (3) is characterized by four parameters, which are often difficult to determine. While maintaining sufficient accuracy, one can use a special case (3), a distribution of this type [1]:

$$f(x) = \begin{cases} \frac{12}{(t_2 - t_1)^4}(x - t_1)(t_2 - x)^2, x > t_1, x < t_2; \\ 0, x \leq t_1, x \geq t_2, \end{cases} \tag{4}$$

The log-normal distribution was used as the second hypothesis:

$$f(x) = \begin{cases} \frac{\exp\left[-\{\ln(x - t_1) - m\}^2 \frac{1}{2\delta^2}\right]}{\delta(x - t_1)\sqrt{2\pi}}, x > t_1; \\ 0, x \leq t_1, \end{cases} \tag{5}$$

where $m = m_x^2\sqrt{\frac{1}{\delta_x^2 + m_x^2}}$; $\delta = \sqrt{2\ln\frac{m_x}{m}}$; m_x and δ_x is parameters of the normal distribution.

The management of the information output to the display devices was carried out from the experimenter's control panel. The obtained histograms of empirical distributions and the verification of their agreement with theoretical ones (1, 2, 4, 5) was carried out using the Pearson goodness test χ^2.

3 Results and Discussions

All tasks solved by the operator, in accordance with the investigated devices, can be divided into four groups.

Te first group includes tasks in which the operator performed command input. The tasks of this group differed in the number of dialed numbers.

The results of checking the agreement between the empirical and theoretical distributions are shown in Table 1.

The histogram and distribution densities of the time of solving the problem of this group by the operator are shown in Fig. 2.

The second group includes tasks in which the operator, while continuously monitoring airborne objects, recorded changes in the state of individual objects. The tasks differed in the number of simultaneously appearing changes and the quality of the background (constant or variable). The results of checking the agreement of empirical and theoretical distributions are shown in Table 2. Figure 3 shows the histogram and theoretical distributions for this problem.

Table 1 The results of checking the agreement between empirical and theoretical distributions

No in order	Number of dialed digits	Meaning χ^2			
		Truncated normal	Logarithmic normal	Beta	Gamma
1	1	43	627	12.9	36.5
2	2	11.7	21.3	9.8	460
3	3	6.7	36.6	21.0	1023
4	4	25.7	149	7.7	1023
5	5	20.1	88.5	7.2	52.6

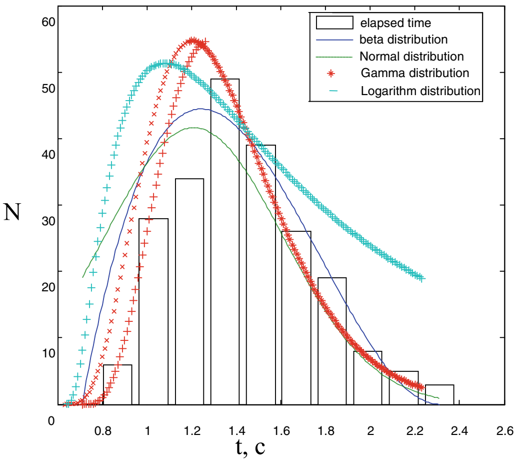


Fig. 2 Histogram and distribution density of the time of the operator's solution to the problem of dialing numbers

Table 2 The results of checking the agreement between empirical and theoretical distributions

No in order	Background	Number of dialed digits	Meaning χ^2			
			Truncated normal	Logaritmic normal	Beta	Gamma
1	Constant	from 1 to 4	13.0	13.0	134	7.7
2	Variable	from 1 to 4	23.9	15.1	74.5	11.9
3	Constant	from 1 to 10	5.6	67.7	143	6.4

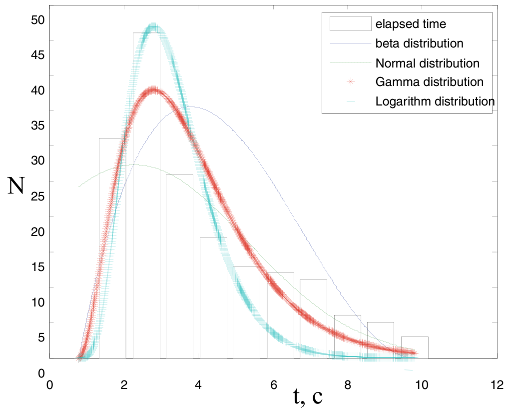


Fig. 3 Histogram and distribution density of the time of the operator's solution to the observation problem

The third group includes tasks in which the operator, on the instructions of the experimenter, determined one characteristic of the goal from the scoreboard. The tasks differed in the way they presented information.

The results of checking the agreement of empirical and theoretical distributions for the problems of this group are shown in Table 3, and Fig. 4 shows the distributions for this problem.

The fourth group includes tasks in which the operator discovered new targets on the Big Screen or counted or determined a given target characteristic. The tasks differed, in addition, in the type of goal forms and the use of the scoreboard.

The research results are shown in Table 4 and Fig. 5.

Analysis of the above results shows that, as a rule, there is a good coincidence of the empirical distribution with any two theoretical ones. It is difficult to give

Table 3 The results of checking the agreement between empirical and theoretical distributions

No in order	Method of presenting information	Meaning χ^2			
		Truncated normal	Logarithmic normal	Beta	Gamma
1	With line highlight	9.7	68.3	7.5	38
2	No selection	7.9	325	8.0	89
3	Only 1 line	5.6	16.5	9.5	414

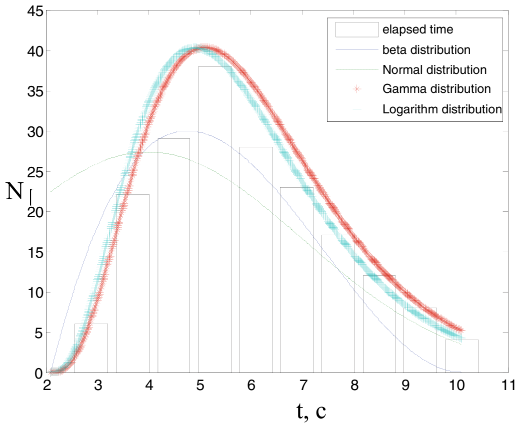


Fig. 4 Histogram and distribution density of the time of the operator's solution to the search problem

Table 4 The research results

No in order	Type of form	Use of score-board	Task type	Meaning χ^2			
				Truncated normal	Logarit-mic normal	Beta	Gamma
1	Short	No	Detection	54.3	13.0	68.6	12.9
2	Short	No	Reading characteristic	27.9	19.1	185	9.3
…	…	…	…	…	…	…	…
3	Full	No	Characterisation	14.7	3.2	24.3	2.8

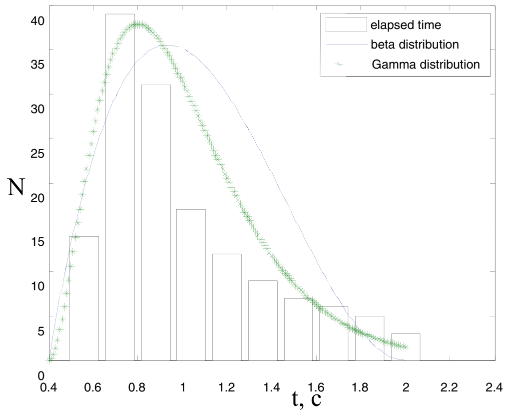


Fig. 5 Histogram and distribution density of the time of the operator's solution to the search problem with the condition

preference to any of the a priori distributions considered, since the estimate of the probability of their coincidence with the empirical ones is approximately the same:

- truncated normal – $P_{coincidence} \approx 0.4$;
- logarithmically normal – $P_{coincidence} \approx 0.3$;
- beta distribution – $P_{coincidence} \approx 0.35$;
- gamma distribution – $P_{coincidence} \approx 0.4$.

In many cases, it is of great interest to obtain a p-quantile estimate of the time for solving various problems t_p. This raises the question; what is the error in the determination t_p for various a priori distribution laws.

Comparison of quantile estimates of the time for solving the considered problems shows that at a level of 0.9, the maximum error in determination $t_{0.9}$ for the considered theoretical distributions is on average:

$$\Delta t_{0.9} = 0.04 t_{0.9} \tag{6}$$

Moreover, $\Delta t_{0.9}$ was determined regardless of the coincidence of a priori distributions with empirical ones.

In many cases, the preferred distribution is that which has a simpler analytical expression and whose parameters can be easily determined. From this point of view, a particular case of beta distribution (4) stands out for the better, which is widely used to set time estimates included in a network project of works.

Distribution (4) is characterized by only two parameters t_1 and t_2 (i.e., the minimum and maximum time for solving the problem), while the other considered distributions are characterized by three parameters.

The results of many studies, for example, [2], indicate that $f(\tau)$ is, as a rule, a truncated ($\tau \geq t_1, \tau \leq t_2$) asymmetric unimodal distribution.

In [17], based on the analysis of statistical material, preference is given to a simplified mathematical model of the beta distribution of the following form:

$$p(\tau) = \frac{12}{(t_2 - t_1)^2}(\tau - t_1)(t_2 - \tau)^2 \tag{7}$$

This distribution is asymmetric and bounded from below (t_1) and from above (t_2), which fully corresponds to real processes of performing operations. To set the prior density of the beta distribution, it is enough to set t_1 and t_2. The expectation of the operation execution time and the variance are equal

$$\bar{t} = \frac{3t_1 + 2t_2}{5}, \; D = 0.04(t_2 - t_1)^2 \tag{8}$$

To estimate t_{ea}, we use distribution (7).

Analysis of the operator's activity with the construction of a plane model [5, 6] allows to more accurately determine the structure of the operator's activity, to highlight the main material (objects of control, information display facilities, control elements, etc.) and intangible (intellectual, volitional, sensual) aspects of the activity operator. In this case, the main actions of the same type of the operator are distinguished within a single plane and can be studied independently and separately, and with the subsequent refinement of the characteristics of the activity, they will again be included in the developed model. In this case, the plane model can be:

– simple: there are many separate operations of the same type that are not related to each other. Each operation has connections only with elements or groups of elements of other planes;
– complex: there are many separate operations of the same type related to each other. There can be connections, both of an individual operation, and a group of operations with elements or groups of elements of other planes;
– mixed: there are many separate operations of the same type, both related and unrelated. There can be connections, both of an individual operation, and a group of operations with elements or groups of elements of other planes.

When using the proposed approach to building a model of the operator's activity, all the advantages of the step-by-step modeling method can be realized, both with respect to the entire model as a whole, and with respect to each plane separately. This will improve the accuracy and adequacy of building a model of the operator's activity as a whole.

4 Conclusions

The proposed approach to modeling the operator's activity is empirical and requires the development of a mathematical description and research of the proposed model. In this model, the operator's activity is distributed (a more figurative concept will be dissolved) between the planes. At the same time, a sufficient level of detailing of individual aspects of the operator's activity is preserved while maintaining a holistic understanding of the process of his activity and the characteristic features of the psychological and motor activity of the operator.

One of the most important tasks of modeling the operator's activity has been solved – the choice of the law of distribution of the time for performing elementary actions by the ACS operator. The choice of the truncated beta distribution corresponds to real processes and has a reasonable restriction on the left and right of the operator's actions.

Acknowledgements This work was supported by the National Academy of National Guard of Ukraine (Kharkiv, Ukraine).

References

1. Herasimov, S., Pavlenko, M., Roshchupkin, E., Lytvynenko, M., Pukhovyi, O., Salii, A.: Aircraft flight route search method with the use of cellular automata. IJATCSE **9**(4), 5077–5082 (2020). https://doi.org/10.30534/ijatcse/2020/129942020
2. Herasimov, S., Borysenko, M., Roshchupkin, E., et al.: Spectrum analyzer based on a dynamic filter. J Electron. Test 357–368 (2021). https://doi.org/10.1007/s10836-021-05954-0
3. Venda, V.F.: Inzhenernaya psikhologiya i sintez sistem otobrazheniya informatsii. Moscow. Mashinostroyeniye, vol. 398 (1975)
4. Kozhushko, Y., Karlov, D., Klimishen, O., Bortsova, M., Herasimov, S., Hrichanuk, O., Bykov, V.: Comparison of the efficiency of some images superposition algorithms used in aircraft map-matching navigation systems. In: IEEE International Conference on Mathematical Methods in Electromagnetic Theory, pp. 282–285 (2018)
5. Herasimov, S., Belevshchuk, Y., Ryapolov, I., et al.: Characteristics of radiolocation scattering of the SU-25T attack aircraft model at different wavelength ranges. Information and controlling systems. Eastern-Eur. J. Enterp. Technol. **6/9**(96), 22–29 (2018). https://doi.org/10.15587/1729-4061.2018.152740
6. Karlov, V., Kuznietsov, O., Artemenko, A., Karlov, A.: Evaluation of the accuracy of measuring the radial velocity of a target with an exponential and alternating decrease in phase correlation of the burst radio signal. Adv. Inf. Syst. **4**(1), 71–75 (2019)
7. Sereda, G.K., Bocharov, G.V., Repkina, G.V.: Inzhenernaya psikhologiya. Vishcha shkola, Kyiv, vol. 307 (1976)
8. Dushkov B.A., Lomov B.F., Smirnov B.A.: Reader in Engineering Psychology. Higher school, Moscow, vol. 287 (1991)
9. Yevseiev, S., Kuznietsov, O., Herasimov, S., et al.: Development of an optimization method for measuring the Doppler frequency of a packet taking into account the fluctuations of the initial phases of its radio pulses. Eastern-Eur. J. Enterp. Technol. **2/9**(110), 6–15 (2021). https://doi.org/10.15587/1729-4061.2021.229221
10. Herasimov, S., Tymochko, O., Kolomiitsev, O., et al.: Formation analysis of multi-frequency signals of laser information measuring system. EUREKA. Phys. Eng. **5**, 19–28 (2019). https://doi.org/10.21303/2461-4262.2019.00984
11. Kulakov, O., Katunin, A., Kozhushko, Y., et. al.: Usage of lidar systems for detection of hazardous substances in various weather conditions. In: 2020 IEEE 6th International Symposium on Microwaves, Radar and Remote Sensing (MRRS), 2020 IEEE Ukrainian Microwave Week, Kharkiv, Ukraine, pp. 360-363 (2020)
12. Herasimov, S., Pavlii, V., Tymoshchuk, O., et al.: Testing signals for electronics: criteria for synthesis. J. Electron. Testing. **35**(148), 1–9 (2019). https://doi.org/10.1007/s10836-019-05798-9
13. Besnard, D., Gacek, C., Jones, B.: Cliff Structure for Dependability: Computer Based Systems from Interdisciplinary Perspective. vol. 304, Springer, London (2006)
14. Computer Safety, Reliability and security. In: Proceedings 24th International Conference SAFECOMP 2005. Friedrikstadt. Norway, vol. 409 (2005)
15. Fatrell, R.T., Shafer, D.F., Best man, L.I.: Management of Software Projects: Achieving Optimal Quality at a Minimum Cost, Williams, Moscow, 1136 p. (2003)
16. Lipaev, V.V.: Reliability and Functional Safety of Real-Time Software Complexes. Moscow, vol. 207 (2013)
17. Listrova, S.V., Listrovaya, E.S., Kurtsev, M.S.: Ranked approach to solving linear and nonlinear Boolean programming problems for planning and control in distributed computing systems. Electron. Model. **1**, 19–38 (2017)

Improving the Method of Virtual Localization of Near-Net-Shaped Workpieces for CNC Machining

Sergiy Plankovskyy, Oleksandra Kapinus, Kateryna Maiorova, and Svitlana Myronova

Abstract The study considers the method of virtual localization of near-net-shaped workpices, which are to be processed on CNC machines. It is shown that virtual localization based on a combination of CMM inspection of the shape and workpiece position and computer-aided placement of the part CAD model into workpice model is an actual way to increase the economic efficiency of CNC machining. The resulting mathematical model of virtual localization showed that the most effective virtual localization is when the surfaces of the workpiece and the placed part are close to equidistant. As a test problem it was considered, the virtual localization of the bracket into workpiece of near-net-shape, which conditionally corresponds to the case of workpiece obtained by additive manufacturing. The workpiece localization in the machine coordinate system was simulated with the calculation of the centers of gravity and by finding out the position of the main axes of the moments of inertia, which allowed to place the part model inside the workpiece model. The results of the test problem solving confirmed that the proposed approach based on the mathematical model of virtual localization allows to meet the conditions for the part accurate locating inside the workpiece. It is noted that further study should be aimed at improving the accuracy of numerical calculations with the use of more accurate algorithms of nonlinear optimization to ensure a uniform distribution of allowance for subsequent processing CNC machining.

Keywords Workpiece localization · CNC machining · Near-Net-Shape · Optimization criterion · Aerospace components

S. Plankovskyy (✉)
O.M. Beketov National University of Urban Economy in Kharkiv, 17, Marshal Bazhanov Street, Kharkiv, Ukraine
e-mail: k.majorova@khai.edu

O. Kapinus
The Ukrainian-American Joint Venture "KODA", 22, Bahaliia Street, Kharkiv, Ukraine

K. Maiorova · S. Myronova
National Aerospace University "KhAI", 17, Chkalova Street, Kharkiv, Ukraine

S. Shkarlet et al. (eds.), *Mathematical Modeling and Simulation of Systems*, Lecture Notes in Networks and Systems 667,
https://doi.org/10.1007/978-3-031-30251-0_10

1 Introduction

In the advanced aerospace manufacturing today there are widely used the heat-resistant alloys and high-performance alloys (superalloys), which have an excellent resistance to plastic deformation and destruction at high temperatures and in oxidizing environment – nickel-based and cobalt-based ones, or with more complex composition, like iron-chromium-nickel based superalloys [1]. Up-today they are commonly used in manufacturing of the aircraft engine turbine rotor blades, the blades of ship propellers and turbines, monolithic aircraft panels, etc. [2]. The main problem manufacturers faced with in the such parts production is the high cost of the material itself and its processing on CNC machines. The workpieces with a specified geometry and proper allowance for the final processing on CNC machines can be produced by two different strateges – using conventional machining technology, for example, milling from the metal blocks, and by additive manufacturing (AM) technologies. In Fig. 1 it is schematically shown the stages of of the same part manufacturing using conventional approach (like milling) and AM technologies.

The analysis of material waste, time of production and the manufacturing cost shows that when using the additive manfacturing technologies (Fig. 1) these indicators are significantly less than when using conventional milling, which is in line with traditional approach [3, 4]. In both cases (conventional machining or additive manufacturing), the most significant factor for increasing the economic efficiency of manufacturing a part made of superalloy from a preliminary shaped workpiece is the minimization of the allowance for the subsequent machining. This, in turn, resulted in the need of the workpieces exact localization in the coordinate system of a CNC machine. Application of the traditional approaches to workpieces positioning on the machine table with using the special jigs requires a lot of time for such a tooling manufacturing (according to JSC “Turboatom” (Ukraine) – up to several months), which significantly extents the overall product manufacturing cycle and reduces its economic efficiency.

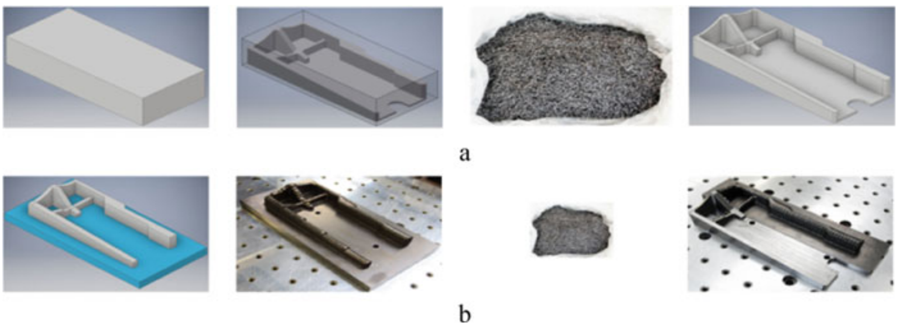


Fig. 1 Part manufacturing by using traditional technologies like milling (a) and additive manufacturing (b)

An alternative way is to implement the parts virtual localization approach during CNC machining, wherein the workpiece is fixed on the machine table in primitive fixtures. Thereafter, the workpiece position in the machine coordinate system is determined by machines in-built measuring device, for example, by a laser scanner. Next, the CAD model of the part is virtually placed in the workpiece in such a way as to distribute the machining allowance as even as possible, and the origin (datum point) of the part coordinate system in the machine coordinate system is determined, as well as the angles between axes of these two coordinate systems. After that, the CNC machining program, which is usually generated with respect to the part coordinate system, is automatically recalculated by the affine transformations into the coordinate system of the CNC machine itself (Fig. 2).

The most complex case for virtual localization is the processing of parts with a partially restored shape – by using, for example, laser surfacing. In such a case, the workpiece shape is not only close to the part itself (so called near-net-shape), but even coincides with its surfaces in some areas. Likewise the aforementioned case of the workpiece virtual localization, at first it's measured in a coordinate measuring machine (CMM) for creating a 3D model. But, in this case the errors of the workpiece shape determining can lead to the "overcoming" of the finished part beyond the specified dimensional tolerances. Therefore, in such cases, the requirements to the workpiece shape measuring accuracy and the accuracy of its virtual localization are the most stringent.

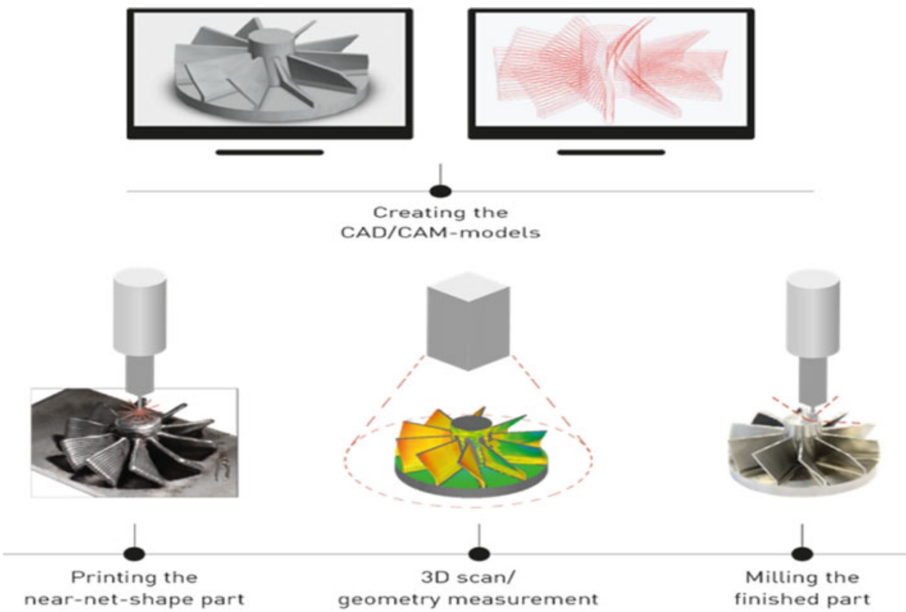


Fig. 2 The sequence of part manufacturing from an AM near-net-shaped workpiece when using virtual localization

Thus, the workpiece virtual localization by means of the combining the computer-aided workpiece shape CMM inspection (measuring) and the workpiece localization on the machine table along with the computer-aided "matching" of the part CAD model inside the workpice is an actual way to increase the economic efficiency of CNC machining. The most promising areas of this process application are the manufacturing of large-sized parts of complex shape and high-precision parts, produced by AM technologies and surfacing (build-up) technologies while the repair of such parts with partial restoration of their shape. And, the one of the main conditions for ensuring of the manufactured parts accuracy is the using of effective algorithms for automatic positioning of parts CAD models in the area limited by the workpiece surfaces.

2 Purpose and Objectives of Research

The purpose of the study is to improve the virtual localization method of workpieces, which are close in shape to the final part (near-net-shaped workpieces), during their processing in the CNC machines.

To achieve this, the following tasks were solved:

- analysis of existing algorithms of virtual localization;
- developing the mathematical model of virtual localization;
- implementation of procedures for numerical solution of the virtual localization test problem.

3 Analysis of the Virtual Localization Existing Algorithms

The relevance of the virtual localization problem caused a significant interest of researchers in its solution. In the earliest works, the problems of prismatic bodies virtual localization were considered by minimizing the distance between the workpiece and the part. Mostly, they were about symmetrical localization of parts with pre-processed elements and elements to be cut. For example, Gou et al. [5] found the efficient localization solutions for symmetric parts (such as cylinders, cones, and spheres) by reducing the space of problem from Euclidean to one-dimensional one. Zhao and Li [6] defined an approach to solving the problem in case when the part localization was failed and the workpiece could not "cover" the elements to be processed. The obvious solution in such a case was the workpiece "additional" material increasing. Gessner and Stanik [7] used an optical device for scanning the casts and applied a conventional method of workpieces localization. Chatelain and Fortin [8] assigned the weight to a measured point, considering the difficulties of material adding to its element, and quickly determined the places of the workpiece material shortage by calculating the oriented Euclidean distance from the measured points to the part. Later, there were works that dealt with the problem of virtual localization of parts of complex shapes. Xiong [9] solved the problem of virtual surface localization in a free form using the method of Lagrange multipliers and Euclidean distance. Shen and co-authors in [10] proposed a hierarchical algorithm for the localization of large workpieces with free-form surfaces. According to the proposed algorithm, during the rough alignment, the distance between the part center of gravity and the measured points at the surface corners is minimized, and than, during final precise alignment, the different weights were assigned to the oriented distances from the measured points to the design surface, and then added. Sijie Yan and co-authors in [11] developed an algorithm of rough and precise localization for surface processing and inspection. Mehrad [12] roughly localized the design surface by minimizing the distance between the measured and design surfaces, which reduced the uncertainties of the precize localization.

Some studies were concerned with calculations of the distance between the measured points and the design surface. In [13], Zhu and co-authors determined the function of the distance from the point to the surface and studied the properties of the complex error of the surface. The authors of the work [14] created a novel distance function to improve the accuracy of localization of strongly curved surfaces. Xu et al. [15] tried to use the bisection method using multidimensional Bernstein polynomials to calculate the point projection onto a free-form surface. He et al. [16] constructed an efficient and reliable distance function, and Flory and Hofer [17] extended the distance function to surface fitting and cloud of points registration. The authors of the work [18] created algorithm of starting localization (positioning) of part CAD model while virtual localization.

The problem of increasing the accuracy and reliability of parts localization and of workpiece measurements have being discussed by many researchers. All in all, it should be said that the problem of virtual localization is far from its final solution, and the existing methods are grounded on empirical assumptions. The most promising is the approach built on the implementation of a two-stage localization problem solution that use preliminary (rough) and final (precize) localization. It should be noted that researchers do not consider yet the problems of virtual localization in the case of workpiece “prohibited zones”, for example, when in the workpiece defects are detected by non-destructive testing methods. In addition, the problem of geometric information processing of 3D-models in the formats of up-today CAD systems remains insufficiently covered in the most studies. Thus, the task of building the effective algorithms for virtual localization of arbitrary shaped parts remains open and relevant.

4 Mathematical Model of the Problem

Using the approaches of works [19, 20] when formulating the problem of virtual localization, let’s consider the part as a three-dimensional geometric object $D \in R^3$ of an arbitrary geometric shape, and the workpiece as the localization area $Z \in R^3$. Let’s also assume that for object D it is allowed translation by vector $u(x, y, z) \in$

R^3 and rotation by angles $\theta = (\theta_x, \theta_y, \theta_y)$. The motion vector that determines the location of object D in the area $Z \in R^3$ will be designated as $\vartheta = (u, \theta)$. The distance between the part and the boundary of the localization area will be denoted as $d_{D,Z} = \min\|O_Z - O_D\|$, where O_Z, O_D are points which meet to the conditions $O_Z \in Z$, $O_D \in D$.

Let's state the problem of virtual localization as follows: to find such a vector $\vartheta = (u, \theta)$, for which $D \in Z$ i and at the same time the conditions of optimal localization are meet. Such conditions can be as follows:

- the mathematical expectation of the distance between the part and the workpiece surfaces maximization $M(d_O) \to \max$, where d_O is the distance between the points O_D and O_Z of the part and the workpiece surfaces correspondingly;
- the minimum distance between the part and the workpiece surfaces maximization $\min\|OZ - OD\| \to \max$;
- the maximum distance between the part and the workpiece surfaces minimization $\max\|OZ - OD\| \to \min$;
- a combination of the aforespecified conditions.

At the first let's consider the optimization problem of the material point localization in the area, and as a criterion for the optimality of its localization it let's use the minimization of the mathematical expectation of the square of the distance between the point itself and of the localization area boundary $M\left(\|O_Z - O\|^2\right) \to \min$. To simplify the explanations, we will consider the problem in a two-dimensional formulation. For this case, the considered criterion can be written in the form:

$$M\left(\|O_Z - O\|^2\right) = \frac{1}{L_Z}\int G_Z\left((x_Z - x_O)^2 + \left(y_Z - y_O\right)^2\right)dG_Z \to \min, \quad (1)$$

where G_Z – the localization area boundary, and L_Z – is a lenght of curve G_Z.

When unit density of curve G_Z the following formulas are correct.

$$\begin{gathered}\int\limits_{G_Z} x^2 dG_Z = J_y^Z, \int\limits_{G_Z} y^2 dG_Z = J_x^Z, \\ \int\limits_{G_Z} x dG_Z = L_Z x_{c.w.}^Z, \int\limits_{G_Z} y dG_Z = L_Z y_{c.w.}^Z\end{gathered} \quad (2)$$

Let's assign axes OX and OY passing through the center of gravity of curve G_Z. Then, taking into account the formulas (2), the condition (1) is as follows:

$$M\left(\|O_Z - O\|^2\right) = \left(J_x^Z + J_y^Z\right)\Big/L_Z + \left(x_o^2 + y_o^2\right) \to \min \quad (3)$$

Considering that the sum of the axial moments of inertia of the curve with respect to the axes passing through its center of gravity is a constant value, it follows from

(3) that $M\left(\|O_Z - O\|^2\right) = \frac{(J_x^Z + J_y^Z)}{L_Z} \to min$ when $x_o = y_o = 0$, so the optimal location of the material point under condition (1) will be the center of gravity of the curve G_Z, which is a boundary of localization area. For the case of localization in the area Z of a two-dimensional object D with a boundary limited by curve G_D, the condition (1) has the for:

$$M\left(\|O_Z - O_D\|^2\right) = \frac{1}{L_Z L_D}\int_{G_Z}\left[\int_{G_D}\left((x_Z - x_D)^2 + \left(y_Z - y_D\right)^2\right)dG_D\right]dG_Z \to \min \tag{4}$$

and it is met when centers of gravity of localization area G_Z and the curve curve G_D coincide.

The same results of localization optimization when it is used criterion of maximization the difference of the squares of the central radii of gyration of the boundaries of workpiece Z and of the part to be placed D:

$$r_Z^2 - r_D^2 = \frac{1}{2L_Z}\int_{G_Z}\left(x_Z^2 + y_Z^2\right)dG_Z - \frac{1}{2L_D}\int_{G_Z}\left(x_D^2 + y_D^2\right)dG_D \to \max. \tag{5}$$

The disadvantage of criteria (4) and (5) is that they do not allow determining the orientation of object D with respect to the axes OX and OY, since their extrema do not depend on the angle of rotation ϑ. To solve this problem, instead of the difference of the squares of the central radii of gyration as optimization criteria it can be used the difference of the squares of the gyration radii of the workpiece and the part boundaries about the main central axes of inertia of the curve G_Z. For definiteness, let's assume that the maximum main central inertia moment $J_{\max}^Z$ of the curve, which coincides with the boundary of the workpiece, corresponds to the moment of inertia about the axis OY, and the minimum one $J_{\min}^Z$ - about the axis OX correspodingly. And it is required to meet the following conditions:

$$r_{\max Z}^2 - r_{yD}^2 = \frac{J_{\max}^Z}{L_Z} - \frac{\left(J_{\max}^Z \cos^2\theta + J_{\min}^D \sin^2\theta\right)}{L_D}, \tag{6}$$

$$r_{\min Z}^2 - r_{xD}^2 = \frac{J_{\min}^Z}{L_Z} - \frac{\left(J_{\min}^Z \cos^2\theta + J_{\max}^D \sin^2\theta\right)}{L_D}, \tag{7}$$

where J_{min}^D, J_{max}^D – main central inertia moment of G_D.

Both criteria are fulfilled when $\theta = 0$, which leads to the requirement of the main central axes of the curves G_Z and G_D coincidance. In the three-dimensional case, these conditions should be formulated for thin shells Ω_Z and Ω_D, stretched over the outer surface of the workpiece and the part to be placed.

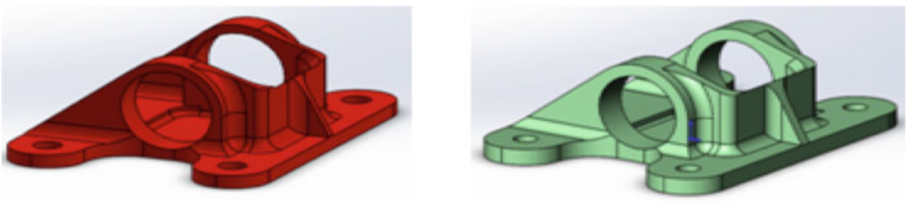
It should be noted that the greatest efficiency of using the proposed virtual localization method is expected in cases when the surfaces of the workpiece and of the part to be placed are close to equidistant. When the differencies in their forms are significant, the application of this **criterion** cannot guarantee not only a uniform distribution of allowance for processing, but even the main condition realization – the part positioning inside the workpiece. However, within the variety of cases when workpieces and parts are close in shapes and sizes, it is expected that the proposed criterion allows at least to determine a first approximation of the solution to the virtual localization problem with a high quality.

5 Experimental Procedures of the Virtual Localization Test Problem Numerical Solution

As a test task, it was considered the bracket virtual localization in the workpiece which shape is rather close to the equidistant one – as if the workpiece shape was obtained by additive manufacturing technologies as they are the most effecient ones (Fig. 3).

Files with digital geometry in Parasolid format were imported into the Design-Modeler geometry module of the AnSys software, where they were translated and rotated with respect to the global coordinate system by arbitrary distances and angles. It was the way to simulate the process of the workpiece localization in the machine coordinate system. After that, according to the aforegiven formulas there were calculated the both centers of gravity and the position of the moments of inertia main axes. In accordance to the coordinates of the gravity centers and the directions of the main central axes of inertia, the model of the part has being placed in the model of the workpiece (Fig. 4).

As a result of the part positioning according to the aforedescribed algorithm, it was completely located inside the workpiece, but the allowance distribution for processing turned out to be somewhat uneven. Such a result can be explained by numerical calculation errors. Nonetheless, in general, the task that was stated at this stage of study can be considered as solved, at least because obtained result is a satisfactory solution to the problem of part preliminary localization, which can be than refined using nonlinear optimization algorithms.



a b

Fig. 3 The geometry of the part (a) and the workpiece (b) of the bracket in the test problem

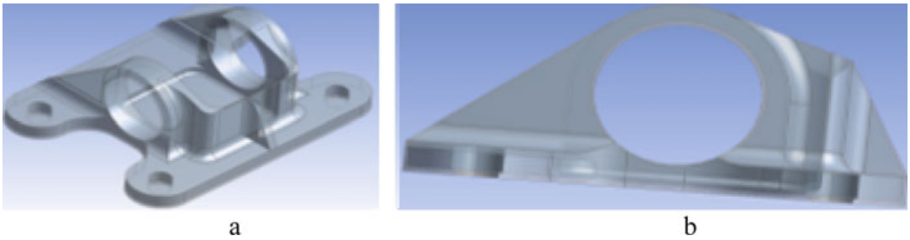


Fig. 4 The part localization in the workpiece (a) and the general view along the XY axes (b)

6 Conclusion

The method of virtual localization of workpieces, which are close in shape to the final part, during their processing by CNC machining has been improved by means of the proposed approach based on a mathematical model of virtual localization. It is shown that such localization can be carried out by combining the centers of mass and the main central axes of inertia of thin shells that coincide with the outer surfaces of the workpiece and the part. The results of solving the test problem have confirmed that the proposed approach based on the mathematical model of virtual localization allows to meet the conditions for the localization of the part inside the workpiece. It should be taken into account that the processing allowance distribution is somewhat uneven. The results can be improved by increasing the accuracy of numerical calculations, as well as by using more accurate algorithms of nonlinear optimization at the stage of refining the result of the preliminary localization problem.

References

1. Pellg, J.: Basic compounds for superalloys. 1st ed., Elsevier, Cham. 624 p. (2018).
2. Soares, C.: Gas turbines. 2nd ed., Butterworth-Heinemann. 1020 p. (2014)
3. Bhaskar, D., Sudarsanam, B., Bradley, J. Science, Technology and application in additive manufacturing. 1st ed., Elsevier, Cham. 354 p. (2019)
4. Manjaiah, M., Raghavendra, K., Balashanmugam, N., Paulo Davim, J.: Additiva manufacturing. Woodhead Publishing. 324 p. (2021)
5. Gou, J., Chu, Y., Li, Z., et al.: On the symmetric localization problem. IEEE Trans. Robot. Autom. **14**(4), 533–540 (1998)
6. Zhao, Y., Sun, J., Li, J., et al.: Remanufacturing of compressor blade by laser direct metal deposition and subsequent multi-axis milling. In: ICALEO: 34th International Congress on Laser Materials Processing, Laser Microprocessing and Nanomanufacturing, vol. 2, no. 73, pp. 1–11. Laser Institute of America (2018)
7. Gessner, A., Staniek, R.: Computer-aidedalignment of castings and machining optimization. J. Mech. Eng. Sci. **229**(3), 485–492 (2015)
8. Chatelain, J.-F., Fortin, C.: A balancing technique for optimal blank part machining. Precis. Eng. **25**, 13–23 (2001)

9. Sun, Y., Xiaoming, W., Dongming, G., Jian, L.: Machining localization and quality evaluation of parts with sculptured surfaces using SQP methods. Int. J. Adv. Manuf. Technol. **42**, 1131–1139 (2009)
10. Shen, B., George, Q. Huang, Mak, K.L., Wang, X.C. et al.: A best-fitting algorithm for optimal location of large-scale blanks with free-form surfaces. J. Mater. Process. Technol. **139**, 310–314 (2003)
11. Sijie, Y., Yunfei, Z., Peng, F., Lai, X.: Research on the localization of the workpieces with large sculptured surfaces in NC machining. Int. J. Adv. Manuf. Technol. **23**(5), 429–435 (2004)
12. Mehrad, V., Xue, D., Gu, P.: Robust localization to align measured points on the manufactured surface with design surface for freeform surface inspection. Comput. Aided Des. **53**, 90–103 (2014)
13. Zhu, L.M., Zhang, X.-M., Ding, H., Xiong, Y.-I.: Geometry of signed point-to-surface distance function and its application to surface approximation. J. Comput. Inf. Sci. Eng. **10**(4), 041003 (2010)
14. Li, W.-I., Yin, Z.-p., Xiong, Y.-I.: Adaptive distance function and its application in free-form surface localization. In: 2009 International Conference on Information and Automation, pp. 24–28. IEEE, China (2009)
15. Sun, Y.-W., Xu, J.-T., Guo, D.-M., Jia, Z.-Y.: A unified localization approach for machining allowance optimization of complex curved surface. Precis. Eng. **33**(4), 516–523 (2009)
16. He, G., Zhang, M., Song, Z.: Error evaluation of free-form surface based on distance function of measured point to surface. Comput. Aided Des. **65**, 11–17 (2015)
17. Flory, S., Hofer, M.: Surface fitting and registration of point clouds using approximations of the unsigned distance function. Comput. Aided Geom. Des. **27**(1), 60–77 (2010)
18. Plankovskyy, S., Tsegelnyk, Y., Myntiuk, V., Zadorozhniy S., Kombarov, V.: Method of parts virtual localization for near shape blanks. Aerospace Tech. Technol. **4**(164), 74–82 (2020). https://www.researchgate.net/publication/343999199_Metod_virtualnogo_bazuvanna_detalej_z_formou_nablizenou_do_formi_zagotovok
19. Lasaruk, A., Sturm, T.: Automatic verification of the adequacy of models for families of geometric objects. In: Sturm, T., Zengler, C. (eds.) Automated Deduction in Geometry. ADG 2008. Lecture Notes in Computer Science, vol. 6301, pp. 116–140. Springer, Berlin (2011). https://doi.org/10.1007/978-3-642-21046-4_6
20. Stoyan, Y.G., Semkin, V.V., Chugay, A.M.: Optimization of the layout of three-dimensional objects in the multi-link region taking into account the shortest states. Cybern. Syst. Anal. **50**(3), 58–70 (2014). http://www.kibernetika.org/volumes/2014/numbers/03/articles/06/6.pdf

Mathematical Modeling and Simulation of Systems in Information Technology and Cybersecurity

Shared Modeling and Simulation Environment for Online Learning with Moodle and Jupyter

Volodymyr Kazymyr, Dmytro Horval, Oleksandr Drozd, and Anatolijs Zabašta

Abstract In regard to online learning, unrestricted access to course materials is paramount hence we have seen active developments of e-learning platforms, with Moodle being the most popular one. Benefits of such platforms are obvious: abundance of features to facilitate course development and, subsequently, grading on behalf of the teacher, as well as providing students with a concise and straight-forward way of taking the said course. Yet, Moodle on its own does not provide a straight-forward way for the students to carry out hands-on assignments, hence requiring any one taking the course to have a learning environment set up on their computer. A solution to this problem also exists in the form of Project Jupyter - a web application which can be described as a tweakable platform for "running" code in the browser. This article describes a technological solution that integrates Moodle with the JupyterLab to solve modeling problems of teaching modelling in a distributed virtual environment.

Keywords Online leaning · Moodle · Jupyter · Jupyter Notebook

1 Introduction

Remote online learning has become a staple in modern education, allowing not only better connectivity and student engagement in the learning process [1], but also making it more resilient to disruptions, caused by unforeseen circumstances.

Currently Moodle is the most widespread online learning platform, having multiple options for course creation, management and grading. However, this platform allows students to submit their practical assignments, but not to complete them in an online environment, hence requiring every participant to set up a personal

V. Kazymyr (✉) · D. Horval · O. Drozd
Chernihiv Polytechnic National University. 95, Shevchenka Street, Chernihiv 14035, Ukraine
e-mail: vvkazymyr@stu.cn.ua

A. Zabašta
Riga Technical University, Riga, Latvia
e-mail: Anatolijs.Zabasta@rtu.lv

S. Shkarlet et al. (eds.), *Mathematical Modeling and Simulation of Systems*, Lecture Notes in Networks and Systems 667,
https://doi.org/10.1007/978-3-031-30251-0_11

working environment on their machines. This not only introduces unnecessary complexity for the students, but also introduces the possibility of such issues as licensing or students not being able to complete the course due to inadequate hardware [2].

Such issues can be eliminated with relative ease by deploying an institution-wide virtualized solution, that would provide users with on-demand pre-packaged workspaces that could be accessed individually [3]. In addition to that, this approach also ensures that all course participants have access to a homogeneous educational and working environment, hence preventing the infamous "It works on my computer" issue from occurring.

While the desired result may be archived in multiple ways, Project Jupyter [4] presents the most flashed-out solution, with its benefits being: scalability, modular design, mature and rich ecosystem, web-based user-facing interface and an ability to combine lecture material and runnable examples in a single notebook.

2 Background

The idea of creating a new Shared Modeling and Simulation Environment (SMSE) for educational purposes was established in the European project "Development of practice-oriented student-centered education in the field of modeling cyber-physical systems" (CybPhys) [5]. The project began as part of a consortium of universities in Europe, Belarus and Ukraine within the framework of the Erasmus + Capacity Building in Higher Education program. However, over time, it was been transformed into a national project with the participation of three Ukrainian universities (Chernihiv Polytechnic National University - CPNU, Kharkiv National Automobile and Highway University - KNAHU, and Kryyi Rih National University - KNU) and three European universities (Riga Technical University, KU Leuven from Bruges, University of Cyprus - UCY). The task of creating the SMSE was entrusted to Chernihiv Polytechnic with the substantive support of the partners. The result was the creation of a learning ecosystem for the practical modeling of cyber-physical systems for innovative physical, mathematical and engineering topics [6].

The creating of the SMSE was based on the experience of using the Jupiter platform in educational and scientific fields. In particular, it concerned such questions as using of Jupyter Notebooks in pedagogical practices [7], organization of computational cognitive science labs [8], teaching with Jupyter Notebook in the Web [9] and creation of JupyterLab interface [10].

As a further advancement of existing practices, the objectives of this study were formulated, which were aimed at integrating the learning management system Moodle, that widely used at universities, with the new capabilities of the Jupyter platform to create virtual laboratories. A goal of the paper is to describe main features of developed technological decision which underlie in SMSE to support an on-line training.

3 Modeling Environment Architecture

Architecture of such a solution is represented on Fig. 1, and is made up of three components. Account management system is represented by Moodle platform. Its main purpose, as name suggests, is authentication of existing accounts into Jupyter, while also fulfilling a role as an online learning platform.

Account management system is represented by Moodle platform. Its main purpose, as name suggests, is authentication of existing accounts into Jupyter, while also fulfilling a role as an online learning platform.

All course materials are to be developed by the teachers and uploaded into a Moodle course, which students can access. Such an approach would ensure consistency in access to both existing courses that do not use interactive lecture materials and such that rely on SMSE, while also providing a simple and streamlined workflow for implementing interactive lectures and assignments into legacy Moodle courses.

All course materials are to be developed by the teachers and uploaded into a Moodle course, which students can access. Such an approach would ensure consistency in access to both existing courses that do not use interactive lecture materials and such that rely on SMSE, while also providing a simple and streamlined workflow for implementing interactive lectures and assignments into legacy Moodle courses.

Since Moodle and Jupyter utilize different approaches to storing data, an integration layer was developed.

Assuming that finalized courses are not subject to frequent change, intermediate storage was implemented in order to reduce workload on Moodle server, associated with frequent export of Jupyter notebooks [11] from courses and simplify data import into Jupyter workspace. SMSE data flow diagram is represented on Fig. 2.

Since Moodle exports course files in a *.mbz archive, a middleware service, SMSE Middleman Service, was developed. Its primary purpose is unpacking of the export file, generated by the Moodle platform, restoring original file names, forming a directory structure of the course in accordance with the names of the Moodle course and uploading files into a course specific git repository.

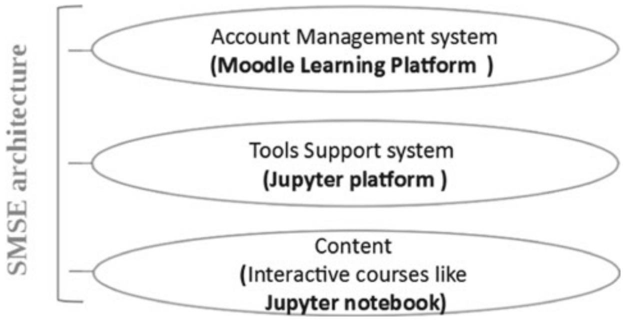


Fig. 1 SMSE architecture overview diagram

Fig. 2 SMSE data flow diagram

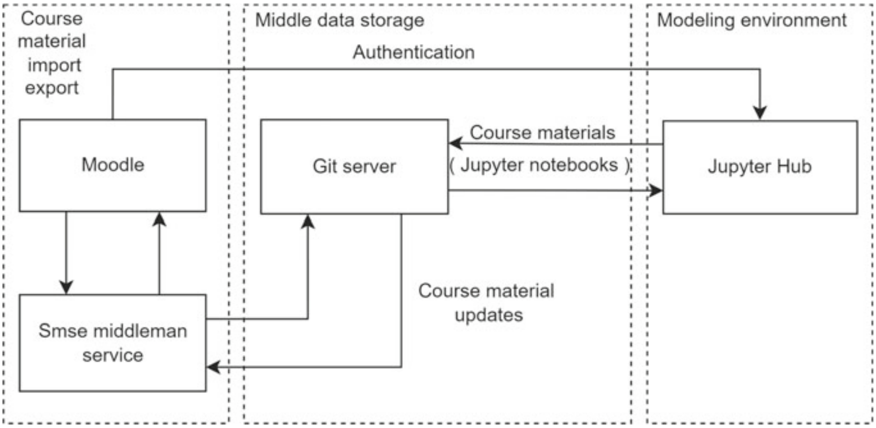


Fig. 3 SMSE deployment diagram

Alternatively, it is capable of uploading changes from the git repository in case updates were pushed from the Jupyter part of the SMSE platform. It should be noted, that in current implementation the Moodle is considered the only source of truth, therefore, this feature is disabled. SMSE deployment diagram is shown on Fig. 3.

4 Jupyter Side

Jupyter platform in SMSE project includes three base component: JupyterHub, JupyterLab and Jupyter Notebook. JupyterHub provides the possibility of using course notebooks to groups of users. It gives users access to common computational environments by way a running a set of images of course's virtual labs. In SMSE the JupyterHub is a set of processes that together provide an own JupyterLab for each person in a group.

In turn, JupyterLab provides web-based user interface for Jupyter Notebooks including consoles, terminal; text editor and file browser to control own file space

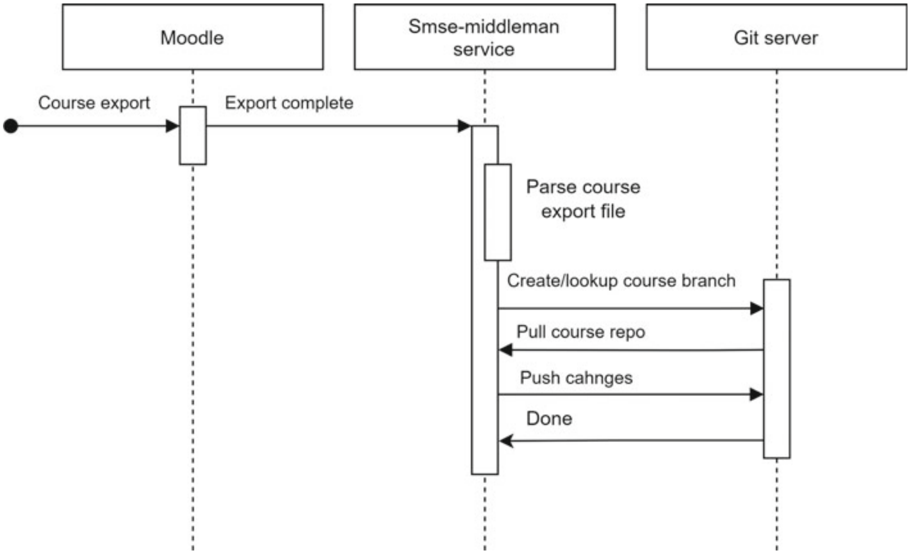


Fig. 4 Document export process diagram

as well as switching between computational kernels. Notebooks themselves are an ideal form of course documents that combines textual information, pictures, links, and sections of code into one document. Thanks to the support of multikernel computing there can be used the examples of code written in different programming languages in one notebook.

Moodle server provides creation of a course and supports authentication of users. Git server executes procedures for creation an archive with Jupyter Notebook files and export of the archive to JupyterLab. Interaction diagram on Fig. 4 reveals the process of export the course document from Moodle to Git server.

The middleware service consists of several components: collection of preliminary scripts, export file parser, course formatter, git adapter and git adapter. Scripts perform preparatory actions on input export files and create initial directory for the course and after performing these steps invoke the Middleman service itself.

Course file parser is the first component in the execution sequence and its purpose is self-descriptive - it parses the course.xml file, generated by moodle and builds a data structure, representing notebooks in course that contains their original name, file path in the unpacked Moodle export file and its association with sections of the course. This data is then passed to the course formatter component which based on the information about desired course structure, creates course directory layout and transfers files from their location in the raw export directory into their respective section directory, restoring original file names in the process.

The last step in uploading course notebooks into the git repository is handled by git adapter, which receives course ID and a path to formatted directory and pushes all changes in the course into a repository.

5 Kernel and Virtual Server Creation

Jupyter notebooks can contain code fragment in different programming languages. Therefore, programming tools must be installed in JupyterLab that will execute this code in the corresponding programming languages. These tools are called as kernels. There are many kernels that can be downloaded and installed in JupyterLab [13]. Installing kernels is the same as installing regular software packages. Often the pip package installer is used. For some programs, for example, MATLAB and OpenModelica, the kernel performs the function of a command translator into the program. In this case, you must first install the program on the same server, and then install the software packages for the kernel.

If JupyterLab is used in a separate server docker image, then it is better to write the sequence of commands for installing kernels in the dockerfile and generate a new docker image.

Using one server with JupyterLab and many kernels may seem like a simple solution. But to ensure flexible management of kernels, environments, software versions, it is better to use several virtual servers with JupyterLab. Here it is very convenient to use Jupyter Docker Stacks [14]. This is a set of docker images of servers that contain JupyterLab, various kernels and addition programs.

There are several base docker images which you can use to create new images with additional kernels and software. But, as it follows from practice, you should carefully choose the basic image. JupyterLab with older versions of images may not interact correctly with the current version of JupyterHub. For this reason, it is sometimes preferable not to take a ready-made base image, but to generate your own image based on the dockerfile that was used to generate the base image [15].

The diagram of using docker images as virtual SMSE servers is shown in Fig. 5.

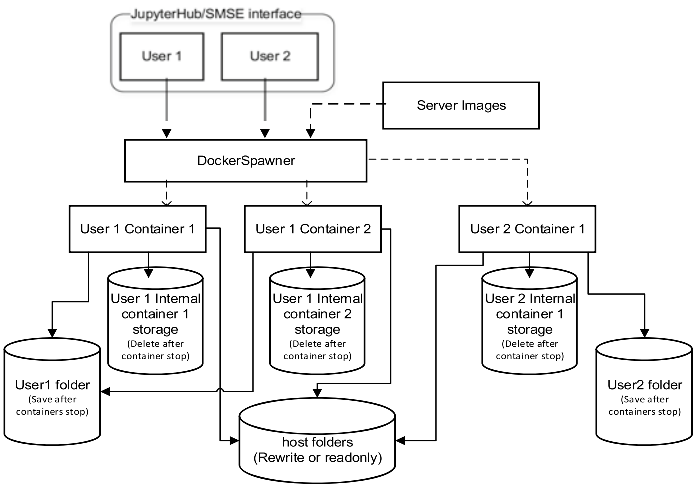


Fig. 5 Diagram of using docker images as virtual SMSE servers

As shown in the diagram, docker images are used to spawn server containers for each user that runs the image. Containers exist only for the duration of their operation. After the user terminates the container, it is deleted. This process raises two important questions:

- How is user data stored in a container?
- How to transfer data from JupiterHub (for example, Moodle heading data) to a container with JupyterLab?

There are three options for data storage in SMSE:

- Container storage, which is removed when the container terminates;
- The user storage that is automatically created for the user on the host server. This storage is not removed with the container, but is only available to the user and is accessible from any container that the user launches.
- Shared storage on the host server that is available to all users and all containers. This storage is convenient for storing some common files, for example ssl certificates.

The first two stores are automatically maintained when containers are spawned. Shared storage is configured via the DockerSpawner.volumes parameter in the "jupyterhub_config.py" - JupyterHub configuration file [16]. In other hand, to pass environment variables from JupyterHub to the container, the c.Spawner.auth_state_hook parameter in the "jupyterhub_config.py" configuration file is very convenient [17].

6 Learning Process

From the user's perspective, SMSE consists just of Moodle, JupyterHub and JupyterLab. The diagram of user interaction with these components is represented on Fig. 6.

During interaction process the actions of a teacher and a student in using the system are basically identical except for the function of creating a course. The course materials are available to students from its directory in the Jupiter lab in read-only mode. At the same time, student can upload these materials to his working directory which is created automatically for each user at startup Labs.

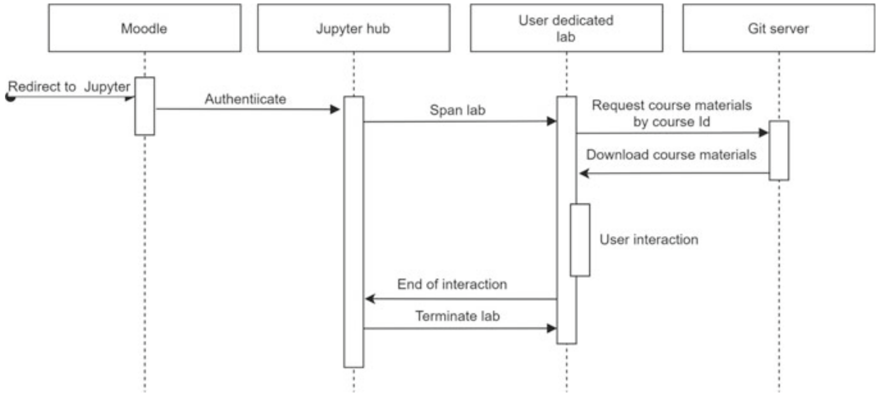


Fig. 6 User interaction with SMSE components

Use case diagram that combines teachers and student action is shown on Fig. 7.

Users can navigate to their personal Jupyter working environment either directly via a URL, if they wish to use it for practice or completing assignments that do not require prerequisite material from the course, or via a link, provided in the Moodle course itself as it is shown on Fig. 8.

The procuring of individual working environments for users is achieved by employing JupyterHub - a highly customizable solution for managing individual note-book servers. JupyterHub also handles user authentication but in the case of SMSE this process is handled via Moodle. After Start the SMSE for course it is possible to select the needed environment from a set of pre-installed in SMSE servers (see Fig. 9).

By starting selected server all course materials will be downloaded into the user's environment. This is achieved by employing a nbgitpuller addon for the Jupyter Note-book\Lab server and requires no further inputs from the user, providing a seamless pathway from the Moodle course page to a pre-configured working environment. In case a user navigates to the Jupyter via a SMSE specific link, the appropriate image will be selected automatically, otherwise a user can select an image of the work environment from the list of existing ones. Such an approach allows seamless learning workflow and allows high flexibility in creation of interactive courses.

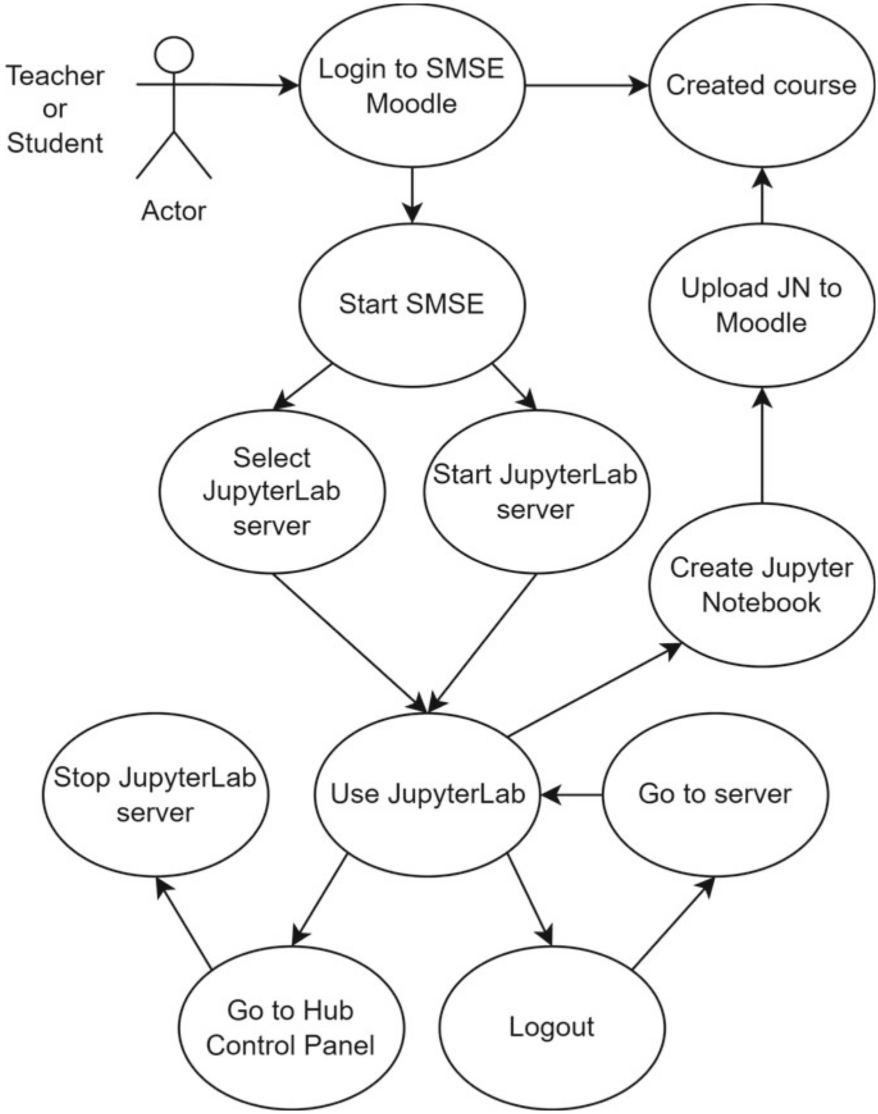


Fig. 7 Teacher/Student use case diagram

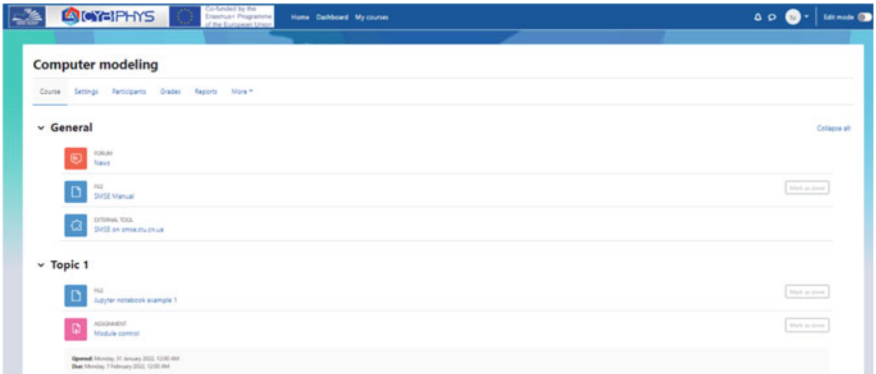


Fig. 8 Moodle course structure

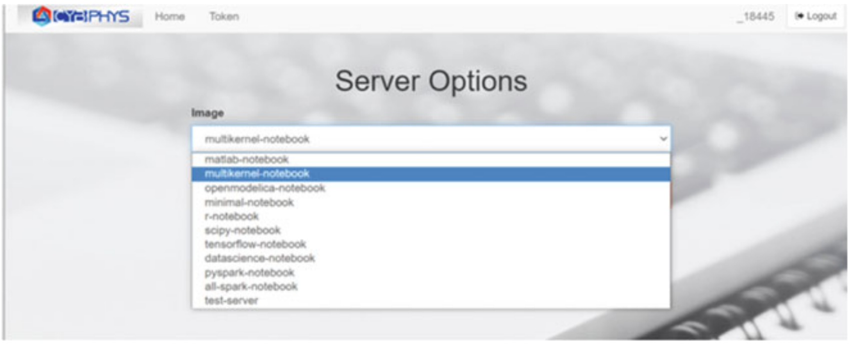


Fig. 9 A set of SMSE servers

Dashboard of running JupyterLab server with kernels pre-installed for the course is shown on the Fig. 10.

Spawning, or, in the other words, spinning up of individual notebook servers is done by DockerSpawner. It also has an additional benefit of having a set of working environments - Docker images, tailored specifically to the needs of either specific course or a group of courses that have considerable amount of shared libraries or kernels. System administrator does creation and\or amendment of these environments. It allows teachers to focus on teaching and treat Jupyter platform as a generic service provider [12]. For example, Fig. 11 shows Jupyter notebook created with Openmodelica kernel that includes code of BouncingBall model and result of simulation process performed by Multikernel server.

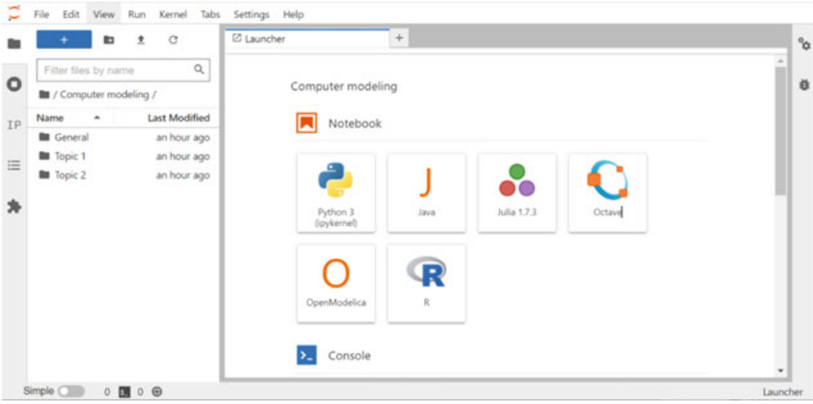


Fig. 10 Dashboard of running JupyterLab

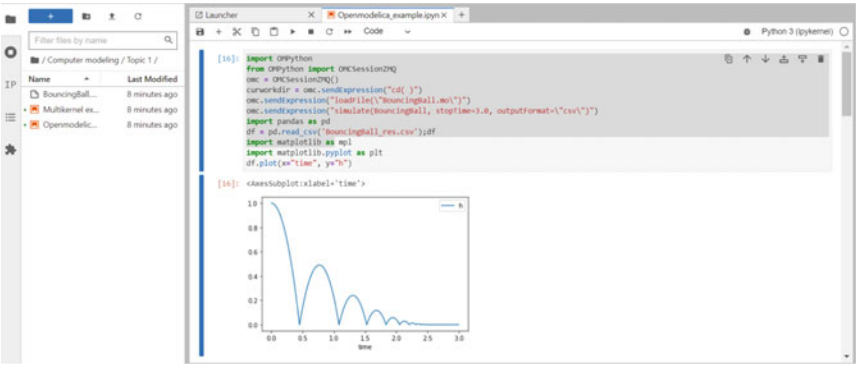


Fig. 11 Example of simulation in SMSE

7 Conclusion

Shared Modeling and Simulation Environment project provides a flexible and highly customizable solution to merging efficiency of online learning platform Moodle and on-demand preconfigured working environments of Jupyter notebooks to provide enhanced and highly interactive learning process for both in class and remote learning programmers and courses. Thanks to the implemented functions of middleman service and the use of the Git server, it is possible to integrate the Moodle environment with the Jupiter platform, providing automatic transfer of course documents and their launch in a virtual laboratory. Together, this implements a convenient tool to support online learning.

References

1. Laskova, K.: 21st century teaching and learning with technology: a critical commentary. In Academia Letters (2021)
2. Zhao, Y., Li, Y., Raicu, I., Lin, C., Tian, W., Xue, R.; Migrating scientific workflow management systems from the grid to the cloud. In: Li, X., Qiu, J. (eds.) Cloud Computing for Data-Intensive Applications. pp. 231–256. Springer, New York (2014). https://doi.org/10.1007/978-1-4939-1905-5_10
3. Ramakrishnan, L.: VGrADS: enabling e-science workflows on grids and clouds with fault tolerance. In: Proceedings of the Conference on High Performance Computing Networking, Storage and Analysis (2009)
4. Jupyter Free software, open standards, and web services for interactive computing across all programming languages. https://jupyter.org/. Accessed 30 Aug 2022
5. Development of practically-oriented student-centred education in the field of modelling of Cyber-Physical Systems (CybPhys). Detailed description of the project. Joint projects, p. 195 (2019)
6. Kazymyr, V., Shkarlet, S., Zabasta, A.: Practical-oriented education in modeling and simulation for cyber-physical systems. In 2020 10th International Conference on Advanced Computer Information Technologies, ACIT 2020 - Proceedings, pp. 691–694, 9208876, (2020)
7. Barba, L., et al.: Teaching and learning with jupyter (2019). https://jupyter4edu.github.io/jupyter-edu-book/. Accessed 30 Sep 2022
8. Blank, D.S., et al.: nbgrader: a tool for creating and grading assignments in the jupyter notebook. J. Open Source Educ. **2**(11), 32 (2019)
9. Pieschel, L., Welten, S., Gleim, L., Decker, S.: Teaching semantic web technologies through interactive jupyter notebooks. In: Companion Proceedings of SEMANTiCS (2021)
10. JupyterLab interface to palmetto. https://www.palmetto.clemson.edu/palmetto/basic/jupyter/. Accessed 30 Sep 2022
11. Toomey, D.: Learning Jupyter. Packt Publishing, Birmingham (2016)
12. Jha, S., Katz, D.S., Luckow, A., Merzky, A., Stamou, K.: Understanding scientific applications for cloud environments. In: Cloud Computing: Principles and Paradigms (2011)
13. Jupyter kernels. https://github.com/jupyter/jupyter/wiki/Jupyter-kernels. Accessed 30 Sep 2022
14. Docker Stacks documentation. Jupyter Docker Stacks. https://jupyter-docker-stacks.readthedocs.io/en/latest/. Accessed 30 Sep 2022
15. Jupyter/docker-stacks. https://github.com/jupyter/docker-stacks. Accessed 30 Sep 2022
16. Jupyterhub dockerspawner. DockerSpawner API. https://jupyterhub-dockerspawner.readthedocs.io/en/latest/api/index.html?highlight=DockerSpawner.volumes#id37. Accessed 30 Sep 2022
17. Spawners-JupyterHub 3.0.0 documentation. https://jupyterhub.readthedocs.io/en/stable/api/spawner.html. Accessed 30 Sep 2022

Automated Analysis of Production Audit with Returnable Waste and Semi-products by Deep Simple Recurrent Network with Losses

Tetiana Neskorodieva, Eugene Fedorov, Anastasiia Neskorodieva, Tetiana Sichko, and Tetyana Utkina

Abstract The article is devoted to the method creating problem for checked indicators estimate to automate the detection of anomalous data in the subject area of production audit, the transformations of which are represented by a mappings sequence. The data transformations model of production audit with returnable waste and semi-products based on a deep simple recurrent network with losses is offered. That allows to scale effectively the DLSRN model (to increase LSRN number without increase in training time of all DLSRN) in case of complications production. It allows to automate the process of the analysis and to use this model for intellectual technology of data analysis creation in the system of audit. The method of parametrical identification of a deep simple recurrent network with losses (DLSRN) reached further development by to use of the proposed one-step training of simple recurrent networks with losses (LSRN). This composition forms DLSRN and provides a representation of neural network weights in the form of raw materials shares, semi-products, finished goods, non-returnable and returnable waste. That allows increasing estimation accuracy by the model of data transformations of production audit with semi-products and returnable waste. It allows using the received estimates for forming the recommended solutions in audit DSS. The algorithm of one-step training of a simple recurrent network with losses (LSRN) due using of CUDA parallel processing technology of information is improved. That allows acceleration determination of values of LSRN neural network weights.

T. Neskorodieva · A. Neskorodieva · T. Sichko
Vasyl' Stus Donetsk National University, 600-Richcha Street, 21, Vinnytsia 21021, Ukraine
e-mail: t.neskorodieva@donnu.edu.ua

A. Neskorodieva
e-mail: a.neskorodieva@donnu.edu.ua

T. Sichko
e-mail: t.sichko@donnu.edu.ua

E. Fedorov (✉) · T. Utkina
Cherkasy State Technological University, Shevchenko Blvd., 460, Cherkasy 18006, Ukraine
e-mail: fedorovee75@ukr.net

T. Utkina
e-mail: t.utkina@chdtu.edu.ua

S. Shkarlet et al. (eds.), *Mathematical Modeling and Simulation of Systems*, Lecture Notes in Networks and Systems 667,
https://doi.org/10.1007/978-3-031-30251-0_12

Keywords audit data · data mapping · a deep simple recurrent neural network with losses · intellectual technology · anomaly · CUDA technology

1 Introduction

The research subject relevance is characterized by global and national trends in digital economy development [1] which are characterized by fast distribution of the multilevel, territorially dispersed computer systems and networks with distributed databases and knowledge of commercial purpose.

Today a current scientific and technical problem of information technologies in the financial and economic sphere is the design methodology forming and creation of audit decision support systems based on intellectual technologies of the large volumes analysis of financial and economic data with the purpose of functionality expansion, increasing of efficiency and universality of audit information technology.

One of the problems during the creation of such systems is the evaluation values methods development for the checked indicators on an inconsistency (data of raw purchase materials and finished goods selling; sales or purchases data in online stores and in financial statements data; data about structure and transportation volumes in logistic network and deliveries or sales in the reporting); abnormal data detection – rare values of characteristics set of an object or audit process.

The work purpose is quality improvement of data analysis in DSS of production audit with intermediate semi-products and returnable waste based on a neural network model of reporting data conversions which reflects production structure.

1. The transformations modeling method of production reporting data with intermediate semi-products and returnable waste due to use of the proposed deep simple recurrent network with losses (DLSRN) which represents the composition of simple recurrent networks with losses (LSRN) is for the first time offered. That allows effective scaling (to increase LSRN number without an increase in training time of all DLSRN) of the DLSRN model in case of production complication. (That allows automating the process of the analysis in DSS of audit).
2. Reached further development parametrical identification method of a deep simple recurrent network model with losses (DLSRN) due to use of the proposed done-step training of simple recurrent networks with losses (LSRN), which composition forms DLSRN. This method provides neural network weights representation in the form of raw materials shares, semi-products, finished goods, non-returnable and returnable waste. That allows increasing estimation accuracy on model of audit data transformations of production intermediate semi-products and returnable waste. (Use the received estimates for forming of the recommended solutions in audit DSS).

2 Literature Review

Let's select the following anomalies main types:

– point (are provided by points in character space) [2];
– contextual (usually a time series point or the rarefied data which depends on the environment) [3];
– collective (the time series section or the rarefied data) [4].

Let's select the following main anomalies detection methods: approach based on rules (logical approach) [5]; approach based on ANN [6]; approach based on a Bayesian inference or its expansion [7]; approach based on a clustering [8]; approach based on the neighborhood (metric approach) [8]; approaches based on distributions [8]; approach based on regression model [8]; approach based on the spectral theory [8]; approach based on information theory [8].

Now the most popular is anomalies detection approach based on neural networks. The disadvantage of the one-class machine of reference vectors is the restriction for reference vectors number. The disadvantage of ANN with classification is the requirement to classify anomalies that is not always possible owing to labor input of obtaining the marked data on each anomaly type. Therefore, ANN with an associative memory and ANN of the forecast were selected in the work.

The neural networks majority with an associative memory possess some or more disadvantages:

- do not possess a hetero associative memory [9];
- do not work material data [10];
- do not have a high capacity of associative memory [11];
- do not have high accuracy [12];
- have high computing complexity [13].

As neural networks of the audit indicators forecast are most often used: Elman's (ENN) neural network or a simple recurrent network (SRN) [9]; the bidirectional recurrent neural network (BRNN) [9]; long short-term memory (LSTM) [10]; the bidirectional recurrent neural network (BLSTM) [11]; the lock recurrent block (GRU) [12]; the bidirectional recurrent neural network (BGRU) [13]; the network an echo status (ESN) [14]; the machine of liquid statuses (LSM) [14]; the neural network of nonlinear autoregression (NARNN) [14]; the neural network of nonlinear autoregression moving average (NARMANN) [14].

Thus, any of networks does not meet all criteria.

For training acceleration and increase of neural network forecasting accuracy models now are used metaheuristics (or modern heuristics). The metaheuristics expands opportunities heuristic, combining heuristic methods based on high-level strategy.

Existing metaheuristics possess one or more of the following disadvantages:

– there is only an abstraction description of a method, or the method description is focused on the solution of a certain task only [15];

– iterations number influence on the solution search process is not considered [15];
– the method convergence is not guaranteed [16];
– there is no opportunity to use not binary potential solutions [16];
– the determination procedure of parameters values is not automated [17];
– there is no opportunity to solve conditional optimization problems [17];
– insufficient method accuracy [18].
– Considered above methods are applied in information intellectual technologies to the solution of management tasks of economic objects and in audit and analysis tasks, in particular [19].

3 Materials and Methods

3.1 Logical-Neural Method of Audit Data Analysis

Valuation and forecasting methods development and forming of the generalized associative connections is described in authors' works of this article [20]. Creation of these methods purposes:

- computing complexity decrease (neural network calculations parallelization possibility);
- accuracy increase (use metaheuristic with random searching; stochastic machines);
- automatic structural identification (use clustering with automatic detection of clusters number);
- methods application possibility for the generalized analysis of elements and subelements of audit data domain.

The model choice in DSS depends on [21]: the characteristics of a data type of audit (given time series, spatial data (in the form of mappings)); identification of anomalies (based on comparison of the actual (checked) indicator value and its expected or evaluation value on the constructed models); check mode: express audit or profound audit and check mode parameters: time, accuracy.

This choice is schematically formalized in the form of a binary decision tree for the choice of neural network model for generally – multiple data analysis of audit data domain (Fig. 1) [22].

At the first level the model choice is carried out depending on data type. If the forecast of data is executed, then the model with a time delay is selected. If data transformation is executed, then the model with an associative memory is selected.

At the second level the model choice based on an associative memory depends on production type: with waste or without them. If production with waste, then is selected model which contains: the layers corresponding to data transformation stages; the layers corresponding to non-returnable losses; cyclic connections for accounting of returnable losses in the layers corresponding to data transformation stages. If production without waste, then is selected model which contains the layers

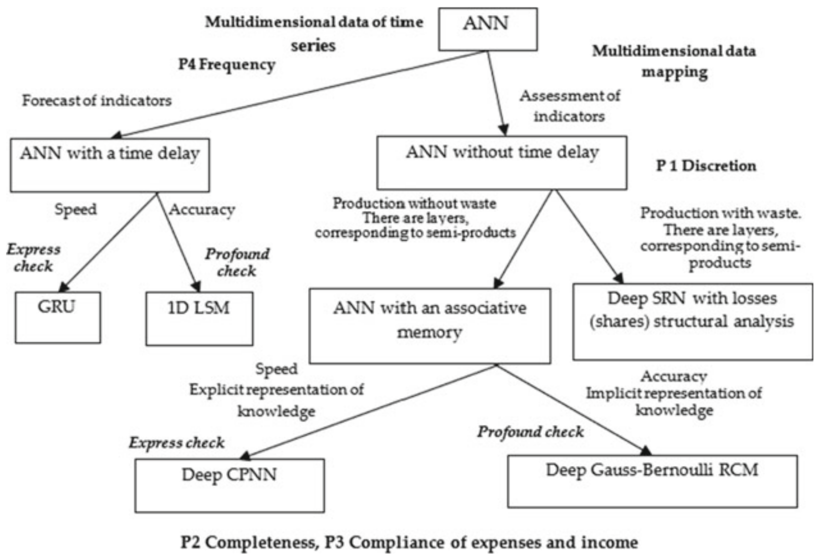


Fig. 1 A binary choice decision tree of neural network model for generality-multiple analysis of audit data domain

corresponding to data transformation stages and auxiliary layers between them for implementation of an associative memory.

At the third level the choice of model for waste-free production depends on the speed of training and accuracy and on explicit or implicit representation of knowledge.

3.2 Neural Network Data Transformation Method of Production Audit with Waste and Intermediate Semi-products

Production can be with returnable and non-returnable waste therefore in this work the method of neural network data transformation of production audit with waste and intermediate semi-products in the mode deepened checks which includes is offered:

- neural network model creation of transformations in the mode of profound check;
- the creation of a parametrical identification method of the neural network model of transformations for reduction of computing complexity (at the expense of a calculations parallelization possibility) that increases training speed.

Among the determined recurrent neural networks working with material data, the simple recurrent network (SRN) has the simplest architecture and the smallest computing complexity therefore it is selected in work as a conversion neural network in the mode of profound check.

3.3 Indicators Forming Method of Production Audit

In the DSS of audit [20] the check of a prerequisite "compliance of expenses and income" to the provisions (standards) of accounting is formalized in the form of check problem of the "settlings with suppliers settlings with customers" mapping.

The equivalence check of atomic elements of mappings is preceded by the analysis of quantitative indices for the check period for the purpose of values identification that deviates from predicted. The predicted values are defined based on regularities that are created on the checked data for others (as a rule, previous) periods.

Let's consider data mapping "settlings with suppliers and settlings with customers". Audit data values when checking this mapping are formed based on a set of quantitative indices of the money characterizing transfers to suppliers of stocks (raw materials and/or components) and receipts of money from customers for the sold products [20]:

$$(V_u, \Delta_u), u \in U, \left(V_g, \Delta_g\right), g \in G, \tag{1}$$

where V – quantitative index (in physical units), Δ – monetary indicator (the amounts transferred by suppliers/customers), u – stocks type, U – stock types set, g – finished goods type, G – set of finished good types.

In the analysis of this mapping (1) (in forward and backward direction) as elements of couples of training set can be selected:

$$\begin{aligned}
&x = \left(V_u(t), V_u(t - \Delta\tau), \ldots V_u(t - M\Delta\tau), u \in U_g\right), y = V_g(t), g \in G, \\
&x = \left(\Delta_u(t), \Delta_u(t - \Delta\tau), \ldots \Delta_u(t - M\Delta\tau), u \in U_g\right), y = \Delta_g(t), g \in G, \\
&x = \left(V_g(t), V_g(t - \Delta\tau), \ldots V_g(t - M\Delta\tau), g \in G_u\right), y = V_u(t), u \in U, \\
&x = \left(\Delta_g(t), \Delta_g(t - \Delta\tau), \ldots \Delta_g(t - M\Delta\tau), g \in G_u\right), y = \Delta_u(t), u \in U,
\end{aligned} \tag{2}$$

where x – input signal, $M\Delta\tau$ – log of delay or advancing, U_g – set of raw materials which is used in production of finished goods of a view g, G_u – set of finished goods by which production view raw materials are used u, y – signal output.

As value of a delay in (2) log or advancing M the value of an indicator of an operational cycle (mean value of time between operation of purchase of raw materials and selling of the finished goods made from these raw materials) and a production cycle can be selected. This indicator depends on the industry, a type of production and the range of products ($1 < M < 30$).

The direct analysis will allow to reveal regularities of finished goods selling depending on purchase of raw materials by finished good types for the periods the checks preceding the period.

The return analysis will allow to reveal regularities of raw materials purchase for the sold finished goods, by raw material types for the periods the checks preceding the period.

The information for the preceding checks periods is considered reliable. Therefore, the relevant data can be selected as a training set.

Comparison of estimated by the model and check values of a will allow revealing finished goods and stock types and the quantization periods by which there are essential deviations and which will be recommended to the decision-maker for detailed study at the bottom level.

To check the mapping (in the forward and reverse direction), the following elements of pairs of the training set can be selected [22]:

$$\begin{aligned} &x = (V_u(t), V_u(t-\Delta\tau), \ldots V_u(t-M\Delta\tau)), y = V_u(t+\Delta\tau), u \in U, \\ &x = (\Delta_u(t), \Delta_u(t-\Delta\tau), \ldots \Delta_u(t-M\Delta\tau)), y = \Delta_u(t+\Delta\tau), u \in U, \\ &x = \big(V_g(t), V_g(t-\Delta\tau), \ldots V_g(t-M\Delta\tau)\big), y = V_g(t+\Delta\tau), g \in G, \\ &x = \big(\Delta_g(t), \Delta_g(t-\Delta\tau), \ldots \Delta_g(t-M\Delta\tau)\big), y = \Delta_g(t+\Delta\tau), g \in G. \end{aligned} \tag{3}$$

For neural network models of the one-dimensional machine of liquid statuses (1D LSM), the lock recurrent block (GRU), a deep simple recurrent network with losses (DLSRN), the deep limited machine of Cauchy (DGBRCM) for increase an accuracy of the forecast the training data of production audit (2), (3) are normalized in a view

$$x_i = \frac{x_i - x_i^{min}}{x_i^{max x_i^{min}}}, y_i = \frac{y_i - y_i^{min}}{y_i^{max x_i^{min}}}. \tag{4}$$

3.4 *Conversion Model Based on a Deep Simple Recurrent Network with Losses*

As well as in a traditional simple recurrent network (SRN) the proposed deep simple recurrent network with losses (DLSRN) the hidden layer is recurrent.

Unlike traditional SRN in the proposed DLSRN: for accounting of non-returnable waste each neuron of a recurrent layer has a connection with a layer of non-returnable waste; for semi-products accounting instead of one simple recurrent network there is a composition of simple recurrent networks, and an output layer of the recurrent network previous idle time is an input layer of the subsequent simple recurrent network, and all non-recurrent layers of composition, except an input layer, correspond to layers of non-returnable waste, semi-products and finished goods.

The block diagram (Fig. 2) of a deep simple recurrent network with losses (DLSRN) fully connected recurrent layers of production (neurons are formed it, designated in continuous black color), not fully connected non-recurrent layers of

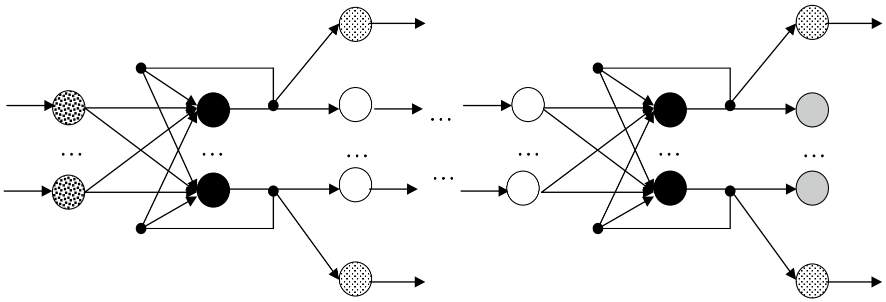


Fig. 2 The block diagram of a deep simple recurrent network model with losses (DLSRN)

non-returnable waste (neurons are formed it, designated in low-fat points), by not fully connected non-recurrent layers of semi-products (neurons are formed it, designated in continuous white color), not fully connected non-recurrent layer of finished goods (neurons are formed it, designated in continuous gray color).

The DLSRN model, the executing mapping of each input sample of raw materials $\boldsymbol{x} = (x_1, \ldots, x_{N^{(0)}})$ output samples of finished goods $\boldsymbol{y} = \left(y_1^{2H}, \ldots, y_{N^{(2H)}}^{2H}\right)$ non-returnable waste $\hat{y}^{(1)} = \left(\hat{y}_1^{(1)}, \ldots, \hat{y}_{N^{(1)}}^{(1)}\right), \ldots, \hat{\boldsymbol{y}}^{(2H)} = \left(\hat{y}_1^{(2H)}, \ldots, \hat{y}_{N^{(2H)}}^{(2H)}\right)$ it is presented in the form:

$$
\begin{aligned}
&y_i^{(0)}(n) = x_i, i \in \overline{1, N^{(0)}},\\
&y_j^{(2h+1)}(n) = f^{(2h+1)}\left(s_j^{(2h+1)}(n)\right), j \in \overline{1, N^{(2h+1)}}, k \in \overline{1, H},\\
&s_j^{(2h+1)}(n) = b_j^{(2h+1)} + \sum_{i=1}^{N^{(2h)}} w_{ij}^{(2h+1)} y_i^{(2h)}(n) + \sum_{i=1}^{N^{(2h+1)}} \overset{(2h+1)}{\underset{ij}{w}} y_i^{(2h+1)}(n-1),\\
&y_j^{(2h+2)}(n) = f^{(2h+2)}\left(s_j^{(2h+2)}(n)\right), j \in \overline{1, N^{(2h+2)}}, k \in \overline{1, H-1},\\
&s_j^{(2h+2)}(n) = b_j^{(2h+2)}(n) + w_j^{(2h+2)} y_j^{(2h+1)}(n),\\
&\hat{y}_j^{(2h+2)}(n) = \hat{f}^{(2h+2)}\left(\hat{s}_j^{(2h+2)}(n)\right),\ j \in \overline{1, N^{(2h+2)}},\ j \in \overline{1, N^{(2h+2)}},\\
&\hat{s}_j^{(2h+2)}(n) = \hat{b}_j^{(2h+2)}(n) + \hat{w}_j^{(2h+2)} y_j^{(2h+1)}(n),
\end{aligned}
\tag{5}
$$

where $N^{(2h+1)}$ – product type number (the neurons in a recurrent layer), $N^{(2h+2)}$ – semi-product type number/finished goods (the neurons number in a non-recurrent layer), $N^{(2h+1)} = N^{(2h+2)}$, $N^{(0)}$ – neurons number of an input layer (raw materials layer), H – recurrent layers number, $b_j^{(2h+1)}$ – bias for production layer neuron j -th, $b_j^{(2h+2)}$ – bias for j -th of semi-products layer neuron ($h < H - 1$)/finished goods ($h = H - 1$), $\hat{b}_j^{(2h+2)}$ – bias for j -th of non-returnable waste layer neuron, $w_{ij}^{(2h+1)}$ – connection weight from i -th neuron of raw materials layer ($h = 0$)/semi-products ($h > 0$) to j -th production layer neuron, $w_j^{(2h+2)}$ – connection weight from i -th production layer neuron to j -th semi-products layer neuron ($h < H - 1$)/finished goods ($h = H - 1$), $\breve{w}_{ij}^{(2h+1)}$ – connection weight from i -th production layer neuron to j -th a production layer neuron, $\hat{w}_j^{(2h+2)}$ – connection weight from i -th production layer neuron to j -th non-returnable waste layer neuron, $y_j^{(2h+1)}(n)$ – j -th production layer neuron output in timepoint n, $y_j^{(2h+2)}(n)$ – output of j -th semi-products layer neuron ($h < H - 1$)/finished goods ($h = H - 1$) timepoint n, $\hat{y}_j^{(2h+2)}(n)$ – output of j -th neuron of non-returnable waste layer in timepoint n, $f^{(2h+1)}$ – activation function of production layer neurons, $f^{(2h+1)}(s) = \exp\left(-\frac{s^2}{2}\right)$, $f^{(2h+2)}$ – activation function of semi-products layer neurons ($h < H - 1$)/finished goods ($h = H - 1$), $f^{(2h+2)}(s) = \exp\left(-\frac{s^2}{2}\right)$, $\hat{f}^{(2h+2)}$ – activation function of non-returnable waste layer neurons, $\hat{f}^{(2h+2)}(s) = \exp\left(-\frac{s^2}{2}\right)$.

3.5 Parametrical Identification Method of a Simple Recurrent Network with Losses Based On One-Step Training

In this work for LSRN model training means the choice of such values of a vector of parameters is selected $\boldsymbol{w} = \left(w_{11}^{(2h+1)}, \ldots, w_{N^{(2h)}N^{(2h+1)}}^{(2h+1)}, w_1^{(2h+2)}, \ldots, w_{N^{(2h+2)}}^{(2h+2)}, \breve{w}_{11}^{(2h+1)}, \ldots, \breve{w}_{N^{(2h+1)}N^{(2h+1)}}^{(2h+1)}, \hat{w}_1^{(2h+2)}, \ldots, \hat{w}_{N^{(2h+2)}}^{(2h+2)}\right)$, which deliver a minimum of a root mean square error (the differences of a sample on model and a test sample)

$$F = \frac{1}{PN^{(2h+2)}} \sum_{\mu=1}^{P} ||y_\mu^{(2h+2)} - d_\mu^{(2h+2)}||^2 + \frac{1}{PN^{(2h+2)}} \sum_{\mu=1}^{P} ||\hat{y}_\mu^{(2h+2)} - \hat{d}_\mu^{(2h+2)}||^2 \rightarrow \min_{w}, \tag{6}$$

where $\boldsymbol{y}_\mu^{(2h+2)}$, $\hat{\boldsymbol{y}}_\mu^{(2h+2)}$ – μ -th output samples on model, $\boldsymbol{d}_\mu^{(2h+2)}$, $\hat{\boldsymbol{d}}_\mu^{(2h+2)}$ – μ - th test output samples, P – power of a test set.

The proposed DLSRN for a possibility of calculations parallelization and explicit representation of knowledge the method a one-step method of training is proposed and it is proposed to train separately each LSRN of composition.

Parametrical identification method of the LSRN model consists of the following blocks.

1. Initialization

The training set is set

$$\{\left(Z_{\mu}^{(2h+1)}, z_{\mu}^{(2h+2)}, \hat{z}_{\mu}^{(2h+2)}, \breve{Z}_{\mu}^{(2h+1)}\right)| \\ \mathbf{Z}_{\mu}^{(2h+1)} \in R^{N^{(2h)} \times N^{(2h+1)}}, \mathbf{z}_{\mu}^{(2h+2)} \in R^{N^{(2h+2)}}, \\ \hat{\mathbf{z}}_{\mu}^{(2h+2)} \in R^{N^{(2h+2)}}, \breve{Z}_{\mu}^{(2h+1)} \in R^{N^{(2h+1)} \times N^{(2h+1)}}\}, \mu \in \overline{1, P}, \tag{7}$$

where $\boldsymbol{Z}_{\mu}^{(2h+1)}$ – μ -th training matrix of associative connections of raw materials/semi-products with production of semi-products/finished goods, $z_{\mu}^{(2h+2)}$ – μ -th training vector of associative connections of production of semi-products/finished goods with semi-products/finished goods, $\hat{z}_{\mu}^{(2h+2)}$ – μ -th training vector of associative connections of production of semi-products/finished goods with non-returnable waste, $\breve{\boldsymbol{Z}}_{\mu}^{(2h+1)}$ – μ the training matrix of associative connections of semi-products/finished goods with returnable waste, P – power of a training set.

2. Setup of synoptic connection weights between neurons of raw materials layer ($h = 0$)/semi-products ($h > 0$) production layer neurons

$$w_{ij}^{(2h+1)} = \frac{1}{P} \sum\nolimits_{\mu=1}^{P} z_{\mu ij}^{(2h+1)}, i \in \overline{1, N^{(2h)}}, j \in \overline{1, N^{(2h+1)}}. \tag{8}$$

3. Setup of synoptic connection weights between neurons of production layer and neurons of semi-products layer ($h < H - 1$)/finished of goods ($h = H - 1$)

$$w_{j}^{(2h+2)} = \frac{1}{P} \sum\nolimits_{\mu=1}^{P} z_{\mu j}^{(2h+2)}, j \in \overline{1, N^{(2h+2)}}. \tag{9}$$

4. Setup of synoptic connection weights between production layer neurons

$$\breve{w}_{ij}^{(2h+1)} = \frac{1}{P} \sum\nolimits_{\mu=1}^{P} \breve{z}_{\mu ij}^{(2h+1)}, i \in \overline{1, N^{(2h+1)}}, j \in \overline{1, N^{(2h+1)}}. \tag{10}$$

5. Setup of synoptic connection weights between neurons of production layer and neurons of a layer of non-returnable waste

$$\hat{w}_{j}^{(2h+2)} = \frac{1}{P} \sum\nolimits_{\mu=1}^{P} \hat{z}_{\mu j}^{(2h+2)}, j \in \overline{1, N^{(2h+2)}}. \tag{11}$$

6. Normalization of synoptic connection weights between neurons of a semi-products layer ($h = 0$)/($h > 0$) production layer neurons

$$w_{ij}^{(2h+1)} = \frac{w_{ij}^{(2h+1)}}{\sum_{l=1}^{N^{(2h+1)}} w_{il}^{(2h+1)}}, i \in \overline{1, N^{(2h)}}, j \in \overline{1, N^{(2h+1)}}. \tag{12}$$

7. Normalization of synoptic connection weights between layer neurons of production and layer neurons of semi-products ($h < H - 1$)/finished of goods

$(h = H - 1)$

$$w_j^{(2h+2)} = \frac{w_j^{(2h+2)}}{\sum_{l=1}^{N^{(2h+1)}} \breve{w}_{il}^{(2h+1)} + \hat{w}_j^{(2h+2)} + w_j^{(2h+2)}}, j \in \overline{1, N^{(2h+2)}}. \tag{13}$$

8. Normalization of synoptic connection weights between production layer neurons

$$\breve{w}_{ij}^{(2h+1)} = \frac{\breve{w}_{ij}^{(2h+1)}}{\sum_{l=1}^{N^{(2h+1)}} \breve{w}_{il}^{(2h+1)} + \hat{w}_j^{(2h+2)} + w_j^{(2h+2)}}, i \in \overline{1, N^{(2h+1)}}, j \in \overline{1, N^{(2h+1)}}. \tag{14}$$

9. Normalization of synoptic connection weights between layer neurons of production and layer neurons of non-returnable waste

$$\hat{w}_j^{(2h+2)} = \frac{\hat{w}_j^{(2h+2)}}{\sum_{l=1}^{N^{(2h+1)}} \breve{w}_{il}^{(2h+1)} + \hat{w}_j^{(2h+2)} + w_j^{(2h+2)}}, j \in \overline{1, N^{(2h+2)}}. \tag{15}$$

10. Setup of layer bias of production

$$b_j^{(2h+1)} = -\left(\sum_{i=1}^{N^{(2h)}} w_{ij}^{(2h)} + \sum_{i=1}^{N^{(2h+1)}} \breve{w}_{ij}^{(2h+1)}\right), j \in \overline{1, N^{(2h+2)}}. \tag{16}$$

11. Setup of layer bias of semi-products $(h < H - 1)$/finished of goods $(h = H - 1)$

$$b_j^{(2h+2)} = -w_j^{(2h+2)}, j \in \overline{1, N^{(2h+2)}}. \tag{17}$$

12. Setup of layer bias of non-returnable waste

$$\hat{b}_j^{(2h+2)} = -\hat{w}_j^{(2h+2)}, j \in \overline{1, N^{(2h+2)}}. \tag{18}$$

Thus, the weights vector of **w** was created.

4 Modeling and Results

LSRN simulation was carried out on a computer with the following characteristics: DELL, Intel Core i5-8300H, 8 Gb RAM. Performance estimates were based on root mean square error (RMSE), and epochs number.

The task of improving the quality of data analysis in the DSS of production audit with sems-products and returnable waste based on a neural network model for transforming reported data was solved in the article. Experimental studies were carried out on a dataset that was used in the audit of a grocery enterprise. This dataset contains 2500 vectors, each of which is characterized by 25 indicators of raw materials (flour, butter, ghee, water, soda, salt, sugar, yeast, eggs, sour cream, vinegar, beef, pork, lamb, chicken, milk, cheese, peppers, potatoes, cottage cheese, cabbage, mushrooms, raisins, onions, carrots), 15 processing centers for raw materials, 15 non-returnable waste types, and 15 types of finished products (pancakes, pies with potatoes and mushrooms, pies with cabbage, carrot cutlets, cutlets cabbage, cheesecakes, beef dumplings, pork dumplings, pork and beef dumplings, beef chebureks, lamb chebureks, cabbage dumplings, cottage cheese dumplings, potato and mushroom dumplings, cabbage dumplings).

During the simulation, 80% of the vectors were randomly selected for training, and 20% for testing procedures. To easily reproduce the results developed by methods of other researchers, the authors chose a well-known and available SRN, which was modified to represent a Gaussian function with zero mean and one standard deviation. Since SRN assumes the implementation of an iterative learning procedure, this article investigated the effect of the number of epochs on the results of the method in the case of the traditional backpropagation learning method. The experiment was carried out by changing the number of epochs from 100 to 3000. The results of this study are shown on Fig. 3.

As can be seen from Fig. 3, an increase in the number of the procedure training epochs reduces the error of the algorithmic implementation of the method in the training mode and in the application mode. The error saturation stage of the algorithm starts from the 2000 epochs. Increase algorithm epochs number the does not increase the accuracy of its operation, it increases the duration of the training procedure. Accordingly, these values are chosen as optimal for the operation of the proposed algorithmic implementation of the developed method when processing the investigated dataset.

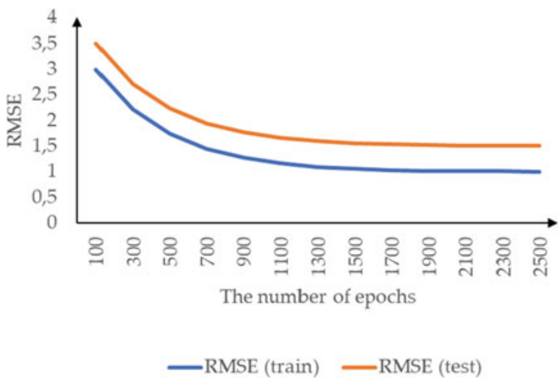


Fig. 3 The error values for training and application modes

Table 1 Comparison of training methods for the proposed LSRN neural network model

Method	Learning characteristics		
	RMSE (train)	RMSE (test)	Computational complexity
Proposed (one-step training)	0.928	1.456	~ P
Traditional (backpropagation)	1.013	1.521	~ $P(N^{(2h)} + N^{(2h+1)})N^{(2h+1)}$

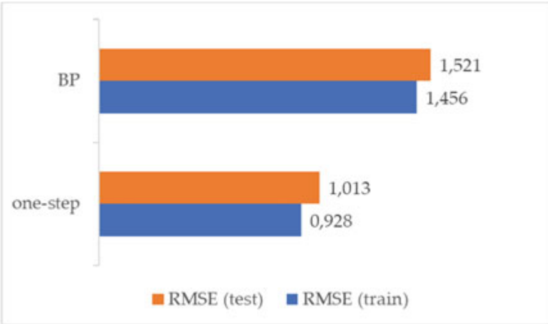


Fig. 4 Error values for all methods investigated

5 Comparison and Discussion

The comparison was based on RMSE and computational complexity, which affects the learning rate. For the traditional learning method and RMSE were calculated at 2000 epochs. The results of experimental modeling of all investigated methods for both training and applied modes are summarized in Table 1, where is the number of raw materials/incoming semi-products for the h -th LSRN, is the number of productions (the number of neurons in the recurrent layer) for the h -th LSRN, P is the cardinality of the training set. For clarity, the results are also shown in Fig. 4.

According to Table 1 and Fig. 4, use of the proposed method (one-step training) increases estimation accuracy on model. According to Table 1, use of the proposed method (one-step training) in combination with GPU reduces computing complexity and by that increases training speed.

6 Conclusions

The mappings modeling method of production audit data with intermediate semi-products and returnable waste based on deep simple recurrent network with losses (DLSRN) is for the first time offered. This model represents composition of simple recurrent networks with losses (LSRN) what effectively to scale (to increase number of LSRN without increase in time of training of all DLSRN) the DLSRN model in case of production complication.

Reached further development a parametrical identification method of a deep simple recurrent network with losses (DLSRN) due to use of the proposed one-step training of simple recurrent networks with losses (LSRN) which composition forms DLSRN which provides representation of a neural network weights in the form of shares of raw materials, semi-products, finished goods, non-returnable and returnable waste. That allows to increase estimation accuracy on model of data mapping of the audited production with intermediate semi-products and returnable waste, i.e. to provide determination coefficient not less than 0.98.

The algorithm of one-step training of a simple recurrent network with losses (LSRN) due to use CUDA information parallel processing technology. That allows to accelerate determination of weights of values a LSRN neural network approximately in $(N^{(2h)} + N^{(2h+1)})N^{(2h+1)}$ times, where $N^{(2h)}$ – raw materials types number/the arriving semi-products for h -th LSRN, $N^{(2h+1)}$ – productions number (the neurons number in a recurrent layer) for h -th LSRN.

Restrictions of the proposed method are as follows. For parametric identification of a deep simple recurrent network with losses (DLSRN), it is necessary to have training matrices of associative links: raw materials/semi-finished products with the production of semi-finished products/finished products; production of semi-finished products/finished products with returnable and non-returnable waste.

Prospects of further research are in checking the proposed methods on broader set of test databases.

References

1. World development report 2016: digital dividends. World Bank, Washington (2016). https://doi.org/10.1596/978-1-4648-0671-1
2. de Sá, A.G., Pereira, A.C., Pappa, G.L.: A customized classification algorithm for credit card fraud detection. Eng. Appl. Artif. Intell. **72**, 21–29 (2018)
3. Jiang, L.: Research on the application of computer aided audit technology. In: International Conference on Applications and Techniques in Cyber Security and Intelligence (ATCI'2018) Proc. Advances in Intelligent Systems and Computing, vol. 842, pp. 921–927. Springer, Cham (2019). https://doi.org/10.1007/978-3-319-98776-7_110
4. Kamiński, B., Jakubczyk, M., Szufel, P.: A Framework for sensitivity analysis of decision trees. Cent. Eur. J. Oper. Res. **26**, 135–159 (2018). https://doi.org/10.1007/s10100-017-0479-6
5. Global center for digital business transformation. https://www.imd.org/dbt/digitalbusiness-transformation
6. Aggarwal, C.C., Sathe, S.: Theory of outlier ensembles. In: Outlier Ensembles. Springer, Cham (2017). https://doi.org/10.1007/978-3-319-54765-7_2
7. Aggarwal, Ch.: Outlier Analysis. Springer, Cham, 2nd ed. (2017). https://doi.org/10.1007/978-3-319-47578-3
8. Aggarwal, C.C., Reddy, C.K.: Data Clustering: Algorithms and Applications. Chapman and Hall/CRC, New York (2018). https://doi.org/10.1201/9781315373515
9. Berglund, M., Raiko, T., Honkala, M., Kärkkäinen, L., Vetek, A., Karhunen, J.: Bidirectional recurrent neural networks as generative models – reconstructing gaps in time series. CoRR. abs/1504.01575, pp. 1–9 (2015). https://doi.org/10.48550/arXiv.1504.01575
10. Potash, P., Romanov, A, Rumshisky, A.: GhostWriter: using an LSTM for automatic rap lyric generation. In: 2015 Conference on Empirical Methods in Natural Language Processing,

pp. 1919–1924. Association for Computational Linguistics, Lisbon, Portugal (2015). https://doi.org/10.18653/v1/d15-1221
11. Kiperwasser, E., Goldber, Y.: Simple and accurate dependency parsing using bidirectional LSTM feature representations. Trans. Assoc. Comput. Linguist. **4**, 313–327 (2016). https://doi.org/10.1162/tacl_a_00101
12. Dey, R., Salem, F.M.: Gate-variants of gated recurrent unit (GRU) neural networks. In: 2017 IEEE 60th International Midwest Symposium on Circuits and Systems (MWSCAS'2017), pp. 1597–1600 (2017). https://doi.org/10.1109/MWSCAS.2017.8053243
13. Khan, S.A., Khalid, S.M.D., Shahzad, M.A., Shafait, F.: Table structure extraction with bi-directional gated recurrent unit networks. In: 2019 International Conference on Document Analysis and Recognition (ICDAR'2019), pp. 1366–1371 (2019)
14. Aggarwal, C.C.: Neural Networks and Deep Learning. Springer, Cham (2018). https://doi.org/10.1007/978-3-319-94463-0
15. Nakib, A., Talbi, El-G.: Metaheuristics for Medicine and Biology. Springer, Berlin (2017). https://doi.org/10.1007/978-3-662-54428-0
16. Yang, X.-S.: Nature-Inspired Algorithms and Applied Optimization. Springer, Charm (2018). doi: https://doi.org/10.1007/978-3-319-67669-2
17. Chopard, B., Tomassini, M.: An Introduction to Metaheuristics for Optimization. Springer, Cham (2018). https://doi.org/10.1007/978-3-319-93073-2
18. Radosavljevic, J.: Metaheuristic Optimization in Power Engineering. Faculty of Technical Sciences, University of Priština, Kosovska Mitrovica, Serbia (2018). https://doi.org/10.1049/pbpo131e
19. Zhu, B.: Research on the application of big data in audit analysis program. In: International Seminar on Automation, Intelligence, Computing, and Networking (ISAICN'2019) Processing, pp. 235–238. Paris (2019). https://doi.org/10.25236/isaicn.2019.047
20. Neskorodieva, T., Fedorov, E.: Method for automatic analysis of compliance of settlements with suppliers and settlements with customers by neural network model of forecast. In: Mathematical Modeling and Simulation of Systems (MODS'2020) Proc. Advances in Intelligent Systems and Computing, vol. 1265, pp. 156–165. Springer, Cham (2021). https://doi.org/10.1007/978-3-030-58124-4_15
21. Neskorodieva, T., Fedorov, E., Izonin, I.: Forecast method for audit data analysis by modified liquid state machine. In: CEUR Workshop Proceedings 2623, pp. 25–35 (2020). http://ceur-ws.org/vol-2631/paper11.pdf
22. Neskorodieva, T., Fedorov, E.: Automatic analysis method of audit data based on neural networks mapping. In: CEUR Workshop Proceedings 2833, pp. 60–70 (2021). http://ceur-ws.org/vol-2833/paper_6.pdf

Model of Knowledge-Oriented Adaptive Enterprise Management System

Yana Savytska, Victor Smolii, and Vitalii Shelestovskii

Abstract Automated control systems are a background of modern production processes. However, various models of system construction are used to manage organizational and technological processes. Such systems are focused generally on managing objects' tasks with a known structure and control algorithms with the difference in data only. If the structure of the control object differs from the implemented one, especially to a large extent, "individual" development is needed. That is rather expensive. For example, the management object in the agricultural sector, with the same structure of management tasks can be changed every year accordingly to the crop rotation. This requires either system redundancy or support for a manual control mode at the recommendations stage and event recording.

The paper solves the problem of developing a model applicable to all automated control systems.

For this, the features of the MVC pattern, the framework organization and the knowledge representation based on ontologies are considered and analyzed. This features makes it possible to propose a model synthesizing the ACS structure based on the knowledge that describes the subject area in the form of ontologies.

For automated control system implementation, a distributed microservice architecture is proposed. These services implement links between ontology entities and corresponding data transformation.

Keywords MVC-pattern · Framework · OWL · RDF · Knowledge · Control system

1 Introduction

The effective management systems development of various kinds is still one of the most urgent tasks, despite the rapid progress in the field of information technologies.

Y. Savytska · V. Smolii (✉) · V. Shelestovskii
National University of Life and Environmental Science of Ukraine, Kiev 03041, Ukraine
e-mail: dr.v.smoliy@gmail.com

S. Shkarlet et al. (eds.), *Mathematical Modeling and Simulation of Systems*, Lecture Notes in Networks and Systems 667,
https://doi.org/10.1007/978-3-031-30251-0_13

Unfortunately, new software and hardware releases are more focused on providing “quantitative” rather than qualitative characteristics of products.

Most often – to reduce the applied problems solving time, increase the accuracy of the data used, and reduce the sampling time of system parameters. However, existing control systems of various classes often can not interact directly with each other, which leads to the following:

– from the technological point of view – the need to create information "layers" between systems to organize the information exchange;
– from the economic point of view – the need to purchase, maintain and operate several tools, instead of the integrated one.

This difference is explained by the principles of their construction and data sources features.

The new universal methods development for constructing control systems will simplify their manufacturing process for enterprises of various kinds. The basis for such applied methods development can be the concepts of developing and designing frameworks, MVC patterns, knowledge representation models, and other mechanisms.

2 Structure of a Management System Model

2.1 Existing Prototypes Analyze

Automated control systems aim to improve the efficiency of managing various processes based on corresponding models. Today, there are two main classes of systems – for managing technological and organizational processes. The main difference between them is how these systems formally implement control processes.

In the first case, the core of the system works on the base of a mathematical model of the technological process. The generation of a control action based on the target function. Data is entered at discrete times, and the control action is generated over a fixed period time.

In the second case, the business model of the enterprise is used as the basis of the links between information components. The data enters the system normally asynchronously without strict restrictions on the time reaction.

According to modern development tools, such systems implement the MVC pattern (model-view-control) [1–4].

The “View” component implements the user interface and data visualization. The “Model” component implements the business logic and data stores management that describes the history of the model states. And the “Control” component is responsible for receiving input data, routing it to the model, and implementing responses. At the same time, these components remain independent of each other within the system and can be implemented in a distributed computing environment.

A distinctive feature of MVC is also the correspondence of multiple views (Views) for one controller. The discretion controller views can change, interacting with the "outside world".

These features allow using the MVC pattern in various systems design – from desktop applications to large corporate systems. Very often MVC pattern is used in WEB-applications development.

It should be noted that the "Control" component logic is prescribed by the developer and remains unchanged during the operation. This indicates the difficulty of scaling and upgrading a software system, requiring rewriting of the source code.

On the other hand, there are software systems that can change their functionality without changing the source code. For example, such products include various IDEs built on the Framework principle [5, 6].

A feature of these structures is the ability to change structure (set of tools) depending on the programming language, the type of application being developed, the target operating system and other factors, having the "fixed" core. However, the functionality of each IDE is limited to the software development task.

2.2 Tools and Background for Models Developing

Implementing an enterprise management system, by analogy with MVC patterns, it is also possible with a high degree of confidence to distinguish three main layers in the architecture – the system interface (View), data flow structure (Control), system business logic and data (Model).

In this paper, we propose to use solutions for the "Control" level development, allowing customizing the functionality according to the framework principle, depending on the scope of the enterprise and its characteristics.

We should take into account the fact that in nature there are no identical objects. Their diversity is determined by intersecting class systems that generalize information about similar implementations and data that characterize their features. For example, pig farms in Ukraine and Germany follow the same technological process. At the same time, their organizational structure and financial accounting principles will be different and dependent on local legislation. They will have different technological equipment, production structure, etc.

To take into account these factors, while creating a management system, we can follow one of the three main directions:

– as before, "manually" set the system configuration, which was previously identified as one of the disadvantages;
– to create specialized structures and databases, which requires preliminary research, design work and long-term accumulation of the necessary data;
– to use the universal mechanisms for describing knowledge about subject areas and using them to "configure" the user system.

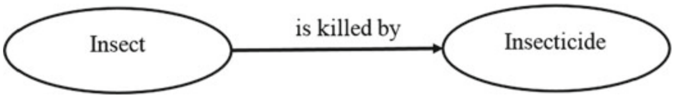


Fig. 1 A simple Ontology example

The third variant is the most rational from the point of universality and flexibility of the decisions made. A positive factor is also the availability of modern tools for knowledge representation that can be used in automated and automatic systems – for example, "Ontologies" [7, 8]. In addition, open resources already contain knowledge descriptions in a variety of subject areas, in particular WEB [9].

To describe the ontology, the OWL language [10, 11] is used. It is one of the applied implementations of XML, as well as OMV – Ontology Metadata Vocabulary [12], used in the description of the ontologies implementations accumulated in ontology (knowledge) databases. OML defines concepts properties, data types, etc. In the definitions of the Ontology, "conceptual entities" are connected by a pairwise directed connection, described by some action. For example, "Insect pest" (concept1) – "is destroyed with" (connection) – "insecticide" (concept2). This is represented by a directed acyclic graph on the Fig. 1.

Information systems obviously use objects to represent concepts. Relationships between concepts are implemented by functions and procedures. The presence of descriptions provided by OWL allows the corresponding classes to be dynamically generated in application systems.

A similar approach is proposed, for example, in [13, 14]. However, in the system proposed, agents are not completely autonomous and must interact with each other in the process of the system configuration dynamical changes and the presence of several agents types. Due to this, in our opinion, management processes become more complicated, requiring the implementation of additional mechanisms for agent interaction.

Normally there is no description in the ontology to generate links that should be implemented as procedures. In [8], for example, it is noted that the concepts act as "agents", and receiving a request, based on their internal structure, know how to transform the received data to generate a response. This position, in particular, indicates the possibility of using a distributed multi-agent system [15] that operates on the basis of an ontological knowledge model. However, for external systems, such a structure will be a "black box" that does not allow adjustment of internal connections.

SOAP [16] and REST [17] systems function on this basis. In order to interact with them, it is necessary to know the characteristics of the input data set. Such a process is based on SDL – Service Description Language [18, 19]. In accordance with such an organization, each agent (concept) in a distributed system is identified by its URI. Accordingly, for the ontology presented on the Fig. 1, the structure of the information system that processes data can be represented by the diagram on the Fig. 2.

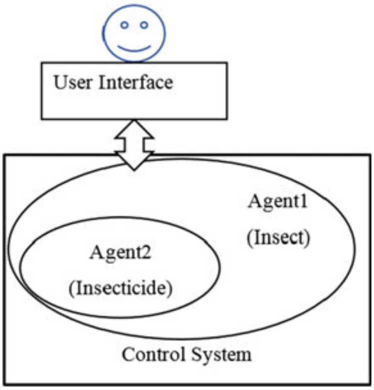


Fig. 2 A simple knowledge-based multiagent control system

The obvious disadvantage of such a system is the impossibility of changing the connection characteristics between concepts 1 and 2.

This disadvantage can be eliminated by changing the structure of multi-agent relationships and redistributing functions – it is proposed that agents make procedures that transform data, and implement concepts in the form of entities. In this case, the structure of the control system will be as shown on the Fig. 3.

With such data organization, the control system has the ability to control the transformation characteristics, the list of which the agent system must provide at the stage of the internal structure forming. For example, to change the tax rate on the amount of excess profits, when changing legislation.

Taking into account the general trend and the need for society processes digitalization [20] the relevance of such solutions is increasing significantly. This can be

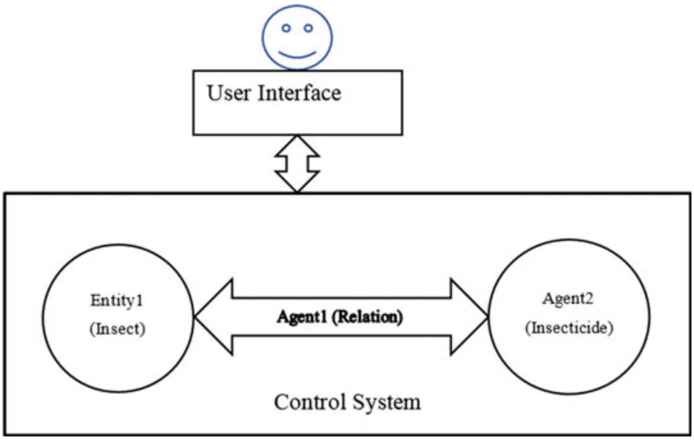


Fig. 3 The modified structure of knowledge-based multiagent Control System

explained with the total information resources integration into a single whole, easily managed and controlled.

This, in turn, poses the problem of mass information distribution and control systems with unified interfaces and data types. In addition, such systems are reliable and easily scalable in both performance and functionality. This position indicates the need to focus in the future on the implementation of most of the control systems in the form of network services.

2.3 Organization of Service Interaction

As written above, the ontology mechanism can be used to describe the entities of the subject area. The LinkedData [21] concept is closely related to this mechanism, in which entities, connected in pairs and described in RDF, have implementations in the WEB. According to this, a graph for the subject area description is formed. Since the vertices represent entities with characteristics described with OWL, it is possible to access the corresponding resource from the outside with a certain set of parameters and obtain the necessary entity implementations. According to this, we can affirm that it is convenient to use REST-API services to implement such systems.

Since, the agents in the proposed system are used as entities for data transformation, the REST mechanism cannot be used, but the SOAP mechanism with a services features description, based on WSDL, is more suitable.

Using the LinkedData analogy, the formed graph describes the HTTP-related transformation procedures - "LinkedProcesses" (LP). On the other hand, such graph describes the algorithm for transforming data in the system in general only. Thus, the resulting graph can be described with a following set:

$$LP = \{R, W, O\}, \text{ where} \tag{1}$$

$R = \{rk\}$ – set of RDF-like rules;

$O = \{on\}$ – ontology describing the subject area;

$W = \{w_{ij}\}$ – a set of WSDL interface descriptions for services that implement agents in the system.

The formation of the LP-graph description draws attention to the fact that the connections between the two western vertices implement the data set transfer that is necessary according to the subject area ontology. This fact allows, among other things, to minimize the amount of meta-information needed to describe the system by replacing the set of < part > elements in the wsdl-description structure for input and output messages with a link to the corresponding element in the ontology.

Each pair in the LP graph can be represented by the following scheme (Fig. 4):

In this scheme, I_{ij} and I_{ji} denote the interfaces between the associated i-th and j-th processes.

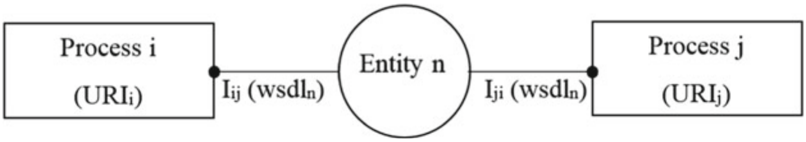


Fig. 4 Base relations in the graph of Linked Processes

If we are guided by standard means of resources and service description, then for each pair of interacting processes it is necessary to use the full WSDL description for the service interfaces used and a link to the corresponding entity description.

To form an LP-graph description, one should pay attention to the fact that the links between some two vertices implement the data set transfer, corresponding to a certain entity from the domain ontology.

The initial, intermediate and resulting data themselves can also be stored in the network, for example, services that implement the REST-API.

2.4 A Multiagent Knowledge-Based Control System Model formation

Based on the previous position and taking into account the orientation towards the MVC model, the generalized structure of such a system can be presented as shown on the Fig. 5.

In such a structure, information about existing Ontologies is stored in common knowledge base centers. Computing resources available for use as Agents implementing data modification procedures are also located in the public network. Clients operating in remote terminal mode act as the user interface.

The "Routing Controller" is located separately in the system. Its main task is to obtain the necessary information about the composition of the entities used in the application and the relationships between them at the stage of control system initial configuration.

There are two main stages in this process. At the first stage, the controller receives a general information about the subject area:

– the controller requests data about the subject area and addresses to the ontology repository;
– having received a link to the corresponding ontology, the controller selects the entities, necessary at the initial stages and their connections with other ontologies;
– the controller repeats this process until the "leaves" in the ontology "tree" are reached, specifying, if necessary, the subject area parameters;
– having received a complete ontology description, the tree optimization process is performed in order to obtain the shortest paths in the graph.

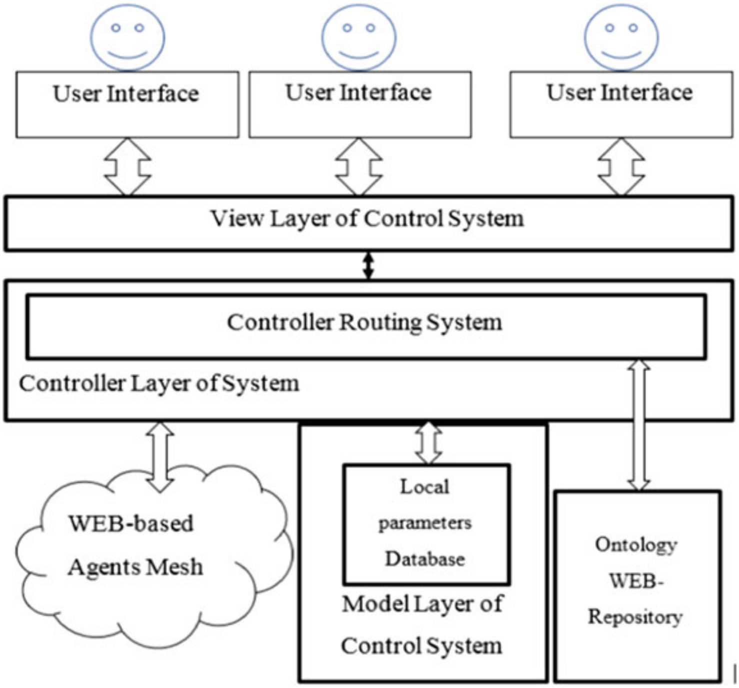


Fig. 5 A knowledge-based multiagent Control System Model

At the second stage, the task of implementing the necessary infrastructure is solved:

– it is determined by selected links which services implement them, requirements for interfaces, based on wsdl-files from services, are formed;
– a description of the necessary object classes for the used entities and their implementation based on the ontologies description and wsdl-files is created;
– a description of the processes sequence is formed with URI specification of the services and entities used.

At subsequent stages, the "Routing Controller" acts as a state machine, receiving requests and data from users and forwarding the necessary data for processing to the appropriate agents. Taking into account the tasks and features described in [22], such a structure should implement a system for describing the domain ontology graph.

2.5 Controller Functioning and Organization Features

From the point of organizing information flows in the proposed multi-agent system, the main information units are “packages” of two types:

- data sent from the Control System Controller to external services that transform them (“request”);
- from WEB-services to the Controller containing transformed data (“response”).

Thus, at the stage of system functioning, the task of the controller is to switch data flows based on received and sent packages, and to register data in the control system.

Based on this, and also taking into account the fact that the system is parallel, the problem of creating a structure that shows not only the configuration of the system, but also allows to register the process of package traffic in the system arises. The parallelism property of the system creates additional difficulties in administration and its nature - synchronous or asynchronous.

The implementation of asynchronous processes in WEB systems is given an advantage today, since in this variant the properties of the communication environment hardware component to a lesser extent affect the performance of the system as a whole.

Given that ontologies are directed non-cyclic graphs, it is logical to describe the internal structure of the system based on data types that allow describing tree structures - for example, matrices or linked lists, which are unidirectional in this case.

To take into account the data processing events chronology, the mechanism of “marking” circulating packages can be used. Similar mechanisms are implemented, for example, in GPSS models or Petri nets. It should also be noted that Petri nets in this case very well describe the process of System Controller functioning.

The characteristic properties of each “vertex” in the described graph are: a container for identifiers of data packages received for processing and their types (request/response), pointers to the corresponding network agent interfaces, and the next level vertices addresses of the hierarchy associated with this vertex.

Based on the proposed organization to describe the system structure, the work of the controller will be to cycle through all the graph vertices with information analysis about the packages in it and further movement control. The implementation of such an organization allows processing data packages in an asynchronous mode and independently of other packages in the system, achieving a high degree of parallelism.

3 Main Work Results

The undoubted advantages of the proposed organization of the system include a high level of parallelism in the data processing. However, the peculiarities should also include the need, in some cases, to set an explicitly fixed sequence of running processes or their synchronization, which requires the development of special forms for describing the graph of the work algorithm and the resources used.

It should be noted that most of the technologies proposed for use today exist and are used in certain problem areas. In addition to the public ontology repositories discussed above, the Universal Description, Discovery, and Integration (UDDI) technology is widely used to disseminate information about available WEB services and their providers [23].

Thus, the proposed model for control systems development has both the properties of frameworks, adjusting its structure depending on the subject area, and has high flexibility and reliability, based on the MVC model. The use of a distributed multi-agent organization makes it easy to achieve scalability of network service resources and a high degree of reliability.

The proposed approach allows to standardize and unify data processing in enterprise management, increase the efficiency of its implementation, as well as improve the availability of the services offered.

As opposed to the approaches in [13, 14], agents remain completely autonomous from external factors and are described in the system in a uniform and unified manner.

References

1. Christensson, P.: MVC Definition. TechTerms. 7 March 2018. https://techterms.com/definition/mvc. Accessed 28 May 2022
2. Mdn web docs. https://developer.mozilla.org/en-US/docs/Glossary/MVC
3. MVC Design Pattern. https://www.geeksforgeeks.org/mvc-design-pattern/
4. Gamma, E., Helm, R., Johnson, R., Vlissides, J.: Design Patterns: Elements of Reusable Object-Oriented Software. Addison-Wesley, Boston (1995)
5. Lutkevich, B.: Definition Framework. TechTarget. (August 2020). https://www.techtarget.com/whatis/definition/framework#:~:text=In%20computer%20zsystems%2C%20a%20framework,tools%20for%20using%20the%20frameworks. Accessed 28 May 2022
6. Batterbee, I.: Design frameworks: what are they and why you should be using them, 15 Nov 2020. https://uxdesign.cc/design-frameworks-what-are-they-and-why-you-should-be-using-them-f480a7a5b349
7. Gruber, T.R.: Toward principles for the design of ontologies used for knowledge sharing? Int. J. Human Comput. Stud. **43**(5–6), 907–928 (1995). https://doi.org/10.1006/ijhc.1995.1081
8. Guarino, N., Oberle, D., Staab, S.: What is an ontology? In: Staab, S., Studer, R. (eds.) Handbook on Ontologies, pp. 1–17. Springer Berlin Heidelberg, Berlin, Heidelberg (2009). https://doi.org/10.1007/978-3-540-92673-3_0
9. Knowledge WEB. http://knowledgeweb.semanticweb.org/
10. Smith, M.K., Welty, C., McGuinness, D.: OWL Web Ontology Language Guide 2004. W3C Rec, 10 February 2004. http://www.w3.org/TR/owl-guide/
11. Lucanu, D.: A logical foundation of the OWL languages. In: Conference: International Symposium on Leveraging Applications of Formal Methods, ISoLA 2004, October 30 - November 2, 2004, Paphos, Cyprus. Preliminary proceedings (2004)
12. Hartmann, J., Palma, R.: OMV - Ontology Metadata Vocabulary for the Semantic Web, v. 1.0 (2005). http://omv.ontoware.org/
13. Skobelev, P.: Multi-agent systems for real time adaptive resource management. In: Industrial Agents: Emerging Applications of Software Agents in Industry, pp. 207–230. Elsevier (2015)
14. Skobelev, P.: Ontology-based open multi-agent systems for adaptive resource management. In: International Conference on Agents and Artificial Intelligence (ICAART 2020), vol. 1, pp. 127–135 (2020)

15. Shoham, Y., Leyton-Brown, K.: Multiagent Systems: Algorithmic, Game-Theoretic, and Logical Foundations. Cambridge University Press (2012). https://doi.org/10.1017/CBO9780511811654
16. SOAP1 SOAP vs REST 101: Understand The Differences. https://www.soapui.org/learn/api/soap-vs-rest-api/
17. Sud, K.: Understanding REST APIs. In: book: Practical HAPI, pp.1–11, June 2020.https://doi.org/10.1007/978-1-4842-5805-7_1
18. Web Services Description Language (WSDL) 1.1. W3C Note, 15 March 2001. https://www.w3.org/TR/wsdl.html
19. Web Services Description Language. http://wiki.gis.com/wiki/index.php/Web_Services_Description_Language
20. The common agricultural policy at a glance https://ec.europa.eu/info/food-farming-fisheries/key-policies/common-agricultural-policy/cap-glance_en. Accessed 24 Apr 2019
21. Heath, T., Bizer, C.: Linked data: evolving the web into a global data space. Synth. Lect. Semant. Web. Theory Technol. **1**(1), 1–136 (2011). https://doi.org/10.2200/S00334ED1V01Y201102WBE001
22. Prev. Agrarian Enterprises Mathematical Ontological Model Development.
23. Uddi Relationship between UDDI and WSDL. https://www.ibm.com/docs/en/rsas/7.5.0?topic=uddi-relationship-between-wsdl

Construction of the Assessment Model of the Information Security System of the Enterprise

Yuliia Tkach, Marina Sinenko, Mykhailo Shelest, Dmytro Mekhed, and Daria Kosareva

Abstract Within this paper, the integrated quality assessment model of the information security system of the enterprise is proposed, based on which it is possible to obtain quantitative characteristics of the information protection system and, accordingly, access its adequacy and effectiveness.

Assessing the quality of the information security system is a rather complex organizational and technological task, the process of which crystallizes an evaluative judgment regarding the suitability, adequacy, and expediency of the applied information security measures to ensure a sufficient level of IS. The purpose of the evaluation of the company's information security system is to create information prerequisites for improving information security.

At the moment, there is no single approach to the assessment of the quality of ISS. Both qualitative and quantitative methods are used to assess the effectiveness of ISS. In our opinion, quantitative methods are more objective and informative, however, for their use, it is necessary to develop a certain set of evaluation criteria.

In the work, general principles of the construction of the quality assessment model of the information security system is analyzed, and based on it, a corresponding model was built that combines the principles of the risk-oriented assessment and the assessment oriented on economic indicators. The proposed assessment allows, depending on the customer's needs of the assessment, to emphasize the protection reliability or, accordingly, economic indicators.

Keywords Information Security · Integrated Model · Quality Assessment System · Enterprise · Risk-Oriented Assessment

1 Introduction

The assessment of the quality (effectiveness) of the information security system of the enterprise is a complex organizational and technological task, the solution

Y. Tkach (✉) · M. Sinenko · M. Shelest · D. Mekhed · D. Kosareva
Chernihiv Polytechnic National University, 95 Shevchenka Str, Chernihiv 14035, Ukraine
e-mail: tkachym79@gmail.com

S. Shkarlet et al. (eds.), *Mathematical Modeling and Simulation of Systems*, Lecture Notes in Networks and Systems 667,
https://doi.org/10.1007/978-3-031-30251-0_14

of which requires a comprehensive systematic approach. Quality assessment of the information security system (ISS) is a process during which an assessment judgement regarding the appropriateness, adequacy, and expediency of the applied information security measures to ensure a sufficient information security (IS) level is crystallized. Both qualitative and quantitative methods are used to assess the information security system (ISS). Today, the following methods are most widely used, namely: statistical, frequency, probabilistic, matrix, expert.

The statistical assessment method is based on the fluctuations analysis of the studied indicator over a certain period of time. The degree of the indicator fluctuations has a mathematically expressed probability of the occurrence of undesired consequences. It should be noted that the regularity of changes in the analyzed quantity extends to the future only for long periods of time, and for the short-term assessment, the extrapolation of past regularities gives significant errors.

The probabilistic method is based on the analysis of random, stochastic events, that is, such events that may or may not occur if a certain set of conditions are met. It provides for the determination of both the assessment of the occurrence of a probable threat, and the calculation of the probability of one or another path of the processes development, in other words, the scenarios of the threat development. The frequency method is a modification of the statistical method.

The methods of expert assessment are a means of forecasting and evaluating the results of actions based on the specialists' forecasts. When implementing the method of expert assessments, a survey of a group of experts is conducted to determine the opinion of specialists regarding certain relative and variable indicators related to the researched issue. An important condition for the correct application of the expert assessment method is the expert's awareness of the researched issue, the ability to determine clear and comprehensive answers. Moreover, the expert should not be interested in a specific solution of the problem presented to him.

In our opinion, quantitative methods are more objective and informative when assessing the ISS quality, however, for their use, it is necessary to develop a certain set of the assessment criteria. In practice, the following types of criteria are used:

– Criterion of the "cost-effectiveness" types, according to which the effectiveness of the funds involved in relation to the obtained results is determined;
– Criteria that allow you to assess the ISS quality according to a given set of indicators. At the same time, multi-criteria optimization methods are used.
– Artificially constructed criteria that allow evaluating the integral effect.

Thus, a necessary component of the establishment of the ISS effectiveness in the selection and substantiation of the assessment criteria and indicators, the selection or development of the methods for finding the appropriate indicators, the selection or development of the methods for finding the appropriate indicators.

The purpose of the assessment of the enterprise's information security system is to create information prerequisites for improving information security. In addition to the main goal, the following tasks can be solved by the ISS assessment:

– Determining the compliance with the established criteria of individual elements of the ISS;
– Identification of the impact of individual critical factors and their combination on the ISS.

2 Analysis of research and publications

Issues of the conceptual apparatus, legal aspects of information activity and ensuring information security are considered in the works of domestic and foreign authors.

The research in [1] is identified four major themes that impact the security issues within organizations. These four factors are identified as security policy documentation, access control, employee awareness, and top-level management support.

The essence of information security management is risk management, which is closely related to each other. Risk management is an important issue in manufacturing companies in today's competitive market. Failure modes and effects analysis (FMEA) method is a risk management tool to stabilize production and enhance market competitiveness by using risk priority numbers (RPN). Although the traditional FMEA approach is an effectively and commonly used method, it has some short comings such as assumption of equal importance of the factors. In [2] study a combined methodology of failure modes and effect analysis and grey relational analysis is used. In the proposed method, the priorities of production failures were determined by GRA approach and these failures were minimized by using FMEA technique. In [3] research builds a novel information security risk management model based on the concept of multidimensional risk management. On the other hand, this research also optimizes and improves the fuzzy mathematical analysis method and proposes a fuzzy comprehensive assessment method as the core algorithm for the risk assessment layer in the model. Work [4] aims to revisit six previously defined challenges in information security risk management to provide insights into new challenges based on current practices and offer a refined set of actionable advice to practitioners, which, for example, can support cost-efficient decisions and avoid unnecessary security trade-offs. Work [5] investigates this potential link between internal control and security risk management. Essentially, the risk of information security needs to be evaluated very well, because applying too detailed procedures can have negative effects. The main objective of [6] is to provide a novel practical framework approach to the development of ISMS, called by the I-Sol Framework, implemented in multimedia information security architecture (MISA). It divides a problem into six object domains or six layers, namely organization, stakeholders, tool & technology, policy, knowledge, and culture. In addition, this framework also introduced novelty algorithm and mathematic models as measurement and assessment tools of MISA parameters.

Mathematical models related to the determination of possible risks are presented in works [7]. In [7] is proposed a stochastic model to quantify the risk associated with the overall network using Markovian process in conjunction with Common Vulnerability Scoring System (CVSS) framework. The model we developed uses host access graph to represent the network environment.

In [8], the risks associated with data processing and transmission are considered, which leads to a violation of integrity, and, therefore, to a decrease in the reliability of information. To solve this problem, the use of corrective codes with multi-level adaptation methods is proposed.

However, a comprehensive concept of assessing the quality of the information security system of the enterprise has not yet been created, which prompts further research in this area.

The purpose of the work is to conduct the analysis of general principles of constructing the model of the quality assessment of the information security system and, based on it, building the integral quality assessment model.

3 Presentation of the Main Material

Depending on the criteria chosen for the ISS assessment, appropriate evaluation methods can be divided into the following groups: assessment by reference, risk-oriented assessment, assessment focused on economic indicators (Fig. 1).

Risk-oriented IS assessment of the enterprise is an assessment method in which the IS risks arising in the information sphere of the enterprise are considered, and existing IS risks and the measures taken to deal with them are compared. Finally, the assessment of the enterprise's ability to effectively manage the IS risks to achieve its goals should be formed.

The basic stages of the risk-oriented assessment of information security include the identification of the IS risks, determination of the adequate processes of the risks management and the key indicators of IS risks, creation on their base the criteria for the IS assessment, collection of the assessment data and measuring risk-factors, creation of the IS assessment.

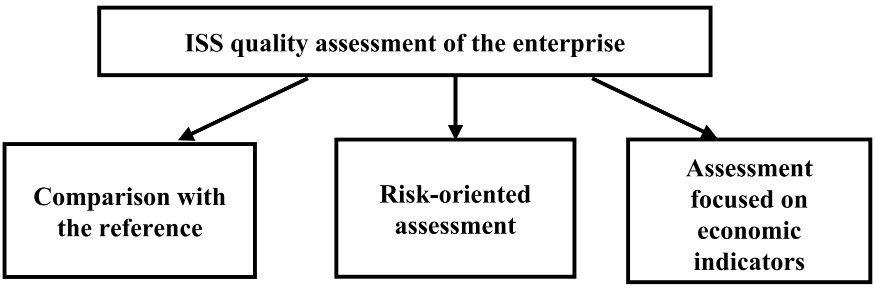


Fig. 1 Means of the ISS assessment

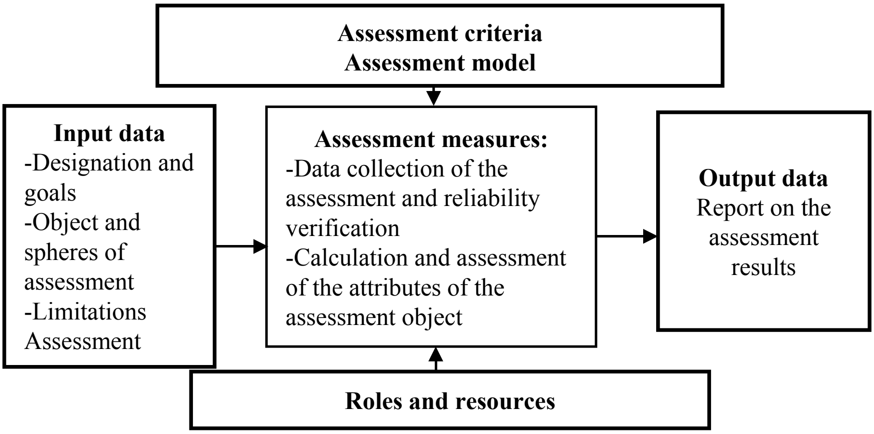


Fig. 2 Basic elements of the IS assessment

Basic Elements of the Assessment Process. The IS assessment includes the following elements of carrying out the evaluation process:

– assessment context, which determines the input data: goals and purposes of the IS assessment, the type of assessment (independent assessment, self-evaluation), the object and areas of the IS assessment, limitation of the assessment and the roles;
– assessment criteria;
– assessment model;
– assessment measures: data collection of the assessment and verification of their reliability, measurement and evaluation of attributes of the assessment object;
– output assessment data.

The basic elements of the IS assessment are presented in Fig. 2 as a process model.

Prior considering the specifics of the IS assessment methods of the enterprise, it is necessary to describe the components common to any IS assessment.

Context of the Information Security Assessment of the Enterprise. Context of the IS assessment includes goals and purposes of the IS assessment, the type of assessment, the object and spheres of the IS assessment, limitations of assessment, roles.

The roles involved in the assessment process include the organizer, analyst, leader of the assessment team, appraiser, asset owner, and a representative of the assessment object.

The organizer (customer) of the IS assessment forms the purpose of the evaluation (improvement of the assessment object, determination of the assessment object's compliance with the established criteria, etc.) and determines the assessment criteria, the object and the area of evaluation. The organizers of the assessment can be a person or companies that is external or internal to the object under the assessment.

The organizer must ensure the access of the assessment team (a leader of the assessment team, an evaluator) to the assets of the object under the assessment for study, to the personnel for conducting surveys, to the infrastructure required during the assessment.

Upon the completion of assessment, the organizer transfers the assessment reporting documents to the interested parties for their use in accordance with the stated purpose of the assessment.

The analyst of the IS assessment chooses the method of the IS assessment, the assessment model, and determines the methodical and information support for the assessment, i.e. methods, attributes of the assessment. The analyst researches assessment results and form a report and recommendations based on the IS assessment outcomes.

The leader of the assessment team and evaluator measure and evaluate the attributes provided by the asset owner and formulate the results. Activities on measurement and assessment are performed exclusively by the leader of the assessment team and the evaluator who are part of the assessment team. Other personnel (a representative of the object under assessment, technical expert) may participate in the work of the assessment group to provide specialized knowledge or advice. They may discuss the formulation of judgments with the evaluator, but will not be responsible for final assessment.

An important aspect in determining the assessment context is the type of the assessment: independent assessment or self-assessment. Depending on the type of assessment, the relationship between the roles of the assessment process and the object under assessment differs.

The independent assessment is achieved by evaluating by the assessment group whose members are independent of the object under assessment. The organizer of the assessment may belong to the same organization to which the object under the assessment is attributed, but not necessarily to the object being evaluated. The independence degree may vary according to the purpose and area of the assessment. In case of the external organizer of the assessment, a mutual agreement is assumed between the assessment organizer and the organization to which the object under the assessment belongs. The representative of the object under the assessment participates in the formation of assessment attributes, ensures the interaction of the assessment group with asset owners. Their participation in the assessment provides an opportunity to determine and take into account the features of the object under the assessment, ensure the reliability of the evaluation results.

Self-assessment is carried out by the organization to evaluate its own ISS. The organizer of self-assessment is usually included in the object under the assessment, as well as the members of the assessment group.

The assessment area may include, for example, one or more processes of the object under the assessment, for instance, the organizer may focus on one or more critical processes and/or protective measures. The choice of the object under the assessment should reflect the organizer's use of the initial evaluation data. For example, if the output data is intended for the use in the improvement of the IS activities, then the assessment scope should correspond to the scope of the planned works on the improvement. The area of the assessment can vary from a separate process to the entire organization. In the assessment context, a detailed description of the object under the assessment should be presented.

Limitations of the assessment include possible unavailability of major assets used in normal business activities of the organization; insufficient time interval allocated for the assessment; the need to exclude certain elements of the objects under the assessment due to the stage of the life cycle. In addition, limitations may be imposed on the amount and type of data that must be collected and analyzed.

The content of the assessment context is agreed by the assessment team leader with the organizer and the authorized representative of the object under the assessment, and is documented prior the assessment begins. Fixing the assessment context is an important stage because it contains initial elements of the assessment process.

Changes in the assessment context may occur during the course of the evaluation. Changes must be approved by the organizer of the assessment and the authorized representative of the object under the assessment. If these changes affect the time schedule and resources of the assessment, the evaluation planning should be revised accordingly.

Quality Assessment Model of Information Security of the Enterprise. Quality assessment model of information security of the enterprise developed by us is a combination of a risk-oriented model and a model focused on economic indicators (Fig. 3).

Risk-Oriented Model of Information Security. The key feature of the risk-oriented assessment is that it is aimed at analyzing the management of the risk assessment organization, that is, it monitors and checks risk management processes.

The risk-oriented assessment gives an objective and most informative picture of the performance level of the organization, effectiveness of managerial decisions, and efficiency of costs for business support and development based on a comparison of the existing risks of the organization's activities and the measures taken by the organization to handle these risks.

The purpose of the risk-oriented assessment is to determine that:

– risks management processes are substantiated and introduced;

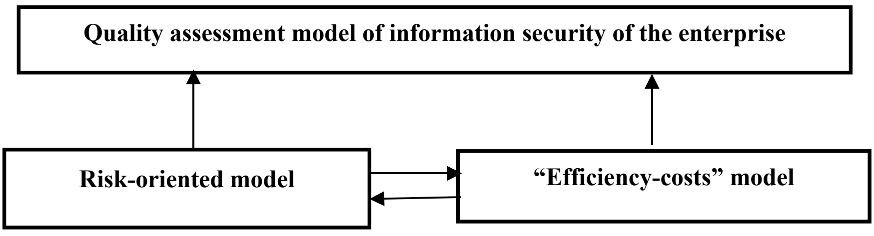


Fig. 3 General quality assessment model of information security of the enterprise

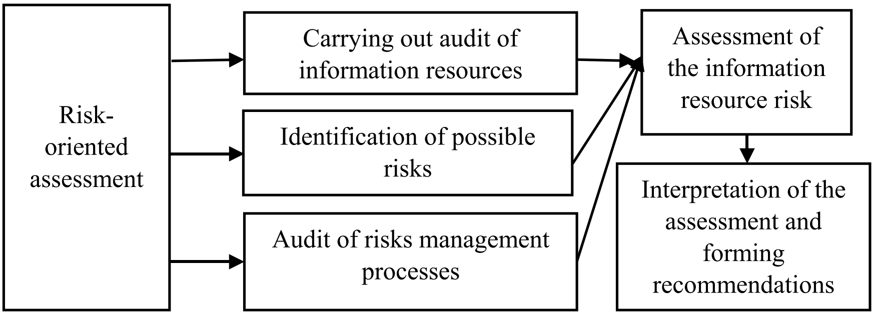


Fig. 4 Algorithm of the risk-oriented model of the assessment

– the final outcome of the assessment should be to provide confidence that risk management is being carried out appropriately and aimed at reducing risks to an acceptable level.

We shall consider elements of the model in detail.

In Fig. 4, the algorithm of the risk-oriented model of the assessment proposed by us is shown. Thus, we shall single out the following steps.

Step 1. Carrying out audit of information resources of the enterprise and their ranging according to the significance level. At this stage, it is necessary to find out what information resources the company has, which are the carriers of information resources, and establish the significance measure of each resource for the company. To establish the significance resource, it is proposed to introduce the significance coefficient (α_i); $\alpha_i \in \{1; 2; 3\}$, that is, the more significant this information resource is, the greater the value α_i. When determining the value α_i, not only possible economic, but also image and other losses are taken into account.

Step 2. Identification of possible risks. At this stage, possible threats to information resources are established, probabilities of the attempts to implement threats are also found. In case when statistical data is insufficient, you can use the expert assessment (Fig. 5).

Here, $p_{ij}-$ probable implementation of the $j-$ threat relating to the $i-$ information resource, summarized probability of the attack on the $i-$ information resource:

$$P_i = \sum\nolimits_{j=1}^{k} p_{ij} - \sum\nolimits_{j=1,l>j}^{k} p_{ij} p_{il} + \cdots + (-1)^{k-1} p_{i1} p_{i2} \ldots p_{ik}. \qquad (1)$$

Step 3. Audit of risk management processes and identification of the vulnerabilities of the ISS. At this stage, the compliance of the adopted protection measures with possible threats is determined. In practice, this means calculations based on statistical data, and in case of their insufficiency, on the basis of expert assessments of the probabilities of repelling possible q_i-attacks on the $i-$ information resource. Thus, for $i-$ information resource, the degree of vulnerability is determined:

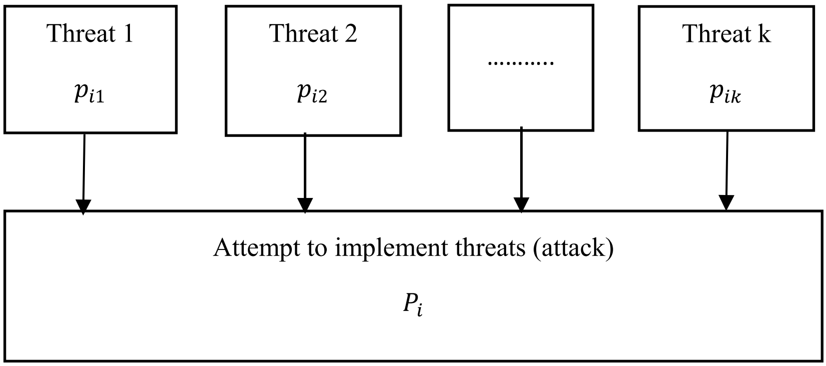


Fig. 5 Probability of attempts to implement possible threats

$$\mu_i = P_i(1 - q_i). \tag{2}$$

Step 4. Risk assessment of the information resource. For each information resource, the risk degree is determined:

$$R_i = \alpha_i \mu_i = \alpha_i P_i(1 - q_i). \tag{3}$$

At the same time, as a measure of the total risk for the ISS, we shall choose the value: $R = \max_i \{R_i\}$.

Step 5. Interpretation of the assessment and forming recommendations concerning the ISS improvement.

After the completion of the ISS assessment according to the adequacy criterion to possible risks, we shall evaluate the ISS from the point of view of economic efficiency.

$$W = \frac{W_{ef}}{W_{sz}}, \tag{4}$$

where $W-$ coefficient of economic effectiveness, $W_{ef}-$ expressed in the monetary equivalent, possible economic loss as a result of damage to available information resources in the aggregate; $W_{sz}-$ total costs of implementing protective measures against possible threats. To calculate W_{ef}, the following formula is proposed:

$$W_{ef} = \sum_i w_i \mu_i, \tag{5}$$

where $\mu_i -$ pre-specified vulnerability of the $i-$ information resource, $w_i -$ economic losses due to the damage of the $i-$ information resource.

Thus, according to the proposed model, the ISS quality assessment of the researched object is characterized by a pair of numbers $(R; W)-$ the degree of the aggregate risk, which is established as a maximum possible risk of loss or damage to information resources and the coefficient of economic efficiency. The assessment will be considered satisfactory, if $R < 0, 1$; $W > 1, 03$. In case of receiving unsatisfactory assessment, it is proposed to review protection means of the established ones during the weak points assessment.

The integrated assessment of the ISS quality can be found in the following form:

$$O = \frac{W^{\beta}}{R^{1-\beta}}, \beta \in (0; 1), \tag{6}$$

that is, the integrated assessment is directly proportional to the economic efficiency ratio and inversely proportional to the aggregate risk. $\beta-$ a parameter set by the enterprise's management depending on the emphasis on the significance of economic security or economic indicators.

4 Conclusions

Within this work, the integrated quality assessment model of information security of the enterprise, which considers both economic indicators and reliability of the protection of available information resources, is proposed. The considered assessment model makes it possible to provide the quantitative characteristic of the quality of the information security system, based on which the conclusions can be drawn regarding the adequacy of this system. In addition, the selection of the parameter β is of great importance, which allows the management of the enterprise, when constructing an integrated assessment of the quality of information security, depending on preferences, to emphasize either information protection or economic efficiency indicators.

References

1. Chaudhry, P.E., Chaudhry, S.S., Reese, R., Jones, D.S.: Enterprise information systems security: a conceptual framework. In: Møller, C., Chaudhry, S. (eds.) Re-conceptualizing Enterprise Information Systems, pp. 118–128. Springer Berlin Heidelberg, Berlin, Heidelberg (2012). https://doi.org/10.1007/978-3-642-28827-2_9
2. Baynal, K., Sarı, T., Akpınar, B.: Risk management in automotive manufacturing process based on FMEA and grey relational analysis: a case study. Adv. Prod. Eng. Manag. **13**(1), 69–80 (2018). https://doi.org/10.14743/apem2018.1.274
3. Yang, M.: Information security risk management model for big data. Adv. Multimed. **3383251**, 10 (2022). https://doi.org/10.1155/2022/3383251
4. Bergstroem, E., Lundgren, M., Ericson, A.: Revisiting information security risk management challenges: a practice perspective. Inf. Manag. Comput. Secur. **27**(3), 358–372 (2019). https://doi.org/10.1108/ICS-09-2018-0106

5. Bilcan, F.R., Ghibanu, I.A., Bratu, I.I., Bilcan, G.A.: The relationship between internal control and security risk management. Acad. J. Econ. Stud. **5**(4), 139–144 (2019). https://zbw.eu/econis-archiv/bitstream/11159/4127/1/1689293888.pdf, https://www.econbiz.de/Record/the-relationship-between-internal-control-and-security-risk-management-bîlcan-florentina-raluca/10012156650
6. Susanto, H., SAlmunawa, M.N., Tuant, Y.C., Aksoy, M.S.: I-SolFramework: an integrated solution framework six layers assessment on multimedia information security architecture policy compliance. Int. J. Elect. Comput. Sci. IJECS-IJENS **12**(1), 20–27 (2012). https://doi.org/10.48550/arXiv.1204.0056
7. Pokhrel, N.R., Tsokos, C.P.: Cybersecurity: a stochastic predictive model to determine overall network security risk using Markovian process. J. Inf. Secur. **8**(2), 91–105 (2017). https://doi.org/10.4236/jis.2017.82007
8. Zaitsev, S.V., Kazymyr, V.V.: Structural adaptation of the turbo code coder and decoder for generating the transmission repeat request under conditions of uncertainty. Radioelectron. Commun. Syst. (Springer) **60**, 18–27 (2017). https://doi.org/10.3103/S0735272717010034

Analysis of the Effectiveness of the Security Incident Response Team Under Intensity Cyber-Attack Increasing

Iryna Dohtieva and Anatolii Shyian

Abstract Recently social groups become a target for the realization of negative information or psychological influence on them, the formation of their intentions to commit harmful behavior to society and so on. Thus, there is a need to study the peculiarities of the Information Security Incident Response Team (SIRT) in the new environment. Paper builds the model that took into account the increase in the intensity of individual incidents in the process of cyber-attacks and the need to restore the efficiency of personnel. The apparatus of queuing systems is used. The methods for managing the effectiveness of countering cyberattacks have been developed, as a result of which it is possible to select those SIRTs that have the desired effectiveness. It is also possible to manage the staff of SIRT, taking into account the individual characteristics of individual employees (for example, the level of efficiency of their work, features of recovery during work, features of their work in high stress with increasing cyberattacks, etc.). A method for managing the effectiveness of SIRT with the use of computer simulation has been developed, which allows to take into account real statistical features already in the process of cyber-attack deployment.

Keywords Cyber-Attack · Security · Incident · Team · Effectiveness Response

1 Introduction

Recently, the use of cyberspace to destabilize the social state of society is growing rapidly. There are more and more cases when social groups become a target for the realization of negative information or psychological influence on them, the formation of their intentions to commit harmful behavior to society and so on. If earlier the target of cyber-attacks was mainly the technical component of socio-technical systems, today cybercriminals keep in mind the social component of such systems.

Here are some examples of successful impact on society, which was carried out using cyberspace, in particular social networks. In [1, 2] examines the impact of social

I. Dohtieva · A. Shyian (✉)
Vinnytsia National Technical University, Khmelnitske Schose 95, Vinnytsia 21021, Ukraine
e-mail: anatoliy.a.shiyan@gmail.com

S. Shkarlet et al. (eds.), *Mathematical Modeling and Simulation of Systems*, Lecture Notes in Networks and Systems 667,
https://doi.org/10.1007/978-3-031-30251-0_15

networks on the implementation of the revolution in Egypt in 2011 and other protests of the “Arab Spring”. The authors [1] in the wake of the revolution in Egypt note that social media played an important role in the anti-government protests. In Article [2], an analysis using databases from twenty Arab countries and the Palestinian Authority revealed the validity of two statements. The first is that the role of the political context must be taken into account when studying the role of social media. Second, the intensity of the use of communication channels monitors the intensity of protest activity. And such activities require a special organization of the Information Security Incident Response Team (SIRT).

Cyberattacks on the social environment often use fakes and disinformation [3–6]. This distribution has its own characteristics. In [3], a large Facebook database was used for comparative analysis of the spread of correct (scientific news) and fake (conspiracy news) information on the network. The authors found that “although consumers of scientific and conspiracy stories present similar consumption patterns with respect to content, cascade dynamics differ. Selective exposure to content is the primary driver of content diffusion and generates the formation of homogeneous clusters, i.e., “echo chambers”. Paper [4] analyzed 14 million messages that distributed 400,000 Twitter articles over ten months in 2016 and 2017. The results of the study revealed a number of important patterns in the use of bots to develop cyberattacks: “social bots played a disproportionate role in spreading articles from low-credibility sources. Bots amplify such content in the early spreading moments, before an article goes viral. They also target users with many followers through replies and mentions. Humans are vulnerable to this manipulation, resharing content posted by bots. Successful low-credibility sources are heavily supported by social bots. These results suggest that curbing social bots may be an effective strategy for mitigating the spread of online misinformation.”

In [5] attention is drawn to the fact that “Social media sites are often blamed for exacerbating political polarization by creating “echo chambers” that prevent people from being exposed to information that contradicts their preexisting beliefs.”. According to the results of a specially conducted field experiment on a survey of supporters of the Republican and Democratic parties of the United States, who actively use Twitter, the authors found: “Republicans who followed a liberal Twitter bot became substantially more conservative posttreatment. Democrats exhibited slight increases in liberal attitudes after following a conservative Twitter bot, although these effects are not statistically significant.” Note that these results resonate with the results of the article [2], which examined the processes in other countries.

In the article [6] simulation modeling is carried out and quantitative data on the possible influence of bots on the formation of public opinion are obtained: “in a highly polarized setting, depending on their network position and the overall network density, bot participation by as little as 2–4% of a communication network can be sufficient to tip over the opinion climate in two out of three cases. These findings demonstrate a mechanism by which bots could shape the norms adopted by social media users.”

Finally, [7] analyzes the state of research on the impact of social networks on the individual, social groups and society as a whole and argues that "What is emerging is a new scientific and engineering discipline - social cybersecurity."

Thus, the work of SIRT in the face of the need to respond to information incidents affecting society, will be in demand for a long time, with a tendency to increase in intensity.

The problem of optimizing the work of SIRT has recently attracted increasing attention of researchers. Thus, in [8] attention is drawn to the fact that the effective work of SIRT requires the formation of specialized databases, which should be provided as the results of reports of such groups on the results of their work. The study draws attention to the fact that: "these teams focus more on eradication and recovery and less on providing feedback to enhance organizational security. This prompts the idea that data collected during security incident investigations may be of insufficient quality for threat intelligence analysis". The authors in [9] argue that "literature on cybersecurity incident detection & response is very rich in automatic detection methodologies, in particular those based on the anomaly detection paradigm. However, very little attention has been devoted to the diagnosis ability of the methods, aimed to provide useful information on the causes of a given detected anomaly. This information is of utmost importance for the security team to reduce the time from detection to response." In [10] the previous opinion is confirmed: "Cultural and security driven sentiments about external observation, as well as publication concerns, limit the ability of researchers to understand the context surrounding incident response. Context awareness is crucial to inform design and engineering. Furthermore, these perspectives can be heavily influenced by the targeted sector or industry of the research. Together, a lack of broad contextual understanding may be biasing approaches to improving operations, and driving faulty assumptions in cyber teams." A study [11] found that "automated incident triage in large cloud services faces many challenges: (1) a highly imbalanced incident distribution from a large number of teams, (2) wide variety in formats of input data or data sources, (3) scaling to meet production-grade requirements, and (4) gaining engineers' trust in using machine learning recommendations."

To increase the efficiency of SIRT in [12] is proposed «integration of information security management and an incident response functions». It will create "learning opportunities that lead to organizational security benefits including: increased awareness of security risks, compilation of threat intelligence, removal of flaws in security defenses, evaluation of security defensive logic, and enhanced security response."

Thus, there is a need to study the peculiarities of the SIRT in the new environment. Such conditions include, for example: requirements for maintaining a high level of attention during work, because in complex cyberattacks some of their important elements may be incorrectly or poorly identified [8–12], increasing the workload of specialists, requirements for their cognitive and creative abilities [13, 14], also take into account various influences of both psychological and physiological nature [13–15], etc.

In general, the main requirement for SIRT activities is the effectiveness of its work in responding to cyberattacks. This is especially critical during of increasing cyber-attacks, when the ability of professionals to perform their duties effectively diminishes over time until they become unable to perform the required activities. It should be emphasized that such a situation can be failure even for the same specialist, even in conditions of intensity increase in a similar cyberattack. This situation occurs at different time intervals [13–15]. In this case, for the effective operation of SIRT, for example, it is either necessary to increase the workload for still able-bodied specialists, or to organize the replenishment of SIRT with new specialists.

Thus, the complex conditions of cyberspace functioning, nomenclature saturation and complexity of cyber-attacks require scientific research of aspects of SIRT activities. In this case, the processes that form such activities are random. Therefore, the work of SIRT can be considered in the statistical approach and be presented as a modified model of queuing, in particular, the operation of SIRT can be modeled in the queuing system (QS) with exponentially distributed time [16, 17]. Such models allow to obtain quantitative indicators and characteristics that are important for managing the effective functioning of SIRT.

2 The Model

Here is an algorithmic description of SIRT during a cyber-attack.

Let the system receive a stream of response requests (parameters that characterize the temporal deployment of one cyberattack, or each individual cyberattack, etc.), and the moment of receipt of the first request is a random variable with an exponential distribution with parameter $\alpha\lambda$, where $\alpha > 1$ is a quantitative component associated with the development of cyberattacks, increasing speed, volume, intensity compared to similar attacks in the past, increasing the complexity of the attack, expanding the area of attacks, which may take into account the multi-vector approach in attacks, the case of complex cyberattacks, new attack technologies.

The request received for service is served a random time (analyzed, decisions are made and commands are given for its execution), which has an indicator distribution with the parameter μ. The time of receipt of the next request is a random variable that has an exponential distribution with the parameter λ. In case of release of the system from servicing of the previous request before receipt of the new application, the intensity of receipt of requests changes. Otherwise, the request is lost if it is received before the completion of the service procedure.

Here is a graphic description of SIRT during a cyber-attack.

The system consists of the source of requests (block 0) and the service subsystem (block 1), the structure of the system is shown in Fig. 1. The studied process and blocks (0, 1) form M/ M/1 with the transition diagram - 01, 10, 11 (for symbols, see [17, 18]). This system is complicated by the load parameter due to the increase in the intensity of requests from block 0, provided the system is unoccupied.

Here is an analytical description of the behavior of the SIRT model.

The functioning of the system is described by the Markov process $\xi(t)$, which is characterized by the set of states $\{e_0, e_1\}$.

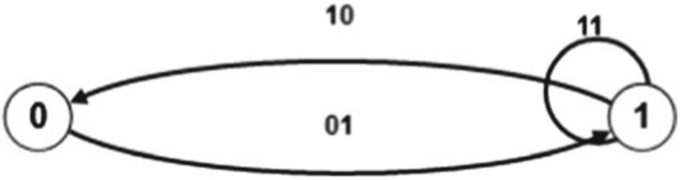


Fig. 1 Graph of SIRT transitions between states

The system is in time τ (τ is exponentially distributed random variable with parameter $\alpha\lambda$) in state e_0 and with probability 1 goes to state e_1, which indicates the moment τ receipt of the request for service.

The stay of the system in the state e_1 is characterized by the time $min(\tau', \eta)$, where τ', η are independent random variables with exponential distribution according to the parameters λ and μ, and τ' is the time of new request, η is the service time. From this state, the system can go back to state e_0 with probability

$$p_{10} = P\left(\eta < \tau'\right) = \frac{\mu}{\lambda + \mu}, \tag{1}$$

or remain in the e_1 state with probability

$$p_{11} = P\left(\tau' < \eta\right) = \frac{\lambda}{\lambda + \mu} \tag{2}$$

Thus, the Markov process $\xi(t)$ in the states e_0, e_1 is respectively time $\zeta_0 = \tau$, $\zeta_1 = min(\tau', \eta)$, and random variables ζ_0, ζ_1 have exponential distributions according to the parameters $\alpha\lambda$, $\lambda + \mu$. Given (1) and (2), the transitions from state-to-state $\xi(t)$ are carried out according to the nested Markov chain, which is given by the matrix of transition probabilities

$$P = \begin{pmatrix} 0 & 1 \\ \frac{\mu}{\lambda+\mu} & \frac{\lambda}{\lambda+\mu} \end{pmatrix}. \tag{3}$$

By determining $Q_{ij}(t)$ are the probabilities that $\xi(t)$ in the i-th state will spend time less than t and move to the j-th state, we obtain:

$$\begin{gathered} Q_{00}(t) \equiv 0, \\ Q_{01}(t) = 1 - e^{-a\lambda t}, \\ Q_{10}(t) = \frac{\mu}{\lambda+\mu}\left(1 - e^{-(\lambda+\mu)t}\right), \\ Q_{11}(t) = \frac{\lambda}{\lambda+\mu}\left(1 - e^{-(\lambda+\mu)t}\right). \end{gathered} \tag{4}$$

As a result of transformations using the integral form of the full probability formula, a system of integral convolution-type equations, Laplace transforms on convolution images, decomposition theorems for simple poles, we obtain:

$$\begin{aligned} P_{00}(t) &= \frac{\mu}{a\lambda+\mu} + \frac{a\lambda}{a\lambda+\mu} e^{-(a\lambda+\mu)t}, \\ P_{01}(t) &= \frac{a\lambda}{a\lambda+\mu} - \frac{a\lambda}{a\lambda+\mu} e^{-(a\lambda+\mu)t}, \\ P_{10}(t) &= \frac{\mu}{a\lambda+\mu} - \frac{\mu}{a\lambda+\mu} e^{-(a\lambda+\mu)t}, \\ P_{11}(t) &= \frac{a\lambda}{a\lambda+\mu} + \frac{\mu}{a\lambda+\mu} e^{-(a\lambda+\mu)t}, \end{aligned} \tag{5}$$

where

$$P_{ij}(t) = P(\xi(t) = j | \xi(0) = i) \tag{6}$$

are the probability that the process $\xi(t)$ at time t is in the j-th state, provided that at the initial moment it is in the i-th state. The probabilistic content of functions (5) is provided by performing: $P_{00}(0) = P_{11}(0) = 1$, $P_{01}(0) = P_{10}(0) = 0$, $P_{00}(0) + P_{01}(0) = 1$, $P_{10}(0) + P_{11}(0) = 1$ for all t.

Within the framework of the studied system, an indecomposable, aperiodic, with return and positive elements Markov chain for which there is a stationary (final) probability distribution is considered. The stationary mode of operation is established with increasing (increasing the number of steps), i.e., $t \to +\infty$, in the studied QS with load (M/M/1 with increasing intensity of the demand under idle system) according to the ergodic theorem for Markov chains with counted (finite) number of states [17, 18]:

$$\begin{aligned} \lim_{t\to+\infty} P_{00}(t) &= \lim_{t\to+\infty} P_{10}(t) = \frac{\mu}{a\lambda+\mu} \\ \lim_{t\to+\infty} P_{01}(t) &= \lim_{t\to+\infty} P_{11}(t) = \frac{a\lambda}{a\lambda+\mu} \end{aligned} \tag{7}$$

In the case of considered SIRT, in the transitions from state to state during countering cyber-attacks, the probabilities of these states no longer depend on the initial probability distribution and the direct step number and are determined only by the transition matrix (3). Each of the states is realized with a certain constant probability (7), and the set of probability data as coordinates form a boundary (or final) vector [17, 18], which simulates, according to systems theory, definition a class and a state of object. In particular, the marginal probabilities in the conditions of SIRT operation during countering cyberattacks allowed to determine the average relative time of SIRT in this i-th state. That is, the results

$$\begin{aligned} \pi_0 &= \frac{\mu}{a\lambda+\mu} \\ \pi_1 &= \frac{a\lambda}{a\lambda+\mu} \end{aligned} \tag{8}$$

allow to describe for a considerable period of time $(0;T)$ the process $\xi(t)$, which will be in the states e_0, e_1 such time, which is estimated by the numbers $\pi_0 T$, $\pi_1 T$, i.e., for a period of time $(0;T)$ on average the system will not occupied about $\pi_0 T$, and busy about $\pi_1 T$.

3 Model Analysis

For to study the behavior of the mathematical model SIRT have been used the software package to implement a simulation model of the studied SIRT, i.e., M/M/1 with increasing the intensity of the application under the downtime condition of the system (QS model with load). Python programming language was chosen for the simulation, which is used to solve a wide range of problems, in particular, for working with data in research, DataMining, DataScience, etc.

A number of library and modules have been used, including the built-in Math module for mathematical operations, the Random module for random number generation, NumPy packets with processing power, Pandas with the object processing data, and the Matplotlib from SciPy stack library.

To visualize the data have been used graphics library Plotly, to build three-dimensional graphs (3D graphics) in the dynamics have been used mplot3d Toolkit from the package Matplotlib [19].

The statistical analysis has been used the Statsmodels, Statistics library, which works with a list of data and has a small number of statistical functions [20, 14].

The software product is divided into modules: dynamics of input data; preliminary output data with forecasted analytically calculated efficiency indicators; an experiment; series of experiments; obtaining graphic data, including 3D graphics; statistical processing according to the list of metrics.

In the graphical part of the simulation environment, it is proposed to model the behavior of the predicted stationary mode of operation, i.e., the limit probabilities that determine the average relative time of SIRT in this *i*-th state, in particular, in its operation during cyberattacks, in three-dimensional space.

Representation of graphs in three spatial dimensions allows to model the behavior of stationary probabilities for attracting of more quantity of dynamic variables, which are depended on the influence of the input parameters for the functioning of the studied system:

α – load parameter (parameter to increase the intensity of applications);

λ – intensity of the flow of applications (intensity of applications);

μ – performance of the channel (server) of application service (intensity of application service).

In Fig. 2–4 the functions of the stationary probability is presented, which describes the employment of the service system (8), in the functions of two variables, in particular: $\pi_1(\lambda, \mu)$, $\pi_1(\alpha, \mu)$ and $\pi_1(\alpha, \lambda)$.

Figure 2 allows to analyze the functions of the graph $\pi_1(\lambda, \mu)$, theoretically calculated by the stationary probability, in relation to the service process extremely from the intensity of receipt and service of requests. For calculations uses the diapasons of the parameter λ within (0; 100), μ within (0; 200) and the constant value of the parameter of the intensity increasing of receipt of request $\alpha = 3$.

Figure 3 shows the calculated graphs for the numerical values of the function $\pi_1(\alpha, \mu)$, i.e., the dependence of this stationary characteristic on the load parameter in the review interval (0; 10) and the performance of the service channel in the

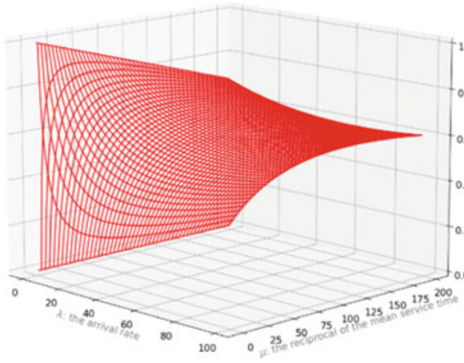


Fig. 2 Graph of functional dependence π1(λ, μ), α = 3

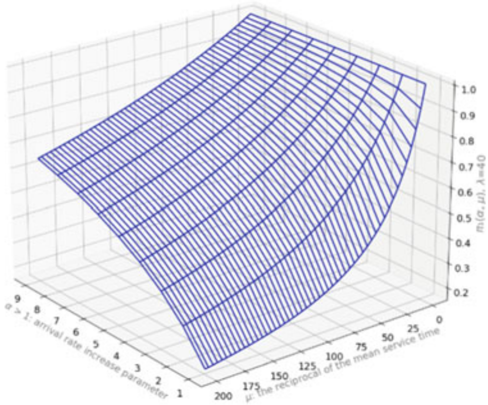


Fig. 3 Graph of functional dependence π1(α, μ), λ = 40

process of cyber-attack. The constant in this case is fixed by the intensity of the flow of requests with a value of λ = 40.

Figure 4 shows the graph of the function π1 (α, λ), which takes into account the dependence on the parameter of increasing the intensity of requests and the actual intensity of the flow of requests, in the above intervals and at a constant value for service intensity: μ = 185.

4 The Example of the Experimental Data Analysis

In the experiments, different values for π_1 will be obtained depending on the variable turbulent environment. Therefore, as an illustration, we carried out a computer simulation of the processing of experimentally obtained values.

In order to obtain the simulated data in the software product, the following procedure is used. Using the input information, the program provides the generation of sequences of random numbers, on the basis of which an array of data is formed,

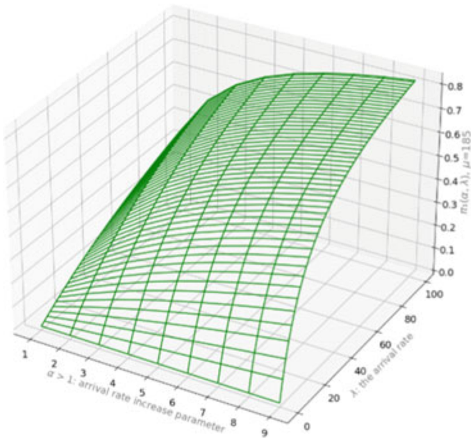


Fig. 4 Graph of functional dependence π1(α, λ), μ = 185

which simulates the behavior of the studied system. According to the results of a separate experiment, the values of experimental statistical characteristics and analytically calculated by formulas (8) theoretical values are obtained. A series of simulating experiments the functioning of M/M/1 with increasing the request intensity under the condition of idleness of the system in one experiment forms a sample of stationary characteristics of the system for further statistical analysis.

Statistical analysis of stationary characteristics of the studied model includes the use of central metrics for the study of data concentration centers and metrics for estimating data variability, which analyze the variance of values.

Graphically, the sample for conclusions on observations is represented by a set of points from the simulation experiment data (1D-sequence of values of stationary characteristics), the boundary of the set of theoretical data of stationary characteristics, which have a constant value on the grid of the number of experiments). The set of data shows emissions that are associated with changes in the behavior of the studied system. Based on the obtained data, the corresponding lines indicate the arithmetic mean, geometric, harmonic, median and mode.

For the theoretical value of π_1, the nearest arithmetic and geometric mean values were. The average arithmetic mean differs most from the average harmonic mean, the value of which in Fig. 5 is the furthest among all the averages from the theoretical result. This average can be used as a cumulative indicator of the evaluation of algorithms and systems, in the case of both values of analytically predicted stationary characteristics of the system.

The main difference between the behaviors of mean and median is related to emissions or extreme values, in particular the median is less sensitive to emissions. It divides a series of values into two parts, each of which houses the same number of population units. For the simulation experiment data, the median value exceeds the values of other metrics.

In order to determine the mode in cases where each experimental value will occur once, data sampling is usually used when constructing histograms, after which the

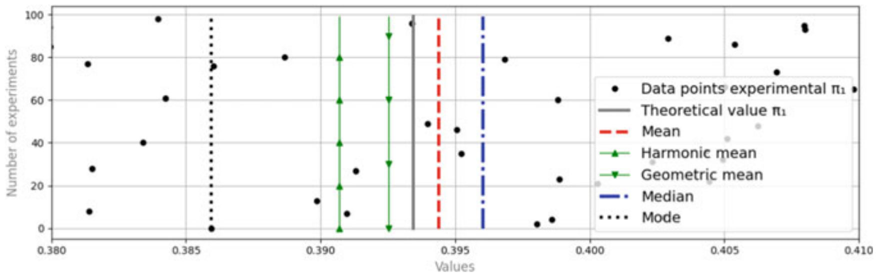


Fig. 5 Graphic data of calculated central metrics for sampling of simulation experiments (SIRT work with load)

value of the frequency of occurrence is determined by intervals, where the interval is assigned the value of its midpoint. Fashion, as a rule, is the value where the histogram reaches its maximum [17]. In the case of our sample, the result of such a procedure is indicated by a dotted line from the points in Fig. 5.

In Fig. 6 presents estimates of data variability in the form of a scale diagram for the scattering of numerical data.

The right side of Fig. 6 shows an example of a set of points obtained in a simulation experiment for π_1. Stationary values are marked in gray, to which experimental results will be close, in red dotted - the value of the average.

Based on a set of points for the selection of random data π_1 for 100 simulation experiments on the left side of Figs. Figure 6 shows a vertical diagram that suggests visualization of a group of numerical data through quantiles, Q1 (25th percentile sample), Q3 (75th percentile sample) and others (0.5 percentile), i.e., medians. Dotted red is similar to a certain number of mean values. The limits of the range of variations of indicators in the left part of Fig. 6 (scale diagrams) - presented in blue according to

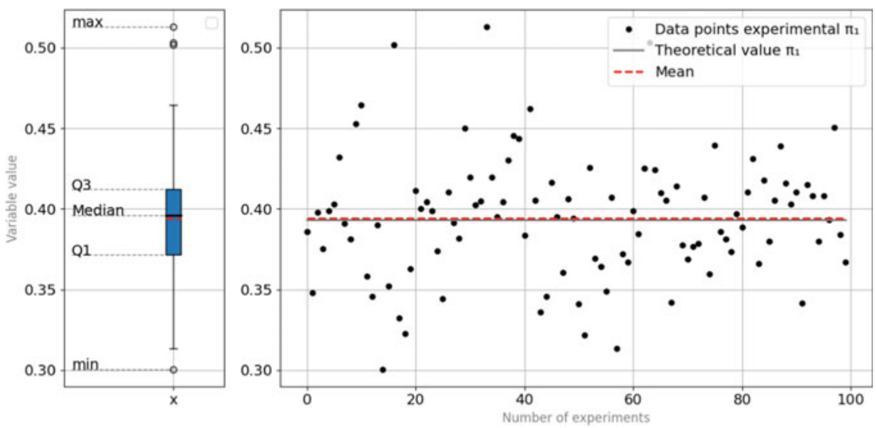


Fig. 6 Graphic data of the calculated metrics for estimating data variability for sampling data of experiments

the minimum and maximum values of the indicator. The distances between the other parts of the variations indicate the degree of variance, asymmetry and the presence of emissions.

5 Discussion and Outlook

Thus, the activities of SIRT in the complex conditions of cyberspace can be represented by a modified model of queuing, the mathematical apparatus of which makes it possible to obtain analytical expressions for indicators of service quality.

Regarding the characteristics of such QS, it should be noted that SIRTs differ from each other both in the number of specialists and indicators that characterize both individual specialists of the group and the group as a whole. For example, these may be such indicators [7–9] as: stress resistance of the specialist, time of retention, ability to work effectively in adverse conditions (including fatigue), time and conditions for recovery, efficiency of communication of specialists in teamwork and more. Therefore, it is possible to divide SIRT by such complex characteristics, which can be characterized by such indicators of the proposed model, which are responsible for the effectiveness of the group. In addition, it is possible to consider SIRT with different personal composition of specialists (quantitative and qualitative - in the sense of the proposed model), which will form a database on the indicator μ.

Similarly, we can talk about the classification of cyberattacks by the characteristics of their intensity (indicator λ in the proposed model) and the rate of increase in intensity (indicator α in the proposed model), which allows to form a database of possible cyberattacks.

Thus, the proposed model with the appropriate characteristics opens the possibility, taking into account the databases described above (one characterizes SIRT, the other - cyberattacks), to optimize the strategy for responding to information security incidents.

It is also important that such optimization can be carried out even in the process of deploying cyberattacks. For example, changing the personal composition of SIRT specialists, replacing one SIRT with another, or doing both at the same time. This makes it possible to increase the reliability of responding to incidents of information security and protection of cyberspace in general.

In the paper on the basis of the proposed model it is proposed to carry out a two-factor analysis of the stationary mode of operation of SIRT in real time, taking into account the characteristics of the system described above. Optimization can be carried out on the basis of, for example, the maximum effectiveness of serviced or lost SIRT requests for cyberattacks or its elements. This allows you to respond quickly to changes in the specifics of cyber-attacks (or a set of cyber-attacks), and the requirements of management on the need for increased attention to serviced or lost cyberattacks or their elements.

Thus, based on the results of the proposed model, the following method can be proposed to increase the effectiveness of SIRT in combating cyber-attacks with increasing intensity.

Stage 1. The classification of SIRT by the characteristics of the restoration of their work in particular from the intensity of the cyber-attack. Such characteristics will depend only on the peculiarities of perception and processing of information by SIRT specialists. They should also have special features of each specialist in terms of effectiveness to deal with stress, as well as features that characterize the restoration of his ability to work. In addition, it is important to develop specialist-specific methods of psychological recovery and the availability of such methods during the SIRT.

The characteristics of SIRT recovery should be determined through the characteristics of the proposed model of work.

Stage 2. Methods for identifying the current intensity of a cyber-attack are being developed. As in the previous stage, these characteristics would be expressed through the characteristics described in the model.

Stage 3. Critical values of characteristics at which SIRT activity ceases to satisfy the set conditions of efficiency of counteraction to cyber-attacks with increasing intensity are revealed.

Such values can easily be expressed through the parameters of the proposed model. Note that critical values can also be determined from special experiments. For example, specially organized SIRT trainings can be used for this purpose.

Step 4. Using the results of the developed model, the current efficiency of this SIRT in the process of counteracting cyber-attacs is monitored, taking into account the increase in their intensity.

Stage 5. Provided that the SIRT activity reaches critical values of efficiency parameters, a decision is made either to continue the work of the SIRT, or to attract a new SIRT (already with those parameters that will meet a given level of effectiveness) for to continue implementation of countering cyberattacks. It is also possible to consider involving additional specialists in SIRT so that after such an update, SIRT can continue to effectively combat cyberattacks. It is also possible to suggest the replacement of some specialists so that, as a result of the characteristics of the new SIRT, they allow to effectively counter cyber-attacks.

The use of the described method requires a specially organized set of exercises and training that can be developed taking into account the results of the proposed model. It is necessary to take into account not only the characteristics of each specialist in this SIRT. It should also be borne in mind that the characteristics of the joint work of a team of specialists are often not additive. Therefore, there is a need to study SIRT with different staff.

The results obtained in this work show that simulation opens the possibility to take into account the statistical patterns of cyberattacks in the analysis of the effectiveness of SIRT. As a result, it allows us to propose a method for calculating the expected effectiveness of SIRT in the face of changes in the stationary characteristics of cyberattacks. In particular, this method may include the following points.

Firstly, there would be a preparatory stage, which is in preparation the statistical characteristics, using a common database for cyber-attacks that have occurred in the past. The next steps are as follows.

1. Calculate statistical characteristics for the time intervals between the onset of individual incidents during a cyber-attack. As the main characteristic, you can choose the average value of the waiting time between individual incidents, which will calculate the numerical value of the indicator λ.
2. Statistical indicators are calculated to increase the intensity of individual incidents during a cyberattack, which will allow to obtain the indicator $\alpha\lambda$.
3. Calculate statistical indicators for the processing time of an individual incident during a cyber-attack μ.
4. The classification of cyberattacks is formed. Moreover, such a classification depends on the task set before SIRT and it must take into account the increase in the intensity of individual incidents during a cyber-attack.
5. A database is formed for the indicators $\alpha\lambda$, λ and μ and their statistical characteristics for each of the classes of cyber-attacks.

The database generated in step 5 can be used to optimize protection against cyber-attacks. It is necessary to take the following steps.

Using information about the cyber-attack class and statistics $\alpha\lambda$ and λ for this class, simulation is performed. As a result of this simulation will be obtained the statistical characteristics (see Fig. 5–6) for the expected effectiveness of SIRT (which is characterized by the indicator μ) in the conditions of this class of cyberattacks. For example, standard values and standard deviation. This allows you to predict the choice of a response group that will provide the required level of protection against cyber-attacks belonging to a given class.

Finally, by tracking the statistical characteristics of incidents during the deployment of the current cyber-attack, it is possible to replace during the cyber defense process one SIRT with another, the statistical characteristics of which will provide the required level of protection.

Thus, activities to optimize the organization of protection against cyberattacks with increasing intensity can be briefly described in the following sequence: classification of cyberattacks → analysis of statistical characteristics of incidents during cyberattacks → analysis of statistical characteristics of individual SIRT → simulation to calculate statistical characteristics of SIRT for individual classes of cyber-attack → forecast of the expected level of protection (depending on the characteristics of cyber-attack and SIRT) → preliminary forecast of SIRT selection to achieve the required level of protection → monitoring of statistical characteristics of the current cyberattack → optimization of SIRT selection to achieve the required level of protection.

6 Conclusions

We studied the SIRT efficiency improving by taking into account the increase in the intensity of individual incidents in the process of cyber-attacks and the need to restore the efficiency of personnel. A model was built that took into account the above features of cyber-attacks. The model uses the apparatus of queuing systems.

Managing the effectiveness of SIRT is to find the maximum of the function, which, in the general case, depends on three variables. The first variable λ is the average waiting time for the next incident during a cyberattack. The second variable α is the characteristic that describes the increase in the intensity of incidents during a cyber-attack. Finally, the third variable μ describes the specifics of the recovery of SIRT as a result of intensive work during a cyber-attack. Loss of ability to work occurs due to fatigue during work, which leads to reduced efficiency of staff.

Based on the model, methods for managing the effectiveness of countering cyber-attacks have been developed, as a result of which it is possible to select those SIRTs that have the desired effectiveness. It is also possible to manage the staff of SIRT, taking into account the individual characteristics of individual employees (for example, the level of efficiency of their work, features of recovery during work, features of their work in high stress with increasing cyberattacks, etc.).

A method for managing the effectiveness of SIRT with the use of computer simulation has been developed, which allows to take into account real statistical features already in the process of cyber-attack deployment.

References

1. Eltantawy, N., Wiest, J.: The Arab spring. Social media in the Egyptian revolution: reconsidering resource mobilization theory. Int. J. Commun. **5**, 1207–1224 (2011)
2. Wolfsfeld, G., Segev, E., Sheafer, T.: Social media and the Arab Spring: politics comes first. Int. J. Press/Politics. **18**, 115–137 (2013)
3. Del Vicario, M., et al.: The spreading of misinformation online. Proc. Natl. Acad. Sci. **113**(3), 554–559 (2016). https://doi.org/10.1073/pnas.1517441113
4. Shao, C., Ciampaglia, GLuca, Varol, O., Yang, K.-C., Flammini, A., Menczer, F.: The spread of low-credibility content by social bots. Nat. Commun. **9**(1), 4787 (2018). https://doi.org/10.1038/s41467-018-06930-7
5. Bail, C.A., Argyle, L.P., Brown, T.W., et al.: Exposure to opposing views on social media can increase political polarization. Proc. Natl. Acad. Sci. **115**(37), 9216–9221 (2018)
6. Ross, B., Pilz, L., Cabrera, B., et al.: Are social bots a real threat? An agent-based model of the spiral of silence to analyze the impact of manipulative actors in social networks. Eur. J. Inf. Syst. **28**(4), 394–412 (2019)
7. Carley, K.M.: Social cybersecurity: an emerging science. Comput. Math. Organ. Theory **26**(4), 365–381 (2020)
8. Grispos, G., Glisson, W., Storer, T.: How good is your data? Investigating the quality of data generated during security incident response investigations. In: Proceedings of the 52nd Hawaii International Conference on System Sciences, 2019. pp. 7156–7165. Hawaii International Conference on System Sciences (2019)

9. Camacho, J., García-Giménez, J.M., Fuentes-García, N.M., Maciá-Fernández, G.: Multivariate big data analysis for intrusion detection: 5 steps from the haystack to the needle. Comput. Secur. **87**, 101603 (2019)
10. Nyre-Yu, M., Gutzwiller, R.S., Caldwell, B.S.: Observing cyber security incident response: qualitative themes from field research. In: Proceedings of the Human Factors and Ergonomics Society Annual Meeting, 2019. pp. 437–441. SAGE Publications, Sage, Los Angeles, CA (2019)
11. Phuong, P., Vivek, J., Dauterman, L., Ormon,t J., Navendu, J.: DeepTriage: automated transfer assistance for incidents in cloud services. In: Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining, 2020, pp. 3281–3289. KDD2020 (2020)
12. Ahmad, A., Desouza, K.C., Maynard, S.B., et al.: How integration of cyber security management and incident response enables organizational learning. J. Am. Soc. Inf. Sci. **71**(8), 939–953 (2020)
13. Kotik, M.A., Emelyanov, A.M.: The nature of human operator errors. Moscow, Transport (1993). (in Russian).
14. Yurkov, O.S.: Psychology practice and engineering psychology. Mukachevo, MDU (2018). (in Ukrainian).
15. Human engineering guide for equipment designers. Woodson, W.E., Conover, D.W. (eds.) Berkeley Los Angeles, Univ. of California Press (1966)
16. Matalytsky, M., Khatskevich, G.: Probability Theory and Mathematical Statistics. LitRes, Moscow (2021). (in Russian)
17. Venttsel, E.S., Ovcharov, L.A.: Theory of Random Processes and Its Engineering Applications. Nauka, Moscow (1991). (in Russian)
18. Goloskokov, O.E, Goloskokova, A.A, Moshko, E.A.: Fundamentals of the theory of exponential queuing systems. Kharkiv, NTU "KhPI" (2017). (in Ukrainian).
19. Kopey, V.B.: Python programming language for engineers and scientists. Ivano-Frankivsk, IFNTUNG (2019). (in Ukrainian).
20. Kelton, V., Lowe, A.: Simulation modeling. CS classic. 3rd edn. Kyiv, BHV Publishing Group (2004). (in Russian)

Mathematical Model for Assessing the Quality Level of Formation of Stochastic Competence of Higher Education Acquires

Elena Trunova, Viacheslav Mamchurovskyi, Daryna Pryschepa, Andrii Akymenko, and Iryna Bilous

Abstract The article analyzes the existing approaches to assessing the level of students' competencies (competence) formation, which proves the need to use a differentiated approach in the development of an appropriate mathematical model. In the process of creating a mathematical model for assessing the level of stochastic competence formation of higher education students, criteria for assessing stochastic competence were selected: professional-cognitive; conative; motivational-valuable. The indicators of formation and the basis of the formation of structural components of stochastic competence are highlighted.

The mathematical model for assessing the level of stochastic competence formation is created using the method of implicit modeling based on dimensional analysis analogy. It is an integrated assessment of the educational activity results of the higher education student taking into account his personal and motivational characteristics. The model establishes the dependence between the assessment of the stochastic competence formation level and the following factors: component-wise assessment of the educational activity results of the higher education student; assessment of the personal characteristics of the higher education student; assessment of the motivation of the student to study stochastics. The levels of competence formation: reproductive, productive, heuristic, creative were defined.

To assess the statistical significance of experimental data, the level of stochastic competence formation in higher education among graduates in the specialty 051 «Economics» was evaluated. The Mann-Whitney U test was used. The results of the experiment confirmed that the application of the developed model provides an objective and reliable assessment of the stochastic competence level formation of higher education students. According to the results of statistical analysis, the proposed model is adequate.

Keywords Stochastic Competence · Assessing the Quality Level of Stochastic Competence Formation · Competency

---

E. Trunova (✉) · V. Mamchurovskyi · D. Pryschepa · A. Akymenko · I. Bilous
Chernihiv Polytechnic National University, 95 Shevchenko str., Chernihiv 14035, Ukraine
e-mail: e.trunova@gmail.com

S. Shkarlet et al. (eds.), *Mathematical Modeling and Simulation of Systems*, Lecture Notes in Networks and Systems 667,
https://doi.org/10.1007/978-3-031-30251-0_16

1 Introduction

The changes taking place in modern society require effective solutions to problems, most of which are stochastic in nature. Today, the entire cycle of natural and socio-economic sciences is based and develops on the basis of laws of probability, and without appropriate stochastic training, adequate perception and correct interpretation of social, political, technical and economic information is impossible. In the constantly changing modern world, a large number of people encounter problems in their lives that are mostly related to the analysis of the impact of random factors and require decision-making in situations that have a probabilistic basis.

Stochastics is a complex of modern professionally significant disciplines. These include «Theory of Probability», «Theory Stochastic Processes», «Mathematical Statistics», «Mathematical Modeling», etc.

The goal of studying stochastics at the university is the formation of the stochastic competence of future specialists as a component of their professional competence.

In our study, stochastic competence will be understood as a characteristic of a specialist's personality, which reflects readiness to study stochastics, the presence of deep and solid knowledge of stochastics, and the ability to use stochastic methods in professional activities, not always in standard situations [1].

Competence is a generalized concept that includes a set of competencies.

Today, the competency-based approach has been recognized as the most effective means of reforming education, within which the learning process is understood as a complex activity aimed at forming a set of general cultural and professional competencies in students [2].

The competency-based approach in education involves changing, first of all, the system for assessing the educational achievements of students, therefore, the attitude of students to educational activities, their cognitive motivation generally depends on the organization of the assessment process and the objectivity of the grades given.

The problem of measuring the level of competencies formed in the educational process still does not have a generally recognized solution either in Ukraine or abroad. One of the reasons for this is the complexity and versatility of the concept of competency, as well as the lack of a unified methodology for measuring the level of competency of applicants for higher education, that is, the solution of this problem requires a differentiated approach.

2 Related Works

The process of forming the necessary competences among applicants for higher education and the implementation of the competency-based approach in the higher professional school as a whole was considered by V.V. Stynska, et al. [3], which only emphasizes the necessity of its use for thorough training of specialists as well as M. Blaskova, et al. [4] propose. In the example of these two articles, we can see the inconsistency of the terms «competency» and «competence», constantly appearing in scientific works related to this topic. The reasons for the terminological inconsistency, the modern approach and a strict distinction between the concepts of competency, competence and the system of competencies were defined by Y. Poskrypko and O. Danchenko [5], and also discovered by M. El Asame and M. Wakrim [6], which not only provides the basic definitions of the concept of competence, but also highlights the main specifications for the presentation and understanding of competence.

The classification of single-criteria models that allows evaluating qualification characteristics from the standpoint of their application to assess the level of formation of professional competencies of future university graduates is shown in Fig. 1.

In order to determine the level competency formation, there are a plenty of models that offer to calculate the integrated indicator. These models can be categorized as:

- mean-value models: simple; additive; multiplicative; linear-piecewise approximation [7];
- fuzzy logic models: Sugeno; Mamdani; Larsen; Tsukamoto [8]; M. Malyar and A. Shtymak [9] propose an algorithm for determining the level of competence of a graduate using the theory of fuzzy sets and fuzzy logic procedures, which makes it possible to determine the level of competence of a university graduate.

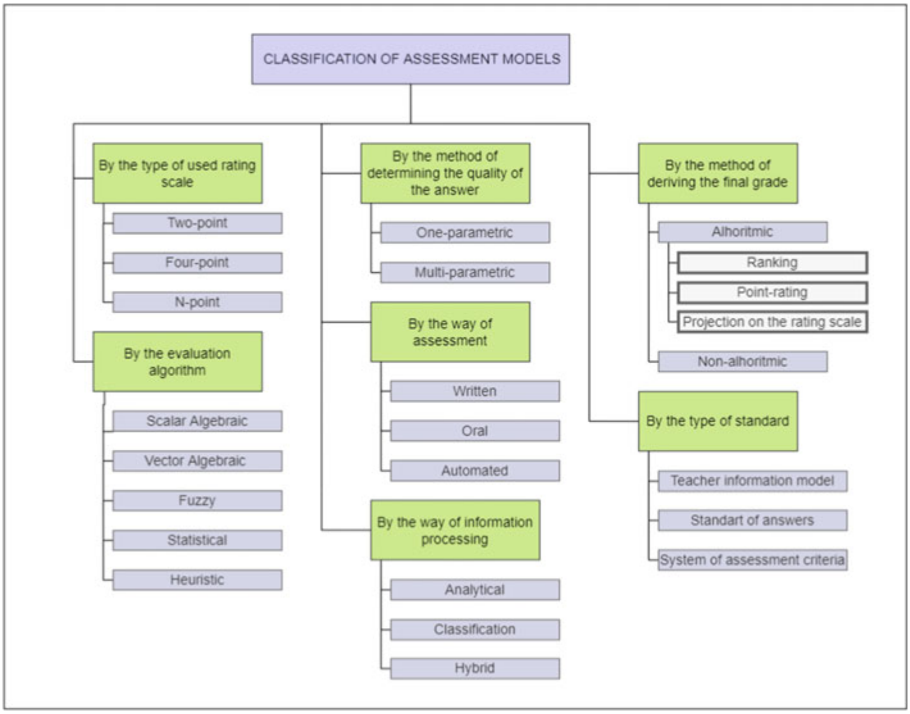


Fig. 1. Classification of single-criteria models for assessing the level of formation of professional competences

The use of the apparatus of fuzzy sets for building a model and an algorithm for determining the level of competence makes it possible to express qualitative characteristics (competencies) by numerical values.

- probabilistic models (Item Response Theory): for dichotomous tasks (Rush, Birnbaum model); for polytomous tasks (Rating Scale Model, Partial Credit
- Model) [10].

It is fair to assume that stochastic competence is one of the most important competences for graduated student. A thorough understanding of the laws of probability theory and mathematical statistics makes it possible to solve most problems of a stochastic nature. That is why it is important to get acquainted with the theoretical foundations, solve practical problems of a stochastic nature, search for effective methods, forms and means of teaching stochastics for future specialists of both non-mathematical [11] and mathematical specialties (in particular, financial and economic ones [12]).

In modern educational standards, competency includes not only knowledge, abilities, skills of various levels, but also personal goals and cognitive qualities, many of which are difficult to form in the conditions of education and even more difficult to diagnose their genesis. This leads to the disunity of the proposed techniques and the impossibility of their systematic application.

All of the above determines the relevance of developing a mathematical model for determining the level of formation of stochastic competence of applicants for higher education.

3 Material and Methods

To solve the research tasks set, the following methods were used: the method of expert assessment, the method of analyzing hierarchies, the method of «imaginary modeling», graph theory, statistical methods.

3.1 Criteria of Formation of Stochastic Competence

Based on the analysis of the scientific and pedagogical literature, the following components of stochastic competence formation were selected for the assessment of professional competencies: cognitive; conative; motivational and valuable.

Table 1 presents the indicators of formation and the basis of the formation of structural components of stochastic competence (see Table 1).

As a result of the theoretical analysis of the problem of development of stochastic competence, the stochastic competence formation levels of the graduate were determined: reproductive; professionally adaptive; professionally competent; creative.

Table 1. Indicators of formation and the basis of formation of stochastic competence

Components	Indicators of formation	Basis of formation
Cognitive	the outlook level on the application of stochastic knowledge in the professional field; to know and to be able to use the stochastic knowledge system; knowledge of stochastic models and related problem solving methods.	Mastering the program material
Conative	the ability to apply stochastic knowledge, skills and abilities in solving professionally oriented problems. independence and quality of task performance; the ability to apply acquired knowledge in a new situation;	The use of stochastics in practical and professional activities
Motivational and valuable	interest in stochastics; awareness of the place, role and significance of stochastics in professional and personal aspects, the need to master stochastic knowledge; striving for current and future learning of stochastics.	Acceptance of the requirements of the qualification characteristics

The model for assessing the level of competence, including the stochastic one, should be defined as a set of assessed characteristics of a higher education applicant, which includes:

- a set of grades received for mastering the educational components, grouped by assessment components;
- a set of assessed results of extracurricular activities of a higher education applicant;
- an professional motivation assessment for learning and future professional activity.

The analysis of existing approaches to the competences (competencies) assessment, the multi-component content, and the interdisciplinary nature of the concepts «competence» and «competency» made it possible to create a multi-component model for assessing the stochastic competence formation level, which fully meets the requirements of modern educational standards:

$$C_T^i = f\left(L_T^i, Q_T^i, M_T^i\right), \tag{1}$$

where C_T^i – the overall assessment of the i-th competency in the T-th period, which forms the stochastic competence; L_T^i – the overall educational achievements assessment of the i-th competency in the T-th period, which forms the stochastic competence; Q_T^i – the overall personal qualities assessment of the i-th competency in the

T-th period, which forms the stochastic competence; M_T^i – the overall professional motivation assessment of the i-th competency in the T-th period, which forms the stochastic competence.

Thus, the task of assessing the stochastic competence formation level of the future specialist is reduced to a step-by-step presentation of the educational results and other activity types through a competencies summary assessment that form stochastic competence.

3.2 Assessment of the Educational Activity Results of the Higher Education Student

In the educational system, according to the traditional approach the result of the student's educational activity is provided in the form of final grades received by him for studying the relevant educational components. Under the conditions of the competence approach, the results of the educational achievements of higher education students should be presented in the form of formed competencies (general, professional, etc.). By the requirements of the educational standards, each competency is included in the set of competencies of the corresponding field of training $C = \{C^i\}$,

$i = 1, \ldots, m$ that determines the graduate competence model, should be divided into the components «know – be able to – have a skill» $C^i = \{K^i, S^i, P^i\}$, де $K^i = \{k_1^i, k_2^i, \ldots, k_l^i\}$ – a subset of «knowledge» of the i-th competency, $S^i = \{s_1^i, s_2^i, \ldots, s_m^i\}$ – a subset of «skills» of the i-th competency, $P^i = \{p_1^i, p_2^i, \ldots, p_r^i\}$ – a subset of «practice» of the i-th competency. The results of developing educational components are presented in educational programs in the form: knowledge, skills, and practices.

At the same time, competencies do not exclude knowledge, practices and skills, however, they fundamentally differ from those that were before internally: from knowledge – by the fact that the current descriptors exist in the form of activity and not only in the information form about the activity; from skills – the fact that competencies assume the various professional and general cultural tasks solving; from practices – by the fact that they are not automated, but are realized by a person and allow to act in non-standard situations in any activity type [13].

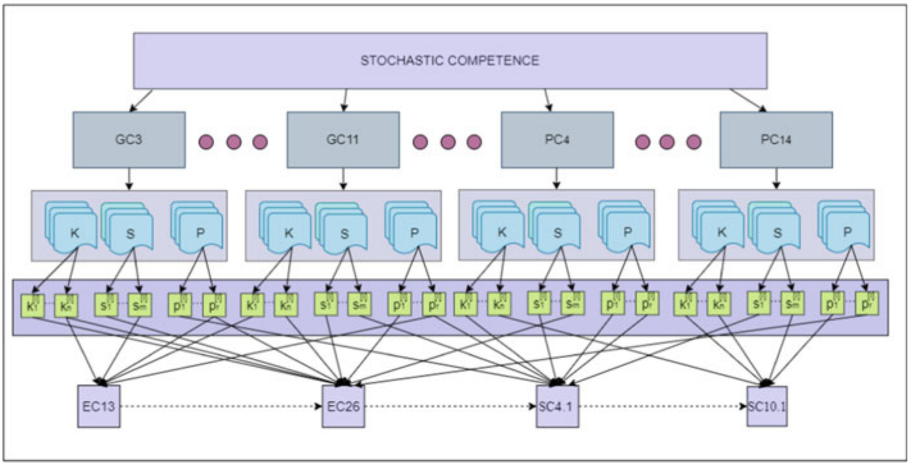


Fig. 2. The cognitive component model of stochastic competence

The cognitive component model of stochastic competence is presented in Fig. 2.

The student educational activity assessment of the results is a competency assessments convolution that is learned and measured in various educational components determined by the educational program for the relevant field, which forms stochastic competence:

$$\begin{cases} L_T^i = \sum_{j=1}^{n} L_{EC_j}^i, \left(i = \overline{1, m},\ j = \overline{1, n}\right), \\ L_{EC_j}^i = \sum_{j=1}^{n}\sum_{l=1}^{a}\alpha_l^i k_l^i + \sum_{j=1}^{n}\sum_{l=1}^{b}\beta_l^i s_l^i + \sum_{j=1}^{n}\sum_{l=1}^{c}\gamma_l^i p_l^i, \\ \alpha_l^i = \frac{q(k)_l^i}{n}, \beta_l^i = \frac{q(s)_l^i}{n}, \gamma_l^i = \frac{q(p)_l^i}{n}, \end{cases} \tag{2}$$

where $L_{EC_j}^i$ – the assessment of the set of components «knowledge – skill – practice» of the j-th educational component, that participates in the formation of the i-th competency; L_{max}^i – the maximal assessment of the all student educational activities results; α_l^i, β_l^i, γ_l^i – the coefficients of significance of the components «knowledge – skill – practice» of the j-th educational component, that participates in the formation of the i-th competency; k_l^i, s_l^i, p_l^i – the assessment of the relevant components «knowledge –skill – practice» of the j-th educational component, that participates in the formation of the i-th competency; $q(k)_l^i$, $q(s)_l^i$, $q(p)_l^i$ – the number of the educational components, that participates in the formation of the relevant components «knowledge – skill – practice» of the i-th professional competency; n – the number of all educational components forming the stochastic competence determined by the appropriate educational program; m – the number of all competencies of the direction forming stochastic competence; a, b, c – the number of control elements of the j-th educational component.

The coefficients of components significance, which is defined as the ratio of the number of educational components forming this component to all ECs forming stochastic competence, is applied into the calculation formula (1) in order to give greater significance to the components formed with the participation of a plenty of ECs.

3.3 Assessment of the Student's Personal Qualities

Under the conditions of the competence approach, the development of the personal qualities of the higher education acquirer is a necessary condition and a result of the successful formation of competencies, therefore its assessment is a mandatory component in assessing the competence formation level and is considered in the mathematical assessment model as one of the arguments of the competence assessment function.

The assessment of both the personal qualities of the higher education student and his educational achievements is an expert assessment and is carried out directly during the training process. In addition to professors and teaching staff, experts may be persons responsible for various types of activities (creative, intellectual, scientific research, etc.). Among the personal qualities of the applicant, the following are distinguished: responsibility, creativity, independence, activity, self-criticism, purposefulness, etc. All these personal qualities of a student determine his abilities in relation to educational activities. The assessment of the student's personal qualities is determined as follows:

$$\begin{cases} Q_T^i = \frac{1}{n}\sum_{j=1}^{n}\sum_{l=1}^{r}\varphi_l Q_{j_l}^i, \left(i = \overline{1,m},\ j = \overline{1,n}, l = \overline{1,r}\right) \\ \left[\begin{array}{l} Y^{(N)} = Y \cdot H^{(N-1)}, \\ H^{(N)} = \frac{1}{\lambda^{(N)}} W' \cdot Y^{(N)}, \\ \lambda^{(N)} = \left|W' \cdot Y^{(N)}\right|, \end{array}\right. \end{cases}, \quad (3)$$

where $Q_{j_l}^i$ – the assessment of factors characterizing the student's personal qualities that form the i-th competency in the T-th period; $Y^{(N)} = (y_l)$ – a column matrix of the N-th iteration coefficients of significance; $W = \left(w_{ij}\right)$ – a matrix of the individual weights (expert assessment) [14]; W' – the transposed individual weights matrix of factors; $H^{(N)}$ – a column matrix of experts competence after N iterations; $\lambda^{(N)}$ – the magnitude of vector of expert competence (normalizing factor); n – the number of all educational components that form the stochastic competence determined by the corresponding educational program; r – the number of assessing factors.

3.4 Assessment of the Student's Professional Motivation for Future Professional Activity

The simplest analysis of a student's professional motivation carried out on time, and timely measures taken to increase it can affect the learning process and, accordingly, the development and formation of professional competencies.

There are various methods of diagnosing the motivation for professional activity. In our study, the motivational component is determined using the theoretical-multiple model of assessing the higher education students' professional motivation:

$$M_T^i = \left(M^{int,ext_p,ext_n}, \tilde{M}_T^{int}, \tilde{M}_T^{iext_p}, \tilde{M}_T^{iext_n}, T\right), \quad (4)$$

where $M^{int,ext_p,ext_n} = \{M_\eta\}, \left(\eta = \overline{1, g}\right)$ – a set of questionnaire questions aimed at identifying the internal, external positive, and external negative motivation of the higher education applicant for future professional activity; $\tilde{M}_T^{int}$ – linguistic assessment of the student's internal motivation (undefined, low, average, above average, high); $\tilde{M}_T^{iext_p}$ – linguistic assessment of external positive motivation (undefined, low, average, above average, high); $\tilde{M}_T^{iext_n}$ – linguistic assessment of internal negative motivation (undefined, low, average, above average, high); T – academic period (year, semester etc.);

The condition for the assessing professional motivation possibility:

$$\mu\left(\tilde{M}_T^{int}\right) \geq \mu\left(\tilde{M}_T^{iext_p}\right) > \mu\left(\tilde{M}_T^{iext_n}\right), \quad (5)$$

where $\mu(\tilde{M}_T^{iext_p})$, $(\mu(\tilde{M}_T^{iext_p})$, $\mu(\tilde{M}_T^{iext_n})$ an aggregated quantification obtained using membership functions.

Criterion for evaluating professional motivation:

$$M_T^{int,ext_p,ext_n} = \sum\nolimits_{i=1}^{5} \sum\nolimits_{j=1}^{n} \mu\left(\tilde{M}_{Tj}^i\right) \to max, \; M_T^i = \sum\nolimits_{j=1}^{r_i} v_{ij} \cdot \mu\left(\tilde{M}_{Tj}^i\right), \quad (6)$$

where $\mu\left(\tilde{M}_{Tj}^i\right)$ – quantitative assessment of the *j*-th question of the *i*-th motivation assessment; r_i – the number of questions from the *i*-th assessment of motivation; v_{ij} – the weight of the *j*-th question concerning the *i*-th motivation assessment;

The overall assessment of professional motivation in the *T*-th academic period:

$$M_T = \sqrt[3]{M_T^{iext_p} \cdot M_T^{iext_p} \cdot M_T^{iext_n}} \quad (7)$$

The result will be a quantification on the interval $0 \leq M_T \leq 1$.

3.5 Assessment of the Level of Stochastic Competence Formation

Thus, the task of assessing the student's stochastic competence formation level is reduced to a step-by-step presentation of the results of educational and other types of activities, and the mathematical model of the assessment of stochastic competence formation level is an integral assessment of the student's educational activities results, taking into account his individual personal and motivational characteristics.

To determine the functional dependence between the analyzed quantities, that is, to present the analytical expression of the model, the approach of «conditional» modeling was used, which consists in replacing the original model according to a certain agreement attributed to this model, with a «conditional» model, which is based on the theory of similarity and dimensionality analysis, which most adequately describes this process. Based on the adoption of an ideal physical model, a corresponding mathematical model was created, namely a mathematical description of the competence assessment process [15].

The assessment of competence corresponds to the physical values characterizing the work, namely:

$$C_T^i = \frac{L_T^i \cdot Q_T^i}{M_T^i}, \tag{8}$$

where C_T^i – work; L_T^i – force; Q_T^i – speed; M_T^i – acceleration.

Finally, the level of stochastic competence formation in the T-th period is determined by the formula:

$$C_T^{st} = \sum\nolimits_{i=1}^{m} V_i \cdot C_T^i, \left(i = \overline{1, m}\right), \tag{9}$$

where C_T^{st} – the level of stochastic competence formation in the T-th period; V_i – the weight of the i-th competency in the process of stochastic competence forming;

C_T^i – the overall assessment of the i-th competency in the T-th period.

The measurement range of the level of stochastic competence formation is defined from 0 to 1. Thus, the assessment interval corresponding to a certain level is

denoted as:

$$\left[I_t, I_{t+1}\right](t = 0, 1, 2, 3, 4) \tag{20}$$

the limits of interval estimates are presented as: reproductive – $[I_0,\ I_1)$; professionally adaptive – $[I_1,\ I_2)$; professionally competent – $[I_2, I_3]$; creative – $[I_3, I_4]$.

Table 2. Interrelation of educational components with program competencies to determine the level of formation of stochastic competence of 051 «Economics»

Cycle		Educational component	ECTS Loans	Semester	Program Competencies
Mandatory components	General training	EC13. Probability Theory and Mathematical Statistics	6	3	GC3, GC4, GC7, SC4, SC-6, SC-7
	Professional training	EC26. Economic activity forecasting	7	7	SC4, SC6, SC7, SC8, SC9
Selective components	General training	SC4.1. Economic and mathematical methods and models	6	5	GC3, GC4, GC7, GC8, SC4, SC6, SC7
		SC4.2. Econometrics	6	5	GC3, GC4, GC7, GC8, SC4, SC6, SC7
	Professional training	SC10.1. Justification of business decisions and risk assessment	6	8	GC7, GC11, SC7, SC14
Practical training		EC29. Internship	6	8	GC11, SC4, SC6, SC7, SC8, SC9, SC14
Final state attestation		EC30. Graduation qualification work	14	8	GC3, GC4, GC7, GC8, GC11, SC4, SC6, SC7, SC8, SC9, SC14

4 Discussion

The proposed mathematical model for assessing the stochastic competence formation was tested during an experiment in which the level of stochastic competence formation was assessed among graduates of higher education in the field of training 051 «Economics» for the entire period of study at the time of graduation.

During the research it was found that stochastic competence is formed as a result of mastering a number of compulsory and selective disciplines, completing the tasks of industrial practice, completing and defending the final qualification of work (see Table 2) [16].

Among the general and special (professional, subject) competences, those directly related to the educational components forming stochastic competence were selected. Also, their weight was determined by the method of hierarchy analysis [17] (Table 3).

To conduct a statistical study as an empirical base at the stage of preliminary preparation, the initial data of the educational, research and creative activities of higher education students from I to IV course of the control group study, the results

Table 3. Significance coefficients of the components of stochastic competence of the 051 «Economics» competence

Marking	Competence	Weight	Educational components
GC3	Ability to think, analyze and synthesize	0,1	EC13, SC4.1, EC30
GC4	Ability to apply knowledge in practical situations	0,06	EC13, SC4.1, EC30
GC7	Skills of using information and communication technologies	0,015	EC13, SC4.1, SC10.1, EC30
GC8	Ability to search, process and analyze information from various sources	0,06	SC4.1, EC29, EC30
GC11	Ability to make informed decisions	0,015	SC10.1, EC30
SC4	Ability to describe economic and social processes and phenomena on the basis of theoretical and applied models, analyze and comprehensively interpret the obtained results	0,175	EC13, EC26, SC4.1, EC29, EC30
SC6	Ability to apply economic and mathematical methods and models for solving economic problems	0,15	EC13, EC26, SC4.1, EC29, EC30
SC7	Ability to use computer processing technologies for solving economic problems, analyzing information and preparing analytical reports	0,025	EC13, EC26, SC4.1, SC10.1, EC29, EC30
SC8	Ability to analyze and solve tasks in the field of regulation of economic and social-labor relations.	0,15	EC26, EC29, EC30
SC9	Ability to predict socio-economic processes on the basis of standard theoretical and econometric models	0,1	EC26, EC29, EC30
SC14	Ability to deeply analyze problems and phenomena in one or more professional areas, taking into account economic risks and possible socio-economic consequences	0,15	SC10.1, EC29, EC30

of expert surveys were used. Potential employers, professors and teaching staff, graduates and students of university acted as experts.

The basis of the statistical analysis was the data obtained as a result of the experiment. Since the assessment of the formation of stochastic competence is an integral assessment of the results of the student's educational activity taking into account his individual personal and motivational characteristics, the preliminary calculations included: the assessment of the results of the educational activity of the student of higher education (L_T^i), obtained on the basis of the mastering of selected educational components, which in a significant measures form stochastic competence according to (2); assessment of the student's personal qualities (Q_T^i) – on the basis of educational, scientific and social activities, according to (3); assessment of the student's professional motivation for future professional activity (M_T^i) – on the basis of the conducted questionnaire of higher education applicants according to formulas (6, 7).

Calculations were made both for each of the program competences separately (8) and for the integral indicator $\left(C_T^{st}\right)$ according to the formula (9).

The Mann-Whitney U test was used to assess the statistical significance of the experimental data [18].

The histogram presented in Fig. 3 demonstrates that the empirical characteristics of the stochastic competence level assessment model coincide with the critical value of the Mann-Whitney U test $W_{0,05} = 1,96$.

The results of the numerical experiment confirmed that the application of the developed model provides an objective and reliable assessment of the level of stochastic competence formation of higher education students. Therefore, according to the results of the statistical analysis, the proposed model is adequate.

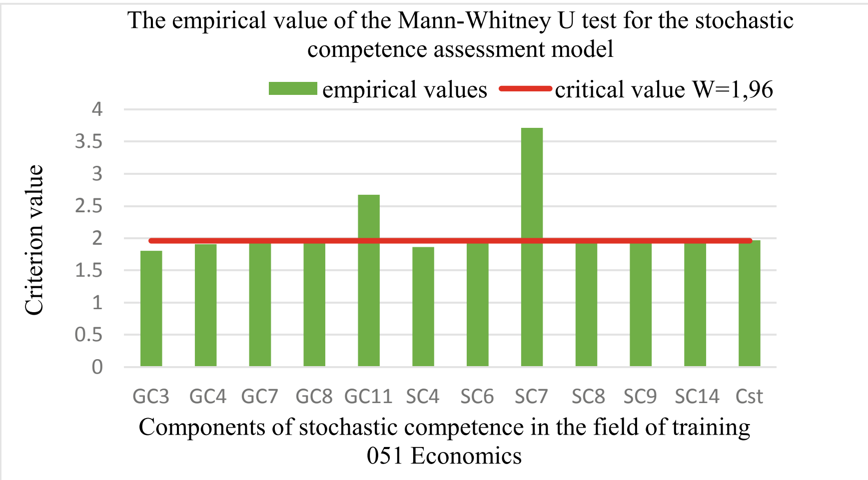


Fig. 3. Results of an experimental study

5 Conclusions and Further Research

The article analyzes the existing approaches to the assessment of the level of formation of competences (competencies), based on which the need to develop a mathematical model for the assessment of the level of formation of stochastic competence of a higher education student was revealed. Based on the selected components of stochastic competence, the components of the complex of assessments were determined, which determine the main characteristics of the student of higher education, which allows giving an objective assessment of the quality results of the educational process. The indicators of formation and the basis of the formation of structural components of stochastic competence are highlighted.

The model is built using the approach of «conditional» modeling, which is based on the analogy of dimensional analysis, while establishing the relationship between the assessment of the level of formation of stochastic competence and the factors: component-wise assessment of the educational activity of the higher education acquirer; assessment of the personal characteristics of the higher education applicant; evaluation of the motivation of the higher education acquirer.

The defined levels and corresponding limits of competence formation are: reproductive; professionally adaptive; professionally competent; creative. The Mann-Whitney U test was used to assess the statistical significance of the experimental data. The results of the experiment confirmed that the application of the developed model provides an objective and reliable assessment of the level of formation of stochastic competence of students of higher education. Therefore, according to the results of the statistical analysis, the proposed model is adequate.

Note that the process of stochastic competence assessment requires a high qualification of specialists participating in the expert assessment process and a large number of operations related to data processing, calculation of the level of competence formation, and visualization of the obtained results. This necessitates the need for further research, a promising direction will be the development of a software complex, which allows, on the one hand, to simplify the calculations made and, on the other hand, to create a convenient toolkit for analyzing the results obtained when making decisions about improving the quality of training of higher education applicants.

References

1. Trunova, O.: The model of the life cycle of stochastic competences. In: Scientific Notes Kirovohrad State Pedagogical University after V. Vynnychenko. Series: The Problems of the methodology of Physical, Mathematical and Technological education, vol. 8(3), pp. 78–89 (2015).
2. Lytvynov, V.V., Kharchenko, V.S., Lytvyn, S.V., Saveliev, M.V., Trunova, E.V., Skiter, I.S.: Tool-Based Support of University-Industry Cooperation in IT-Engineering. Chernihiv National University of Technology (2015)

3. Stynska, V.V., Yashchyshyn, Z.M., Klishch, I.P.: Competency-based approach in Ukrainian vocational training. In: Scientific journal of M.P. Dragomanov National Pedagogical University. Series 5, Pedagogical Sciences: Realities and Perspectives, vol. 79(2), pp. 139–142 (2021). https://doi.org/10.31392/NPU-nc.series5.2021.79.2.30.
4. Blaskova, M., Blasko, R., Kucharcikova, A.: Competences and competence model of university teachers. In: Procedia – Social and Behavioral Sciences, vol. 159, pp. 457–467 (2014). https://doi.org/10.1016/j.sbspro.2014.12.407.
5. Poskrypko, Y., Danchenko O.: Competence and competency: consensus. Sci. Notes «KROK» Univ. **3**(55), 117–127 (2019). https://doi.org/10.31732/2663-2209-2019-55-117-127.
6. El Asame, M., Wakrim, M.: Towards a competency model: a review of the literature and the competency standards. Educ. Inf. Technol. **23**, 1–12 (2018). https://doi.org/10.1007/s10639-017-9596-z
7. Kuzminska, O., Mazorchuk, M., Morze, N., Pavlenko, V., Prokhorov, A.: Study of digital competence of the students and teachers in Ukraine. In: Ermolayev, V., Suárez-Figueroa, M.C., Yakovyna, V., Mayr, H.C., Nikitchenko, M., Spivakovsky, A. (eds.) Information and Communication Technologies in Education, Research, and Industrial Applications, vol. 1007, pp. 148–169. Springer, Cham (2019). https://doi.org/10.1007/978-3-030-13929-2_8
8. Dzwigol, H., DzwigolBarosz, M., Miskiewicz, R., Kwilinski, A.: Manager competency assessment model in the conditions of industry 4.0. Entrepreneursh. Sustainabil. Issues **7**(4), 2630–2644 (2020). https://doi.org/10.9770/jesi.2020.7.4(5)
9. Malyar, M., Shtymak, A.: Model of determining the competence of graduate with the use of fuzzy sets. In: Management of Development of Complex Systems, vol. 22(1), pp. 151–157 (2015). https://doi.org/10.13140/RG.2.1.2630.8567.
10. Zendler, A.: cpm.4.CSE/IRT: compact process model for measuring competences in computer science education based on IRT models. Educ. Inf. Technol. **24**(1), 843–884 (2019). https://doi.org/10.1007/s10639-018-9794-3
11. Dzyamko, V., Mesarosh, L.: Formation of stochastic competence through the implementation of interpersonal relationships. Sci. Bull. Uzhhorod Univ. Ser. «Pedagogy. Soc. Work» **0**(1(42)), 59–63 (2018). https://doi.org/10.24144/2524-0609.2018.42.59-63
12. Zadorozhna T.M., Rudenko I.B.: Development of stochastic competence in the context of preparation of specialists of finance - economic direction. In: Collection of Research Papers «Pedagogical Sciences», vol. 66, pp. 305–310 (2014)
13. Kolosok, I.O, Demin, O.A.: «Knowledge», «skills» and «skills» as pedagogical category and methodological factors of its formation. In: National University of Life and Environmental Sciences of Ukraine. Series «Machinery & Energetics», vol. 275, pp. 216–227 (2017)
14. Lukianova, V., Shutyak, Y., Polozova, V.: Expert assessment method in socio-economic research and Scales transformations. In: Economics Proceedings of the 2019 7th International Conference on Modeling, Development and Strategic Management of Economic System (MDSMES 2019) (2019). https://doi.org/10.2991/mdsmes-19.2019.67.
15. Kuneš, J.: Similarity and Modeling in Science and Engineering. Cambridge International Science Publishing Ltd, Cambridge (2012). https://doi.org/10.1007/978-1-907343-78-0
16. The higher education standard to the specialty 051 «Economics» of the first (bachelor's) level of higher education. https://mon.gov.ua/storage/app/media/vishchaosvita/zatverdzeni%20standarty/12/21/051-ekonomika-bakalavr.pdf. Accessed 10 May 2022
17. Saaty, R.W.: The analytic hierarchy process—what it is and how it is used. Math. Modelling **9**(3–5), 161–176 (1987). https://doi.org/10.1016/0270-0255(87)90473-8
18. Bergmann, R., Ludbrook, J., Spooren, W.P.J.M.: Different outcomes of the Wilcoxon—Mann—Whitney test from different statistics packages. Am. Statist. **54**(1), 72–77 (2000). https://doi.org/10.1080/00031305.2000.10474513

Method for Assessing the Reliability of Information Under Conditions of Uncertainty Through the Use of a Priori and a Posteriori Information of the Decoder of Turbo Codes

Sergei Zaitsev, Yevhen Ryndych, Natalia Sokorynska, Liliia Zaitseva, Pavel Kurbet, and Borys Horlynskyi

Abstract The article researched and described the rapid development of wireless technologies, their advantages and issues. A central problem, which is explored in the article, complained tasks of evaluating the channel, increasing the reliability of information transmission and using parametric adaptation. The article proposes a method for assessing the reliability of the information in high-speed transmission of information over radio channels in conditions of increased noise. The method is based on the calculation of the decoding inaccuracy index (error rate) by the maximum a posteriori probability in systems with multi-parameter adaptation in terms of parameters of error-correcting codes, in particular, turbo codes and codes with low parity check (LDPC). The using of the sign change of a priori and a posteriori LRR during iterative decoding and taking into account the noise dispersion values in the channel reliability parameter, the method allows obtaining information reliability values (error coefficient) without using an additional service channel. Simulations let to understand that the accuracy of the information approaches the reliability estimate when using an additional service channel .

Keywords Information reliability · Wireless networks · Coding · Decoding · Turbo codes · LDPC · Critical infrastructure · Smart Grid · Parametric and structural adaptation

---

S. Zaitsev (✉) · Y. Ryndych · N. Sokorynska
Chernihiv Polytechnic National University, 95 Shevchenka Str., Chernihiv 14035, Ukraine
e-mail: serza1979@gmail.com

N. Sokorynska
e-mail: sokor-nata@ukr.net

S. Zaitsev
University of Technology (Politechnika Świętokrzyska), Kielce, Poland

S. Zaitsev · L. Zaitseva · P. Kurbet · B. Horlynskyi
Institute of Telecommunications and Global Information Space of the National Academy of Sciences of Ukraine, Kyiv, Ukraine
e-mail: lili5990n@ukr.net

S. Shkarlet et al. (eds.), *Mathematical Modeling and Simulation of Systems*, Lecture Notes in Networks and Systems 667,
https://doi.org/10.1007/978-3-031-30251-0_17

1 Relevance

Currently, there is a trend of rapid development of wireless technologies.

Last years the next generations of networks are developed and implemented including security aspects [1] and new approaches in coding. The main advantages of the fifth generation of wireless communication technology are [2, 3]:

- low signal delay;
- increased bandwidth;
- increased user mobility;
- higher data transfer speed (peak speed of 20 Gbit/s);
- increased transmission speed.

These advantages allow using such wireless networks in systems which have special requests for quality. Implementing such technologies in systems of critical infrastructure will provide a higher quality of the system in common. Especially in energy systems and other 24/7 systems which transfer control signals. To transfer correctly and fast these signals are very important in extraordinary situations like emergencies, natural disasters, and damage to infrastructural objects during hostilities. Also, it is important to telemedicine and other real-time systems. Because modern distributed infrastructure, like Smart Grid, are always include wireless channels which are often functioning in difficult conditions (Fig. 1). In such systems, wireless networks take a central part of the data transmission mechanism: information and control at the same time.

In this regard, one of the main tasks is to evaluate the channel, increase the reliability of information transmission and use parametric adaptation [3].

Using interference-resistant codes allows to increase reliability and quality. This effect can be achieved by using such codes like LDPC codes [4, 5] and turbo codes (TC) [5–7].

TC and LDPC codes are adopted by the fifth-generation mobile communication standards 4G LTE and 5G, respectively.

4G and 5G systems use adaptive modulation, power, and coding techniques. The main point of our research is the adaptation of coding. In modern standards oding rate R is adjusted in the range from 1/5 to 2/3. At the same time, the use of TC is more expedient at low coding rates, and LDPC codes—at high ones.

The high efficiency of turbo codes is due to the iterative decoding algorithms developed for them. Decoding algorithms developed for turbo codes use «soft» solutions at the input and output of the decoder. In this connection, they received the name of algorithms with «soft» input—«soft» output SISO (soft input—soft output). These algorithms include the Viterbi algorithm with a «soft» output SOVA (soft output Viterbi algorithm), the decoding algorithm based on the maximum a posteriori probability MAP (maximum a posteriori probabilities) or, as mentioned in some sources, the BCJR algorithm (Bahl-Cocke-Jelinek-Raviv), as well as less complex Max Log MAP and Log MAP algorithms [7, 8].

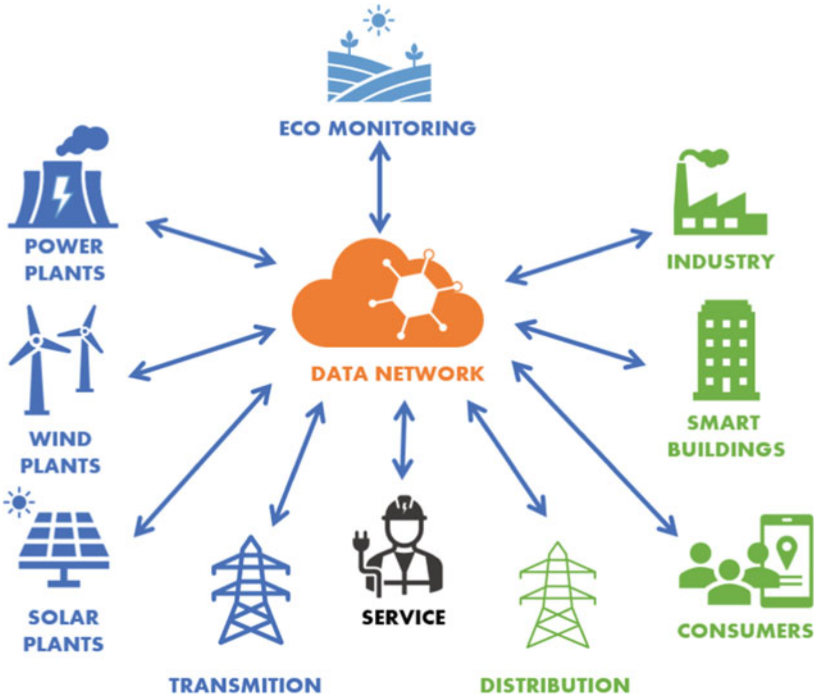


Fig. 1 Smart electrical grids

According to the 3rd Generation Partnership Project (3GPP) TS 38.212, LDPC is recommended for the fifth-generation due to its high throughput, low latency, low decoding complexity and rate compatibility.

2 Analysis of Studies and Publications

According to [8], a method presented there solves two problems, namely, estimates of the logarithmic ratio of the likelihood function and quantization. This method is focused on high-performance computing units with low latency, achieved using deep neural networks.

The paper [9] presents the development of a turbo receiver based on the Bilinear Generalized Approximate Message Transfer (BiG-AMP) algorithm. In this turbo receiver, all received symbols are used to estimate the channel state, user activity, and program data symbols, which effectively exploit the common sparsity pattern. The extrinsic information from the channel decoder is used for joint channel estimation and data discovery.

In [10] proposes the use of a compression sounding (CS) channel estimator in a system using orthogonal frequency division multiplexing (OFDM) and software defined radio (SDR) devices. The application of compression sounding theory is enabled by using sparse reconstruction algorithms such as orthogonal match search (OMP) and compression sample match search (CoSaMP) to take advantage of the sparse nature of the pilot subcarriers used in OFDM, optimizing system throughput.

Paper [11] proposes a new method for iterative channel estimation and data decoding. In the proposed method, the probability of occurrence of transmitted symbols is shifted. The a priori information about the offset is used for the initial channel estimation. The proposed scheme is based on the parallel concatenation of two shifted convolutional codes, which are constructed as systematic recursive convolutional codes with state-dependent puncturing.

Paper [12] presents an iterative receiver for a channel with phase-coherent block fading. The receiver jointly estimates the channel and decodes a low density parity check (LDPC) code using a sum product algorithm.

3 Formulation of the Problem

The purpose of the article is to develop an adaptive method for assessing the reliability of information under conditions of uncertainty through the use of a priori and a posteriori information of the decoder. The method makes it possible to adapt to changing the parameters of the encoder and decoder of the turbo code (LDPC code) by using the logarithmic ratios of the likelihood functions (LLRs) and the calculated values of the noise dispersion.

4 Presentation of the Main Material

On Fig. 2 shows a block diagram of a two-component TC encoder.

The TC encoder consists of a cascaded construction of recursive systematic convolutional codes (RSCC) connected in parallel, separated by an interleaver (I).

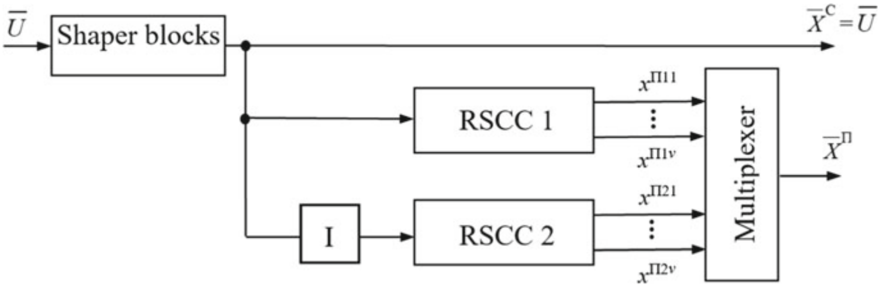


Fig. 2 Structural diagram of the encoder TC

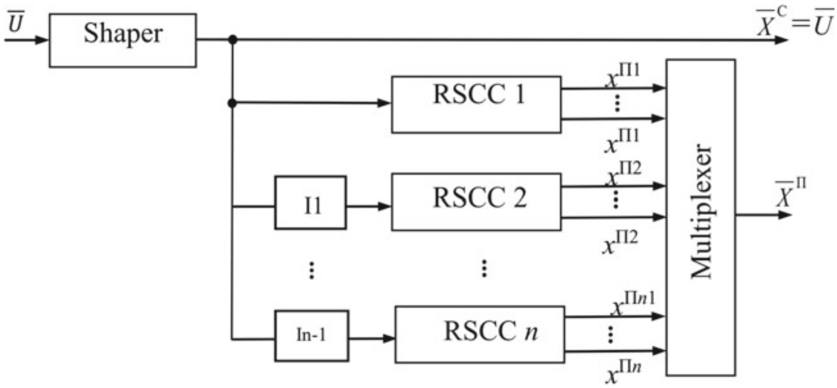


Fig. 3 Structural diagram of the TC multi-component encoder

In Fig. 3 shows the structural diagram of the multi-component TC encoder.

At the moment of time t, an information bit u_t, $t \in \overline{1, N}$ of a block of size N, is received at the RSSC input. The RSSC of the turbo code, depending on the value of the input bit, forms systematic c_t^C and check bits c_t^Π, $t \in \overline{1, N}$, $c_t^C, c_t^\Pi \in (0, 1)$. To implement the procedure of phase modulation of the PSK-2 signal, systematic c_t^C and check bits c_t^Π are converted into systematic x_t^C and check symbols, x_t^Π, $t \in \overline{1, N}$, $x_t^C, x_t^\Pi \in (-1, 1)$. The code word of the turbo code is formed by the parallel connection of two RSSCs separated by an interleaver. As a result of turbo coding, each systematic bit c_t^C will correspond to two check bits $c_t^{\Pi 1}, c_t^{\Pi 2}$, which are then converted into symbols $x_t^C, x_t^{\Pi 1}, x_t^{\Pi 2} \in (-1, 1)$.

The effective representation of the "soft" solution or the logarithmic ratio of the likelihood functions (LLR) outside the decoder is defined by the expression [7]:

$$L(x_t|y_t) = \ln \frac{P(y_t|x_t = +1)}{P(y_t|x_t = -1)} + \ln \frac{P(x_t = +1)}{P(x_t = -1)} = L_a(x_t) + L(y_t|x_t),$$

where $L(y_t|x_t)$ is the LLR y_t, which is obtained by measuring y_t at the output of the channel during alternating conditions, which can be transmitted $x_t = +1$ or $x_t = -1$, and $L_a(x_t)$ is the a priori LLR of the data bit x_t. To simplify notation, equation can be rewritten as follows [7]:

$$L'(x_t) = L_c(y_t) + L_a(x_t).$$

Here $L_c(y_t)$ it means that the LLR member is obtained as a result of channel measurements made in the receiver. For systematic codes, the LLR at the output of the decoder is equal to the following [7]:

$$L(x_t) = L'(x_t) + L_e(x_t).$$

In this expression, $L'(x_t)$ is the LLR outside the demodulator (at the decoder input), and $L_e(x_t)$ is the «external» LLR, which represents external information resulting from the decoding process. The output LLR of the decoder will take the form:

$$L(x_t) = L_c(y_t) + L_a(x_t) + L_e(x_t).$$

The sign $L(x_t)$ is a firm decision about the symbol x_t, and the modulus $|L(x_t)|$ is the degree of reliability (plausibility) of this decision.

Decoder 1, in accordance with its algorithm, produces «soft» decisions about decoded symbols (output LLR), which consist of three parts [7]:

$$L^1\left(x_t^C\right) = L_c \cdot y_t^{C1} + L_a^1\left(x_t^C\right) + L_e^1\left(x_t^C\right),$$

where x_t^C is the systematic symbol of the TC encoder.

At the same time, the "external" information of decoder 1 about the symbol x_t^C, which is a priori for decoder 2 (taking into account the interleaving operation), will take the form [7]:

$$L_e^1\left(x_t^C\right) = L_a^2\left(x_t^C\right) = L^1\left(x_t^C\right) - L_a^1\left(x_t^C\right) - L_c \cdot y_t^{C1}.$$

The second elementary decoder, having received a priori information about the information symbols, makes similar calculations, determining its «external» information about the symbol x_t^C [7]:

$$L_e^2\left(x_t^C\right) = L_a^1\left(x_t^C\right) = L^2\left(x_t^C\right) - L_a^2\left(x_t^C\right) - L_c \cdot y_t^{C2},$$

which enters the decoder 1 input of the next decoding iteration.

After performing the necessary number of iterations or in the case of a forced stop of the iterative decoding procedure, decisions are made about the decoded symbols:

$$x_t^C = \begin{cases} 1, & \text{if } L\left(x_t^C\right) \geq 0 \\ 0, & \text{if } L\left(x_t^C\right) < 0 \end{cases}.$$

As is known, the decoding of TC symbols takes place according to the diagram of the corresponding RSCC, while the transition recursion, direct recursion, reverse recursion, LLR at the output of the decoder and the parameter of «external» information are calculated [7]. Let's consider the features of calculating the output LLR for decoder 2, using the Map decoding algorithm.

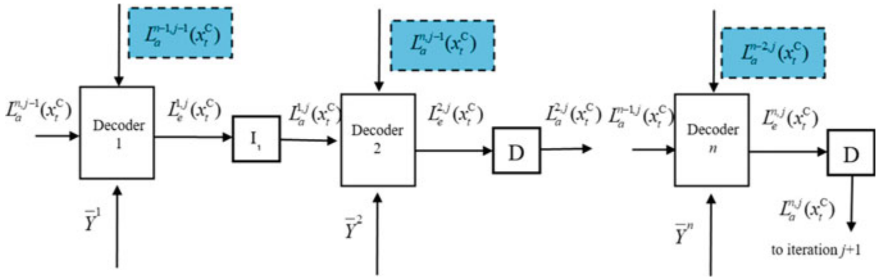


Fig. 4 Structural diagram of a multi-component TC

The structural diagram of the three-component TC decoder model is shown in Fig. 4.

As in the case of two-component TC, three-component decoders work in series. A feature of the decoding of a three-component TC, in contrast to a two-component one, is that the a priori information for the component encoder is formed as the sum of not two, but three components: the channel reading of the systematic bit, as well as the values of the LLR obtained by the two previous component decoders (if necessary, with previous iteration, including interleaving (I)/deinterleaving (D) procedures).

The first decoder, using the "output" LLR, a priori LLR from the second and third decoders of the previous iteration and information from the channel, determines the "external" information about the symbol x_t^C:

$$L_e^{3,j}\left(x_t^C\right) = L^{3,j}\left(x_t^C\right) - L_a^{2,j}\left(x_t^C\right) - L_a^{1,j}\left(x_t^C\right) - L_c \cdot y_t^{C2}.$$

The second decoder uses the "source" LLR, the a priori LLR from the third decoder of the previous iteration and the a priori LLR from the first decoder of the current iteration, as well as information from the communication channel, to determine the "external" information about the symbol:

$$L_e^{2,j}\left(x_t^C\right) = L^{2,j}\left(x_t^C\right) - L_a^{3,j-1}\left(x_t^C\right) - L_a^{1,j}\left(x_t^C\right) - L_c \cdot y_t^{C2}.$$

The third elementary decoder, having received a priori information about the information symbols from the first and second decoders, as well as using the original LLR and the information received from the channel, determines its "external" information about the symbol x_t^C:

$$L_e^{3,j}\left(x_t^C\right) = L^{3,j}\left(x_t^C\right) - L_a^{2,j}\left(x_t^C\right) - L_a^{1,j}\left(x_t^C\right) - L_c \cdot y_t^{C2}.$$

There are three events about decision-making during decoding by the decoder d,

$d \in \overline{1,2}$, iterations of decoding j, $j \in \overline{1,I}$, bits of information:

1) Event A_1.

Changed sign in values $L_a^{d,j}\left(x_t^C\right)$ and $L_e^{d,j}\left(x_t^C\right)$ iteration j does not occur $\left(\text{sign}\left(L_a^{d,j}\left(x_t^C\right)\right) = \text{sign}\left(L_e^{d,j}\left(x_t^C\right)\right)\right)$, $L\left(x_t^C\right) \geq 0$. They will make a "firm" decision that the bit $x_t^C = +1$ was passed.

2) Event A_2.

Changed sign in values $L_a^{d,j}\left(x_t^C\right)$ and $L_e^{d,j}\left(x_t^C\right)$ iteration j does not occur $\left(\text{sign}\left(L_a^{d,j}\left(x_t^C\right)\right) = \text{sign}\left(L_e^{d,j}\left(x_t^C\right)\right)\right)$, $L\left(x_t^C\right) < 0$. They will make a "firm" decision that the bit $x_t^C = -1$ was passed.

3) Event A_3.

The sign of the value of the a priori $L_a^{d,j}\left(x_t^C\right)$ and the sign of the value of the a posteriori information $L_e^{d,j}\left(x_t^C\right)$ of iterations j is not equal to zero $\left(\text{sign}\left(L_a^{d,j}\left(x_t^C\right)\right) \neq \text{sign}\left(L_e^{d,j}\left(x_t^C\right)\right)\right)$. Decoding errors may occur.

The uncertainty index (error rate) for a two-component decoder d, $d \in \overline{1,2}$ decoding iteration j, $j \in \overline{1,I}$ is calculated using the following procedure:

$$\tilde{U}^{d,j} = \Sigma_{d=1}^{2} R^{d,j}(t+1) = R^{d,j}(t) + 1, \text{ if } \text{sign}\left(\mathrm{L}_{\mathrm{a}}^{\mathrm{d,j}}\left(\mathrm{x}_{\mathrm{t}}^{\mathrm{C}}\right)\right) \neq \text{sign}\left(\mathrm{L}_{\mathrm{e}}^{\mathrm{d,j}}\left(\mathrm{x}_{\mathrm{t}}^{\mathrm{C}}\right)\right), \mathrm{t} \in \overline{1,\mathrm{N}}.$$

In the case of applying adaptation and reconstruction of the decoder from two-component to multi-component, the uncertainty index for the decoder d, $d \in \overline{1,n}$, decoding iteration j, $j \in \overline{1,I}$, is calculated using the following procedure:

$$\tilde{U}^{n,j} = \Sigma_{d=1}^{n} R^{d,j}(t+1) = R^{d,j}(t) + 1, \text{ if } \text{sign}\left(\mathrm{L}_{\mathrm{a}}^{\mathrm{d,j}}\left(\mathrm{x}_{\mathrm{t}}^{\mathrm{C}}\right)\right) \neq \text{sign}\left(\mathrm{L}_{\mathrm{e}}^{\mathrm{d,j}}\left(\mathrm{x}_{\mathrm{t}}^{\mathrm{C}}\right)\right), \ \mathrm{t} \in \overline{1,\mathrm{N}}.$$

The more often the values of the uncertainty index R increase, the more often incorrectly decoded bits appear, which leads to a deterioration in the reliability of information reception.

The total uncertainty index R_Σ is determined by the sum of the uncertainty indexes for all decoding iterations:

$$R_\Sigma = \Sigma_{j=1}^{I} R^{d,j}.$$

For the convenience of calculations and adaptation, we will normalize the uncertainty index:

$$\tilde{R}_\Sigma = \frac{R_\Sigma}{B \cdot \tilde{N} \cdot I} = \frac{\sum_{j=1}^{I} R^{d,j}}{B \cdot \tilde{N} \cdot I},$$

where B – is the number of data blocks of some observation window, $\tilde{N}$ – is the variable size of the data block, I – is the number of turbo code decoding iterations.

When calculating the channel reliability parameter for the LLR, information about the value of the noise dispersion in the channel is used. We obtain analytical expressions for calculating the noise variance for a multicomponent decoder. This information will be used to improve the accuracy of calculating the decoding uncertainty index (error rate).

Let L_e – be a random variable, the values of which are the results of decoding by the i-th decoder, namely: calculation of LLR about transmitted bits in n-blocks of length N: $L_e^i\left(x_{kt}^C\right)$, $t \in \overline{1, N}$, $k \in \overline{1, n}$. The mathematical expectation and variance of the random variable L are defined by the following expressions:

$$M_{L_e} = \frac{\sum_{k=1}^{n}\sum_{t=1}^{N} L_e^i\left(x_{kt}^C\right)}{nN}, \; D_{L_e} = \frac{\sum_{k=1}^{n}\sum_{t=1}^{N}\left(L_e^i\left(x_{kt}^C\right) - M_{L_e}\right)^2}{(n-1)(N-1)}.$$

In this case, with a two-component decoder, the interference variance for the n-th iteration of decoding of each channel of the OFDM system, taking into account the selected decoding algorithm, will be determined as follows (for the n-th iteration of decoding):

$$\hat{\sigma}_{n1}^2 = \frac{1}{N_1}\Sigma_{t=0}^{N_1-1}\left(\left(L^{1,n}\left(x_{1t}^{\mathrm{C}}\right) + L^{2,n}\left(x_{1t}^{\mathrm{C}}\right)\right) - \hat{y}_{1t}\right)^2, \ldots,$$
$$\hat{\sigma}_{nv}^2 = \frac{1}{N_v}\Sigma_{v=0}^{N_v-1}\left(\left(L^{1,n}\left(x_{vt}^{\mathrm{C}}\right) + L^{2,n}\left(x_{vt}^{\mathrm{C}}\right)\right) - \hat{y}_{vt}\right)^2.$$

With three-component decoding:

$$\hat{\sigma}_{n1}^2 = \frac{1}{N_1}\Sigma_{t=0}^{N_1-1}\left(\left(L^{1,n}\left(x_{1t}^{\mathrm{C}}\right) + L^{2,n}\left(x_{1t}^{\mathrm{C}}\right) + L^{3,n}\left(x_{1t}^{\mathrm{C}}\right)\right) - \hat{y}_{1t}\right)^2, \ldots,$$
$$\hat{\sigma}_{nv}^2 = \frac{1}{N_v}\Sigma_{v=0}^{N_v-1}\left(\left(L^{1,n}\left(x_{vt}^{\mathrm{C}}\right) + L^{2,n}\left(x_{vt}^{\mathrm{C}}\right) + L^{3,n}\left(x_{vt}^{\mathrm{C}}\right)\right) - \hat{y}_{vt}\right)^2,$$

Accordingly, with four-component decoding:

$$\hat{\sigma}_{n1}^2 = \frac{1}{N_1}\Sigma_{t=0}^{N_1-1}\left(\left(L^{1,n}\left(x_{kt}^{\mathrm{C}}\right) + L^{2,n}\left(x_{kt}^{\mathrm{C}}\right) + L^{3,n}\left(x_{kt}^{\mathrm{C}}\right) + L^{4,n}\left(x_{kt}^{\mathrm{C}}\right)\right) - \hat{y}_{1t}\right)^2, \ldots,$$
$$\hat{\sigma}_{nv}^2 = \frac{1}{N_v}\Sigma_{v=0}^{N_v-1}\left(\left(L^{1,n}\left(x_{kt}^{\mathrm{C}}\right) + L^{2,n}\left(x_{kt}^{\mathrm{C}}\right) + L^{3,n}\left(x_{kt}^{\mathrm{C}}\right) + L^{4,n}\left(x_{kt}^{\mathrm{C}}\right)\right) - \hat{y}_{vt}\right)^2,$$

where $\hat{y}_{1t}, \hat{y}_{vt}$ – are the estimated transmitted symbols for the 1st iteration of decoding the first and v-th channels, respectively; $\hat{y}_{1t} = 1$, if $L_c \cdot y_{1t}^C > 0$ and $\hat{y}_{1t} = -1$, if $L_c \cdot y_{1t}^C < 0$, respectively $\hat{y}_{vt} = 1$, if $L_c \cdot y_{vt}^C > 0$ and $\hat{y}_{vt} = -1$, if $L_c \cdot y_{vt}^C < 0$.

And so on, depending on the number of component encoders (decoders).

A block diagram of a two-component turbo code decoder with a decision block, which contains modules for calculating uncertainty indicators $R^{d,j}$, R_Σ, $\tilde{R}$, is shown in Fig. 5.

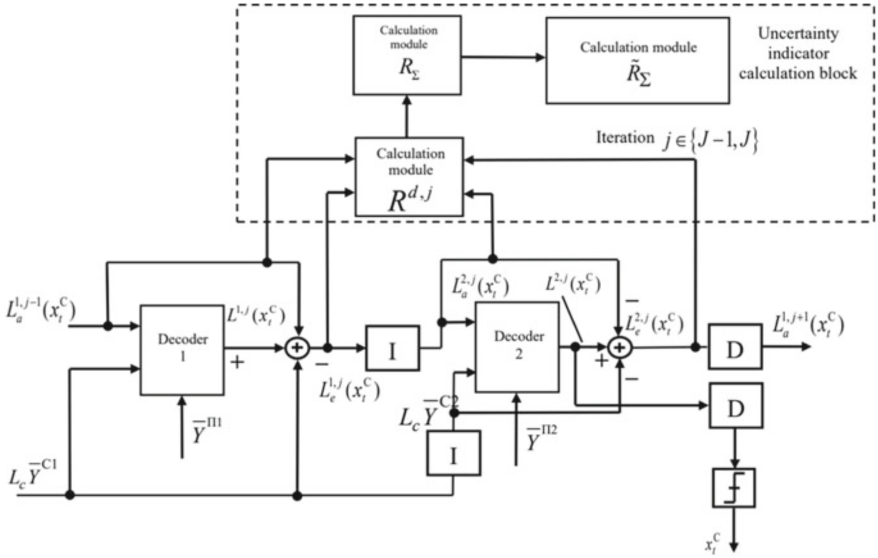


Fig. 5 Block diagram of a TC decoder with a decoding uncertainty evaluation module

5 Analysis of the Results

Simulation modeling was used to analyze the effectiveness of the method. For comparison, the fourth-generation mobile communication standard LTE-Advanced was chosen. The simulation was carried out in the Visual Studio 2019 environment. A data transmission system was simulated with turbo codes, an OFDM modulator (demodulator), a channel with additive white Gaussian noise, modules for calculating the decoding error probability (the transmitted sequence was compared with the transmitted one - imitation of the service channel through which test information is transmitted to assess the reliability of information) and the indicator of decoding uncertainty (error rate). The values of the decoding uncertainty index (error rate) were calculated only on the basis of the decoding results. The simulation results were obtained based on the reliability $= 0, 95$, $t = 0, 95$ (Laplace function argument), relative accuracy $d = 0.1$.

Turbo code was used with generators (1, 23/21), Log Map decoding algorithm, redundancy $R = 1/3$, pseudo-random and regular interleaver (de-interleaver), number of bits in the block $N = 400$, 1000. The signal-to-noise ratio changed from 0 up to 1.6 dB.

In Fig. 6 shows the graphs of the dependence of the probability of a bit error $P_{B\ \mathrm{dec}}$ and the uncertainty index (error coefficient) on the signal-to-noise ratio E_b/N_J, calculated by the standard method of simulation modeling, using the proposed method at 8 iterations of turbocode decoding, compared to the known approximate calculation method.

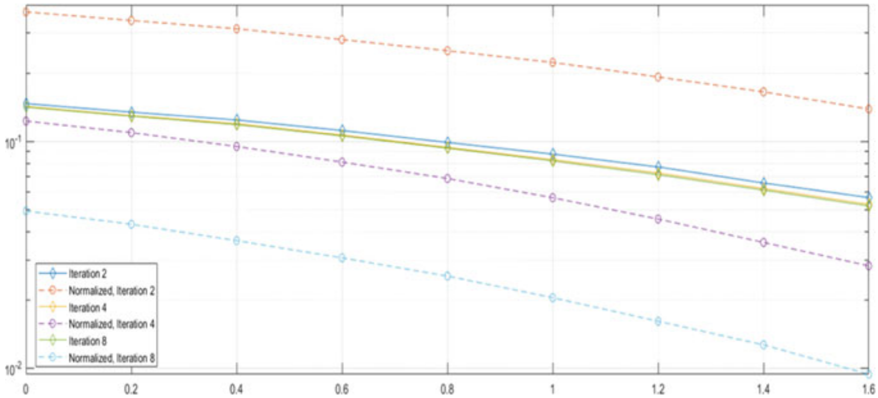


Fig. 6 Graph of the dependence of the average probability of a bit error and the decoding uncertainty index (error rate) on the signal-to-noise ratio in the channel

The analysis shows that the proposed method provides greater accuracy in assessing the reliability of information in comparison with the closest analogue.

On Fig. 7, 8 shows graphical dependences of the average probability of a decoding bit error $P_{B\ \mathrm{dec}}$ and the decoding uncertainty index (error rate) on the signal-to-noise ratio E_b/N_J, where E_b is the bit energy and $N_J = \sigma^2/2$ is the spectral density.

Analysis of the simulation results shown in Fig. 7, 8 shows that as the data block size increases from $N = 400$ to 1000, the decoding uncertainty index (error rate) curve approaches the decoding error probability curve. For example, for $N = 1000$, 8 decoding iterations with a signal-to-noise ratio of 1.4 dB, the value of the decoding error probability is $5 \cdot 10^{-5}$, and the value of the decoding uncertainty index (error rate) is $9 \cdot 10^{-5}$. For N = 1000, 4 decoding iterations with a signal-to-noise ratio of

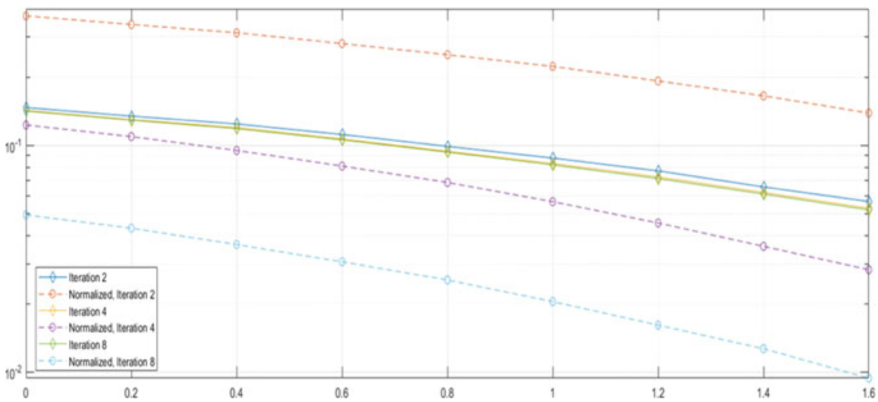


Fig. 7 Graph of the dependence of the average probability of a bit error and the decoding uncertainty index (error rate) on the signal-to-noise ratio in the channel for $N = 400$ and various decoding iterations

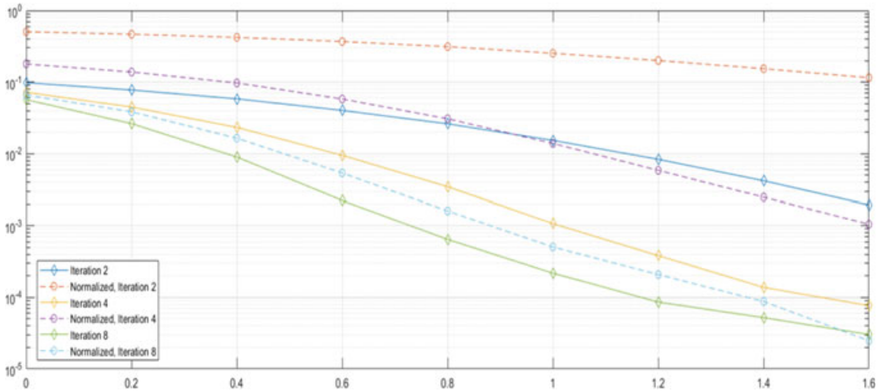


Fig. 8 Graph of the dependence of the average probability of a bit error and the decoding uncertainty index (error rate) on the signal-to-noise ratio in the channel for $N = 1000$ and various decoding iterations

1.4 dB, the value of the decoding error probability is $1, 5 \cdot 10^{-4}$, and the value of the decoding uncertainty index (error rate) is $2, 5 \cdot 10^{-3}$.

An analysis of these graphical dependencies shows that with an increase in decoding iterations, the accuracy of estimating the reliability of information increases (the decoding uncertainty (error rate) curves approach the decoding error probability modelling curves).

The degree of similarity between decoding uncertainty indicators and decoding results will be estimated using the correlation function. The following figure shows graphical dependences of the correlation coefficient on the signal-to-noise ratio in the channel for $N = 1000$ and various decoding iterations (Fig. 9).

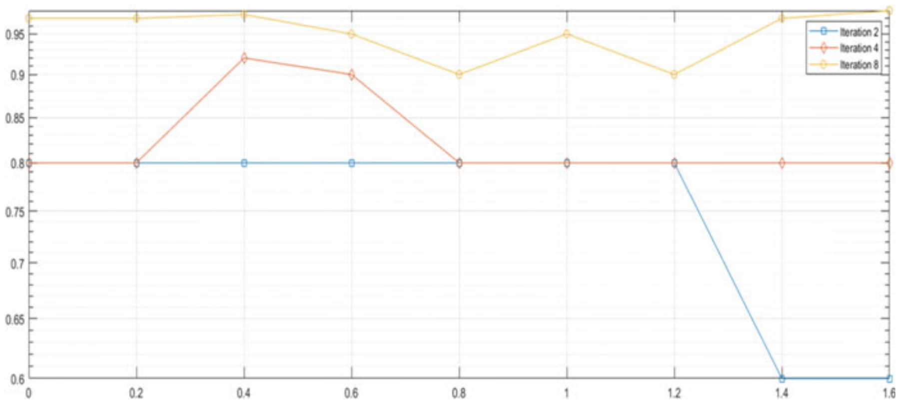


Fig. 9 Graph of the dependence of the correlation coefficient on the signal-to-noise ratio in the channel for $N = 1000$ and various decoding iterations

The analysis shows that with an increase in decoding iterations, the accuracy of assessing the reliability of information increases, so for 8 decoding iterations, the values of the correlation coefficient change from 90 to 98%, for 4 iterations – from 80 to 92%, for 2 decoding iterations – from 60 to 80%.

6 Conclusions

The result of the research includes *n* adaptive method for assessing the reliability of information under conditions of uncertainty through the use of a priori and a posteriori information of the decoder, which allows adapting to changing the parameters of the encoder and decoder of the turbo code (LDPC code) through the use of LRR and the calculated values of the noise dispersion. The using of the sign change of a priori and a posteriori LRR during iterative decoding and taking into account the noise dispersion values in the channel reliability parameter, the method allows obtaining information reliability values (error coefficient) without using an additional service channel. Simulations let to understand that the accuracy of the information approaches the reliability estimate when using an additional service channel. This method can be used in conjunction with other methods of parametric and structural adaptation under conditions of a priori uncertainty.

References

1. Bae, J.H., Abotabl, A., Lin, H.P., Song, K.B., Lee, J.: An overview of channel coding for 5G NR cellular communications. APSIPA Trans. Actions Sig. Inf. Process. 1–14 (2019).https://doi.org/10.1017/ATSIP.2019.10.
2. Shafi, M., et al.: 5G: a tutorial overview of standards, trials, challenges, deployment and practice. IEEE J. Sel. Areas Commun **35**(6), 1201–1221 (2017). https://doi.org/10.1109/JSAC.2017.2692307.
3. Adebusola, J.A., Ariyo, A.A., Elisha, O.A., Olubunmi, A.M., Julius, O.O.: An overview of 5G technology. In: 2020 International Conference in Mathematics, Computer Engineering and Computer Science (ICMCECS), pp. 1–4 (2020). https://doi.org/10.1109/ICMCECS47690.2020.240853.
4. MacKay, D.J., Neal, R.M.: Near Shannon limit performance of low density parity check codes. Electron. Lett. **32**(18), 457–458 (1996)
5. Arora K., Singh J., Randhawa Y.S.: A survey on channel coding techniques for 5G wireless networks. Telecommun. Syst. **73,** 637–663 (2020)
6. Kazymyr, V., Zaitsev, S., Ryndych, Y., Zarovskyi, R.: Technology for improve cyber security using turbo codes. In: 2015 Information Technologies in Innovation Business Conference, ITIB 2015 - Proceedings, pp. 85–88, (2015). https://doi.org/10.1109/ITIB.2015.7355060
7. Berrou, C. Codes and Turbo Codes, p. 415. Springer, Cham (2010)
8. Arvinte M., Tewfik A.H., Vishwanath S.: EQ-Net: joint deep learning-based log-likelihood ratio estimation and quantization. https://arxiv.org/abs/2012.12843v2
9. Bian, X., Mao, Y., Zhang, J.: Joint activity detection and data decoding in massive random access via a turbo receiver. In: 2021 IEEE 22nd International Workshop on Signal Processing

Advances in Wireless Communications (SPAWC), .pp. 361–365 (2021). https://doi.org/10.1109/SPAWC51858.2021.9593149.
10. Yanza-Verdugo, A., Pucha-Cabrera, C., Ortega, J.I.: Compressive sensing based channel estimator and LDPC theory for OFDM using SDR. Ingenius, no.23, pp.74–85 (2020).https://doi.org/10.17163/ings.n23.2020.07.
11. akeuchi, K., Muller, R.R., Vehkapera, M.: A construction of turbo-like codes for iterative channel estimation based on probabilistic bias. In: 2011 IEEE Global Telecommunications Conference – GLOBECOM 2011, pp. 1–5 (2011). https://doi.org/10.1109/GLOCOM.2011.6133738.
12. Jin, X., Eckford, A.W., Fuja, T.E.: Analysis of joint channel estimation and LDPC decoding on block fading channels. In: International Symposium on Information Theory and its Applications, ISITA2004, pp. 679 – 684, Parma, Italy, 10–13 October 2004

Study of the Workspace Model in Distributed Structures Using CAP Theorem

Oleksandr Khoshaba, Viktor Grechaninov, Tetiana Molodetska, Anatoliy Lopushanskyi, and Kostiantyn Zavertailo

Abstract The article proposes a workspace representation model for real-time systems in distributed structures using the CAP theorem. This model reflects the principles of operation of real-time systems in distributed systems, where it becomes possible to determine the size of the workspace. The workspace representation model has a feature that consists of choosing some compromise solutions between two boundaries. This allows developers to design a real-time system, taking into account the "margin of safety" (or reliability) of the operation of a distributed structure where property P will be observed (i.e., obtaining undistorted answers). The practical implementation of the object of study is based on the algorithm for creating a software environment, divided into two stages: creating and deploying a Docker image for a distributed structure. The development of a multi-stage project implementation made it possible to create research stages using real-time systems in distributed structures. The article also explores issues and problems associated with using real-time systems in distributed structures, where the properties of the CAP and PACELC theorems are mainly considered. A literature review was conducted using the CAP and PACELC theorems. A demo that violates Consistency properties (CAP or PACELC theorems) in a distributed structure is shown. Some modern solutions to the problem of using real-time systems in distributed structures have been provided.

Keywords real-time systems · distributed structures · theorem CAP · theorem PACELC · Docker Image · Kubernetes

O. Khoshaba (✉) · V. Grechaninov · A. Lopushanskyi · K. Zavertailo
Institute of Mathematical Machines and Systems Problems of the Ukraine National, Academy of Science, 42 Academician Glushkov Avenue, Kyiv 03187, Ukraine
e-mail: Oleksandr.Khoshaba@gmail.com
URL: http://khoshaba.vk.vntu.edu.ua/

O. Khoshaba · T. Molodetska
Vinnytsia National Technical University, 95 Khmelnytske shose, Vinnytsia 21021, Ukraine

S. Shkarlet et al. (eds.), *Mathematical Modeling and Simulation of Systems*, Lecture Notes in Networks and Systems 667,
https://doi.org/10.1007/978-3-031-30251-0_18

1 Introduction

There are compromise solutions in research on modern distributed systems based on models and theorems. One of the most famous in distributed structures is the CAP [2, 3] theorem. However, there are many criticisms of the CAP theorem. One of them includes doubts about its legitimacy and explainability for working in distributed structures. Other remarks are devoted to choosing the most effective possible alternatives, which allow it to be corrected or supplemented with some provisions.

Besides, in 2012, Eric Brewer wrote a follow-up paper entitled "CAP Twelve Years Later: How the' Rules' Have Changed" [3]. In it, Brewer explained the misuse of the CAP theorem by many designers and researchers over the years. The typical CAP definition states that "a shared-data system can't have all 3". This definition has stirred misunderstandings about the CAP theorem and its meaning because it implies that some systems are unavailable or inconsistent 100% of the time.

Brewer explains in [3] that the easiest way to understand CAP is to think of two nodes on opposite sides of a partition. For example:

– scenario 1: Allowing at least one node to update its state will cause the nodes to become inconsistent, thus forfeiting Consistency;
– scenario 2: If the choice is to preserve Consistency, one side of the partition must act as if it is unavailable, thus forfeiting Availability.

Let us consider in more detail the disadvantages of the CAP theorem.

So, there are quite a few criticisms of the CAP theorem. For its seemingly clear concept of triple constraint, the CAP theorem is criticized for oversimplifying important concepts in models for distributed structures, leading to a misunderstanding of its original meaning.

The choices between consistency and availability in distributed structures may change depending on some conditions (system requirements in nodes) or the user's specific data (for applications or databases). Moreover, these choices can often happen within the same distributed structure on which real-time systems operate. For example, such a choice is only made when partitioning or failure occurs in the structures. At other times, no tradeoffs for creating the distributed structures are required.

Furthermore, the CAP theorem ignores time delays (latency) in its original definition, although latency and availability are deeply connected in practice. In practice, the CAP nature occurs during a deadline (in real-time systems), the period when the system must make a fundamental decision.

In general, the CAP theorem can be perceived as some abstract philosophy, which indicates that there is an object that has contradictory properties. These properties do not have to be three. However, they can be two or more than three. Consider examples of using the models for distributed structures in scientific research.

2 Related Works

2.1 Models of Distributed Structures

The first version of the CAP principle appeared as $ACID$ vs. $BASE$ [9, 10, 13]. But then she changed her definition and got proof that turned it into a theorem. Thus, the works [9, 10] attract attention, where scientists studied the need for real-time accessibility restrictions for processing web transactions. Also, two models, $ACID$ (Atomicity, Consistency, Isolation, and Durability) and $BASE$ (Basically Available, Soft state, Eventual consistency), were analyzed in the same place, where scientists carried out their comparative characteristics.

The CAP theorem states that a distributed system can simultaneously have at most two of the three properties, such as.

$Consistency$ means that the data in all distributed nodes do not contradict each other at each time. The consistency here is very different from the consistency guaranteed in model $ACID$ transactions. In formal proof, this property is called atomicity. It is expressed in the presence of a standard order of operations in a distributed system. This order of operations is similar to those that exist in a non-distributed system. For example, each sequence that contains a read operation must return the value set in the last sequence of the write operation. Essentially, this is linearizability.

$Availability$ means that every query received by a healthy node entails an answer. Author Eric Brewer [2] originally put forward a softer requirement: "almost all queries must get answers," but the scientist used a strict version of this requirement to prove the theorem.

In practice, very important, a time constraint should be added to this definition since lack of it can lead to a result equivalent to no answer.

$Partition$ tolerance means that splitting a distributed system into several isolated sections does not lead to an incorrect response from each section.

In practice, those messages whose delivery has exceeded a specific time limit are also considered lost. This feature is a consequence of the requirements for the functioning of a distributed system.

At the same time, forfeiting partition tolerance is not feasible in realistic environments since we will always have network partitions. Thus, we need to decide between Availability and Consistency, which can be represented by $ACID$ (Consistency) and $BASE$ (Availability). The same problem is choosing between Availability (A) and Consistency (C) when developers build real-time systems in distributed structures.

The CAP theorem postulates that [15] in a "shared-data system," only two of the three CAP characteristics can be achieved simultaneously. For a growing number of applications and use cases (including web applications, especially on large and ultra-large-scale, and even in the e-commerce sector), availability and partition tolerance are more important than strict consistency.

These applications must be reliable, implicating availability and redundancy (consequent distribution among two or more nodes, which is necessary as many systems run on cheap, commoditized, and unreliable machines and provide scalability). These properties are difficult to achieve with ACID properties. Therefore, approaches like BASE are applied. Other properties relating to NoSQL technologies include, among others, sharding (i.e., Horizontal partitioning by some key and storing records on different servers to improve performance), Horizontal scalability (i.e., Distributing both data and load over many servers), and Vertical scaling (i.e., Use of multiple cores and/or CPUs by a DBMS) [15].

The BASE approach, according to Brewer [2] forfeits the ACID properties of consistency and isolation in favor of "availability, graceful degradation, and performance." The acronym BASE comprises the following characteristics: Available, Soft-state, and Eventually consistent.

With regards to database management systems, Brewer [2] concludes that current databases are better at consistency than availability and that wide-area databases can't have both-a notion widely adopted in the NoSQL community and has influenced the design of non-relational data stores. Systems that the BASE properties can characterize include Amazon's Dynamo [11], which is available and partition-tolerant but not strictly consistent, i.e., the writes of one client are not seen immediately after being committed to all readers. Google's BigTable [5] chooses neither ACID nor BASE, but the third CAP alternative is a consistent and available system and cannot fully operate in network partitions [15].

2.2 Implications of the CAP theorem

A consequence of theorems for various models for distributed structures is that only the following three combinations of consistency, accessibility, and partition resistance are possible.

AP. Distributed structures of this type respond to queries, but the data returned may not always be up-to-date, with slower data updates but "always" available. Service DNS or distributed databases DynamoDB and Cassandra are examples of such a model.

CP. Distributed structures of this type always return up-to-date data, but some, or even all, nodes in the system may not respond if partitioned. Such systems give atomic updates but can lead to timeouts. NoSQL databases such as Google BigTable, MongoDB, HBase, and Redis are all systems of this type.

CA. In theory, distributed structures of this type always return up-to-date data when there are no partitions. Because of the last limitation, usually, such systems are used only within one machine. Examples are classical relational databases.

Usually, developers choose between CP and AP because CA is a monolith without partitions. For large-scale systems, designers cannot abandon P and therefore have a difficult choice between C and A.

In CA, node failure means complete unavailability of the service. However, this does not exclude scalability in many cases since we can clone independent monoliths and distribute the load over them.

Also, all three properties are best represented quantitatively rather than in binary terms. At the same time, the partition has some controversial features. These features are associated with disagreements in the very definition and size of the distributed structure.

Now, the requirements of the CAP theorem apply to distributed systems in general, not only to databases. Today, however, there is no need for such a categorical division of systems into $CA/CP/AP$ classes as the CAP theorem provides.

Therefore its application to these distributed structures and real-time systems is not entirely correct. Moreover, such classification may even be considered erroneous since the broad interpretation of CAP does not consider the needs of modern distributed applications and databases.

2.3 The $PACELC$ theorem

Modern solutions to the problem of using real-time systems are based on studying the theory of CAP (and its derivative $PACELC$ [6]) and conducting experimental studies to find compromise (optimal) options for the number of pods in nodes in distributed structures. Such solutions are necessary for building real-time systems.

The $PACELC$ extends and clarifies the CAP theorem, regulating the need to find a compromise between latency and data consistency in distributed application systems.

The $PACELC$ theorem described by Daniel J. Abadi [1] is considered an alternative approach to distributed systems design. It is based on the CAP model, but in addition to consistency, availability, and partition tolerance, it also includes latency and logical exclusion between combinations of these concepts.

Abadi [1] divides distributed system operation into two modes: operation without and partitioning. Accordingly, the design choices for a distributed system are as follows: in the case of partitioning, one must choose between availability and consistency. Otherwise, in the case of regular system operation, one must choose between reduced response time and consistency. Also, properties such as availability, scalability, and elasticity are often used in cloud technologies when building databases [7, 8, 12, 14].

At the same time, the $PACELC$ theorem should be considered a more specific implementation of the CAP theorem, aimed at solving specific issues of using distributed applications or databases. Let us consider examples of the use of the $PACELC$ theorem in scientific research.

The $PACELC$ states that in the case of partition (P), the system tradeoffs between availability and consistency (A and C); else (E), in the absence of partitions, the system tradeoffs between latency (L) and consistency (C).

3 Purpose of Work

The aim of this work is:

- to develop and investigate a workspace representation model for the use of real-time systems in distributed structures;
- design the object of study based on the algorithm for creating a software environment;
- consider some modern solutions to the problem of using real-time systems in distributed structures;
- consider the CAP theorem and its derivative - PACELC;
- to conduct preliminary experimental research in real-time systems in distributed structures.

4 Scientific Novelty

The paper proposes and investigates a workspace model of a real-time system in distributed structures using the CAP theorem and its derivative (PACELC) as an example.

A multi-stage project implementation was proposed based on the research carried out, which made it possible to create stages for studying the workspace model using real-time systems in distributed structures.

5 Problem Formulation

The purpose of the work considered in the article focuses on studying the workspace model, which is used in real-time systems for distributed structures.

For this, it is created:

- a multi-stage project to solve research problems;
- a software environment for experimental research;
- software tools that allowed to performance of experimental studies.

5.1 Input Parameters (data)

Input parameters (data) are:

- a research object, which is a service operating based on $CRUD$ technology;
- a research subject, which is a real-time system;

- research tools, which are applied software and are written in Bash and Python languages to conduct sketch studies to determine restrictions on the use of resources;
- software environment, which is used in the study of the use of real-time systems in distributed structures and consists of the Centos 8 operating system, the distributed Podman, and Kubernetes structure;
- hardware consisting of the Dell server, Poweredge R620, 2 x Intel Xeon E5-2690/3GGC/192 GB of RAM;
- a communication environment consisting of a corporate network and data transfer means at a speed of 1 Gb/s;
- the research site is software and hardware for research located at the National Technical University, Vinnytsia, Ukraine.

5.2 *Description of the Process of Using Models for Real-Time Systems in Distributed Structures*

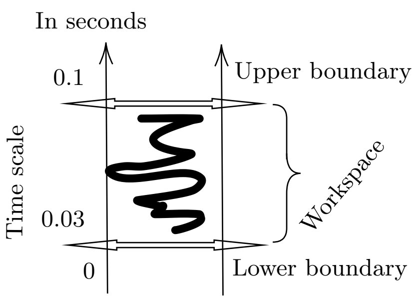
Today's most urgent problem is using models for real-time systems in distributed structures. This problem's importance lies in ensuring real-time systems' reliability. Reliability, in this case, corresponds to property P from the CAP or $PACELC$ theorem discussed above.

At the same time, there is a need to ensure the accuracy and consistency of the data that constitute property C of the CAP or $PACELC$ theorem. At the same time, it becomes necessary to consider and conduct experimental studies using the property L (latency) from the $PACELC$ theorem.

However, the operation of real-time systems in distributed structures has a feature, which consists of a choice of some compromise solutions between two boundaries (Fig. 1).

The figure (Fig. 1) shows a workspace representation model for using real-time systems in distributed structures. This model reflects the principles of operation of real-time systems in distributed systems, where it becomes possible to determine the size of the workspace. This allows developers to design a real-time system, taking

Fig. 1 The workspace representation model



into account the "margin" (or reliability) of the distributed structure, where property P will be observed (i.e., obtaining undistorted answers).

Such a choice of compromise solutions, on the one hand, is based on obtaining a latent period. On the other hand, in a distributed structure, developers should ensure that accurate (consistent) answers are obtained according to the CAP theory from objects to subjects of research.

The values of the latent period, which correspond to the lower limits, are obtained in the course of experimental studies and should be located in the workspace (Fig. 1).

The upper boundary of the workspace corresponds to the period of the real-time system, which corresponds to the deadline. Most often, such a period is 0.1 s.

Thus, based on the peculiarities of the two boundaries (Fig. 1), a compromise appears in distributed systems in choosing the optimal number of pods in nodes. On the one hand, this feature is provided by the property P from the PACELC theorem.

On the other hand, it is necessary to ensure the property of L from the PACELC theorem, where we should preserve property C. This requirement is associated with the operation of real-time systems.

Therefore, some size of the working area remains, which we must determine during experimental studies. Also, the volume of resources in the distributed structure may likely be such that the working area may not exist. This feature is because the lower boundary of the working area (which corresponds to the latent period in the distributed structure) will intersect with the upper boundary (which corresponds to the deadline in the system in real-time).

Thus, an increase in the number of pods in the nodes of the distributed structure leads to an increase in the reliability of the operation of real-time systems on the one hand. On the other hand, an increase in the number of pods in the nodes leads to an increase in latency in a distributed structure, which interferes with the work of real-time systems due to the operation of deadlines. Conducting studies using real-time systems in distributed structures allows developers to determine such a workspace (Fig. 1) and develop compromise solutions regarding the required number of pods in nodes.

6 The Multi-stage Implementation of the Project

The multi-stage implementation of the project is to design and develop stages for research in the field of using real-time systems in distributed structures and consists of the following solution to problems.

1. Analysis and formalization of the subject area. At this stage, the formation of the goals and objectives of the study, an overview of theoretical information, and literary sources on the necessary topics are performed.
2. Development of the scenario of experimental research and design of a software environment.
3. Work with the subject of research:

Table 1 Comparative characteristics of some basic configurations for obtaining pessimistic and optimistic estimates for the workspace model

Some basic configuration components	Parameters for pessimistic estimation	Parameters for an optimistic estimate
CRUD Structure: Variable Localization	Database	Memory
Localization of nodes and pods	Local or corporate network	One computer
Number of nodes	Two or more	One
Quantity of pods	Three or more	Two

   - selection of software tools and systems;
   - determination of the structure of the research object;
   - conducting comparative characteristics of software that we will use for research;
   - develop a program code for conducting sketch experiments to make restrictions on the use of information resources.

4. Work with the object of research:

   - design of the CRUD structure of the object of research;
   - development and testing of the program code of the object of research;

5. Production and processing of primary experimental data.

Such tasks can be reviewed, changed, and supplemented in the future.

Accomplishing these points in the practical implementation of the object of study requires excellent effort. All of them are important and necessary in implementation. For example, the second point allows us to determine optimistic and pessimistic estimates for the workspace model based on some components of the software environment.

Pessimistic estimates mean overestimated values of the lower boundary of the model (Fig. 1) at which this boundary rises, where the workspace decreases. The conditions (or configuration) in which developers can obtain pessimistic and optimistic estimates are shown in (Table 1).

In the opposite case, creating a configuration in which optimistic estimates can be obtained increases the likelihood of the successful use of real-time systems in distributed structures.

7 Practical Implementation of the Research Object

The practical implementation of the research object is:

- a creation of the research object itself with the help of program code and modern information technologies;
- creating a software environment for the research object in the form of a container;
- implementation of the container in a distributed structure.

The creation of the software environment of the research object as a container with its subsequent implementation into a distributed structure is carried out according to the algorithm (Algorithm 1).

Algorithm 1: Algorithm for creating a software environment for the research object

```
//
// Create Docker Image for Kubernetes
//
Input: Docker Image Requirements
Result: Docker Image
Data: Parameters for the operating system, programming language, compilers, system and
      application utilities, etc
repeat
    Dockerfile creation;
    Building an image using the Dockerfile;
    Docker image verification;
until the docker image does not correspond to the input requirements;
Upload docker image to hub.docker.com;
//
// Deploy Docker Image to Kubernetes
//
Input: Requirements for working with a distributed structure
Result: Create nodes and pods, and launch services for execution
Data: Parameters for nodes, pods, and services
repeat
    Create a manifest file for Kubernetes;
    Build and create pod from the manifest file;
    Validate the pod creation and find more information about system parameters;
until the pod does not correspond to the input requirements;
Launching the required number of nodes and pods.
```

Algorithm 1 shows the sequence of creating a software environment for the research object in a distributed structure. This sequence consists of two stages subject to verification: creating a docker image and deploying it into a distributed structure.

8 Demonstration Example of a Draft Study to Check Violation of Property C (CAP of $PACELC$ theorem) in distributed systems

As an example, consider executing a simple script ($scriptTest.sh$):

```
    $ cat scriptTest.sh
#!/bin/bash
  curl -H "Content-Type: application/json" -X DELETE
    http://localhost:8080/data/$2 &
  sleep $1
  curl -H "Content-Type: application/json" -X GET
    http://localhost:8080/allData
```

This script describes the execution of a command to delete a record, which is specified in the second argument, for the delay time specified in the first argument.

Next, we start the service that corresponds to the $CRUD$ technology using the maven collector:

```
    $ mvn jetty:run
```

As a result of executing the command, the following messages will appear in the terminal:

```
[INFO] Scanning for projects...
[INFO]
[INFO] ----------< org.vntu.crudDataProcessing:
    crudDataProcessing >----------
[INFO] Building crudDataProcessing Maven Webapp
        0.0.1-SNAPSHOT
[INFO] --------------------------------[ war ]----------
[INFO]
[INFO] >>> jetty-maven-plugin:9.4.12.v20180830:run
    (default-cli) > test-compile @ urudDataProcessing
        >>>
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources
    (default-resources) @ crudDataProcessing ---
    ...
[INFO] Started o.e.j.m.p.JettyWebAppContext@26865482
    {Archetype Created Web Application,/,
    file:///home/khoshaba/Documents/projects/crud01/
    src/main/webapp/,AVAILABLE}{file:///home/khoshaba
    /Documents/projects/crud01/src/main/webapp/}
[INFO] Started ServerConnector@2429002c{HTTP/1.1,
    [http/1.1]}{0.0.0.0:8080}
[INFO] Started @20440ms
```

```
[INFO] Started Jetty Server
```

To ensure the simultaneous launch of two commands with a given delay, we will enter the first command to delete the record in the background. After that, the second command to read all records should start first. Then, the second command to delete the record will start with a given delay. Let's see the output of this script.

Thus, this script runs a script that deletes the second record with a specified delay of 0.05 s (in the background) after the command to read all records:

```
$ ./scriptTest.sh 0.05 2
[{"id":1,"timeStamp":"String 1","responseTime":10},
{"id":2,"timeStamp":"String 2","responseTime":20},
{"id":3,"timeStamp":"String 3","responseTime":30},
{"id":4,"timeStamp":"String 4","responseTime":40},
{"id":5,"timeStamp":"String 5","responseTime":50},
{"id":6,"timeStamp":"String 6","responseTime":60},
{"id":7,"timeStamp":"String 7","responseTime":70},
{"id":8,"timeStamp":"String 8","responseTime":80},
{"id":9,"timeStamp":"String 9","responseTime":90},
{"id":10,"timeStamp":"String 10","responseTime":100},
{"id":11,"timeStamp":"String 11","responseTime":110}]
```

Next, we perform a check, where we set command on the command line to check whether record two has been deleted:

```
$ curl -H "Content-Type: application/json" -X
    GET \url{http://localhost:8080/allData}
[{"id":1,"timeStamp":"String 1","responseTime":10},
{"id":3,"timeStamp":"String 3","responseTime":30},
{"id":4,"timeStamp":"String 4","responseTime":40},
{"id":5,"timeStamp":"String 5","responseTime":50},
{"id":6,"timeStamp":"String 6","responseTime":60},
{"id":7,"timeStamp":"String 7","responseTime":70},
{"id":8,"timeStamp":"String 8","responseTime":80},
{"id":9,"timeStamp":"String 9","responseTime":90},
{"id":10,"timeStamp":"String 10","responseTime":100},
{"id":11,"timeStamp":"String 11","responseTime":110}]
$
```

As a result of the command, we do not see record two. This result means that we indeed deleted record two. Therefore, performing a record delete between two requests to read all records produces a different response. Thus, property C is violated according to the CAP (or $PACELC$) theorem in distributed systems.

Further research should be directed towards increasing the latency and changing the number of pods in the nodes of a distributed system.

9 Some Modern Solutions to the Problem of Using Real-Time Systems in Distributed Structures

Some modern solutions to the problem of using real-time systems in distributed structures are as follows.

1. The use of transactions in the case of designing distributed applications or databases. In some cases, to improve the availability and quality of real-time systems, the deadline is reduced, which leads to the loss of responses. Although, this increases the latency period (property L of the $PACELC$ theorem) and leads to an increase in property A (accessibility) of the $PACELC$ theorem. However, it still manages to preserve property C (accessibility).

Also, for distributed databases, long-running transactions lead to negative consequences. Therefore, the practicality of using transactions in distributed databases depends on users' tasks.

2. To reduce the time of transactions, an analysis of the subject area is often performed for optimization, i.e., reducing the number of operations (if it is possible) to solve specific problems.
3. Use of modern hardware: server hardware and communication tools. This will increase the performance of the distributed structure and real-time systems, increase the deadline values, reduce the latency value, and increase the workspace (in the workspace representation model).
4. Use of modern software: enterprise operating systems, tools for supporting distributed structures, programming languages, systems, and application utilities.

10 Conclusions

1. The workspace representation model is proposed for using real-time systems in distributed structures. This model reflects the principles of operation of real-time systems in distributed systems, where it becomes possible to determine the size of the workspace. The workspace representation model has a feature that consists of choosing some compromise solutions between two boundaries. This allows developers to design a real-time system, taking into account the "margin of safety" (or reliability) of the operation of a distributed structure where property P will be observed (i.e., obtaining undistorted answers).
2. The development of a multi-stage implementation of the project made it possible to create stages of research using real-time systems in distributed structures.
3. Based on the created service using CRUD technology and a script, the study showed a violation of property C of the CAP theorem.
4. A demo that violates C properties (CAP or PACELC theorems) in a distributed system (Kubernetes) is shown.

5. The practical implementation of the research object is described based on the algorithm for creating a software environment, divided into two stages: creating and deploying a Docker image for a distributed structure.

References

1. Abadi, D.J.: Consistency tradeoffs in modern distributed database system design: CAP is only part of the story. Computer **2**, 37–42 (2012)
2. Brewer, E.A.: Towards robust distributed systems. In: Proceedings of the XIX Annual ACM Symposium on Principles of Distributed Computing – Portland, OR, vol. 19, no. 7, ACM (2000). https://doi.org/10.1145/343477.343502
3. Brewer, E.: CAP Twelve years later: How the "Rules" have Changed. Computer. **45**(2), 23–29 (2012). https://doi.org/10.1109/MC.2012.37
4. Browne, J.: Brewer's CAP theorem. Accessed 30 May 2022, January 2009. http://www.julianbrowne.com/article/viewer/brewers-cap-theorem
5. Chang, F., et al.: Bigtable: a distributed storage system for structured data. ACM Trans. Comput. Syst. (TOCS) **26**(2), 4 (2008)
6. Diack, B.W., Ndiaye, S., Slimani, Y.: CAP theorem between claims and misunderstandings: what is to be sacrificed? Int. J. Adv. Sci. Technol. **56**, 1–12 (2013)
7. Elzeiny, A., Elfetouh, A., Abo, Riad, A.: Cloud storage: a survey. Int. J. Emerg. Trends Technol. Comput. Sci. **2**(4), 342–349 (2013). ISSN 2278–6856
8. Kossmann, D., Kraska, T., Loesing, S.: Anevaluation of alternative architectures fortransaction processing in the cloud. In: SIGMOD Conference, pp. 579–590 (2010). https://doi.org/10.1145/1807167.1807231
9. Machado, K., Kank, R., Sonawane, J., Maitra, S.: A comparative study of ACID and BASE in database transaction processing. Int. J. Sci. Eng. Res. **8**(5), 116–119 (2017)
10. Machado, K., Kank, R., Sonawane, J., Maitra, S.: A comparative study of ACID and BASE in database transaction processing. Accessed 30 May 2022. https://www.ijser.org/researchpaper/A-Comparative-Study-of-ACID-and-BASE-in-Database-Transaction-Processing.pdf
11. Sadalage, P.J., Fowler, M.: NoSQL Distilled: a Brief Guide to the Emerging World of Polyglot Persistence. Pearson Education, London (2013)
12. Sakr, S., Liu, A., Batista, D., Alomari, M.: A survey of large scale data management approaches in cloud environments. IEEE Commun. Surv. Tutorials. **13**(3), 311–336 (2011). https://doi.org/10.1109/SURV.2011.032211.00087
13. Shim, S.S.: Guest editor's introduction: the CAP theorem's growing impact. Computer **45**(2), 21–22 (2012)
14. Siba, M., Breß, S., Schallehn, E.: Cloud data management: a short overview and comparison of current approaches. Grundlagenvon Datenbanken (2012)
15. Strauch, C.: Nosql databases. Stuttgart Media University, Lecture selected topics on software-technology ultra-large scale sites. Manuscript (2011)

Mathematical Modeling and Simulation of Cyber Physical Systems

Realistic Thermal Infrared Aerospace Image Simulation Backed by Observed Spectral Signatures

Sergey A. Stankevich and Anna A. Kozlova

Abstract The paper describes a thorough technique for simulating a thermal infrared image of the land surface using an available multispectral image of the visible, near-infrared and short-wave infrared bands and the spectral library of typical land covers. The technique is based on spectra nonlinear translation from reference spectral bands to the target one in proportion to corresponding pixel fractions, taking into account the radiative transfer model. To determine the reference spectra fractions inside a mixed pixel, the TCMI (target-constrained minimal interference) matched filtering under the NCLS (non-negatively constrained least squares) physical constraints was applied, which is more efficient than other known ones. The fast radiative transfer model is used for TIR image synthesis simulation. At that, the model is taken into account the additional heat transfer from short-wave solar irradiation complementary to the land surface steady temperature. The structural similarity metric (SSIM) was estimated between the reference and simulated images for objective assessment of the simulation's quality. Experimental simulations of real thermal infrared images demonstrated a reasonably realistic output. The developed end-to-end technique will be useful in the preliminary design of infrared remote sensing systems.

Keywords Remote sensing · Multispectral image · Thermal infrared · Visible and near-infrared · Short-wave infrared · Spectral signature · Spectral library

1 Introduction

Infrared remote sensing plays an important role in the study of the Earth and other planets. It is widely and efficiently applied for geological mapping, anthropogenic impact assessment, droughts monitoring and prediction, climate change modelling, etc. [1].

S. A. Stankevich (✉) · A. A. Kozlova
Scientific Centre for Aerospace Research of the Earth, NAS of Ukraine, Kiev, Ukraine
e-mail: st@casre.kiev.ua

S. Shkarlet et al. (eds.), *Mathematical Modeling and Simulation of Systems*, Lecture Notes in Networks and Systems 667,
https://doi.org/10.1007/978-3-031-30251-0_19

Infrared radiation carries unique and useful information about the land surface temperature as well as about the emissivity of objects and covers [2]. These characteristics make it possible to distinguish objects from backgrounds and from each other, retrieve the thermal regime of the land surface and identify the geosystems' condition [3].

In Ukraine, satellite data is most in-demand in such areas as agriculture, natural resources, ecology and defense. In all of these applications, infrared aerospace imaging is quite important. At the same time, now there are certain problems with obtaining high-quality satellite infrared imagery. First of all, this is the insufficient spatial resolution of infrared imaging, due to the fundamental physical limitations of infrared radiation: long wavelength and low quantum energy [4].

The best available satellite imagery of the long-wave (thermal) infrared (TIR) spectral range provides a spatial resolution scaled to the Earth's surface, approximately 30–60 m [5]. At the same time, a large number of remote sensing images are acquired in the visible/near-infrared (VNIR) and short-wave infrared (SWIR) spectral bands, with much higher spatial resolution and signal-to-noise ratio (SNR) [6]. This circumstance allows engaging the VNIR-SWIR spectral bands to enhance resolution and refine the TIR image [7], and it is not so necessary for VNIR-SWIR and TIR images to be recorded by the same satellite [8].

Many researchers are enhanced remotely sensed infrared imagery by fusion with more competitive images [9, 10], by invoking existent external information [11], or by applying artificial intelligence techniques [12]. However, our intent in this paper is to develop a model for generating an infrared image ab initio, backed by a VNIR-SWIR multispectral image only. Obviously, to solve such a problem, alone image information is not enough, so we assume the spectral library involvement for a typical land cover spectra [13].

2 State-of-the-Art

Spectral signatures translation in remote sensing is used mainly to enhance the spatial resolution of images [14]. By doing so, it is important to enforce the physical rigor over the applied models of spectral transformation [15]. Although sometimes the spectral signatures support is also served for the additional spectral bands' synthesis [16] and thereby for the imagery spectral resolution enhancement [17]. Physically based simulation of band images is particularly useful in the design and the performance evaluation of future remote sensing systems that do not exist yet [18].

A wide variety of methods are used for the resolution enhancement of infrared images. They are: improving the quality of infrared photodetectors [19], developing new high-resolution TIR instruments [20], miscellaneous modifications of thermal downscaling and thermal sharpening based on statistical models [21], refining the modulation transfer function (MTF) in the frequency domain, applying deep learning techniques [22] and many others.

Additional spectral bands synthesizing can be performed by smart interpolation based on such advanced methods as random forest regression [23], iterative fuzzy weighting of true physical spectral bands [24], cross-translation of spectral bands between different satellite systems by pre-classification of spectral signatures and generative adversarial network (GAN) [25], etc. Physical simulations of radiation transfer processes, such as Rayleigh scattering inside the atmosphere [26], the land surface reflectance and emittance properties [27], and radiometric and geometric parameters of irradiance and sensor [28], must be involved to obtain realistic results.

The issue of synthesizing infrared imagery in the thermal spectral band requires a particular consideration. The main problem here is the need to take into account and simulate the land surface thermophysical properties [29], specifically, the correct separation of temperature and emissivity coupled to each other [30]. In this case, the radiative transfer model should be supplemented by models of heat transfer and secondary heating of lands surface [31]. The thermal conductivity and heat-sink capacity of different types of land surface, as well as the heat transfer in the atmosphere should be taken into account to estimate heat fluxes [32]. The main source of thermal energy for the land surface is its own heating up to some steady temperature T and solar irradiation, which provides an additional increase in temperature up to T_+ [33].

Multiple methods are used for separation of the emissivity and the observed brightness temperature of the land surface: deriving the land surface emissivity from another VNIR sensor based on the relationship with the normalized difference vegetation index (NDVI) [34]; iterative correction of firstly assumed as 0.99 emissivity based on the empirical relationship between its minimum and maximum values under the emissivity spectral shape keeping constraint [35]; Bayesian nonparametric deconvolution the material emissivity spectra from the temperature Planck curve [36]; emissivity estimation from the VNIR imagery using a neural network trained on typical land surface reflectance spectra [37]; linearization of the land surface emissivity spectral shape [38]. Adaptation of the most famous TES (temperature and emissivity separation) algorithm, originally developed by Gillespie et al. (1998) [39] for the ASTER (advanced spaceborne thermal emission and reflection radiometer) satellite sensor, was carried out for other TIR sensors, including TIRS (thermal infrared sensor), as well as for the future Ukrainian infrared spectroradiometer with superresolution.

In addition to classical methods for the joint determination of land surface temperature and emissivity, modified approaches are also known: unmixing of the pixel temperature components using linearization of the Planck formula [40] and the land surface emissivity restoration from the observed bands brightness temperatures using a pre-trained deep learning neural network [41]. It should be noted that even non-classical approaches initially rely on physical models of the thermal radiative transfer.

Combining the classical model of reflection, thermal radiation, and temperature prediction with element-wise analysis of the land surface allows to simulate a realistic high-resolution TIR image of a given scene [42]. To improve the simulation accuracy, any model tricks are used, taking into account additional parameters of the land surface and atmosphere, for example, the anisotropy of optical reflection, the structure of vegetation canopies and undergrowth, the water vapor content in the air and even the soil granularity and chemical composition [43]. To simulate the TIR image of a large scene, physical models of dynamics of both vertical and horizontal heat fluxes are additionally required, as well as three-dimensional geometric models of major objects, taking into account their materials of construction [44].

Anyone approaching the TIR image simulation definitely needs to know the spectral distributions of the reflectance ρ and the emissivity ε of the land surface covers, so there must be a special-purpose database of these distributions, i.e. typical land cover reflectance and emissivity spectral library [45].

It should be noted that for opaque bodies, Kirchhoff's law is true, which establishes the numerical equality of spectral emissivity and absorptance [46]. The consequence of Kirchhoff's law for infrared radiation is the following equality

$$\rho(\lambda) + \varepsilon(\lambda) \equiv 1 \tag{1}$$

where (λ) denotes the spectral distribution. Equation (1) eliminates the need to establish distinct spectral libraries for the reflectance and for the emissivity of typical land covers separately [47].

In recent years, plenty of such spectral libraries, including the open access ones, have been developed [48]. They contain spectra of vegetation species, soils and open grounds, rocks and minerals, water surfaces, ice and snow, construction materials, artificial pavements, etc. Some spectral libraries include tens and hundreds of thousands of spectra [49]. Such a huge number and detailing requires a special technique for georeferencing a specific pure or mixed spectrum from the library to the land surface point and, accordingly, to the remote sensed image. These operations are convenient to perform in a geographic information system (GIS) environment. And another special technique is required for spectral unmixing within the analyzed scene and for the correct endmembers' selection [50].

3 Materials and Methods

The overall workflow of a TIR image simulation can be decomposed into several weakly dependent processes. The main one is the indeed TIR image restoration and the secondary ones are providing the necessary input data for the main one. The main process constitutes a determining at sensor radiance value in each element of the TIR image. Secondary processes are designed to form distributions of the land surface characteristics inside the entire image scene. These characteristics include the land surface temperature and emissivity, as well as external solar irradiation. Additional processes, in its turn, need their own input data: this is primarily the basic VNIR-SWIR image of the land surface, as well as the technical specifications of the sensors used, atmospheric parameters and imaging conditions.

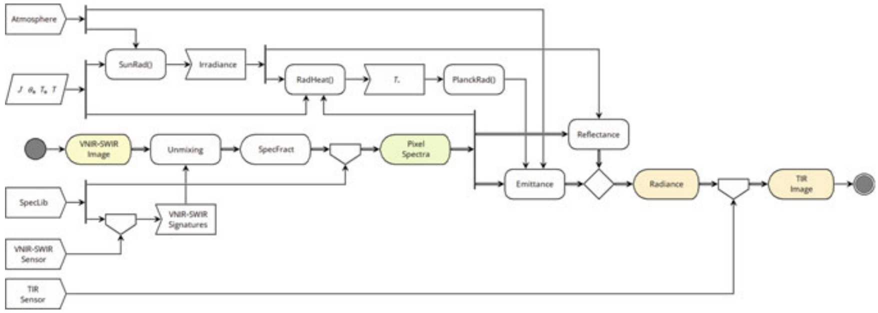


Fig. 1 Workflow diagram of TIR image simulation

A UML-like diagram of the TIR image simulation detailed workflow is shown in Fig. 1.

As a result of the Fig. 1 flowchart execution input VNIR-SVIR image is transformed to output TIR image by sophisticated data processing.

3.1 VNIR-SWIR Image Transformation

The central idea of VNIR-SWIR image transformation is the TIR signatures restoration through the land surface spectra translation into the target spectral bands of the TIR image. This requires, firstly, to determine the spectra present on the land surface, secondly, to retrieve them from the spectral library, and thirdly, to calculate at sensor radiance in the pre-specified TIR spectral bands.

Spectra Determining. The only source of information about the land surface spectra is the available VNIR-SWIR image. However, it does not contain exactly spectra, but spectral signatures, i.e. discrete sets – optical signal vectors in the operating spectral bands of the VNIR-SWIR sensor. If the spectral signature is $\rho = \{\ \rho_i\} = (\rho_1, \rho_2, \ldots, \rho_m)^{\mathrm{T}}$ inside a pixel of an atmospherically corrected and converted into land surface reflectance VNIR-SWIR image, then it was obtained by convolution of the reflectance spectrum $\rho(\lambda)$ with the relative spectral response $R_i(\lambda)$ of the i-th band of the sensor:

$$\rho_i = \frac{\int R_i(\lambda) \cdot \rho(\lambda)\, d\lambda}{\int R_i(\lambda)\, d\lambda} = R_i(\lambda) \otimes \rho(\lambda) \tag{2}$$

To compare the image's signatures with the spectra contained in the spectral library, the latter ones must be converted into spectral signatures with (2) too. From now it becomes possible to estimate which of the spectral library's spectra belongs to the image's spectral signature.

Spectra Unmixing. Unfortunately, spectral signatures of actual images contain, as a rule, mixtures of libraring typical spectra. Therefore, all spectra of the mixture in appropriate proportions should be involved into translation. And to do so it is necessary to evaluate the pixel fractions of these spectra [51].

A fairly large number of methods for linear unmixing of spectral data are known [52]. In this research, an efficient method TCMI (target-constrained minimal interference) matched filtering under the NCLS (non-negatively constrained least squares) physical constraints on the spectra fractions was applied [53].

Every m-dimensional pixel signature ρ of VNIR-SWIR image consists of a set of p libraring signatures weighted by its fractions $\alpha = (\alpha_1, \alpha_2, \ldots, \alpha_p)^{\mathrm{T}}$ also of dimension m each:

$$\rho = \boldsymbol{S} \times \alpha + \zeta \tag{3}$$

where **S** is libraring spectral signatures matrix of $m \times p$ dimensions, and ζ is a residue vector that can be considered as additive noise. The TCMI-NCLS method consists in finding a vector α as a solution of the problem

$$\begin{cases} (\rho - S \times \alpha)^{\mathrm{T}} \sum^{-1} (\rho - S \times \alpha) \to \ \min \\ \alpha \succ 0 \end{cases} \tag{4}$$

The issue of the method, like any other matched filtering, is the possible presence within the scene the spectra, that are unrecorded in spectral library.

TIR Image Synthesis. TIR image synthesis is simulated by the fast radiative transfer model with high spectral sampling [54]. The land surface spectral radiance $E(\lambda)$ is composed of reflected solar irradiation $\rho(\lambda)\cdot E_*(\lambda)$ and its own thermal radiation $\varepsilon(\lambda)\cdot M(\lambda, T_+)$:

$$E(\lambda) = \rho(\lambda) \cdot E_*(\lambda, T_*) + [1-\rho(\lambda)] \cdot M(\lambda, T_+) \tag{5}$$

where $T_* \approx 5800$ K is a Sun's surface temperature, T_+ is land surface temperature [55]. The spectral distribution of solar irradiation on the land surface is expressed by the equation

$$E_*(\lambda) = M(\lambda, T_*) \left(\frac{r_*}{D_* \cdot e_*(J)} \right)^2 \cos\theta_* \ \tau(\lambda, \theta_*) \tag{6}$$

where

$$M(\lambda, T) \ = \ \frac{2\pi h c^2}{\lambda^5 \left(\exp\left(\frac{h c}{k \lambda T}\right) - 1\right)} \tag{7}$$

is a Planck's spectral radiance of a black body, $h = 6{,}626 \bullet 10^{-34}$ W$\bullet$s^2 is Planck constant, $c = 2{,}998 \bullet 10^5$ km/s is a light speed in a vacuum, $k = 1{,}38 \bullet 10^{-23}$ W$\bullet$s^2/K is the Boltzmann constant [56], $r_* = 6{,}96 \bullet 10^7$ km is the Sun radius, $D_* = 1{,}496 \bullet 10^8$ km is the Sun to Earth average distance, e_* is the Sun to Earth distance variation factor depending on the current date's Julian day J,

$$e_* = 1 - 0.01674 \cdot \cos[0.9856 \cdot (J - 4)] \tag{8}$$

θ_* is the Sun zenith angle, $\tau(\lambda, \theta_*) = \exp\left(\frac{\ln \tau(\lambda)}{\cos \theta_*}\right)$ is the spectral transmittance of atmosphere along the inclined optical path [57].

The land surface temperature T_+ is determined by its own internal steady temperature T and by the additional heat transfer due to solar irradiance (6). The total heat flow will be

$$\int_0^\infty M(\lambda, T_+)\, d\lambda = \sigma T^4 + \int_0^\infty E_*(\lambda)\, d\lambda = \sigma T_+^4 \tag{9}$$

where $\sigma = 5.67 \cdot 10^8$ W/(m^2·K^4) is the Stefan–Boltzmann constant [58]. Then the land surface total temperature can be expressed from (9) as

$$T_+ = \sqrt[4]{\frac{1}{\sigma} \int_0^\infty E_*(\lambda)\, d\lambda + T^4} \tag{10}$$

For now, after obtaining all prescribed parameters of infrared emittance, it becomes possible to calculate the spectral radiance at TIR sensor in the i-th working spectral band E_i:

$$E_i = R_i(\lambda) \otimes [E(\lambda) \cdot \tau(\lambda)] \tag{11}$$

Signature Translation. Thus, through the foregoing, it is possible to implement a general approach to the VNIR-SWIR signature translation into the TIR spectral band. This approach is explained by Fig. 2.

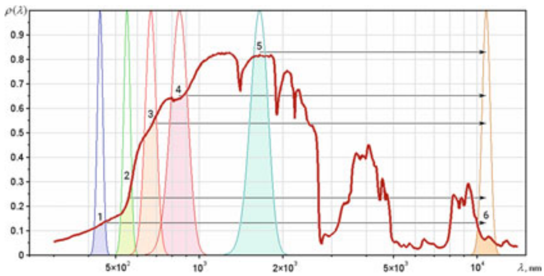


Fig. 2 Signature translation from VNIR-SWIR to TIR band

First, spectral unmixing based on VNIR 1, 2, 3, 4, and SWIR 5 bands is performed and the vector of libraring spectra fractions is found. Then each of the non-zero fraction spectra is convolved with the spectral response of the target TIR band 6. Finally, all TIR signatures obtained are weighed in proportion to their fractions. The described procedure applies to all pixels of the input VNIR-SWIR image. That's done it; the target image is formed.

3.2 Auxiliary Processing

Auxiliary processing provides input data to the main workflow. Conventionally, it can be subdivided into aerospace and ground segments.

Aerospace Segment. The aerospace segment includes annotation data of aerospace imaging – Julian day, the Sun zenith angle, etc.; technical specifications of the reference primary VNIR-SWIR and virtual target sensors, first of all – spectral responses of working bands; environmental conditions of imaging, namely – the spectral transmittance of the atmosphere. Figure 3 shows a plot of the upright spectral transmittance of the standard atmosphere that was used for simulation in the current research.

Ground Segment. The ground segment includes preeminently a reference spectral library that should fit as best as possible with the covers' reflectance spectra within the scene. Modern open access spectral libraries contain an inherently abundant number of spectra, so their preliminary selection for the study area is mandatory. Also, the ground segment refers to the value of the internal steady temperature of the land surface.

4 Results and Discussion

The presented method was applied to the TIR image simulation in two spectral bands of the TIRS sensor of the Landsat-8 remote sensing satellite system backed by 12 spectral bands of actual VNIR-SWIR imagery of the MSI sensor of the Sentinel-2 remote sensing satellite system.

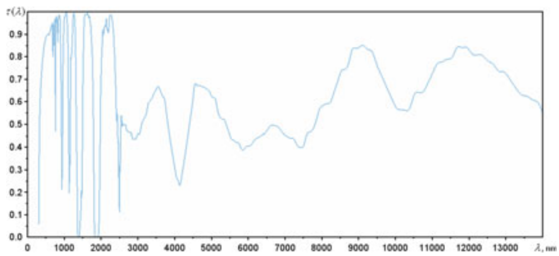


Fig. 3 Standard atmosphere spectral transmittance

4.1 Input Data

Spectral Signatures. The spectral responses of the MSI sensor bands are plotted in Fig. 4, and the spectral responses of the TIRS sensor bands are plotted in Fig. 5. The MSI sensor has enough VNIR-SWIR bands for confident unmixing of the land surface spectral signatures.

Land surface spectral signatures were formed by convoluting the MSI spectral responses with quasi-continuous spectra of typical land covers: vegetation, soil, rocks, water surfaces, artificial materials, etc. Most of the spectra in 0.3–14 μm range were obtained from the ECOSTRESS open access spectral library (speclib.jpl.nasa.gov).

Study Area. A restrained urban location of the Kyiv city within the Zhuliany airport vicinity was chosen as a study area. A Sentinel-2A level 2A 12 bands multispectral image (Fig. 6) for March 21, 2022 was used as a reference. The Sentinel-2A satellite image was downloaded from the Ukrainian mirror of Copernicus Open Access Hub (sentinel.spacecenter.gov.ua).

A totally 20 spectra, including 6 artificial pavements, subjectively most suitable for the study area were selected from the full spectral library for the experiment.

Ground-Truth Data. The best way to verify the simulation results is to compare ones with independently obtained actual data. For this purpose, the Landsat-8 level 1T actual TIR image of the study area (Fig. 7) for March 24, 2022 was used in the current research. The Landsat-8 satellite image was downloaded from the U.S. Geological Survey's EarthExplorer portal (earthexplorer.usgs.gov).

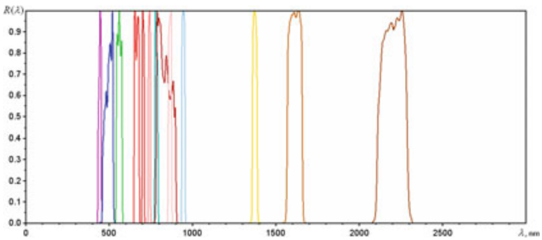


Fig. 4 Spectral responses of 12 VNIR-SWIR reference spectral bands of the Sentinel-2 MSI sensor

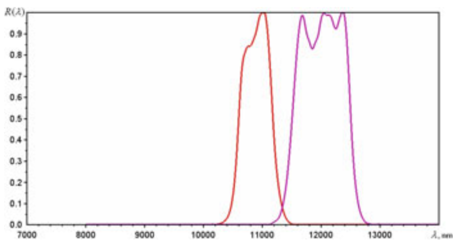


Fig. 5 Spectral responses of 2 TIR target spectral bands of the Landsat-8 TIRS sensor

Fig. 6 Sentinel-2A multispectral image of study area: natural color RGB composite (*a*), false-color CIR composite (*b*), and grayscale SWIR band image (*c*)

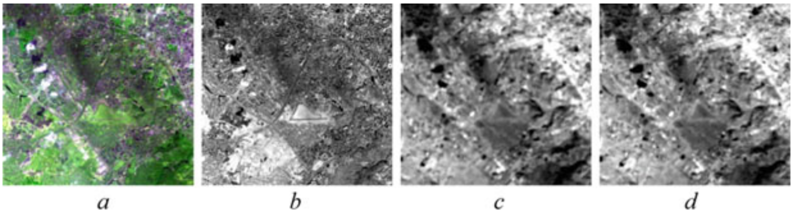


Fig. 7 Landsat-8 multispectral image of study area: false-color CIR composite (*a*), SWIR band image (*b*), TIR first band (10.9 μm) image (*c*), and TIR second band (12 μm) image (*d*)

A level 1 image (at sensor radiance) as opposed to level 2 image (surface reflectance) was engaged for data validation because the precisely at sensor radiance was simulated.

4.2 Simulation Results

The workflow of Fig. 1 running over the VNIR-SWIR multispectral image of Fig. 6 with the involvement of all other necessary data ensures the simulated images synthesizing in both spectral bands of the TIRS sensor. The key operation in processing is the fractions of libraring signatures calculation. In addition to the previously proposed TCMI-NCLS method (4), several well-known methods were used for comparison: linear spectral unmixing (LSU), mixture tuned matched filtering (MTMF), spectral information divergence (SID) and generalized likelihood ratio test (GLRT) [59]. Typically, we utilize the software implementation of key algorithms in the SciLab computational environment (www.scilab.org), but some intermediate generic processing was performed with the ESA SNAP (step.esa.int/main/toolboxes/snap/) and QGIS (www.qgis.org) open source software.

Figure 8 comprises the obtained simulated TIR images in a 10.9 μm spectral band. The TIR images in the other 12 μm spectral band are very similar, so the ones are not shown here. The radiative transfer model was used the same for all simulations.

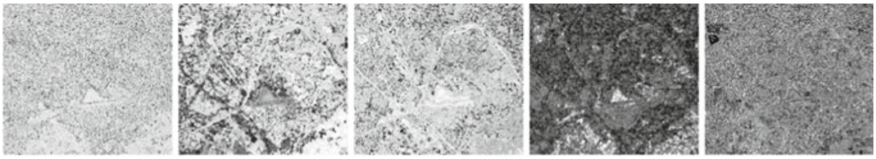


Fig. 8 Simulated Landsat-8 TIR 10.9 μm band images obtained with signature unmixing methods: LSU (*a*), MTMF (*b*), SID (*c*), GLRT (*d*) and proposed TCMI-NCLS (*e*)

Table 1 SSIM values between simulated and reference TIR images

Simulated TIR image	Reference TIR image	
	Landsat 8 TIRS 10.9 μm band	Landsat 8 TIRS 12 μm band
LSU	0.763	0.806
MTMF	0.762	0.820
SID	0.818	0.857
GLRT	0.813	0.854
TCMI-NCLS	**0.883**	**0.910**

The similarity between the images simulated using generic unmixing methods and the reference image Fig. 7 *c* is not so well; in some cases, such as Fig. 8 *b* and *d*, even contrast inversion is observed.

4.3 Results Evaluation

However, the visual similarity is not an objective assessment of the simulation's quality. Therefore, a widespread structural similarity metric (SSIM) [60] was estimated between the reference image and each of the simulated ones. Estimates are in Table 1.

As follows from Table 1, the proposed TCMI-NCLS method for spectral unmixing provides the best similarity of simulated images with reference ones. Among generic methods, the best simulation accuracy is inherent in statistical spectral unmixing.

5 Conclusions

Thus, a reasonably realistic simulation of TIR image is feasible backed by the VNIR-SWIR multispectral image. The basic tools for such simulation are the spectral signatures convolution, the radiative transfer model in the full optical spectral range 0.3–14 μm, the reference spectral library of typical land covers, the method for spectral signatures mixing, as well as the radiometric specification of the input VNIR-SWIR sensor and the target TIR one. The key prerequisites of successful TIR image simulation are a complete and adequate selection of reference spectral signatures as well as the correct method for ones' spectral unmixing. The essence of the proposed solution is the spectra translation between the observed and simulated spectral bands.

The availability of end-to-end technique for TIR image simulation will be quite useful in the preliminary design of infrared remote sensing systems, as well as in the spatial resolution enhancement of TIR imagery backed by higher resolution VNIR-SWIR imagery.

Future research should be aimed at relevant reference spectra automatic selection from a comprehensive spectral library over a specific scene of aerospace imaging; to invent some method for handling the signatures not registered in the spectral library; to impose stricter physical constraints on the spectral signature fractions; to large-scale testing the developed technique over remote sensed imagery of various landscapes.

References

1. Sobrino, J.A., Del Frate, F., Drusch, M., Jiménez-Muñoz, J.C., Manunta, P., Regan, A.: Review of thermal infrared applications and requirements for future high-resolution sensors. IEEE Trans. Geosci. Remote Sens. **54**(5), 2963–2972 (2016). https://doi.org/10.1109/TGRS.2015.2509179
2. Cheng, J., Liang, S., Meng, X., Zhang, Q., Zhou, S.: Land surface temperature and thermal infrared emissivity. In: Liang, S., Wang, J. (eds.) Advanced Remote Sensing, pp. 251–295, 2nd edn. Academic Press, London (2020). https://doi.org/10.1016/B978-0-12-815826-5.00007-6
3. Quattrochi, D.A., Luvall, J.C.: Thermal infrared remote sensing for analysis of landscape ecological processes: current insights and trends. In: Weng, Q. (ed.) Scale Issues in Remote Sensing, pp. 34–60. Wiley, Hoboken (2014). https://doi.org/10.1002/9781118801628.ch03
4. Rees, W.G.: Physical Principles of Remote Sensing, 3rd edn. Cambridge University Press, New York (2012). https://doi.org/10.1017/CBO9781139017411
5. Li, Z.-L., Tang, B.-H., Wu, H., Ren, H., Yan, G., Wan, Z., Trigo, I.F., Sobrino, J.A.: Satellite-derived land surface temperature: current status and perspectives. Remote Sens. Environ. **131**, 14–37 (2013). https://doi.org/10.1016/j.rse.2012.12.008
6. Al-Wassai, F.A., Kalyankar, N.V.: Major limitations of satellite images. J. Glob. Res. Comput. Sci. **4**(5), 51–59 (2013)
7. Aiazzi, B., Baronti, S., Garzelli, A., Santurri, L., Selva, M.: Spatial enhancement of TIR ASTER data via VNIR images and generalized Laplacian decomposition. In: Zagajewski, B., Sobczak, M., Wrzesień, M. (eds.) Proceedings of 4th EARSeL Workshop on Imaging Spectroscopy. New quality in environmental studies, pp. 489–500. EARSeL, Warsaw (2005). 10.1.1.381.7996
8. Piestova, I., Lubskyi, M., Svideniuk, M., Golubov, S., Sedlacek, P.: Satellite imagery resolution enhancement for urban area thermal micromapping. Cent. Eur. Res. J. **4**(1), 35–39 (2018)
9. Aghamaleki, J.A., Ghorbani, A.: Infrared and visible image fusion based on optimal segmenting and contour extraction. SN Appl. Sci. **3**, 369 (2021). https://doi.org/10.1007/s42452-020-04050-w
10. Meher, B., Agrawal, S., Panda, R., Dora, L., Abraham, A.: Visible and infrared image fusion using an efficient adaptive transition region extraction technique. Eng. Sci. Technol. Int. J. **29**, 101037 (2022). https://doi.org/10.1016/j.jestch.2021.06.017

11. Stankevich, S.A., et al.: Knowledge-based multispectral remote sensing imagery superresolution. In: van Gulijk, C., Zaitseva, E. (eds.) Reliability Engineering and Computational Intelligence. Studies in Computational Intelligence, vol. 976, pp. 219–236. Springer, Cham (2021). https://doi.org/10.1007/978-3-030-74556-1_13
12. Sisodiya, N., Dube, N., Thakkar, P.: Next-generation artificial intelligence techniques for satellite data processing. In: Hemanth, D.J. (ed.) Artificial Intelligence Techniques for Satellite Image Analysis. Remote Sensing and Digital Image Processing, vol. 24, pp. 235–254. Springer, Cham (2020). https://doi.org/10.1007/978-3-030-24178-0_11
13. Zhang, W., Shi, M., Zhang, B., Li, Y., Dai, H.: Image simulation for GF-5 visual and infrared multispectral sensor. In: Proceedings of the 40th Asian Conference on Remote Sensing (ACRS 2019), vol. 3, pp. 1999–2005. KSRS, Daejeon (2019)
14. Stankevich, S.A., Piestova, I.O., Lubskyi, M.S.: Remote sensing imagery spatial resolution enhancement. In: Abbasov, I.B. (ed.) Recognition and Perception of Images: Fundamentals and Applications, pp. 327–360. Wiley, Hoboken (2021). https://doi.org/10.1002/9781119751991.ch9
15. Restaino, R., Vivone, G., Addesso, P., Picone, D., Chanussot J.: Resolution enhancement of hyperspectral data exploiting real multi-platform data. In: Kwan, C. (ed.) Recent Advances in Image Restoration with Applications to Real World Problems, pp. 17–38. IntechOpen, London (2020). https://doi.org/10.5772/intechopen.92795
16. Thomas, C., Ranchin, T., Wald, L., Chanussot, J.: Synthesis of multispectral images to high spatial resolution: a critical review of fusion methods based on remote sensing physics. IEEE Trans. Geosci. Remote Sens. **46**(5), 1301–1312 (2008). https://doi.org/10.1109/TGRS.2007.912448
17. Sun, X., Zhang, L., Yang, H., Wu, T., Cen, Y., Guo, Y.: Enhancement of spectral resolution for remotely sensed multispectral image. IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens. **8**(5), 2198–2211 (2015). https://doi.org/10.1109/JSTARS.2014.2356512
18. Segl, K., Richter, R., Küster, T., Kaufmann, H.: End-to-end sensor simulation for spectral band selection and optimization with application to the Sentinel-2 mission. Appl. Opt. **51**(4), 439–449 (2012). https://doi.org/10.1364/AO.51.000439
19. Chen, J., Wang, J., Li, X., Chen, J., Yu, F., He, J., Wang, J., Zhao, Z., Li, G., Chen, X., Lu, W.: Recent progress in improving the performance of infrared photodetectors via optical field manipulations. Sensors **22**(2), 677 (2022). https://doi.org/10.3390/s22020677
20. Buffet, L., Gamet, P., Maisongrande, P., Salcedo, C., Crebassol, P.: The TIR instrument on TRISHNA satellite: a precursor of high resolution observation missions in the thermal infrared domain. Proc. SPIE **11852**, 118520Q (2021). https://doi.org/10.1117/12.2599173
21. Mao, Q., Peng, J., Wang, Y.: Resolution enhancement of remotely sensed land surface temperature: current status and perspectives. Remote Sens. **13**(7), 1306 (2021). https://doi.org/10.3390/rs13071306
22. Bhutto, J.A., Tian, L., Du, Q., Sun, Z., Yu, L., Soomro, T.A.: An improved infrared and visible image fusion using an adaptive contrast enhancement method and deep learning network with transfer learning. Remote Sens. **14**(4), 939 (2022). https://doi.org/10.3390/rs14040939
23. Xie, Y., Han, X., Zhu, S.: Synthesis of true color images from the Fengyun advanced geostationary radiation imager. J. Meteorol. Res. **35**, 1136–1147 (2021). https://doi.org/10.1007/s13351-021-1138-3
24. Ma, L., Zhenhong, J., Yang, J., Kasabov, N.: Multi-spectral image change detection based on single-band iterative weighting and fuzzy C-means clustering. Eur. J. Remote Sens. **53**(1), 1–13 (2020). https://doi.org/10.1080/22797254.2019.1707124
25. Vandal, T.J., McDuff, D., Wang, W., Duffy, K., Michaelis, A., Nemani, R.R.: Spectral synthesis for geostationary satellite-to-satellite translation. IEEE Trans. Geosci. Remote Sens. **60**, 4702611 (2022). https://doi.org/10.1109/TGRS.2021.3088686
26. Shanmugam, V., Shanmugam, P., He, X.: New algorithm for computation of the Rayleigh-scattering radiance for remote sensing of water color from space. Opt. Express **27**(21), 30116–30139 (2019). https://doi.org/10.1364/OE.27.030116

27. Qiu, X., Zhao, H., Jia, G., Li, J.: Atmosphere and terrain coupling simulation framework for high-resolution visible-thermal spectral imaging over heterogeneous land surface. Remote Sens. **14**(9), 2043 (2022). https://doi.org/10.3390/rs14092043
28. He, X., Xu, X.: Physically based model for multispectral image simulation of Earth observation sensors. IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens. **10**(5), 1897–1908 (2017). https://doi.org/10.1109/JSTARS.2017.2658678
29. Vostikolaei, F.S., Akhoondzadeh, M.: A comparison of four methods for extracting land surface emissivity and temperature in the thermal infrared hyperspectral data. Earth Obs. Geomatics Eng. **2**(1), 56–63 (2018). https://doi.org/10.22059/eoge.2018.239666.1011
30. Schmugge, T., French, A., Ritchie, J.C., Rango, A., Pelgrum, H.: Temperature and emissivity separation from multispectral thermal infrared observations. Remote Sens. Environ. **79**(2–3), 189–198 (2002). https://doi.org/10.1016/S0034-4257(01)00272-3
31. Friedl, M.A.: Forward and inverse modeling of land surface energy balance using surface temperature measurements. Remote Sens. Environ. **79**(2–3), 344–354 (2002). https://doi.org/10.1016/S0034-4257(01)00284-X
32. Bateni, S.M., Liang, S.L.: Estimating surface energy fluxes using a dual-source data assimilation approach adjoined to the heat diffusion equation. J. Geophys. Res. Atmos. **117**(D17), D17118 (2012). https://doi.org/10.1029/2012JD017618
33. Cao, B., et al.: A review of Earth surface thermal radiation directionality observing and modeling: historical development, current status and perspectives. Remote Sens. Environ. **232**, 111304 (2019). https://doi.org/10.1016/j.rse.2019.111304
34. Sobrino, J.A., Jiménez-Muñoz, J.C., Sòria, G., Romaguera, M., Guanter, L., Moreno, J., Plaza, A., Martínez, P.: Land surface emissivity retrieval from different VNIR and TIR sensors. IEEE Trans. Geosci. Remote Sens. **46**(2), 316–327 (2008). https://doi.org/10.1109/TGRS.2007.904834
35. Liu, H., Zhang, C., Ye, F., Zhang, J.: Temperature and emissivity separation methods based on TASI data. In: Proceedings of the International Conference on Remote Sensing, Environment and Transportation Engineering (RSETE 2013), pp. 685–688. Atlantis Press, Nanjing (2013). https://doi.org/10.2991/rsete.2013.166
36. Berrett, C., Williams, G.P., Moon, T., Gunther, J.: A Bayesian nonparametric model for temperature-emissivity separation of long-wave hyperspectral images. Technometrics **56**(2), 200–211 (2014). https://doi.org/10.1080/00401706.2013.869262
37. Vanhellemont, Q.: Combined land surface emissivity and temperature estimation from Landsat 8 OLI and TIRS. ISPRS J. Photogramm. Remote. Sens. **166**, 390–402 (2020). https://doi.org/10.1016/j.isprsjprs.2020.06.007
38. Lan, X., Zhao, E., Li, Z.-L., Labed, J., Nerry, F.: An improved linear spectral emissivity constraint method for temperature and emissivity separation using hyperspectral thermal infrared data. Sensors **19**(24), 5552 (2019). https://doi.org/10.3390/s19245552
39. Gillespie, A.R., Rokugawa, S., Matsunaga, T., Cothern, J.S., Hook, S., Kahle, A.B.: A temperature and emissivity separation algorithm for Advanced Spaceborne Thermal Emission and Reflection radiometer (ASTER) images. IEEE Trans. Geosci. Remote Sens. **36**(4), 1113–1126 (1998). https://doi.org/10.1109/36.700995
40. Xie, F., Shao, H., Liu, Z., Liu, C., Zhang, C., Yang, G., Wang, J., Cai, N.: Retrieval of the pixel component temperatures from multi-band thermal infrared image using Bayesian inversion technique. Proc. SPIE **9880**, 98802A (2016). https://doi.org/10.1117/12.2227579
41. Wang, H., Mao, K., Yuan, Z., Shi, J., Cao, M., Qin, Z., Duan, S., Tang, B.: A method for land surface temperature retrieval based on model-data-knowledge-driven and deep learning. Remote Sens. Environ. **265**, 112665 (2021). https://doi.org/10.1016/j.rse.2021.112665
42. Poglio, T., Mathieu-Marni, S., Ranchin, T., Savaria, E., Wald, L.: OSIrIS: a physically based simulation tool to improve training in thermal infrared remote sensing over urban areas at high spatial resolution. Remote Sens. Environ. **104**, 238–246 (2006). https://doi.org/10.1016/j.rse.2006.03.017
43. Carson, T., Salvaggio, C.: Soil signature simulation in the thermal infrared. Opt. Eng. **54**(10), 104102 (2015). https://doi.org/10.1117/1.OE.54.10.104102

44. Kottler, B., Burkard, E., Bulatov, D., Haraké, L.: Physically-based thermal simulation of large scenes for infrared imaging. In: Proceedings of the 14th International Joint Conference on Computer Vision, Imaging and Computer Graphics Theory and Applications (VISIGRAPP 2019), pp. 53–64. SciTePress, Prague (2019). https://doi.org/10.5220/0007351400530064
45. Bandfield, J.L., Rogers, D.A.: Thermal infrared spectral modeling. In: Bishop, J.L., Bell III, J.F., Moersch, J.E. (eds.) Remote Compositional Analysis Techniques for Understanding Spectroscopy, Mineralogy, and Geochemistry of Planetary Surfaces, pp. 324–336. Cambridge University Press, Cambridge (2019). https://doi.org/10.1017/9781316888872.017
46. Riedl, M.J.: Optical Design Fundamentals for Infrared Systems, 2nd edn. SPIE Press, Bellingham (2001). https://doi.org/10.1117/3.412729
47. Salisbury, J.W., Wald, A., D'Aria, D.M.: Thermal-infrared remote sensing and Kirchhoff's law 1. Laboratory measurements. J. Geophys. Res. **99**(B6), 11897–11911 (1994). https://doi.org/10.1029/93JB03600
48. Kerekes, J.P., Strackerjan, K.-E., Salvaggio, C.: Spectral reflectance and emissivity of man-made surfaces contaminated with environmental effects. Opt. Eng. **47**(10), 106201 (2008). https://doi.org/10.1117/1.3000433
49. Haraké, L., Burkard. E.: Towards physically based material appearance in the thermal infrared spectrum: a short survey. In: Proceedings of the Eurographics Workshop on Material Appearance Modeling (EG MAM 2018), pp. 37–40. Eurographics Association, Karlsruhe (2018). https://doi.org/10.2312/mam.20181198
50. Degerickx, J., Okujeni, A., Iordache, M.-D., Hermy, M., Van der Linden, S., Somers, B.: A novel spectral library pruning technique for spectral unmixing of urban land cover. Remote Sensing **9**(6), 565 (2017). https://doi.org/10.3390/rs9060565
51. Stankevich, S.A., Shklyar, S.V.: Land cover classification in hyperspectral aerospace imagery by separation of spectral components mixtures. J. Autom. Inf. Sci. **38**(12), 31–41 (2006). https://doi.org/10.1615/JAutomatInfScien.v38.i12.40
52. Wei, J., Wang, X.: An overview on linear unmixing of hyperspectral data. Math. Probl. Eng. **2020**, 3735403 (2020). https://doi.org/10.1155/2020/3735403
53. Stankevich, S.A., Shklyar, S.V.: Advanced algorithm for endmembers unmixing on hyperspectral image (in Ukrainian). In: Proceedings of the 1st Ukrainian Conference with International Participation "Earth Observations for Sustainable Development and Security", pp. 85–89. Naukova Dumka, Kiev (2008)
54. Liu, X., Yang, Q., Wu, W., Zhou, D.K., Larar, A.M.: Fast radiative transfer model from IR to UV for remote sensing applications. Proc. SPIE **10424**, 1042407 (2017). https://doi.org/10.1117/12.2277897
55. White, O.R. (ed.): Colorado Associated University Press, Boulder (1977)
56. Hollands, K.G.T.: Thermal Radiation Fundamentals. Begell House, New York (2004)
57. Goswami, D.Y.: Principles of Solar Engineering, 3rd edn. CRC Press, Boca Raton (2015). https://doi.org/10.1201/b18119
58. Minkina, W.: Theoretical basics of radiant heat transfer – practical examples of calculation for the infrared (IR) used in infrared thermography measurements. Quant. InfraRed Thermogr. J. **18**(4), 269–282 (2021). https://doi.org/10.1080/17686733.2020.1738164
59. Keshava, N.: A survey of spectral unmixing algorithms. Lincoln Lab. J. **14**(1), 55–78 (2003)
60. Brunet, D., Vrscay, E.R., Wang, Z.: On the mathematical properties of the structural similarity index. IEEE Trans. Image Process. **21**(4), 1488–1499 (2012). https://doi.org/10.1109/TIP.2011.2173206

Physical Modeling and Analysis of Gate Drive DC-DC Converter Impact on Photovoltaic Inverter Efficiency

Serhii Stepenko, Anatoliy Prystupa, Roman Yershov, Oleksandr Bondarenko, Aleksandr Kosmach, and Oleksandr Hlushko

Abstract This paper presents results of the physical modeling and experimental investigations of the photovoltaic system based on full-SiC quasi-z-source inverter. The goal of the current research is to analyze the impact of the gate drive DC-DC converters on the photovoltaic inverter efficiency change based on experimental measurements. There are different modulation indices and different shoot-through duty cycle values applied within inverter performance analysis. It was obtained that inverter demonstrates over 97% peak efficiency in significant power range within the nominal range of operation. At the same time, the voltage level of the gate drive DC-DC converter increased from 15 to 20 V. It makes a notable impact on the efficiency of the inverter power stage in certain points within the high-efficiency power range. The total power consumption by gate drivers increased by few watts and remains relatively constant in the whole operation range of the system. Due to the proven high efficiency, the investigated solution is feasible for photovoltaic applications.

Keywords Dc-ac converter · Efficiency analysis · Gate drive dc-dc converter · MOSFET · Performance analysis · Physical modeling · Photovoltaic inverter · Power semiconductor switch · Quasi-z-source inverter · SiC semiconductors

1 Introduction

The concept of Z-Source (Impedance-Source) Inverter (ZSI) is proposed as a compromised alternative [1] for voltage-source inverters (VSI) and current-source inverters (CSI). ZSI allow providing both the buck and boost functionality within the single switching stage, moreover, have the natural immunity against short circuit, which

S. Stepenko (✉) · A. Prystupa · R. Yershov · A. Kosmach · O. Hlushko
Chernihiv Polytechnic National University, Shevchenko Str. 95, Chernihiv 14030, Ukraine
e-mail: serhii.stepenko.ua@ieee.org

O. Bondarenko
National Technical University of Ukraine "Igor Sikorsky Kyiv Polytechnic Institute", Prosp. Peremohy 37, Kyiv 03056, Ukraine

S. Shkarlet et al. (eds.), *Mathematical Modeling and Simulation of Systems*, Lecture Notes in Networks and Systems 667,
https://doi.org/10.1007/978-3-031-30251-0_20

leads to the improved reliability. The above-mentioned advantages make it an attractive an efficient solution for photovoltaic (PV) applications. The recent researches devoted to ZSI in PV resulted in many dc-dc and dc-ac topologies for a single-phase (1-ph) [2] and three-phase (3-ph) applications [3].

Quasi-Z-Source Inverter (QZSI) topology [4] derived from the traditional ZSI has become a suitable solution for PV installations due to its inheritance of all ZSI's advantages enhanced by lower component ratings and continuous input current mode (CCM) availability [5–7].

The urge of current research is caused by means of interest in the development of the combined power supply systems based on renewable energy sources [8] and necessity of the investigation of the elements' parameters [9]. The efficiency of the systems constructed from semiconductors based on Silicon Carbide (SiC) technology [10–15] was reported significantly higher than based on typical Silicon (Si) semiconductors. However, there are still enough points and stages to improve the overall efficiency as it was shown in for PV system rated for nominal power of 1800 W, which corresponds to 5…10 PV panels string (depending on rated PV panel power) [12].

The goal of current research is to perform a physical modeling and experimental investigations of the isolated gate drive DC-DC converter impact on full-SiC QZSI efficiency based on comparative numeric analysis.

The paper is structured as follows. In Sect. 2 the literature review is performed. Section 3 is devoted to description of methods and equipment, particularly the prototype used as an experimental platform. Section 4 shows the obtained results. Section 5 provides evaluation and discussion of the results followed by Sect. 6 Conclusions.

2 Literature Review

The usage of multilevel inverter is reasonable in high-power designs, where the high voltage stresses on inverter's switches could be avoided by those series cascading inside a single half-bridge arm. Numerous publications devoted to ZSI- and QZSI-derived solutions for PV, wind, Microgrids applications are recently appeared [5–7, 12, 14, 16, 17]. Extremely high efficiency of 99.4% achieved for 3-ph 50 kW full-SiC based PV string inverter as it is reported in [10]. Another full-SiC solution for 25 kW 3-ph PV string inverter has demonstrated over 97% peak efficiency [11]. These can be examples of extra-high efficiency, which could be achieved much easier in high-power systems. The latter also includes a detailed step-by-step explanation and design guidelines for all the system components. A comprehensive analysis of power losses, temperature field and efficiency is provided in [13] for solution based on CSI constructed from SiC MOSFETs as well as comparative analysis of power losses for full-SiC and hybrid approaches, but experimental results are not shown there.

A valuable and demonstrative experimental comparison was presented in [14], devoted to three topologies: a quasi-Z-source inverter, a VSI with a boost converter and a VSI with an interleaved boost converter. Since modern requirements both for standalone and grid-connected kW-scale PV systems comprise high efficiency, high reliability and high power density, all these features could be achieved through the lowering the ripple of input current and DC-link voltage, resulting in high quality of output current together with sparing requirements to the output filter quality.

Another approach that allows improving the system reliability and to achieve a higher power density by the reduction of redundant passive elements, is interleaving that is often applied in VSI. It allows reducing both the inductors current ripple significantly and DC-link voltage ripple. One of the main parameters which are extremely important for modern power supply systems is a high quality of the voltage and current [18]. This could be estimated by means of Total Harmonic Distortion (THD).

Another important factor impacting overall efficiency is power losses in semiconductor switches of the converter [19]. There are numerous techniques aimed to improve these parameters in power systems.

An interleaved quasi-z-source inverter (IQZSI) with simple boost control (SBC) topology is a promising solution for usage in PV applications. It has several benefits, such as reduced output THD and low ripple at QZS-stage components, which potentially lead to lower power losses and higher power density of the overall system. In order to improve DC-link utilization, modulation index and higher gain achieving [20], the maximum boost control (MBC) [21] with appropriate modification is required. It smooths variation in the ST duty cycle. The application of MBC in QZSI revealed its urgency and proved the power decoupling necessity [22].

There are several methods and approaches to estimate the characteristics of the processes occurred in power systems [23–25] by means of calculation and modeling. Some significant results have been obtained earlier to ensure the necessary output voltage for the three-level QZSI [26]. There are also some interesting combinations for classical topologies as Cuk & SEPIC to get an efficient DC-DC converter for photovoltaic systems [27]. The most recent discussions bring us the broad comparison of multilevel inverter topologies operating in hostile environments [28] which are rather important for the future investigations in the frame of current research.

The aim of this work, devoted to physical modeling and experimental investigations of the isolated gate drive DC-DC converter impact on full-SiC QZSI efficiency, is important in order to improve the overall efficiency of such systems [29, 30].

3 Methods and Equipment

In the current work, the QZSI (Fig. 1) is being applied for 1-ph PV system with the main parameters, which are collected in Table 1. The main stages of the topology include: QZS-network, represented by components L_1, D_1, C_1, L_2, C_2; full-bridge 2-level inverter based on MOSFET switches S_1, S_2, S_3, S_4; and output asymmetric LCL-filter represented by L_{F1}, C_F, L_{F2} components, feeding the local load or the grid.

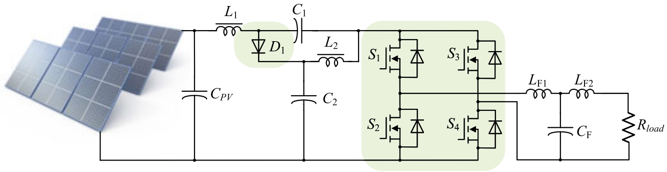


Fig. 1 Quasi-Z-Source Inverter [12]

Table 1 Main system parameters

Parameter	Description	Value
Pnom	Nominal power	Pnom = 1800 W
Vnom	Nominal voltage	Vnom = 200–400 V
Inom	Nominal current	Inom = 5 A
Vload	Load RMS voltage	Vload = 230 V
Pmin	Minimal power	Pmin = 90 W
Pmax	Maximal power	Pmax = 2000 W
Vmin	Minimal voltage	Vmin = 180 V
Vmax	Maximal voltage	Vmax = 480 V
Imax	Maximal input current	Imax = 10 A

The detailed discussions and explanations of the QZSI for 1-ph PV application as well as the control approaches, including simple boost control (SBC), maximum boost control (MBC), constant boost control (CBC) and their modifications, were provided [12, 22, 26]. The selected nominal values for power stage passive and semiconductor components are collected in Table 2.

Table 2 Nominal values of power components

Component	Description	Value
L_1, L_2	QZS-stage inductors	1800 mH
C_1	QZS-stage capacitor	1200 μF
C_2	QZS-stage capacitor	680 μF
L_{F1}	Output LCL-filter input inductor	560 μH
L_{F2}	Output LCL-filter output inductor	200 μH
C_F	Output filter capacitor	15 μF
V_{RRM} D_1	QZS-stage diode voltage	1200 V
V_{DS} S_1-S_4	Inverter stage power switches voltage	1200 V
Imax	Gate drivers Broadcom ACPL-H342 current	2.5 A

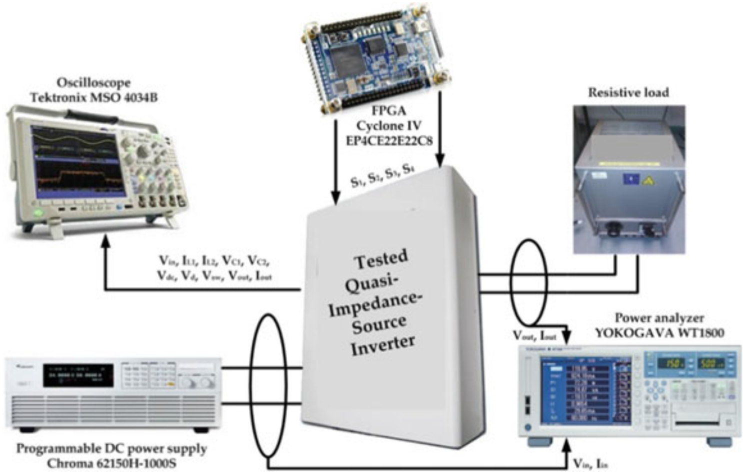


Fig. 2 The experimental setup structure for physical modeling

The functional circuit diagram of the experimental setup for physical modeling and measurements is depicted in Fig. 2. It shows general facilities of the experimental verification. The experimental setup includes the following equipment:

- Programmable DC power supply (PV array simulator) Chroma 62150H-1000S;
- QZSI as a PV inverter incorporating output LCL-filter;
- Control board FPGA Cyclone IV EP4CE22E22C8;
- Resistive load for up to 3 kW output power;
- Oscilloscope Tektronix MSO 4034B;
- Power analyzer YOKOGAWA WT1800;
- LCR Bridge/Meter HM8118;
- Infrared thermal camera Fluke Ti10.

The QZS-stage was built using the electrolytic aluminum capacitors of a model EKMS3B1VSN122MA50S with parameters 105 °C, 315 V, Ir = 3.25 A, ESR = 100 mOhm and ALC10(1)681DL500 with parameters 85 °C, 500 V, Ir = 3.65 A, ESR = 244 mOhm were applied. For high-power semiconductor devices diode of a model C4D02120A at the QZS-stage and MOSFET switches of a model C2M0080120D at the inverter stage both built on SiC technology were applied.

4 Physical Modeling Results

The experimental results are shown in Fig. 3 and Fig. 4 below. The QZS-stage input voltage, inductors current, output current and voltage are shown there.

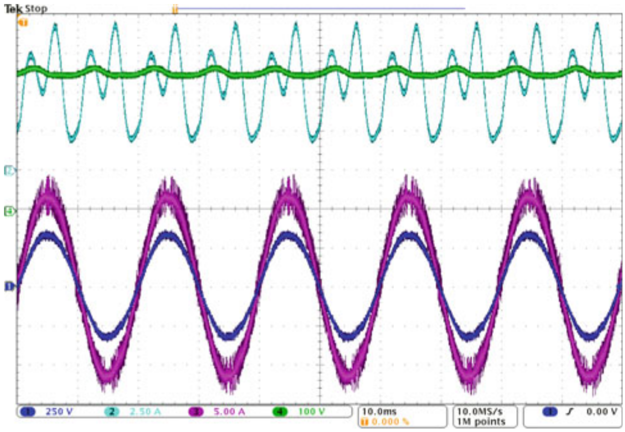


Fig. 3 Experimental waveforms for the installed Gate Drive DC-DC Converters for 15 V

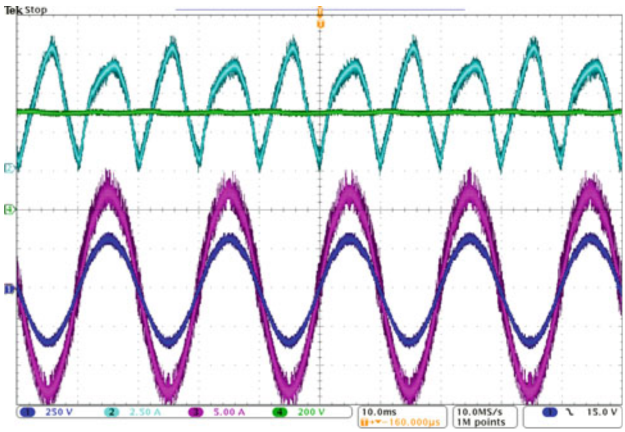


Fig. 4 Experimental waveforms for the installed Gate Drive DC-DC Converters for 20 V

Experimentally measured peak efficiency for different modulation indices M is shown in Fig. 5 and Fig. 6. For the operation mode without or with small ST duty cycles, there is certain flat efficiency ranges at 1100…1700 W. The efficiency of the converter power stage at each data point is obtained as a relation of output to input instantaneous powers (i.e. current–voltage products) which are directly measured and digitally processed by YOKOGAWA WT1800 power analyzer.

The important and interesting phenomenon could be observed, that deep modulation (lower modulation index) decreases efficiency in the whole operation power range. The most probable explanation is based on that under lower M values, power losses in QZS-stage diodes will decrease faster, while power losses in MOSFETs increase slower. This difference matters for the whole efficiency.

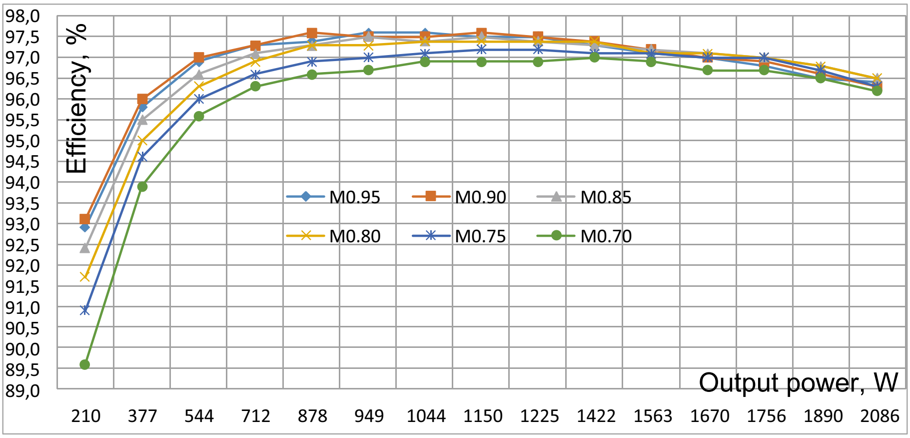


Fig. 5 Experimentally measured efficiency of 2L QZSI for 15 V gate drive DC-DC voltage

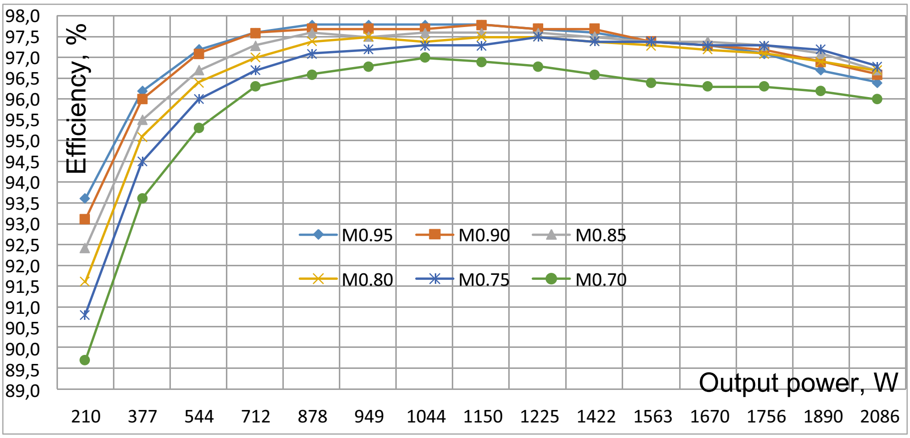


Fig. 6 Experimentally measured efficiency of 2L QZSI for 20 V gate drive DC-DC voltage

5 Evaluation and Discussion of the Results

The most essential result is that experimental investigations are shown that increasing of the nominal positive output voltage of gate drive dc-dc converter form 15 V (isolated single-channel dc-dc converter module RKZ-1215S manufactured by Recom) to 20 V (isolated dual-channel dc-dc converter module RKZ-122005D also manufactured by Recom) could increase overall system efficiency by 0.2–0.5%.

It is due to the transient process dealing with the gate charging is forced by applying of a higher positive voltage. Respectively, the gate discharging is forced by applying of negative −5 V potential from the second channel of 20 V output dc-dc converter model. Thus, both the open and closed SiC switch thresholds could be achieved more rapidly and switch being in active state within shorter time duration.

Also in certain application the gate charge Qc behavior of selected power switch should be validated due to its dependency on gate-applied voltage, in our case gate charge i.e. charging time constant, is increased together with the applied gate voltage around of 5–20%. Moreover, a 20 V-output dc-dc converter module has higher conversion efficiency (86% versus 84% in 15 V-output one) that is multiplied to all isolated control channels.

Experimental measurements have shown that by replacing all gate drive dc-dc converters from 15 V-output to 20 V ones, the power consumption increased from 2.0 W to 3.5 W while all other gate-driving components were remained unchanged.

It should be noted that significant part of static power losses in each gate-driving circuit are belonged to gate-source pulling down resistors those have under 10 kOhm resistance range. And under applying of relatively small values of pulling down resistors (under 1 kOhm) the absolute power losses in gate drive subsystem are increased more than absolute value in Watts given by system efficiency increasing could be brought.

Despite all positive fact, descried above, the gate voltage stress is also increased. It is due to fact that the actual voltage value is being close to the maximum possible value applied to the gate electrode according to datasheet. Therefore, to prevent gate damages, gate-driving chain must be equipped with the full-featured set of passive protection components – limiting and pulling down resistors, force-closing Shottky diode in parallel with the current limiting resistor and fast Zener diode in parallel to the gate-source port.

6 Conclusions

The physical modeling of the system based on the quasi-z-source inverter fully constructed with silicon carbide semiconductors has been performed in this work. The goal of research to analyze the impact of gate drive DC-DC converters on the full-SiC QZSI efficiency was achieved by experimental measurements.

It was obtained that QZSI demonstrates over 97% peak efficiency within the different modulation indices and different shoot-through duty cycle values.

At the same time, the increasing of the nominal output voltage level of the gate drive DC-DC converter from 15 to 20 V could improve the overall efficiency in the certain points from 0.2% up to 0.5% while total power consumption by gate drivers is increased on 1.5–2.0 W and remains almost constant in whole operation range.

Due to its proven high efficiency and relative simplicity, the developed solution is feasible for usage in autonomous electrical power systems, especially highly-efficient photovoltaic applications.

Acknowledgements This research was basically supported by the European Regional Development Fund and the program Mobilitas Pluss under the project MOBJD126 awarded by the Estonian

Research Council and also supported by the Ministry of Education and Science of Ukraine, research grants #0120U101554 and #0122U001522.

References

1. Peng, F.Z.: Z-source inverter. IEEE Trans. Ind. Appl. **39**, 504–510 (2003)
2. Kalaiarasi, N., Paramasivam, S. Kuntu, S.: Comparison of Z-source Inverter with DC-DC Boost converter fed VSI for PV applications. In: IEEE 2nd International Conference on Electrical Energy Systems, pp. 87–91 (2014)
3. Kumar, A., Bhattacharya, A.: Three level Z source inverter based photovoltaic power conversion systems. In: 41st Annual Conference of the IEEE Industrial Electronics Society, pp. 001357–001362 (2015)
4. Li, Y., Anderson, J., Peng, F.Z., Liu, D.: Quasi-Z-source inverter for photovoltaic power generation systems. In: 24th Annual IEEE Applied Power Electronics Conference and Exposition, Washington, DC, pp. 918–924 (2009)
5. Huang, S., Zhang, Y., Hu, S.: Stator current harmonic reduction in a novel half quasi-Z-source wind power generation system. Energies **9**, 770 (2016)
6. Priyadarshi, N., Padmanaban, S., Ionel, D.M., Mihet-Popa, L., Azam, F.: Hybrid PV-wind, micro-grid development using quasi-Z-source inverter modeling and control—experimental investigation. Energies **11**, 2277 (2018)
7. Choi, W.-Y., Yang, M.-K.: Transformerless quasi-Z-source inverter to reduce leakage current for single-phase grid-tied applications. Electronics **8**, 312 (2019)
8. Qawaqzeh, M., et al.: Development of algorithm for the operation of a combined power supply system with renewable sources. In: 3rd KhPI Week on Advanced Technology. IEEE (2022)
9. Halko, S., Halko, K., Suprun, O., Qawaqzeh, M., Miroshnyk, O.: Mathematical modelling of cogeneration photoelectric module parameters for hybrid solar charging power stations of electric vehicles. In: 3rd KhPI Week on Advanced Technology. IEEE (2022)
10. Mookken, J., Agrawal, B., Liu, J.: Efficient and compact 50kW Gen2 SiC device based PV string inverter. In: PCIM Europe 2014; International Exhibition and Conference for Power Electronics, Intelligent Motion, Renewable Energy and Energy Management, Nuremberg, pp. 1–7 (2014)
11. Öztürk, S., Canver, M., Çadırcı, I., Ermiş, M.: All SiC grid-connected PV supply with HF link MPPT converter: system design methodology and development of a 20 kHz, 25 kVA prototype. Electronics **7**, 85 (2018)
12. Stepenko, S., Husev, O., Vinnikov, D., Roncero-Clemente, C., Pires Pimentel, S., Santasheva, E.: Experimental comparison of two-level full-SiC and three-level Si–SiC quasi-Z-source inverters for PV applications. Energies **12**, 2509 (2019)
13. Fernández, E., Paredes, A., Sala, V., Romeral, L.: A simple method for reducing THD and improving the efficiency in CSI topology based on SiC power devices. Energies **11**, 2798 (2018)
14. Wolski, K., Zdanowski, M., Rabkowski, J.: High-frequency SiC-based inverters with input stages based on quasi-Z-source and boost topologies—experimental comparison. IEEE Trans. Power Electron. **34**(10), 9471–9478 (2019)
15. Moradpour, M., Pirino, P., Losito, M., Franke, W.-T., Kumar, A., Gatto, G.: Multi-objective optimization of the gate driver parameters in a SiC-based DC-DC converter for electric vehicles. Energies **13**, 3720 (2020)
16. Wang, B., Tang, W.: A new CUK-based Z-source inverter. Electronics **7**, 313 (2018)
17. Wang, B., Tang, W.: A novel three-switch Z-source SEPIC inverter. Electronics **8**, 247 (2019)
18. Zhuk, A.K., Zhuk, D.A., Krivoruchko, D.V.: Voltage harmonic distortion in autonomous electric power system with an adjustable power line conditioner. Tekhnichna Elektrodynamika **5**, 26–30 (2018)

19. Gorodnyi, A.N.: Analyzing of transistor switch dissipation power in sequential type switched-mode and quasi-resonant zero current switch converters. Tekhnichna Elektrodynamika **3**, 75–76 (2012)
20. Nguyen, M., Choi, Y.: PWM control scheme for quasi-switched-boost inverter to improve modulation index. IEEE Trans. Power Electron. **33**(5), 4037–4044 (2018)
21. Nguyen, M.-K., Choi, Y.-O.: Maximum boost control method for single-phase quasi-switched-boost and quasi-Z-source inverters. Energies **10**, 553 (2017)
22. Roncero-Clemente, C., Stepenko, S., Husev, O., Romero-Cadaval, E., Vinnikov, D.: Maximum boost control for interleaved single-phase Quasi-Z-Source inverter. In: 43rd Annual Conference of IEEE Industrial Electronics Society, Beijing, China, pp. 7698–7703 (2017)
23. Al-Issa, H.A., Qawaqzeh, M., Kurashkin, S., Halko, S., Kvitka, S., Vovk, O., Miroshnyk, O.: Monitoring of power transformers using thermal model and permission time of overload. Int. J. Electr. Comput. Eng. **12**(3), 2323–2334 (2022)
24. Gorodny, A., Dymerets, A., Kut, Y., Denisov, Y., Natalia, D.: Generalized method of commutation processes calculation in high-frequency switched-mode power converters. In: Palagin, A., Anisimov, A., Morozov, A., Shkarlet, S. (eds.) Mathematical Modeling and Simulation of Systems, vol. 1019, pp. 71–80. Springer, Cham (2020). https://doi.org/10.1007/978-3-030-25741-5_8
25. Zhuk, O., Zhuk, D., Kryvoruchko, D., D'yakonov, O.: Control of improved hybrid power line conditioner. In: IEEE 38th International Conference on Electronics and Nanotechnology, pp. 605–610 (2018)
26. Roncero-Clemente, C., Husev, O., Stepenko, S., Vinnikov, D., Romero-Cadaval, E.: Output voltage control system for a three-level neutral-point clamped quasi-Z-source inverter. Przegląd Elektrotechniczny **89**(5), 76–80 (2013)
27. Nathan, K., Ghosh, S., Siwakoti, Y., Long, T.: A new DC–DC converter for photovoltaic systems: coupled-inductors combined cuk-SEPIC converter. IEEE Trans. Energy Convers. **34**(1), 191–201 (2019)
28. Silva, C.E., Alzamora, A.M., Paula, H.D.: Broad comparison of multilevel inverter topologies operating in hostile environments: DC transmission applicability and feasibility. IEEE Trans. Ind. Appl. **58**(5), 6852–6863 (2022)
29. Darbas, C., Olivier, J.-C., Ginot, N., Poitiers, F., Batard, C.: Cascaded smart gate drivers for modular multilevel converters control: a decentralized voltage balancing algorithm. Energies **14**, 3589 (2021)
30. Suh, J.-D., Yun, Y.-H., Kong, B.-S.: High-efficiency DC–DC converter with charge-recycling gate-voltage swing control. Energies **12**, 899 (2019)

Simulation of Processes in the Problem of Blind Signal Identification

Dmytro Kucherov and Natalia Khalimon

Abstract The paper deals with the processes modeling for the problem of blind identification of signals arising from multi-channel reception or multi-sensor processing of signals using linear algebra methods. A brief review of existing algorithms has been given, in which we show the current state of the methods used, their main capabilities, and existing limitations. Based on the presented problem statement of blind signal identification, we considered the prototype of solutions. We focused on the popular algorithms MUSIC, AMUSE, and SOBI using second-order signal statistics. Based on the results of the presented algorithms analysis, the advantage of preliminary signal whitening is established to eliminate unfavorable trends in the signal, the effect of shifting the signal matrix eigenvalues by the noise variance value, and the joint diagonalization of the set of signal matrices obtained because of correlation shifts relative to zero. Modeling of typical processes is proposed, including both a mixture of sinusoidal signals, and sinusoidal signals with noise, as well a time series of normally distributed point sequences to represent the effectiveness of the proposed algorithms. Finally, in the course of the obtained results discussion, the general stages of the algorithm are formulated, which provide effective results for the blind identification of signals. The presented illustrative examples are confirmed by a program code developed for the Matlab environment, which allows researchers to be used to reproduce the main results of the article.

Keywords Blind source separation · Simulation · Joint diagonalization

1 Introduction

There is a need to detect the emitted signal in radar, radio communications, and radio navigation. In the conditions receiving a set of radio signals, we are forced to accept the one mixing with interfering signals and further estimate its parameters. A similar task in radar detection occurs, for example, when it is necessary to distinguish the

D. Kucherov (✉) · N. Khalimon
National Aviation University, 1 L. Huzara Ave., Kyiv 03058, Ukraine
e-mail: d_kucherov@ukr.net; dmytro.kucherov@npp.nau.edu.ua

S. Shkarlet et al. (eds.), *Mathematical Modeling and Simulation of Systems*, Lecture Notes in Networks and Systems 667,
https://doi.org/10.1007/978-3-031-30251-0_21

number of targets in the group. When searching for a signal, it is required to find one and determine its parameters, namely the type, amplitude, phase, and direction of arrival. When establishing radio communication, exist a similar problem associated with detecting the initial signal from multiple copies due to the multipath propagation in a heterogeneous space.

Among the blind methods of estimating the parameters of radio signals, the methods MUSIC, AMUSE, SOBI, and others have become widespread. It is advisable to use the agreed filtering and threshold processing at the primary processing stage. Unlike other known methods of identification based on search-gradient approaches, methods of linear algebra are used here, namely decomposition by eigenvalues or singular values. In this case, it also exists the task of filtering the emitted signals from their additive mixture and interference noise.

The paper aims to model the processes used in existing approaches to the problem of blind identification of signals to build efficient research algorithms.

2 Related Works

The works [1–4] offer a detailed review of blind identification methods that exist to date. The main algorithms presented in the above overviews are based on second-order signal statistics, although their sufficiency for solving this problem until proven yet. In works [1, 2], one can also find the main signal extraction approaches used in a broad range of systems, including systems for digital communications. Paper [3] presents modern techniques used for blind channel equalization used in signal processing in digital communication channels. Here the primary efforts are directed at the iterative approach for algebraic solutions, where each step optimizes some contrast function for searching direction for the next iteration. This approach makes it possible to ensure robustness, has a high convergence rate, and allows to overcome the difficulties of finding solutions. The work [4] is devoted to a broad class of problems.

One of the first works, inspired by the results of the ESPRIT and MUSIC algorithms (see [1]), were formally set the task of the input signal frequency estimating, but also a detailed description of the procedure for restoring the received signal and the channel, is the work [5]. Here, an algorithm for multiple unknown signals extraction (AMUSE) has been proposed; this algorithm uses the second-order statistic of stationary signals. The results of this work were refined in [6], where the proposed algorithm for blind identification has also based on second-order statistics (SOBI). Following this algorithm, the obligatory stage of processing is the whitening of the noisy signal and the use of joint diagonalization of the correlation matrices obtained as a result of orthogonal transformations of the observed sequences.

The practical orientation of blind signal separation in the problem of antenna beamforming has been represented by works [7–9]. The SCORE algorithm based on adaptive blind extraction of signals received from a narrowband antenna array has been presented by authors [7]. This approach makes significant use of the spectral properties of the received signals and is based on the weight processing of the input signal. The convergence criterion is the proximity of the input signal to training one that is measured by the minimum distance square between these signals. A distinctive feature of [8] is the use of 4th order signal statistics, however, here, too, there are problems in choosing cumulant matrices, and noise also has a strong influence. The approach proposed in [9] essentially relies on some properties of signals, which include a constant signal amplitude, also known as the constant modulus algorithm, and the direction of signal arrival.

In [10], the modeling of blind signal separation is proposed based on the combined use of the recursive least squares method, which assumes the use of general and natural gradients, which allows the authors to avoid the signal whitening procedure. An empirical mode decomposition algorithm filters the leakage of noise. The presented results do not allow evaluating the effectiveness of the proposed approach. The quality of signal recovery is also unclear.

The authors of [11] used a neural network to separate colored signal sources with a small kurtosis. Despite the satisfactory results of the simulation, the authors miss the issues related to setting up and fitting the network to the required results.

The main positive result based on the algebraic approach of blind separation of several signals is the recently presented algorithm for simultaneous diagonalization of a set of matrices.

It is based on an iterative algorithm of plane rotations, where a result is a unitary matrix. Its multiplication provides a common diagonalization of a set of matrices. The essence of this algorithm is described in [12], and the influence of small additive perturbations on a unitary matrix that takes place in calculations, is presented in [13].

In [14], the statistical properties of weakly correlated processes are studied, namely, the effect of delay on the correlation properties of covariance matrices. When studying these properties, the authors of [14] found that the SOBI algorithm is better than AMUSE, but it is far from perfect. They also confirmed the conclusion made in [4] about the choice of delay for calculating covariance data based on a preliminary study of the estimated processes or some other additional information.

An improvement of the SOBI method based on its addition to the sliding window method is presented in [15]. The efficiency of such a procedure essentially depends on the length of the window and, therefore, is not a completely solved problem.

This work is continuing research of [16, 17], where the problem of detecting a signal and measuring its parameters was solved by alternative approaches, namely, in [16] it was proposed to identify the signal frequency after preliminary frequency filtering, and in [17] an analysis of the parameters of the chirp signal was proposed by using time–frequency analysis.

3 Problem Statement

We consider a system whose inputs receive a signal

$$x(t) = As(t) + n(t), \tag{1}$$

where $x(t) \in R^n$ is the observed vector, which is formed from the $s(t) \in R^m$ emitted vector by m unknown signal sources; $A \in R^{n\times m}$ is the transfer matrix from m sources to n sensors; $n(t)$ is interfering signal or noise; t is the moment of observation.

In traditional signal processing, the receiving system identifies the signal $s(t)$ by the vector $x(t)$, using additional information, for example, about the known parameters of the signal $s(t)$ or, for example, the direction of its arrival. Here, another problem is considering when a priori information about the signal $s(t)$ and the transfer matrix A is almost non-exist.

The task of the researcher is to restore the signals $s(t)$, to establish their number under conditions of unknown noise, on which restrictions can be imposed in the form

$$|\xi_i(l)| \le N, \tag{2}$$

where N is the permissible noise level. An assumption was made about the independence of signal sources. In addition, we assume that the system is stationary and signals are mixing linear.

A1. Signals $x(t)$ and $n(t)$ are stationary and have zero averages;

A2. Signals $x(t)$ and $n(t)$ are statistically independent and do not depend on each other;

A3. The number of sensors is not less than the number of signal sources, i.e. $n \ge m$.

A4. Signal sources have unit variance.

A5. It is also proposed that the system without memory, i.e. the output signal of the previous system's unit is the input for the next one.

The task is to set the number of signals m, to obtain estimates of input signals using some matrix V from the observation vector $x(t)$, i.e. define

$$\hat{s}(t) = Vx(t) \tag{3}$$

4 Problem Solution

As in [4–6], to solve the problem, it is supposed to use one of the classes of spectral methods based on the analysis of the eigenvalues of the covariance or correlation matrix of data, which provides better estimation characteristics than methods, for example, based on autoregression.

4.1 MUSIC

Following the MUSIC method, the solution to the problem is based on the expansion of the covariance matrix $R_x \in R^{n\times n}$ in terms of eigenvalues. In this case, the peak values of the spatial spectrum corresponding to the estimates of the frequency of the input sequence of vectors $x(t)$ are determined by the expression

$$P_{MUSIC}(f) = \frac{1}{\sum\limits_{i=m+1}^{n} \left|a_i(f)\hat{v}_i\right|^2} \tag{4}$$

where $a_i(f)$ are the vectors of the matrix component $A = [a_1(f), a_2(f), \ldots, a_m(f)]^T$, which are weight coefficients for the estimates of the eigenvector $\hat{V} = [\hat{v}_1, \hat{v}_2, ..., \hat{v}_n]$ obtained by decomposing the matrix

$$\hat{R}_x = E\left\{(x-\overline{x})(x-\overline{x})^T\right\} \tag{5}$$

by eigenvalues λ_i, and eigenvectors v_i satisfying the equation

$$A^H v_i = 0, i = m+1, m+2, \ldots, n. \tag{6}$$

In (5) $E\{\cdot\}$ is math expectation, $\overline{x}$ is the average.

The main disadvantage of this approach is the initial uncertainty about the number of signals processed by the algorithm. There are likely several users, such as mobile users, who use one information channel at the same time and therefore interfere with each other, which is called inter-channel interference.

In digital communication systems, the effect of the multipath propagation naturally occurs when the signal is emitted by a single source, but due to reflection from connected surfaces, this one re-reflecting in many directions. This phenomenon in single-channel reception causes so-called inter-symbol interference, which distorts the estimated signal. Thus, the uncertainty regarding the receiving sample is the main problem of blind signal extraction.

4.2 AMUSE

To eliminate this shortcoming, the authors [5] proposed an algorithm for extracting a set of unknown signals AMUSE (Algorithm for Multiple Unknown Signals Extraction). According to this algorithm, a singular decomposition is performed after the procedure (5) is ended

$$R_x = U\Lambda U^T \tag{7}$$

where Λ is a diagonal matrix of dimension $\Lambda \in R^{n\times n}$ with components $(\lambda_1^2+\sigma^2, \lambda_2^2+\sigma^2, ..., \lambda_m^2+\sigma^2, .\sigma^2, ..., \sigma^2)$ on the main diagonal, and U is a unitary matrix of the same dimension.

After defining singular values corresponding to signal sources and noise components of the matrix Λ, the elements of the main diagonal matrix are shifted by subtracting σ^2. Next, a matrix of nonzero eigenvalues is formed, i.e.

$$\Lambda' = diag[\lambda_1^{-1}, \lambda_2^{-1}, ..., \lambda_m^{-1}] \tag{8}$$

The following steps of the algorithm consist of the orthogonal transformation of the observation vector $x(t)$, i.e. $y(t) = \Lambda' x(t)$, choosing the correlation shift τ, and calculating the correlation matrix

$$\hat{R}_y = E\left\{(y-\overline{y})(y-\overline{y})^T\right\} \tag{9}$$

its symmetrization by calculation

$$T = (R_y + R_y^T)/2 \tag{10}$$

and obtaining a unitary matrix V after decomposing T in terms of eigenvalues. Then the estimated signal can be obtained in the form $\hat{S} = VY$, and the estimate of the channel matrix as $\hat{A} = U\Lambda' V$.

4.3 SOBI

By authors [6] were noticed the stability of the second-order statistics to noise components, and the AMUSE algorithm has been strengthened by the procedure of joint diagonalization of correlation matrices. Unlike AMUSE, after the operation (5), the signal is whitened to eliminate trends in receiving one. For operation (8), several constant correlation shifts $\tau = 1, 2, 3,\ldots$ are selected, for which correlation matrices need to calculate. These actions allow the restoration of a narrowband signal, such as a speech signal, by one unitary matrix, which corresponds to the decomposition by singular values of these correlation matrices. The authors [6] named this algorithm Second Order Blind Identification (SOBI).

For a given set of matrices $A = \{A_1,\ldots, A_k\}$ the unitary matrix V is such that diagonalizes the set of matrices A if the condition

$$\sum_{j=1}^{k} off(VA_jV^H) = \min_{a_{ij}}(\sum_{1\le i\ne j\le k}^{n} |a_{ij}|^2) \tag{11}$$

which is achieved by the iterative selection of Jacobi angles [12, 13].

In addition, unlike [6], the algorithm is supplemented by a preliminary estimate of the number of signal sources.

5 Simulation

The problem of identification of three harmonic signals operating at frequencies $f_1 = 0.1\ \mathrm{s}^{-1}, f_2 = 0.5\ \mathrm{s}^{-1}, f_3 = 0.8\ \mathrm{s}^{-1}$ is considering. These signals are summed and mixed with a noise signal last obeys a standard normal distribution. The sum signal is processed by the previously discussed algorithms. The form of these signals and the sum is shown in Fig. 1.

Figure 2 shows the amplitude spectrum of the sum signal obtained by the MUSIC method for the case when the presence of a) one signal, b) two signals, c) and three signals. Graph built in the Matlab environment, we used the pmusic() function.

On Fig. 3, 4, and 5 show the actions of the AMUSE algorithm in the case of recovering one signal f_1 without noise (see Fig. 3) and in the presence of noise (see Fig. 4), an additive mixture of signals f_1 and f_3 (see Fig. 5).

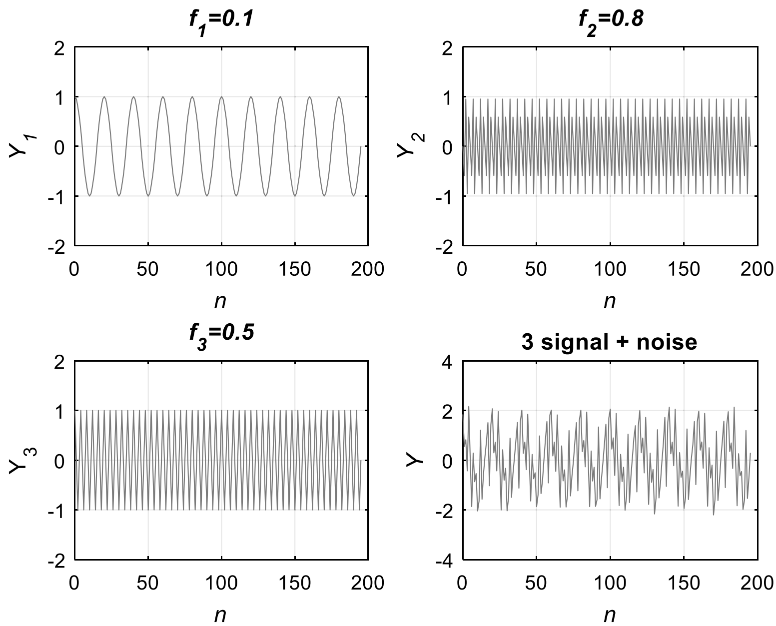
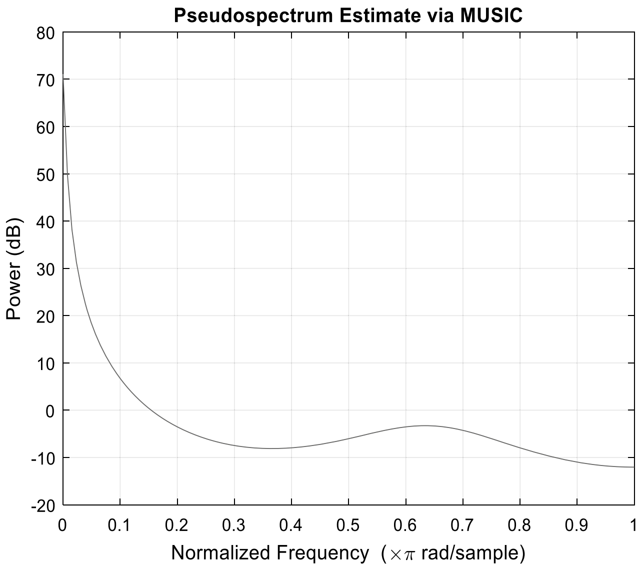
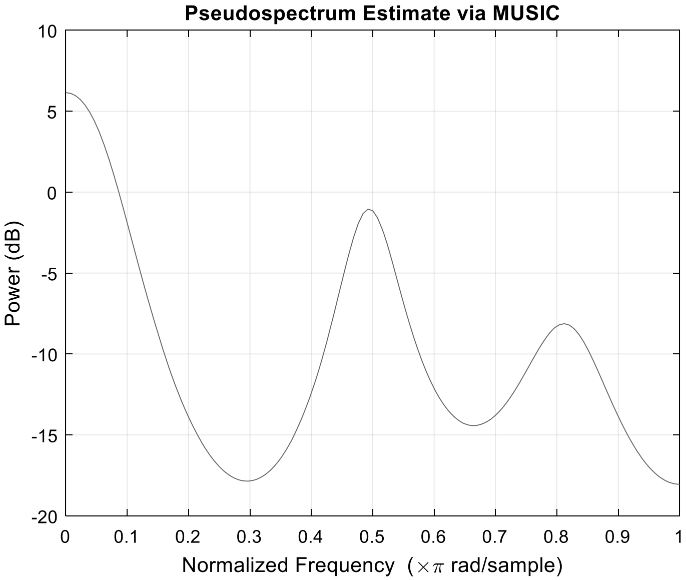


Fig. 1 Three harmonic signals and their additive mixture with noise

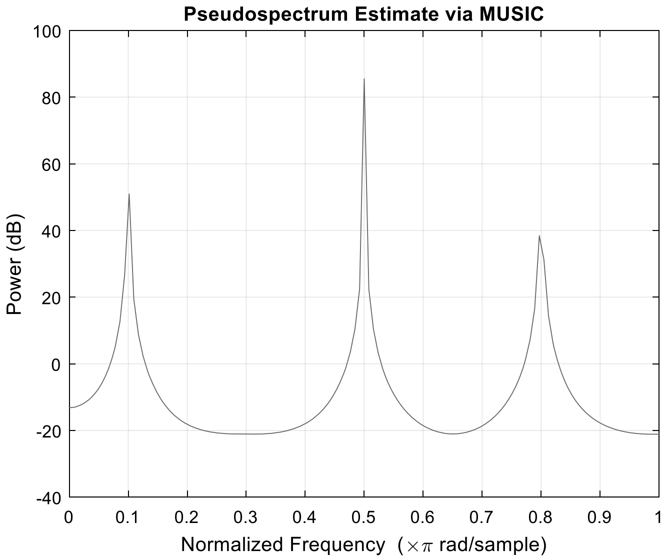


a)

Fig. 2 Graphs of the estimates of frequency for a mixture of three harmonic signals and noise with the standard normal distribution: a) one signal, b) two signals, c) and three signals

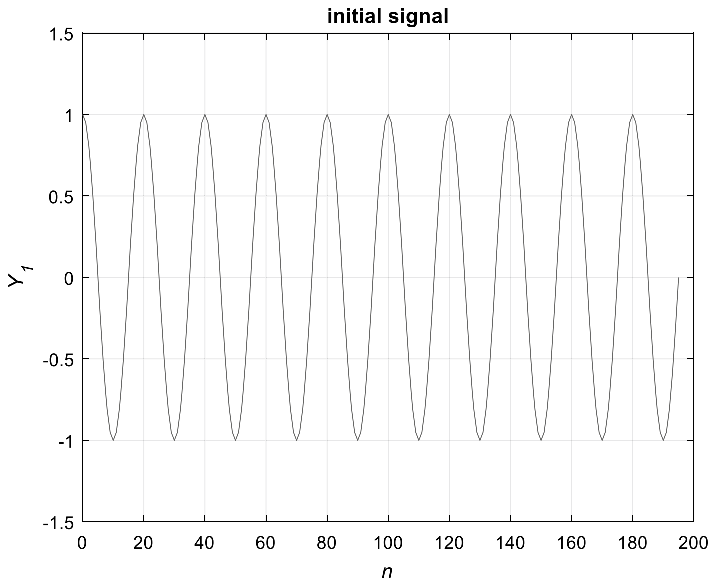


b)

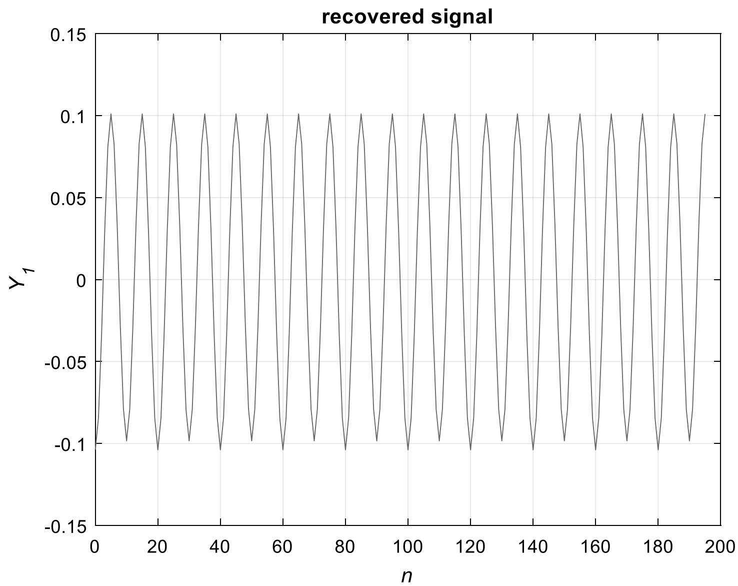


c)

Fig. 2 (continued)



a)



b)

Fig. 3 Graphs of original *a*) and recovered signal *b*) without noise

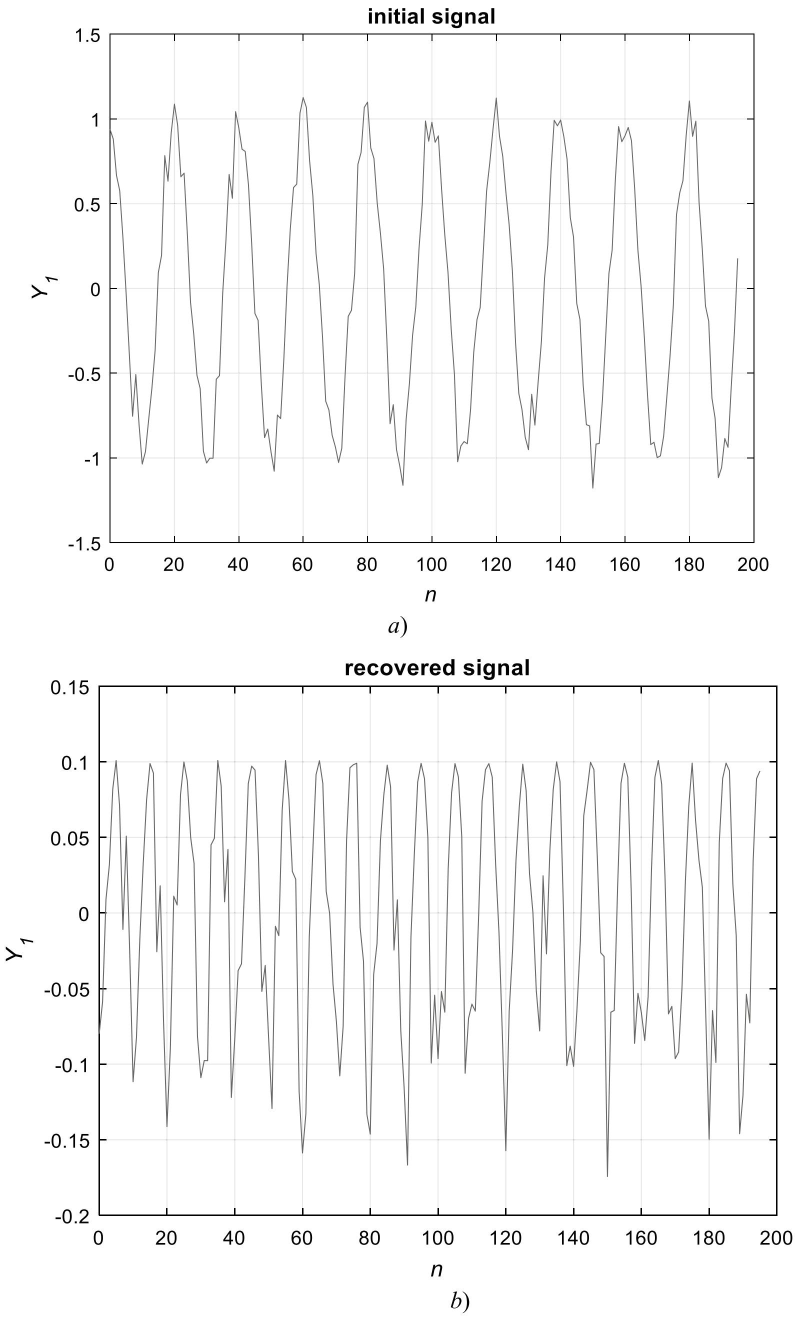
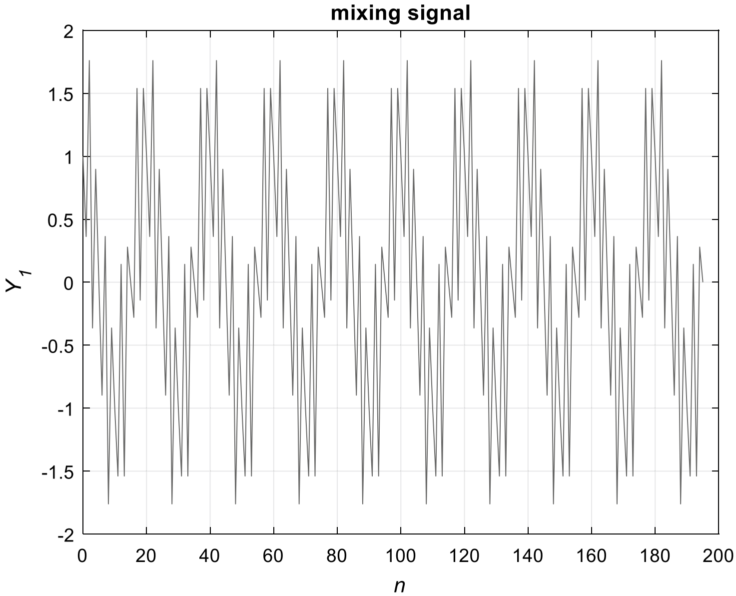
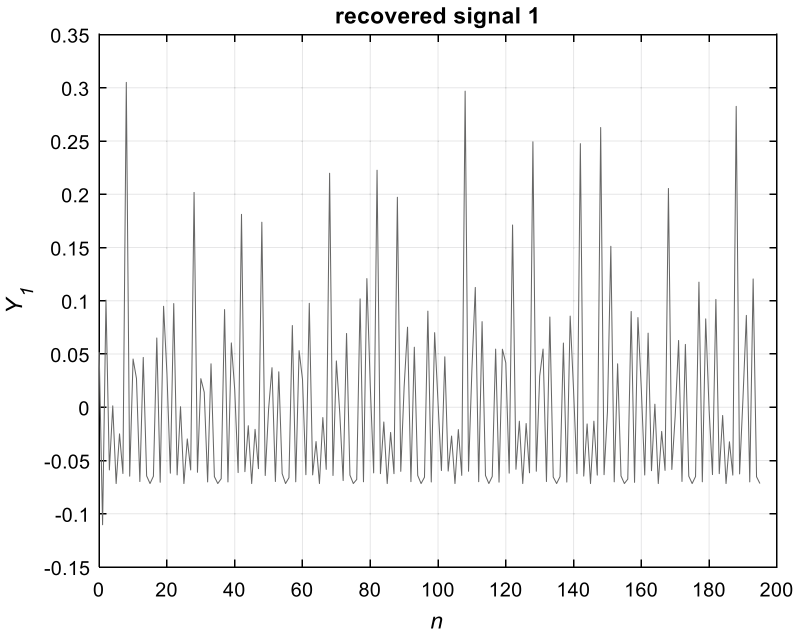


Fig. 4 Graphs of the original *a*) and recovered signal *b*) with noise

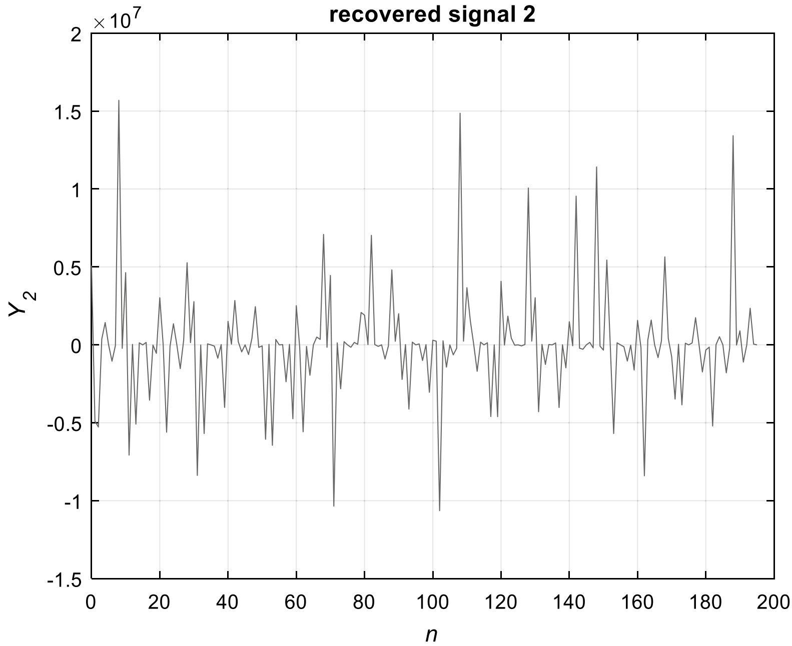


a)



b)

Fig. 5 Graphs of the mixture of signals with frequencies f_1 and f_3 *a*) and the reconstructed signals 1 *b*) and 2 *c*) without noise



c)

Fig. 5 (continued)

Matlab code for the AMUSE algorithm.

```
  % covariance matrix
  M = ones(1,length(Y))*mean(Y); % vector of mean values
  Zc = Y - M;                    % shift
  Rxx = (Zc'*Zc)./length(Y);     % covariance matrix
  % 2. singular value decomposition
  [Ux,Dx,Vx]=svd(Rxx);
   Dx=diag(Dx);
   if n<m, % under assumption of additive white noise and
          %when the number of sources are known or can a priori
estimated
    Dx=Dx-real((mean(Dx(n+1:m))));
    Q= diag(real(sqrt(1./Dx(1:n))))*Ux(:,1:n)';
  %
  else    % under assumption of no additive noise and when the
          % number of sources is unknown
     n=max(find(Dx>1e-199)); %Detection the number of sources
     Q= diag(real(sqrt(1./Dx(1:n))))*Ux(:,1:n)';
  end;
   % data transformation
   Dx=Dx-real((mean(Dx(n+1:m))));
   Q= diag(real(sqrt(1./Dx(1:n))))*Ux(:,1:n)';
  Yb=Q.*Y;
   % correlation matrix tau=1
   N=max(size(Yb));
   Yb=Yb-kron(mean(Yb')',ones(1,N));
   Rybybp=(Yb(:,1:N-1)*Yb(:,2:N)')/(N-1);
   % symmetrization & eigen value decomposition
   Rybybp= (Rybybp+Rybybp')/2;
   [Vyb Dyb]=eig(Rybybp);
   [D1 perm]=sort(diag(Dyb));
   D1=flipud(D1);
   Vyb=Vyb(:,flipud(perm));
   W = Vyb'*Yb;
```

Figure 6 shows the three-time series of normally distributed numerical sequences consisting of 1000 points. The sequence s_1 represents a process of autoregression with coefficients given by numbers (0.3, 0.6), s_2 is a moving average process with coefficients (−0.3, 0.3), and s_3 is also a process of autoregression with coefficients given by coefficients (−0.8, 0.1). Figure 7 shows the result of mixing these processes. The results of the SOBI algorithm are shown in Fig. 8. Analysis of the presented results shows this algorithm is more effective.

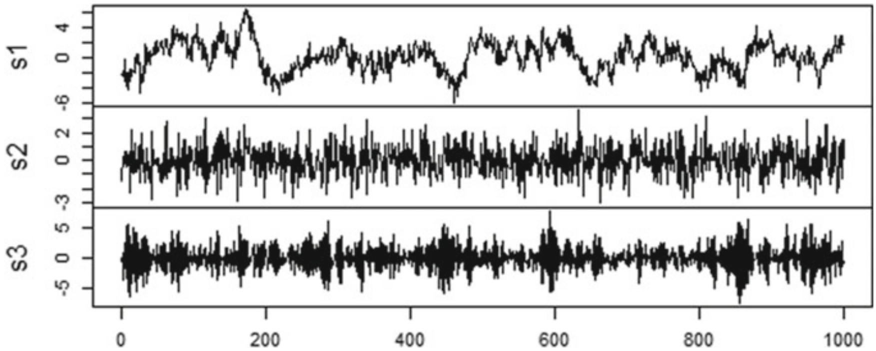


Fig. 6 Three-time series are defined by autoregression and moving average processes

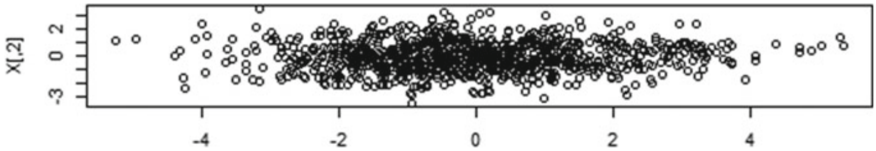


Fig. 7 A mixed number series

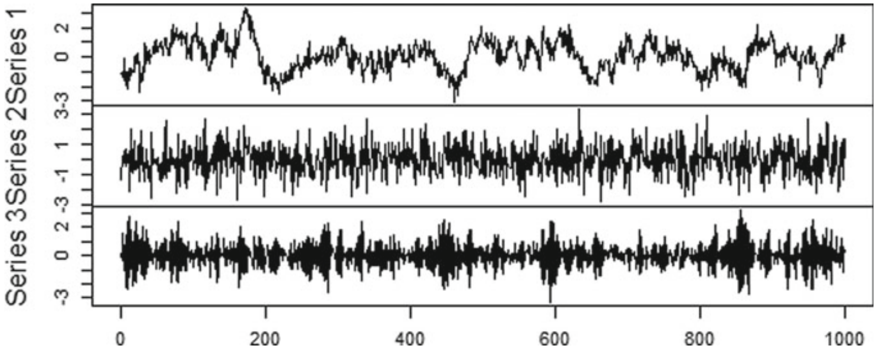


Fig. 8 Recovered time series by the SOBI algorithm

The following Matlab code calculates the joint diagonalization procedure.

```
% Init
V   = eye(m);
encore  = 1;
while encore, encore=0;
 for p=1:m-1, Ip = p:m:nm ;
 for q=p+1:m, Iq = q:m:nm ;
    % Computing the Givens angles
        g       = [ A(p,Ip)-A(q,Iq)  ; A(p,Iq) ; A(q,Ip) ] ;
        [vcp,D] = eig(real(B*(g*g')*Bt));
        [la, K] = sort(diag(D));
        angles  = vcp(:,K(3));
    if angles(1)<0 , angles= -angles ; end ;
        c       = sqrt(0.5+angles(1)/2);
        s       = 0.5*(angles(2)-j*angles(3))/c;
        if abs(s)>jthresh, % updates matrices A and V by a
%Givens rotation
                encore          = 1 ;
                pair            = [p;q] ;
                G               = [ c -conj(s) ; s c ] ;
                V(:,pair)       = V(:,pair)*G ;
                A(pair,:)       = G' * A(pair,:) ;
                A(:,[Ip  Iq])      =  [  c*A(:,Ip)+s*A(:,Iq)  -
conj(s)*A(:,Ip)+c*A(:,Iq) ] ;
     end% if
    end% q loop
   end% p loop
end% while
```

6 Conclusions

The paper considers the possibilities of existing tools for solving the blind identification of signals problem on the example of harmonic signals and time series, including the sum signal with noise. The study presents the results of applying well-known methods based on the algebraic approach of extracting unknown signal components. So, for example, the MUSIC algorithm, which assumes the selection of signal peaks, seems to be more accurate for measuring the signal frequency. The AMUSE and SOBI algorithms are also able to extract the waveform. Some of the obtained results may seem unsatisfactory, for example, the dependence of the MUSIC algorithm on a priori information about the number of signal sources, and the extraction of a signal distorted by noise using the AMUSE algorithm. We also introduced a preliminary estimate of signal sources into the algorithm. The obtained results confirm the necessity of signal preprocessing based on the whitening of the input information, as is done in SOBI, and the efficiency of the joint diagonalization of matrices, which is preferable for separating signals of several types. An analysis of the SOBI algorithm showed that this algorithm is more general for separating different signals. To solve the blind source separation problem, Matlab codes of some programs are presented. Further research focuses on improving the SOBI algorithm by preprocessing the input data and increasing the order of the statistics of the input signals.

References

1. Schmidt, R.O.: Multiple emitter location and signal parameter – estimation. IEEE Trans. Antennas Propag. AP-**34**(3), 276−280 (1986)
2. Yeredor, A.: Second-order methods based on color. In: Handbook of Blind Source Separation, pp. 227–279 (2010). https://doi.org/10.1016/B978-0-12-374726-6.00012-6
3. Zarzoso, V., Comon, P., Slock, D.: Semi-blind methods for communications. In: Handbook of Blind Source Separation, pp. 593–638. Academic Press (2010). https://doi.org/10.1016/B978-0-12-374726-6.00020-5
4. Cichocki, A., Amari, S.-I.: Adaptive blind signal and image processing. Willey, West Sussex (2002). https://www.wiley.com
5. Tong, L., Soon, V.C., Huang, Y.F., Liu, R.: AMUSE: a new blind identification algorithm. In: IEEE International Symposium on Circuits and Systems, New Orleans, LA, USA, vol. 3, pp. 1784–1787 (1990). https://doi.org/10.1109/ISCAS.1990.111981
6. Belouchrani, A., Abed-Meraim, K., Cardoso, J.-F., Moulines, E.: A blind source separation technique using second-order statistics. IEEE Trans. Signal Process. **45**(2), 434–444 (1997). https://doi.org/10.1109/78.554307
7. Agee, B.G., Shell, S.V., Gardner, W.A.: Spectral self-coherence restoral: a new approach to blind adaptive signal extraction using antenna arrays. Proc. IEEE **78**(4), 753–767 (1990). https://doi.org/10.1109/5.54812
8. Cardoso, J.-F., Souloumiac, A.: Blind beamforming for non-Gaussian signals radar and signal processing. IEE Proc. F **140**(6), 362–370 (1994). https://doi.org/10.1049/ip-f-2.1993.0054
9. Van Der Veen, A.-J.: Algebraic methods for deterministic blind beamforming. Proc. IEEE **86**(10), 1987–2008 (1998). https://doi.org/10.1109/5.720249
10. Luo, Z.: Fetal electrocardiogram extraction using blind source separation and empirical mode decomposition. J. Comput. Inf. Syst. **8**(12), 4825–4833 (2012). https://www.semanticscholar.org/paper/Fetal-Electrocardiogram-Extraction-using-Blind-and-Luo/a4b1d28a187a6e9135a7307accc978352b35a7f9
11. Cichocki, A., Thawonmas, R.: On-line algorithm for blind signal extraction of arbitrarily distributed, but temporally correlated sources using second order statistics. Neural Process. Lett. **12**, 91–98 (2000). https://link.springer.com/article/10.1023/A:1009616029367
12. Cardoso, J.-F., Souloumiac, A.: Jacobi angles for simultaneous diagonalization. SIAM J. Matrix Anal. Appl. **17**(1), 1–3 (1996). https://doi.org/10.1137/S0895479893259546
13. Cardoso, J.-F.: Perturbation of joint diagonalization. https://arxiv.org/abs/1703.00591. Accessed 29 May 2022
14. Miettinena, J., Nordhausenb, K., Ojab, H., Taskinen, S.: Statistical properties of a blind source separation estimator for stationary time series. Statist. Probab. Lett. **82**(11), 1865–1873 (2012). https://doi.org/10.1016/j.spl.2012.06.025
15. Wang, C., Wang, J., Zhang, T.: Operational modal analysis for slow linear time-varying structures based on moving window second order blind identification. Signal Process. **133**(4), 169–186 (2017). https://doi.org/10.1016/j.sigpro.2016.10.016
16. Kucherov, D.P., Berezkin, A.L.: Identification approach for determining radio signal frequency. In: XI International Conference on Antenna Theory and Techniques (ICATT-2017), Kyiv, Ukraine, pp. 379–382 (2017). https://doi.org/10.1109/ICATT.2017.7972668

17. Kucherov, D.P., Berezkin, A.L., Onikienko, L.S.: Detection of signals from a LoRa system under interference conditions. In: International Scientific-Practical Conference Problems of Infocommunications. Science and Technology (PICS&T-2108), Kharkov, Ukraine, pp. 437–441 (2018). https://doi.org/10.1109/INFOCOMMST.2018.8632135

Simulation of Voltage Distribution Along a Low-Voltage Power Line with Distributed Generation of Private Households

Vadim Bodunov, Anatoliy Prystupa, Tetiana Kulko, and Alexander Gai

Abstract The current trend in the development of the world energy sector is the rapid growth of the number of distributed generation sources (DG) of private households. The peculiarity of private household DG is that investments are proportional and sometimes even less than the cost of additional reconstruction of the power grid. Therefore, the connection of such DG to low voltage networks is carried out without any additional changes in the cross-sections of power lines. Additional interest in household generating capacities appeared after the large-scale invasion of the Russian Federation into the territory of Ukraine. If earlier the main thing for the owner of DG was primarily the commercial aspect, the possibility of a quick return on capital, now the issue of energy independence, redundancy of power sources, starting with low-power autonomous generator sets for emergency power supply of only critical electrical equipment to DG, which can fully meet the energy needs of the owner of a private household, is increasingly being raised. On the other hand, for distribution power grids, there is limited information support on operating parameters. In the absence of reliable information, there is a need to develop approximate methods for estimating the permissible value of the power of private households' DG depending on the place of connection, the parameters of the existing low-voltage distribution network and its operating modes. In this work, the influence of the DG power and the place of its connection on the voltage value along the length of the distribution electric network was modeled. An analytical method for minimizing the voltage deviation in a line with a uniformly distributed load by selecting the power of the DG depending on the place of its connection is developed.

Keywords Low-voltage distribution network · Voltage deviation · Distributed generation · Optimization

V. Bodunov (✉) · A. Prystupa · T. Kulko
Chernihiv Polytechnic National University, Chernihiv, Ukraine

A. Prystupa
e-mail: vad10979@stu.cn.ua; anatoliy.prystupa@stu.cn.ua

A. Gai
National University of Life and Environmental Sciences of Ukraine, Kyiv, Ukraine

S. Shkarlet et al. (eds.), *Mathematical Modeling and Simulation of Systems*, Lecture Notes in Networks and Systems 667,
https://doi.org/10.1007/978-3-031-30251-0_22

1 Introduction

According to official statistics of the State Agency on Energy Efficiency and Energy Saving of Ukraine [1], at the end of 2021, there were 44,888 solar power plants (SPP) in private households with a total power of 1205 MW in Ukraine. One of the main reasons for such a rapid development of solar energy is state incentives in the production and consumption of alternative fuels and energy through a number of organizational and economic measures [2, 3]. Currently, state stimulation of renewable energy is a global trend. Thus, similar to the Ukrainian "green tariff", in European countries there are so-called Feed-in tariffs (FITs) [4].

The statistics on the dynamics of the development of private household power plants in recent years indicates that their number will continue to grow (see Fig. 1).

In 2022, during the full-scale invasion of the Russian Federation into the territory of Ukraine, as a result of missile and artillery strikes on critical infrastructure facilities of settlements, the owners of private households additionally felt the need for backup power sources. And if earlier the main thing for the owner of generating capacities was primarily the commercial aspect, the possibility of a quick return on capital, now the issue of energy independence, backup power sources is increasingly being raised, starting with low-power autonomous generator sets for emergency power supply of only electronics to installations that can fully meet the energy needs of the owner of a private household, that is, up to units and tens of kilowatts. It should be understood that after some time, after the end of martial law, the owners of these installations will want to make a profit from their use, which is possible only by selling the generated electricity. Thus, these sources will become network connected DG in the future.

The main part of state incentives for the development of renewable energy relates to powerful power plants that affect the operating modes of medium and high voltage power grids, however, the increase in the level of automation and the gradual introduction of the Smart Grid concept in the near future will also involve private household power plants in the regulation of low-voltage power grids.

The peculiarity of low-voltage grid-tied DG is that the investments are proportional and sometimes even less than the cost of additional reconstruction, for example,

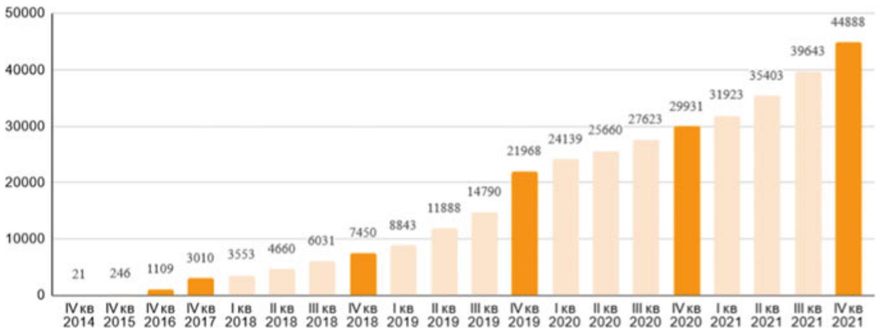


Fig. 1 Dynamics of growth of the number of private households' DG in Ukraine

to increase the bandwidth of the electrical network, so, as a rule, their connection to low-voltage distribution networks is carried out without any additional changes in the cross-sections of power lines.

Another feature of DGs considered in this study is their low power and, as a result, the inability to significantly affect the energy indicators of distribution networks, but at the same time they are able to influence on the voltage level of the low-voltage network to which they are directly connected. Since the voltage levels depend not only on the load on the network, but also on the actual generation of electricity from the DGs [5–10]. To stabilize the voltage, it is necessary to introduce automatic voltage regulation or use energy storage [9]. The use of DGs also affects other indicators of the quality of electricity - it changes the amount of active power losses [11, 25] and distorts the voltage waveform due to the presence of converters that are components of DGs [12, 30, 31], can cause a break in the stability of the power system [32].

Scientists in many countries are exploring the problem of finding effective modes of power generation from renewable sources and their connection points. The paper [13] proposed systematization of the actual models and methods applied to the optimal DG placement problem, based on the analysis of more than 80 works. The studies [18, 23, 24] presented a critical review of different methodologies for optimal placement of distributed generation in power distribution systems. The paper [14] proposed the method for determination of optimal placement and value of installed capacity of renewable energy source by the criterion of minimum losses of active power. The paper [22] proposed the decision of topological task of distributed generation placement using a pareto optimization. The paper [15] considered the main technical problems relating to adoption of distributed generation sources in electric mains of Ukraine. The study [16] presented algorithms for choosing the optimal parameters of distributed generation and places for their interconnection into electrical networks at various nominal voltages. In [17, 25] authors proposed to ensure an effective reduction of electrical losses of energy in distribution networks with local sources of energy generation and accumulation due to possible possibilities of dynamic management of the network configuration. An article [19] presented an optimization approach that employs an artificial bee colony algorithm to determine the optimal DG-unit's size, power factor, and location in order to minimize the total system real power loss. The paper [21] presented a new methodology using particle swarm optimization for the placement of distributed generation in the radial distribution systems to reduce the power loss. The papers [20, 23] proposed to maximize reliability by optimizing of distributed generation location.

As optimization criteria are used:

– total active power losses;
– technological costs of electric energy in the network;
– indicators that characterize the reliability of power supply;
– cost and profit functions;
– voltage quality;
– total generation capacity, etc.

Most of the proposed approaches relate to distributed generation sources with capacities ranging from hundreds of kW to tens to hundreds of MW, while the lack of initial information on the operating modes of existing low-voltage power grids in the private sector makes it impossible to use accurate methods for calculating the modes, so it is advisable to use an approximate model of a transmission line in the form of a line with a uniformly distributed load [26]. At the same time, such a model requires the analysis of the voltage regime along the entire length, or the additional use of discrete methods.

The purpose of this study is to formulate, based on the results of simulation, the criterion for choosing the optimal capacity of a private household's DG, which ensures the minimum voltage deviation along the transmission line and the execution of regulatory requirements for electricity quality.

2 Hypotheses of Research

Due to the fact that adequate measurements of the operating mode of the low-voltage distribution network can be provided only on the main section, it is proposed to use the model of a line with evenly distributed load along the length. This model is quite typical for power lines laid along the streets of the private sector of cities and rural areas. Consider the change of voltage values in a low-voltage distribution electric network of length L with connected DG at a relative distance L_* from the beginning of the line.

The modes are calculated with the following assumptions:

- the distribution network has no branches;
- load is evenly distributed along the line;
- total load power at the beginning of the line is known;
- DG is connected at a relative distance L_* from the start of the line;
- the power of the power system is much larger than the power of the load and the power plant of a private household, which allows to consider the power system as a source of unlimited power with a constant electromotive force;
- voltage loss is determined by the lengthwise component;
- power factor of DG is close to unity.

Under these assumptions, the electrical network model will look as follows (Fig. 2).

The voltage loss from the source to an random point on the area from the power system to the DG $l \leq L_* \cdot L$ will be equal to [17, 18]:

$$\Delta U(l) = \frac{P_L(r_0 + x_0 \cdot tg\phi_L)}{2 \cdot U_{nom} \cdot L} \cdot \left(2L \cdot l - l^2\right) - \frac{P_{DG} \cdot r_0 \cdot l}{U_{nom}} \tag{1}$$

P_L – is the total active power of the transmission line, kW;
$tg\phi_L$ – is the power factor of the total load, a.u.;

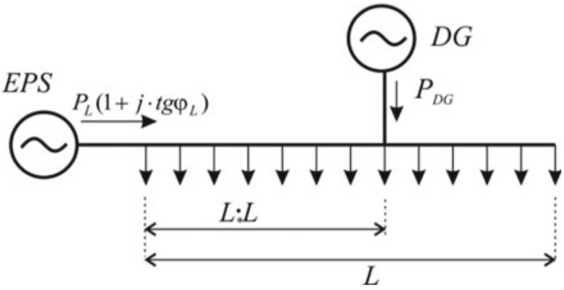


Fig. 2 Electric network model with DG

r_0, x_0 – linear active and inductive impedances of the transmission line with length, Ohm/km;

U_{nom} – is the nominal voltage of the network, kV;

P_{DG} – active power of DG, kW;

L_* – relative distance from the beginning of the line to the DG as a fraction of the total length, in %;

l – coordinate of the current point, km.

and for the section $l > L_* \cdot L$:

$$\Delta U(l) = \frac{P_L(r_0 + x_0 \cdot tg\phi_L)}{2 \cdot U_{nom} \cdot L} \cdot \left(2L \cdot l - l^2\right) - \frac{P_{DG} \cdot r_0 \cdot L_* \cdot L}{U_{nom}} \tag{2}$$

Figure 3 shows an example of modeling the voltage regime along a transmission line with a distributed load when the DG power changes.

The inflection of the characteristic in Fig. 3 corresponds to the distance from the beginning of the line to the DG connection point. The black plane shows the maximum permissible voltage of consumers according to GOST 13,109 [27]. As can be seen, at high values of DG power, the voltage near the DG may exceed the maximum permissible value.

At the same time, in recent years, much attention has been paid to the quality of electrical energy and electromagnetic compatibility, one of the key parameters of which is the effective voltage.

Scientific works devoted to the optimization of operation modes of electric networks take into account this parameter as a technical constraint [6, 8, 10, 12] when using other optimization criteria: power losses, energy, financial indicators.

It is proposed to consider the actual voltage value at the points of common connection as a parameter to be optimized and maximally approximated to the nominal value.

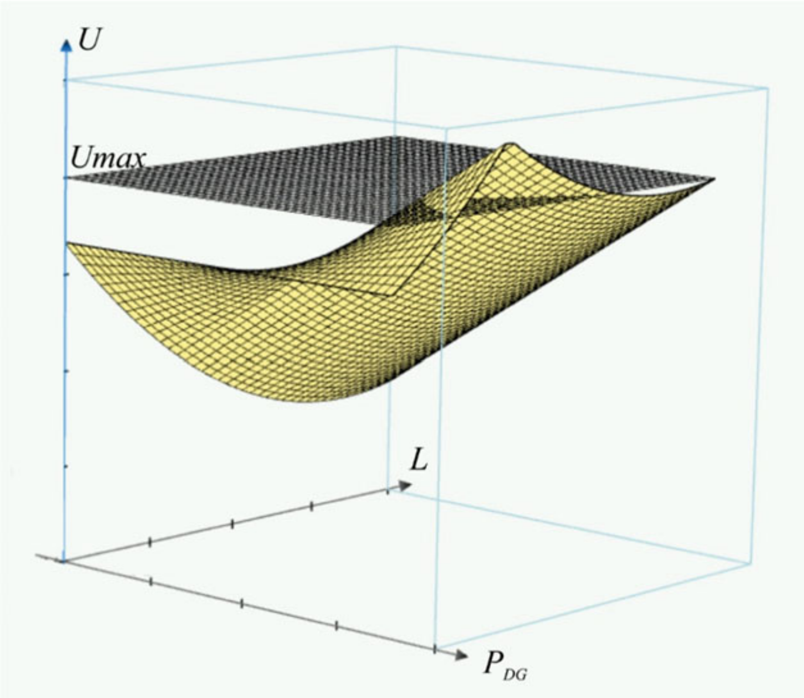


Fig. 3 Voltage distribution along the line when changing the DG power

3 Modeling and Results

Figure 4 shows graphs of voltage distribution along the line for different values of DG power. As it can be seen, all variants correspond to the normally permissible range of voltage deviation [27, 28], but in different points of the network we have different values of voltage deviations. According to the efficiency of the electrical equipment of consumers connected to this network, there is no clear answer, which voltage is better: higher than the nominal or lower. Thus, for different types of electrical receivers (heating elements, induction motors, electronic power supplies, etc.), positive or negative voltage deviations can both improve and worsen the efficiency of their work. Therefore, the optimization problem was set to minimize voltage deviations throughout the network, which corresponds to all points of connection of consumers.

In such a formulation, it is immediately clear that the upper and lower graphs (at generation capacities P_{DG1} and P_{DG4}) are worse than the two middle ones (at generation capacities P_{DG2} and P_{DG3}), since they have larger (by module) voltage deviations on most of the length. At the same time, it is not possible to make a choice between P_{DG2} and P_{DG3} without forming a mathematical description of the proposed criterion.

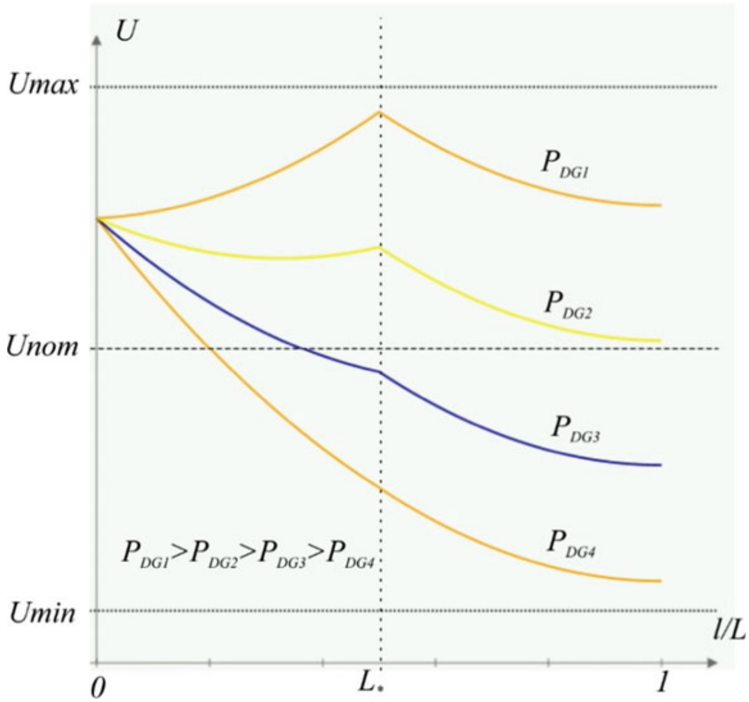


Fig. 4 Voltage distribution along the transmission line for different generation capacities

As such a mathematical apparatus, we propose to use the least squares method, which is widely used in experimental studies [29]. Adapting this method to our problem, we obtain:

$$F = \sum_{i=1}^{N} (U_i - U_{nom})^2 = \sum_{i=1}^{N} (U_0 - \Delta U_i - U_{nom})^2 \rightarrow \min, \tag{3}$$

i – is the number of the consumer connection point, i = 1..N

U_i – voltage at the connection point of the i-th consumer;

U_{nom} – nominal voltage of the network;

U_0 – power supply voltage;

ΔU_i – voltage loss from the source to the connection point of the i-th consumer.

The current coordinate l in formulas (1) and (2) can be represented as

$$l = \frac{L \cdot i}{N} \tag{4}$$

then the voltage loss from the power supply to any point of the network will be

$$\Delta U_i = \begin{cases} \dfrac{P_L(r_0 + x_0 \cdot tg\phi_L)L \cdot i}{2 \cdot U_{nom} \cdot N} \cdot (2 - i) - \dfrac{P_{DG} \cdot r_0 \cdot L \cdot i}{U_{nom} \cdot N}, & i \le k \\ \dfrac{P_L(r_0 + x_0 \cdot tg\phi_L)L \cdot i}{2 \cdot U_{nom} \cdot N} \cdot (2 - i) - \dfrac{P_{DG} \cdot r_0 \cdot L_* \cdot L}{U_{nom}}, & i > k \end{cases} \quad (5)$$

k - DG connection point,

$$k = L_* \cdot N \quad (6)$$

Substituting (5) into (3) and solving with respect to P_{DG}, we obtain:

$$P_{DG} = \frac{3 \cdot U_{nom} \cdot N(U_0 - U_{nom})(N \cdot L_* - 2 \cdot N - 1)}{Lr_0(4N^2L_*^2 - 6N^2L_* - 3NL_* - 1)} + \\ + \frac{P_L(r_0 + x_0 \cdot tg\phi_L)(N^2L_*^3 - 4N^2L_*^2 + 8N^2 - L_* + 6N + 2)}{4r_0(4N^2L_*^2 - 6N^2L_* - 3NL_* - 1)}. \quad (7)$$

Figure 5a shows the surface of optimal P_{DG} values according to formula (7) when varying N and L_*. The rising of the characteristic at small values of L_* is explained by the attempt of the mathematical model to raise the voltage level by reducing losses in the short section between the power source and the DG connection point by changing the direction of generation and obtaining a negative voltage drop in this section.

In reality, this approach is impractical, as it will lead to a change in the direction of energy transmission and the need to modernize the protection and automation system, as well as to check the admissibility of cross-sections of power lines. Therefore, restrictions on the P_{DG} power should be introduced into the mathematical model.

$$0 < P_{DG} \le P_L \quad (8)$$

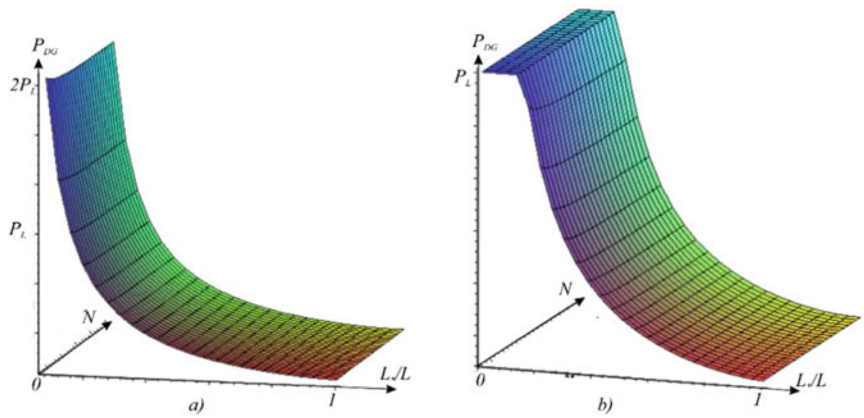


Fig. 5 Graph of optimal values of PDG at variation of N and L_*: without limitation (a); with limitation of the DG maximum power (b)

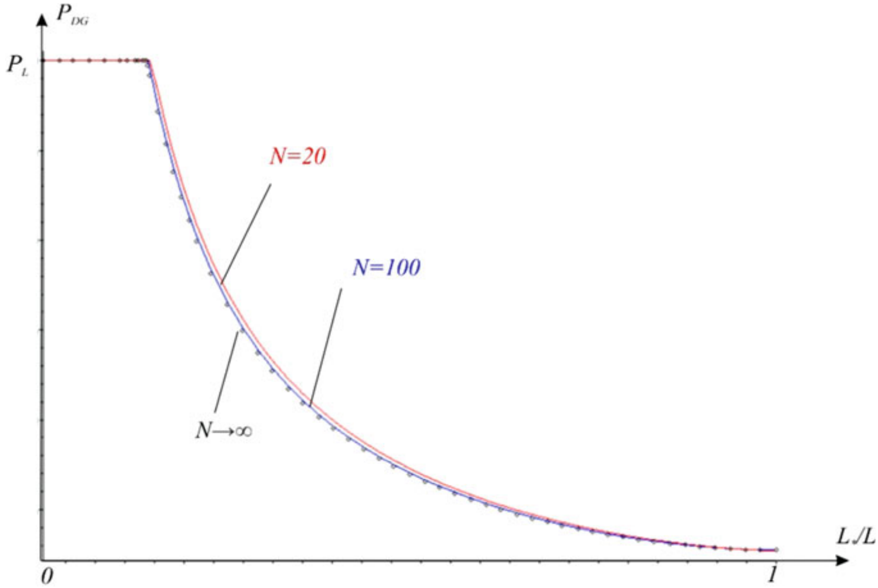


Fig. 6 Graph of optimal P_{DG} values when varying N and L* and limiting the maximum power of DG

In this case, the surface will look as shown in Fig. 5b.

As can be seen on Fig. 5b, the surface is practically independent of N and for the analysis it is sufficient to use the dependence $P_{DG}(L)$ at N = const (Fig. 6).

The value of active power losses in accordance with the model in Fig. 2 will be

$$\Delta P = \int_0^{L_* \cdot L} \frac{\left(P_L \frac{(L-l)}{L} - P_{DG}\right)^2 + \left(P_L \frac{(L-l)}{L} tg\phi_L\right)^2}{U_{nom}^2} r_0 dl + \\ + \frac{(P_L(1-L_*))^2\left(1 + tg^2\phi_L\right)}{3U_{nom}^2} r_0 L(1 - L_*) \tag{9}$$

After integration, we get:

$$\Delta P = \frac{3P_{DG}^2 \cdot L_* - 6L_* \cdot P_L \cdot P_{DG} + 3P_L P_{DG} L_*^2 + P_L^2\left(1 + tg^2\phi_L\right)}{3U_{nom}^2} r_0 L \tag{10}$$

The extremum of this function corresponds to the optimal value of PDG in terms of active power losses. We determine it by equating the derivative to zero, we get:

$$P_{DG}(\Delta P \to \min) = P_L(1 - L_*/2) \tag{11}$$

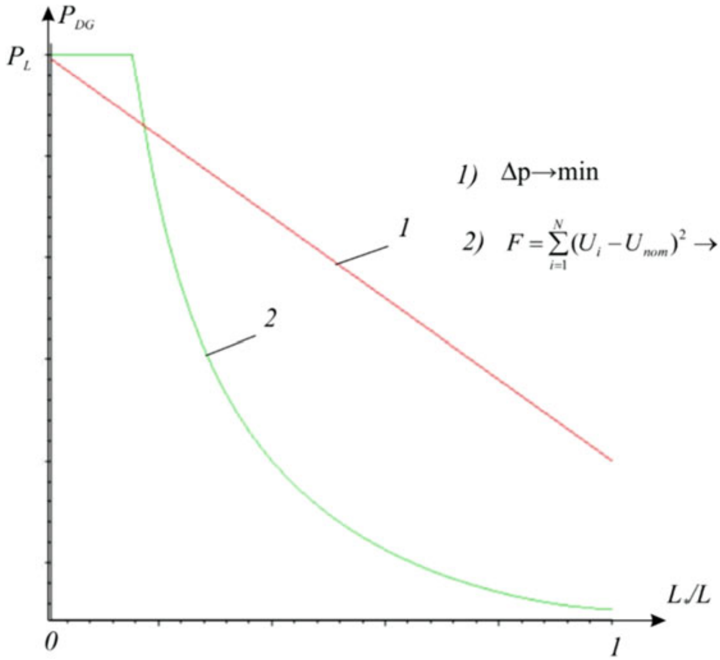


Fig. 7 Graph of optimal P_{DG} values when varying N and L_* and limiting the maximum power of DG

Comparing the obtained results with the power optimization by active power losses in the network (Fig. 7), we can see that when installing DG on most of the length of the power network, the optimal value of P_{DG} by power losses significantly exceeds the optimal value by voltage deviation.

From this it can be concluded that low-power DG, including solar power plants of private households, are unable to significantly affect the optimization of power losses, but can be effectively involved in minimizing voltage deviations in low-voltage distribution networks.

4 Conclusions

The widespread use of low-power DGs connected to the low-voltage network affects the voltage levels in this network. The voltage level can be very high even in nodes far from the mains power transformer. Under adverse circumstances, such a power supply can lead to exceeding the allowable voltage.

Simulation of voltage distribution along a low-voltage power line with distributed generation of private households performed in this study.

A criterion for choosing the optimal DG power of a private household has been formulated, which ensures minimal voltage deviation along the power transmission line and compliance with regulatory requirements for power quality.

Mathematical modeling shown that small DG capacities are able to provide minimal voltage deviation along the low-voltage power transmission lines.

The results obtained can also be applied to the preliminary justification of grid-connected generation of private households and power plants connected to low-voltage distribution networks.

References

1. Sonyachni elektrostanciyi u pryvatnyx domogospodarstvax: dynamika rozvytku. https://saee.gov.ua/uk/content/sesd. Accessed 20 Sept 2022
2. Zakon Ukrayiny «Pro alternatyvni vydy palyva». Vidomosti Verxovnoyi Rady Ukrayiny, № 12, art. 94 (2000)
3. Zakon Ukrayiny «Pro rynok elektrychnoyi energiyi» Vidomosti Verxovnoyi Rady Ukrayiny, № 27–28, art.312 (2017)
4. Feed-in tariffs (FITs) in Europe. https://www.pv-magazine.com/features/archive/solar-incentives-and-fits/feed-in-tariffs-in-europe/. Accessed 20 Sept 2022
5. Collin, A.J., Hernando-Gil, I., Acosta, J.L., Ilie, I.-S., Djokic, S.Z.: Realising the potential of smart grids in LV networks. Part 2: Microgeneration. In: 2nd IEEE PES International Conference and Exhibition on Innovative Smart Grid Technologies 2011, pp. 1–8 (2011)
6. Gawlak, A.: Impact of microgeneration in a low-voltage network on the quality parameters of electrical energy. In: 18th International Scientific Conference on Electric Power Engineering (EPE) 2017, pp. 1–4 (2017)
7. Kowalak, R., Czapp, S.: Improving voltage levels in low-voltage networks with distributed generation - case study. In: Progress in Applied Electrical Engineering (PAEE) 2018, pp. 1–6 (2018)
8. Pinto, R.J.C., Calado, M.R.A., Mariano, S.J.P.S., Espírito-Santo, A.E.V.: Micro-generation with solar energy: power quality and impact on a rural low-voltage grid. In: 9th International Conference on Compatibility and Power Electronics (CPE) 2015, pp. 87–92 (2015)
9. Oliva, P.C., Madureira, A.G., Matos, M.: Advanced voltage control for smart microgrids using distributed energy resources. Electric Power Syst. Res. **146**, 132–140 (2017)
10. Silva, F., Pinto, S., Silva, J. F.: Impact of microgeneration on the quality of power: effect on the voltage waveform. In: 35th Annual Conference of IEEE Industrial Electronics 2009, pp. 3672–3678 (2009)
11. Gawlak, A., Poniatowski, L.: Power and energy losses in low voltage overhead lines with prosumer microgeneration plants. In: 18th International Scientific Conference on Electric Power Engineering (EPE) 2017, pp. 1–4 (2017)
12. Camiloa, F.M., Pires, V.F., Castro, R., Almeida, M.E.: The impact of harmonics compensation ancillary services of photovoltaic microgeneration in low voltage distribution networks. Sustain. Cities Soc. **39**, 449–458 (2018)
13. Georgilakis, P.S., Hatziargyriou, N.D.: Optimal distributed generation placement in power distribution networks: models, methods, and future research. IEEE Trans. Power Syst. **28**(3), 3420–3428 (2013)
14. Lezhniuk, P.D., Komar, V.A., Sobchuk, D.S.: Method for determination of optimal installed capacity of renewable sources of energy by the criterion of minimum losses of active power in distribution system. Energy Power Eng. **6**, 37–46 (2014)

15. Kyrylenko, O.V., Pavlovskyj, V.V., Lukyanenko, L.M.: Texnichni aspekty vprovadzhennya dzherel rozpodilnoyi generaciyi v elektrychnyx merezhax. Texnichna elektrodynamika **1**, 46–53 (2011)
16. Praxovnyk, A.V., Popov, V.A., Yarmolyuk, E.S., Kokoryna, M.T.: Perspektyvi y puty razvytyya raspredelennoj generacyy v Ukrayne. Energetyka: ekonomika texnologiyi ekologiya **2**, 7–14 (2012)
17. Zharkin, A.F., Novskyj, V.O., Popov, V.A., Yarmolyuk, O.S.: Pidvyshhennya efektyvnosti keruvannya rezhymamy rozpodilnyx merezh za umov zastosuvannya rozoseredzhenyx dzherel generaciyi ta zasobiv akumulyuvannya elektroenergiyi. Texnichna elektrody‘namika **3**, 37–43 (2021)
18. Akorede, M.F., Hizam, H., Aris, I.: A review of strategies for optimal placement of distributed generation in power distribution systems. Res. J. Appl. Sci. **5**(2), 137–145 (2010)
19. Abu-Mouti, F.S., El-Hawary, M.E.: Optimal distributed generation allocation and sizing in distribution systems via artificial bee colony algorithm. IEEE Trans. Power Del **26**(4), 2090–2101 (2011)
20. Banerjee, B., Islam, S.M.: Reliability based optimum location of distributed generation. Int. J. Electr. Power Energy Syst. **33**(8), 1470–1478 (2011)
21. Bhumkittipich, K., Phuangpornpitak, W.: Optimal placement and sizing of distributed generation for power loss reduction using particle swarm optimization. Energy Procedia **34**, 307–317 (2013)
22. Bodunov, V., Kulko, T., Prystupa, A., Gai, A.: Topological task of distributed generation placement using a pareto optimization. In: IEEE 3rd International Conference on Intelligent Energy and Power Systems (IEPS) 2018, pp. 183–188 (2018)
23. Hamedi, H., Gandomkar, M.: A straightforward approach to minimizing unsupplied energy and power loss through DG placement and evaluating power quality in relation to load variations over time. Int. J. Electr. Power Energy Syst. **35**(1), 93–96 (2012)
24. Yadav, A., Srivastava, L.: Optimal placement of distributed generation: an overview and key issues. In: International Conference on Power Signals Control and Computations (EPSCICON) 2014, pp. 1–6 (2014)
25. Krasnozhon, A.V., Buinyi, R.O., Pentegov, I.V.: Calculation of active power losses in the grounding wire of overhead power lines. Technical Electrodynamics, 2016(4), 23-25V (2016)
26. Bodunov, V., Kulko, T., Prystupa, A., Gai, A.: Topological task of distributed generation placement using a pareto optimization. In: 2018 IEEE 3rd International Conference on Intelligent Energy and Power Systems (IEPS), pp. 183–188 (2018). https://doi.org/10.1109/IEPS.2018.8559502
27. GOST 13109 Norms of quality of electric energy in power supply systems of general purpose - introduced in Ukraine on 01.01.1999 (National Standard of Ukraine) (1999)
28. DSTU EN 50160:2010 Characteristics of power supply voltage supplied by distribution networks of general purpose - introduced in Ukraine on 01.07.2012 (National Standard of Ukraine) (2010)
29. Wolberg, J.: Data Analysis Using the Method of Least Squares: Extracting the Most Information from Experiments. Springer, Heidelberg (2006)
30. Roncero-Clemente, C., Husev, O., Stepenko, S., Vinnikov, D., Romero-Cadaval, E.: Output voltage control system for a three-level neutral-point clamped quasi-Z-source inverter. Przegląd Elektrotechniczny **89**(5), 76–80 (2013)
31. Pimentel, S.P., Husev, O., Vinnikov, D., Roncero-Clemente, C., Stepenko, S.: An indirect model predictive current control (CCS-MPC) for grid-connected single-phase three-level NPC Quasi-Z-Source PV inverter. In: 2018 IEEE 59th International Scientific Conference on Power and Electrical Engineering of Riga Technical University (RTUCON), pp. 1–6 (2018). https://doi.org/10.1109/RTUCON.2018.8659840
32. Tiutiunnyk, F., Prystupa, A., Bodunov, V.: Improving methods for evaluating the stability of electrical systems with distributed generation. In: 2016 II International Young Scientists Forum on Applied Physics and Engineering (YSF), pp. 37–40 (2016). https://doi.org/10.1109/YSF.2016.7753795

A Conceptual Model for Increasing the Speed of Decision-Making Based on Images Obtained from UAVs

Volodymyr Voytenko, Yuriy Denisov, Roman Yershov, and Maksym Solodchuk

Abstract To reduce the load on the operator of an unmanned aerial vehicle (UAV) during long search and rescue, and monitoring missions, the concept of an automatic system is proposed, which directly on board performs a preliminary analysis of images received from a high-resolution navigation video camera, determines areas of interest, and sets the position of an additional camera with a reduced viewing angle to scale the image of the selected area. This allows the operator to speed up the final decision and reduces the response time to the detection, identification of an object, as well as to the preparation of a mission report. To develop a technical system that will be able to solve the tasks, a complex hierarchical model is considered, consisting of three components: a software system for image pre-processing and analysis, an electromechanical camera positioning system, and a higher-level human–machine complex. It is determined that the model of the first component should be based on the use of a deep learning artificial neural network using inference trees. The Simulink model of the positioning system contains a controller that improves the dynamics of two interconnected electric drives by using three control loops in each of them. The features of information perception by the operator of the UAV control complex are analysed and the need to consider the effect of global precedence is noted. The results of the simulation of the electromechanical link are presented and the complexes of further research are outlined.

Keywords Unmanned Aerial Vehicle (UAV) · Image processing · Pattern recognition · Electric drive · Automatic control · Human-machine interaction

V. Voytenko (✉)
Lund University, Naturvetarvägen 18, 223 62 Lund, Sweden
e-mail: volodymyr.voitenko@control.lth.se

V. Voytenko · Y. Denisov · R. Yershov
Chernihiv Polytechnic National University, Shevchenko Str. 95, Chernihiv 14035, Ukraine

M. Solodchuk
State Scientific Research Institute of Armament and Military Equipment Testing and Certification, Chernihiv, Ukraine

S. Shkarlet et al. (eds.), *Mathematical Modeling and Simulation of Systems*, Lecture Notes in Networks and Systems 667,
https://doi.org/10.1007/978-3-031-30251-0_23

1 Introduction

To carry out search and rescue, surveillance, intelligence, and reconnaissance missions, unmanned aerial vehicles (UAVs) are widely used with sensor units installed on board, and which operate in various ranges of electromagnetic (EM) waves: from infrared to ultraviolet [1]. For navigational and other purposes, the visible EM-range is essential, allowing the UAV operator to rely on a familiar view of the terrain and objects located on it. In this case, images are created by one or more high-resolution video cameras [2], which can be placed on gyro-stabilized platforms [1], and this ensures, in particular, high image quality.

For processing large areas of the earth's surface, aircraft-type devices are more appropriate, since, unlike various rotor structures, they have a significantly higher speed and flight range [3]. The high flight speed of the UAV during the mission leads to an increase in the information flow from the screen of the display device, causing operator fatigue and even the emergence and development of occupational diseases. Reduced attention, in turn, increases the likelihood of mistakes and missed targets. Therefore, the task of building a technical system that automatically performs preliminary processing of video information and helps to reduce the load on the human operator is relevant.

These video sensor units are often equipped with two high-resolution video cameras: the main (navigation) camera with a sufficiently wide viewing angle for reliable orientation of the UAV crew in space, and an additional (spot) camera with a small viewing angle and an enlarged image that this camera can form [1]. Provided that the angle of view of the main video camera remains unchanged, an increase in the UAV flight height allows expanding the area of the terrain displayed on the monitor screen by reducing the detail, however, using the spot camera, an additional enlarged image appears, which makes it easier for the operator to identify the target, reducing the time for making a decision on further actions.

It is essential for modern conditions to implement as many functions as possible, performed directly on board the UAV [4], so that the operator has almost the last step: considering the hint generated by the technical system, complete the image analysis, and perform the prescribed mechanical action. Work on board allows, in particular, to perform (or continue to perform) a task autonomously when it is carried out in search mode along a given route, which increases stealth and stability when exposed to various external factors.

Saving even one second or a certain fraction of it on this path is a task that requires efforts on several levels.

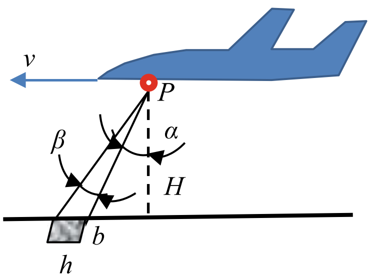
2 The Structure of the System with Advanced Capabilities for Analyzing Images Obtained from UAVs

Figure 1 shows a UAV flying at a height H with a speed v. The gyro-stabilized platform P, on which the camera unit is installed, is directed forward at an angle α relative to the normal to the ground surface, which allows the operator to better navigate the route. The main (navigation) video camera has a fairly wide angle of view β, which determines the size of the terrain area (b is the width, h is the "height", i.e., the size along the direction of the UAV movement), which is converted into an image of the format $K_f = b{:}h$, defined by the video standard.

In the context of solving the task of accelerating the decision-making process, the UAV operator scales the image of a fragment in which an object of interest may be located. This is usually done through the operator's own actions, which, using the appropriate controls, directs the gyro-stabilized platform with the sensor unit installed on it. The next action of the operator should be exactly the increase (zooming) of the image, remotely implemented using the appropriate electric drive, which increases the focal length of the lens, while simultaneously reducing the viewing angle of the camera. In this case, as a rule, an autofocusing procedure is additionally carried out, for which a separate electric drive with its own electromechanical system and a dedicated control algorithm is used. Thus, the operator, having previously assessed the image, must perform several successive control actions, accompanied by the reaction of electromechanical devices. All this increases the time for obtaining a scalable image, which in this case gets to the operator in a single video stream. And this, in turn, reduces the amount of information about the flight, which makes it difficult to document after the mission is completed.

To overcome these shortcomings in practice, an additional video camera (spot camera) with a small viewing angle, which is located on the same platform as the main (navigation) one, next to other necessary sensors, can be used to zoom in on the image. Switching the image on the video monitor can again be performed by the operator as a result of additional actions. However, since the main and spot cameras are located on the same platform, the operator can observe only that fragment of the surface, the direction of which coincides with the center of the original image (or another predetermined point that was set before the start of the mission). But

Fig. 1 Definition of symbols



since the area of interest can be located in another part of the input image, additional positioning of the gyro-stabilized platform will be required, followed by a return to the original position to continue navigation. This, again, wastes valuable time and additional workload and fatigue for the operator.

A solution is proposed (Fig. 2), which provides for the placement of a narrow-angle spot camera (SC) on its own platform (1), which can move relative to the base gyro-stabilized (2) along two coordinates. In this case, two video channels actually work in parallel and independently of each other, which, even if it is impossible to transmit the entire video stream, for example, due to interference, limited bandwidth of the video channel, or other reasons, allows you to accumulate an array of information from both cameras. This, in particular (but not only), reduces the need for repeat missions along the same route, which is the most stressful for the operator.

To implement the idea, you need to add one more remote-controlled electric drive (SCED) of the spot camera (SC) to the existing remote-controlled electric drive (MCED) of the main camera (MC). Now the main camera has a fixed angle of view β_1, and the spot camera has a fixed angle of view β_2, and the zoom factor (magnification) of the image is

$$M = \beta_1/\beta_2.$$

The angle β_2 can be fixed, or it can change with a discrete step, which will further simplify the interaction of two systems in automatic mode. To facilitate the use of digital software and hardware, it is advisable to discretize the input image by dividing its entire plane into zones – rectangular sections that have the same format as the

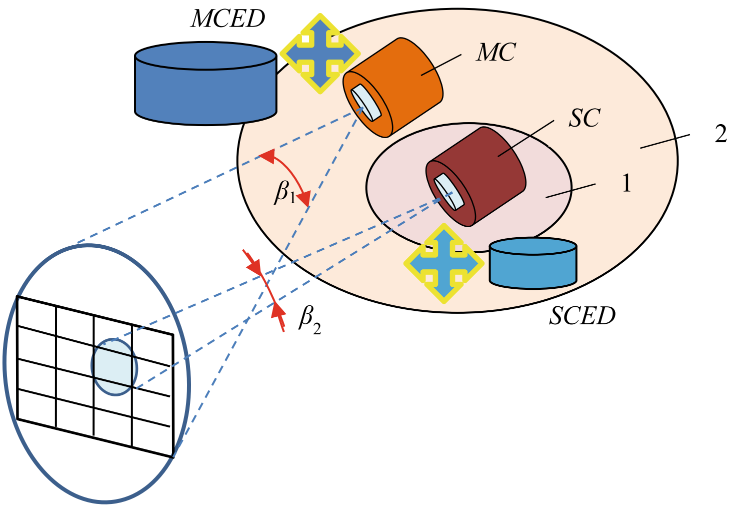


Fig. 2 Constructive model of interaction between two cameras

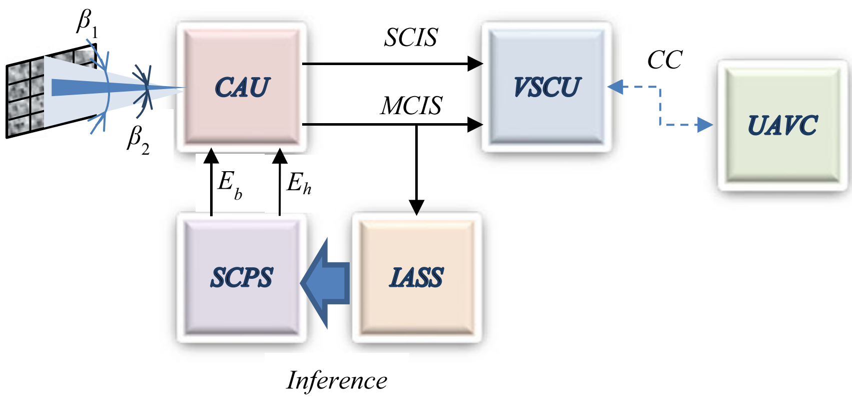


Fig. 3 Advanced system structure

input image. Figure 2 shows the division into $N_z = 16$ zones, providing for the use of a spot camera with a viewing angle $\beta_2 = \beta_1/16$.

The proposed solution has such disadvantages as increased weight, dimensions, and complexity of the entire complex of sensors, since in addition to additional mechanical elements, there is also a two-position electric drive of the spot camera. However, the use of the varifocal lens of the main camera in the basic version also requires two drives (for changing the focal length and for focusing). In addition, such a lens itself is more complex, more expensive, less stable, and reliable compared to a device with fixed parameters. The additional spot camera platform is simpler than the main one (it has fewer degrees of freedom), it is designed for a smaller load, and is therefore less inertial.

Therefore, the proposed structure (Fig. 3) contains:

CAU – Camera Apparatus Unit;

VSCU – Video Signal Conversion Unit;

IASS – Image Pre-processing and Analyzing Software System;

SCPS – Spot-Camera Positioning System;

UAVC – UAVs Control Complex.

This structure uses both the main camera and an additional spot camera with fixed focal lengths and focusing, but with different viewing angles β_1 and β_2.

CAU forms two separate digital streams – Spot-Camera Image Signal (*SCIS*) and Main Camera Image Signal (*MCIS*), which arrive at the *VSCU*, and then through the Communication Channel (*CC*) enter the *UAVC*. During the mission, *CC* is a radio link with the UAV. When setting up the device on the ground and reading information from on-board drives, both wired and wireless technologies can be used. In some cases, there may be no CC (execution of an autonomous flight with subsequent access of the operator to the information stored on the built-in drive during the mission).

IASS, based on the images received from the main camera, forms one of the possible conclusions about the number of the zone to which the spot camera should be positioned. Logical information from this block (*Inference*) enters the *SCPS*, where electrical signals for positioning the spot camera along the yaw angle E_b and pitch angle E_h are generated.

UAVC is the cutting edge of the human-machine system. Here, telemetry and video information are received, visualized, converted, and the response initiated by the operator is carried out by performing predetermined actions: the use of controls for transmitting commands in the opposite direction (to the UAV), as well as further processing of information, documenting the mission, transferring it to other authorities, etc.

3 Image Pre-processing and Analyzing Software System

Image pre-processing may be required to correct visibility conditions, angle, eliminate interference, noise, etc. [5]. In remote sensing applications, where the geometric parameters of the sensor's field of view are known (which is a common situation), knowing the height of the sensor above the observed surface can be sufficient to normalize the image size if the shooting angle is constant. Similarly, rotation normalization requires knowing the angle by which the image (or sample) should be rotated, which again requires spatial hint. In this remote sensing problem, information about the direction of flight may be sufficient to rotate the resulting images to a standard orientation. If additional spatial information is not available, normalizing the size and rotation can be a very difficult task. To solve it, it is necessary, using known methods, to automatically find features in the image that could be a spatial hint.

So, general approaches to solving all the necessary problems listed above are known and will not be the subject of discussion in this article. The final purpose of the analysis within our system is to determine the region of interest and generate commands for positioning the spot camera. Initially, we see the primary task of pattern detection and classification.

The question of choosing and using certain methods of image processing requires a separate study. However, we can say that today this area is dominated by algorithms based on the use of artificial intelligence methods, which are based on machine learning of artificial neural networks. The theoretical foundations of machine learning have been laid down for a long time, and their classification and application features for solving pattern recognition problems are described, in particular, in [6].

If the data is not big data (which could have been solved by simple technology and tools), then the machine-learning techniques like the supervised learning and the dimensionality reduction techniques are useful [7].

A real breakthrough in the field of pattern recognition and computer vision was the development of a convolutional neural network AlexNet [8, 9]. AlexNet implementation is easier after the releasing of so many deep learning libraries like PyTorch, TensorFlow, Keras [10].

When designing our system, one of the main requirements formulated earlier should be kept in mind: most functions must be performed directly on board the UAV. After all, this not only improves the dynamic performance of the man-machine system, but also allows you to perform parts of the mission offline, accumulating information directly on the UAV, which is very important in some cases. However, it should be noted that most embedded systems are based on the use of microcontrollers that are not very high performance. In some cases, this is due not so much to economic considerations as to purely constructive and energy constraints. To an even greater extent, this applies to UAVs, where the requirements for weight and size indicators of any onboard equipment are much more stringent than for ground vehicles.

The issues of solving the problem of applying algorithms based on the use of artificial neural networks of deep learning in the conditions of constructive limitations by means of embedded systems have recently received more and more attention [11, 12].

An accent on hardware electronics, processor architecture, and connections to memory is done in [12]. This shows us that the results are obtained with greater reliability due to the achievements of modern electronic technologies, the structure of the ANN and in combination with more advanced software methods.

In [13], we can see an attempt to use the concept of CNN-based inference for IoT edge clusters. As you know, IoT is a device with a hard resource limit, and often it is stronger than that of UAVs. Here authors offer DeepThings, a special framework for the adaptive-distributed launch of CNN. To minimize the required memory, DeepThings uses a special scalable partitioning of convolutional layers while maintaining parallelism. Perhaps this approach will be useful when organizing a "multi-drone" mission in the future, but not in our case.

Deep Neural Networks (DNNs) are now a mainstream machine learning technology. However, running DNNs on resource-constrained devices is not trivial, as it requires high performance and energy costs. Deploying a DNN to run in the cloud or splitting a DNN with adaptive distribution of computing between the device and the edge to use hybrid computing resources nearby to determine the DNN in real time [14] during the performing of the above missions is impossible even in terms of the limitations of the UAV radio link itself. So, we need to find a more standalone embedded solution.

Advancements in compression of neural networks and neural network architectures, coupled with an optimized instruction set architecture, could make microcontroller-grade processors suitable for specific low-intensity deep learning applications [15]. There, a simple extension of the instruction set was proposed with two main components – hardware loops and dot product instructions.

The influence of the depth of a convolutional network on its accuracy in the large-scale image recognition setting was studied in [16]. Careful evaluation of increasing depth networks using an architecture with very small (3×3) convolution filters has shown that a significant improvement over previous configurations can be achieved by increasing the depth to 16–19 weight layers. These conclusions formed the basis of successes in the localization and classification tracks. It is also shown that the authors' representations generalize well to other datasets, where they achieve state-of-the-art results. The two most powerful ConvNet models are now publicly available to facilitate further research into the use of deep visual representations in computer vision.

Two efficient approximations to standard convolutional neural networks were proposed in [17]. In one case, the filters are approximated by binary values, which reduces the required memory by a factor of 32. In another case, the input to the convolutional layers is also binary. The use of mainly binary operations allows not only saving memory, but also significantly speeding up convolutions. The proposed method in real time allows you to run convolutional networks on processors (not only on graphics). As a result, such networks can solve complex visual problems with simple and energy-efficient devices placed on UAVs. This approach was evaluated in the ImageNet classification problem, where it showed almost the same accuracy as a comparable version of the AlexNet network. The authors claim that comparing the proposed Binary-Weight-Networks and XNOR-Networks with other binarized networks such as BinaryConnect and BinaryNets also shows better accuracy results.

Before we move on to practical machine learning, we need to touch on one aspect regarding the data set we need in "aerial vision" as a new frontier of computer vision [18]. It is noted that the existing UAV data sets are mainly focused on object detection, and not on recognition. These datasets can be used in our case because we assume that more reliable recognition can currently be performed by a human.

DeepCluster [19] may turn out to be a very promising solution in the context of the formulated tasks. This clustering method iteratively groups features using a standard clustering algorithm (K-means method) and uses subsequent assignments as a supervision to update neural network weights. What can be extremely important is the potential effectiveness of this model for learning deep representations in specific domains when no labeled dataset is available. In this case, a detailed description of the input data and domain-specific knowledge is not required.

Considering the very good results of object detection in images obtained by DNN AlexNet [8], we will choose it as the basis for building our future classifier. The results obtained can be further used as a benchmark for comparison with other developed network architectures. The essential differences of our case from traditional problems are that the number of classes can be much less than in the standard problem of pattern recognition. The concept of a class itself has a different meaning than usual: it is just a number – the number of a rectangular area of the image where the probability of the presence of an object of interest exceeds a certain specified value. This opens up the possibility of detecting an object of interest directly on board the UAV with relatively low-performance microcontrollers.

Suppose the main camera's view angle is $\beta = 23°$. Moreover, if the UAV is at a height of $H = 100$ m, the height of the image on the surface is $h = 20$ m (Fig. 1). At a speed $v = 72$ km/h, the UAV travels a distance $s = 20$ m, i.e., the image frame is updated by 100% at $T_r = 1$ s.

Today, the most popular image format formed by video cameras installed on UAVs is HD n_xx $n_y = 1920 \times 1080$ ($K_f = 16{:}9$). This implies the size of one pixel for $H = 100$ m, $\beta = 23°$:

$$\Delta = h/n_y = 2000/1080 \approx 1,85cm.$$

Table 1 The number and format of split zones of the input image

N_1	1	2	3	4	5	6	8	10	12	15
N_z	1	4	9	16	25	36	64	100	144	225
N_h	1920	960	640	480	384	320	240	192	160	128
N_v	1080	540	360	270	216	180	135	108	90	72
h, m	35,5	17,8	11,8	8,9	7,1	5,9	4,4	3,6	3	2,4
b, m	20	10	6,7	5	4	3,3	2,5	2	1,7	1,3

Here n_x, n_y are the number of pixels along the horizontal and vertical lines, respectively.

Table 1 shows the quantitative parameters of image zoning of the main UAV video camera operating in HD format.

The following designations are used in the table:

N_1 – number of image zones along one of the axes;

N_z – total number of image zones;

N_h – the number of pixels in one zone of the image horizontally;

N_v – the number of pixels in one zone of the image vertically;

h, b – the size of the zone on the ground at a flight altitude $H = 100$ m, the angle of view of the main camera $\beta = 23°$.

Compared to the standard parameters of the AlexNet network [8], where the input image format is 224×224, and the number of outputs is 1000, the resource requirements in our network are less, which increases the possibility of its successful deployment on board the UAV using the appropriate element base.

In the case of the parallel mode of detection of the zone of interest, each zone can be processed independently from one another, and the result will be formed at the same time. However, taking into account the time interval for updating the image frame of the main camera, it is also possible to perform zoning in sequential mode. In both modes, of course, we are talking about the work of pre-trained neural networks, but not about their training, the execution time of which can be significantly longer than the longest mission.

4 Spot-Camera Positioning System

Figure 4 shows a structural model of the system designed to position the spot camera to the zone determined at the previous stage using the Image Pre-processing and Analyzing Software System.

The model contains two electric positioning drives:

SCVD – Spot-Camera Vertical Drive;

SCHD – Spot-Camera Horizontal Drive.

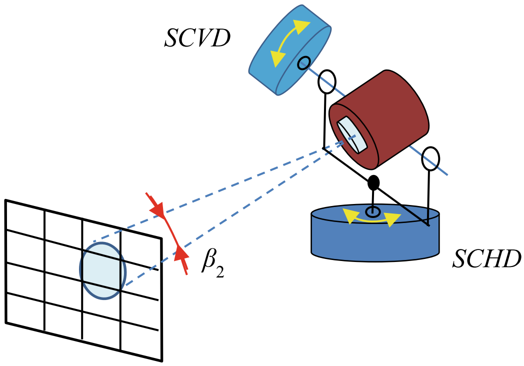


Fig. 4 Constructive model of the spot camera positioning system

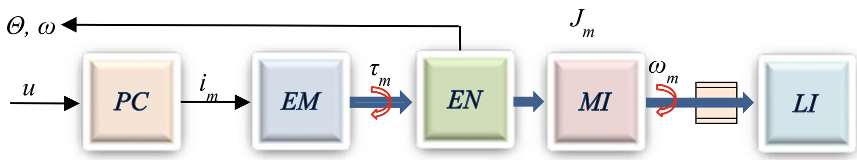


Fig. 5 Key components of one of the electric actuators for positioning a spot camera with mechanical links

To solve the tasks which are set, it is advisable to use rotating brushless electric motors as part of these electric drives. Figure 5 shows the key components of one of the spot camera positioning motors with mechanical links affecting the actuators.

The reference voltage u controls the motor current i_m, which is generated by the power converter *PC* and flows through the motor windings *EM*. The motor generates a torque that accelerates the rotational inertia *MI* and counteracts the friction. The encoder *EN* measures the speed and rotation angle. The inertia of load is modeled using the *LI* block.

The power converter in the first approximation provides current

$$i_m = K_{pc} u,$$

which is linearly related to the applied control voltage u. Here K_{pc} is the conductivity of the converter. The torque generated by the motor is proportional to the current

$$T_m = K_m i_m,$$

where K_m is the motor torque constant. The torque generated by the motor overcomes the rotational inertia of the motor J_m, and the rotating part is accelerated to the rotational speed ω. Friction effects are modeled by the coefficient B_m, and the friction moment itself is proportional to the rotation speed and is approximately $B_m\ \omega_m$.

The current is controlled by an electronic AC voltage source with feedback on the actual motor current. The AC voltage source is implemented using a pulse-width modulated (PWM) pulse converter. When designing the voltage control system on the windings, we will take into account the electrical dynamics of the motor, due to its resistance and inductance, as well as the back EMF.

Spot camera positioning motors do not exist in separate but are connected to mechanical links. These links have two significant effects on the engine: they give additional inertia, and also create torque because of the imperfect balancing of the elements of the spot camera gimbal suspension. Both additional actions can vary depending on the rotation angles. In Fig. 5, the connected mechanical links are modeled using the *LI* block, however, at this stage, we will assume that the suspension of the spot camera is perfectly balanced, and the load inertia is a constant value.

We write the torque balance on the motor shaft as

$$K_m K_{pc} u - B\omega - \tau_c(\omega) = J\dot{\omega}, \tag{1}$$

where B, τ_c and J are the effective total viscous friction, Coulomb friction, and inertia due to the motor, bearings, and load:

$$B = B_m + B_l, \tau_c = \tau_{c,m} + \tau_{c,l}. \tag{2}$$

To analyze the dynamics (1), we neglect the nonlinearities

$$J\dot{\omega} + B\omega = K_m K_a u,$$

and then apply the Laplace transform:

$$sJ\Omega(s) + B\Omega(s) = K_m K_a U(s),$$

where $\Omega(s)$ and $U(s)$ are the Laplace transform of the signals in the time domain $\omega(t)$ and $u(t)$, respectively. The last expression can be turned into a linear transfer function

$$\frac{\Omega(s)}{U(s)} = \frac{K_m K_a}{Js + B}$$

of motor speed relative to the input control signal, having one (mechanical) pole.

Once we have the model in the form above, we can create a step response graph and use standard control system design tools.

To ensure good dynamic and static characteristics in both drives, we use three-loop automatic control systems, including [20]:

- internal current loop;
- speed loop;
- outer angle loop.

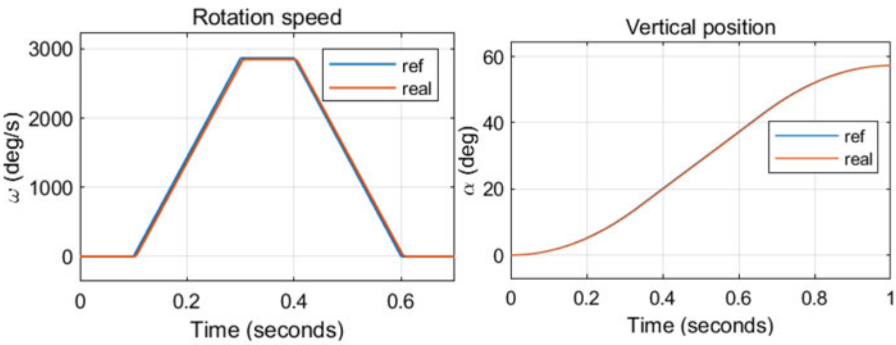


Fig. 6 The transient processes in the vertical angle positioning loop

Figure 6 shows the result of modeling of the speed loop and angle positioning of the spot camera along the direction of the UAV movement for various references.

It should be noted that the specifics of the UAV (limited energy resources and time) require the use of more advanced control algorithms than those used in this simplified model. These can be optimal (or quasi-optimal) controllers, in which it is possible to provide a transient process in a minimum time, without overshoot and with given energy costs [21]. An important issue is also taking into account the dynamics of the power converter.

The difference between the two electric drives (Fig. 4) is, first of all, that the positioning subsystem along the pitch angle is more powerful, since it contains, among other things, the positioning subsystem along the yaw axis together with the corresponding electric motor and power converter as a load. The less powerful yaw angle positioning subsystem is actually loaded only on the spot camera, which allows using lighter and more compact electromechanical components in it and obtaining better dynamic characteristics. At the final stage of system development, the parameters of the corresponding electric motors and power converters need to be clarified.

Additional analysis will require consideration of the features of two-coordinate positioning, considering the specifics of the gimbal on which the spot camera is located [3].

5 Human-Machine UAVs Control Complex

Within the framework of this part of the multidisciplinary study, we will limit ourselves to issues related to interaction with images from two cameras. From this point of view, the first question is to find out what exactly takes more time and strains the operator in the first place? What are the main features of a person's perception of information from a monitor screen that can affect the success and timeliness of a long-term mission with the participation of a UAV?

In [22], the idea is put forward, discussed, and tested that the global structuring of the visual scene precedes the analysis of local features. It has been experimentally established that global signals, which contradict local ones, inhibited reactions at the local level. It turned out that global differences were detected more often than local ones. It is proposed in [22] that perception proceeds from global analysis to more and more fine-grained analysis. The global precedence has several possible advantages such as utilization of low-resolution information, economy of processing resources, and disambiguation of indistinct details. Although evidence from the psychological literature (see references in [22]) supports the notion that global features are extracted earlier and/or better than local ones, in most previous research little attention has been given to the control over the complexity of global and local features.

So, we must have in mind that global processing is a necessary stage of perception. It turned out that global differences were detected more often than local ones. This may be interpreted as a support to the idea that global processing is done before more local analysis is completed.

By shifting this stage to a software system for analysing images from the main camera, we connect a person when he is aware of his abilities without fail to start with a global analysis, but for a scaled scene and more accurately identify an object of interest. Moreover, further reluctance of a person to study local elements promises additional bonuses in the context of the task of reducing reaction time and reducing operator fatigue.

The same "global precedence" effect was confirmed in [23] in contrast experiments with animals. Humans responded faster to global than to local targets, with human reaction time is independent of display size for both local and global processing. Finally, variations in stimulus density did not affect global search slopes in humans. Overall, results suggest that perceptual grouping of operations involved in the processing of hierarchical stimuli does not require human attention.

Human-machine interface (HMI) research related to the use of UAVs [24] highlights the extreme requirements in terms of operator workload. Real-time simulation of a system with a human in the control loop shows the directions for HMI cognitive optimization and equipping UAVs with appropriate levels of autonomy and intelligence.

So, when working on a system for determining the positioning zone of a spot camera, it is advisable to take into account a number of recommendations related to improving human performance and maintaining good awareness of the situation. This concerns the use of appropriate visual cues such as spatial arrangement, clustering, icon design, and category design to help the operator recognize information and events. Since our task so far is not to develop the HMI itself, but only to clarify the needs, capabilities and impact on the Image Pre-processing and Analyzing Software System, let us pay attention to the following.

UAV status not related to the current mission should be hidden unless requested by the operator or required for a critical decision.

The visual search models suggest that, in a fast parallel search, information about the location of a target's visual template among distractions and identification of information are not available simultaneously. That is, the location of the target is recognized at earlier stages of visual processing than target identification [22].

The results of [25] showed that the search time does not increase with an increase in the number of distractors at both scales (local/global). However, target detection on a local scale required significantly more time than on a global scale. This latter finding agrees with the phenomenon of ‘global precedence’ [22].

Therefore, the global precedence (human) model can be used to refine the IASS (machine learning) decision making process.

It should also be noted that the specifics of the automotive industry focus mainly on how to create a secure interaction with technology that will help the driver complete the driving task, as well as give the driver more time to perform other tasks that are not related to driving [26]. However, this formulation of the problem and some other general approaches intensively developed for automobiles are fully consistent with the needs and can be useful for UAVs as well.

6 Conclusion

To reduce the load on the operator of an unmanned aerial vehicle (UAV) during long search and rescue and monitoring missions, the concept of an automatic system is proposed, which directly on board performs a preliminary analysis of images received from a high-resolution navigation video camera, determines areas of interest, and sets the positioning of an additional camera with a reduced viewing angle to scale the image of the selected area. This allows the operator to speed up the final decision and reduces the response time to the detection, identification of an object, as well as to the preparation of a mission report. To develop a technical system that will be able to solve the tasks, a complex hierarchical model is considered, consisting of three components: a software system for image pre-processing and analysis, an electromechanical camera positioning system, and a higher-level human-machine complex. It is determined that the model of the first component should be based on the use of a deep learning artificial neural network using inference trees, the number of inferences of which is equal to the number of zones into which the image of the main video camera is divided. The Simulink model of the positioning system contains a controller that improves the dynamics of two interconnected electric drives by using three control loops in each of them. The reference signal for the external loops is the rotation angles of the additional video camera in one of two directions, determined by the software pre-processing and image analysis system. Signals of the desired camera rotation speeds are formed at the outputs of the external loops. These signals are used to form a reference for the internal current loops of both motors’ windings, which provide the required torques of the rotors. The features of information perception by the operator of the UAV control complex were analysed and the main requirements for the construction of a human-machine interface were determined, considering the effect of global precedence. The results of the simulation of the electromechanical link were presented and the complexes of further research were outlined.

Acknowledgements The authors are highly grateful to Knut and Alice Wallenberg Foundation for providing a grant, as well as to Björn Olofsson for helpful advice.

References

1. WESCAM MX-10 EO/IR. https://www.l3harris.com/all-capabilities/wescam-mx-10-air-surveillance-and-reconnaissance. Accessed 30 Aug 2022
2. FCB-9500 Series – New Colour Camera Block. https://www.image-sensing-solutions.eu/FCB-New-generation.html. Accessed 21 Aug 2022
3. Beard, R., McLain, T.: Small Unmanned Aircraft: Theory and Practice. Princeton University Press (2012)
4. Bashynskyi, V.G., Ragulin, V.V., Solodchuk, M.O., Fomin, A.V., Isachenko, O.O.: Justification of the need to process video information on board the reconnaissance UAV. Scientific works of the State Scientific Research Institute of Armament and Military Equipment Testing and Certification (11), 105–115 (2022). https://doi.org/10.37701/dndivsovt.11.2022.12. Accessed 21 Aug 2022
5. Gonzalez, R.C., Woods, R.E.: Digital Image Processing, 3rd edn. Pearson (2007)
6. Bishop, C.M.: Pattern Recognition and Machine Learning. Springer, New York (2006)
7. Suthaharan, S.: Decision tree learning. In: Machine Learning Models and Algorithms for Big Data Classification. Integrated Series in Information Systems, vol. 36. Springer, Boston (2016). https://doi.org/10.1007/978-1-4899-7641-3_10
8. Krizhevsky, A., Sutskever, I., Hinton, G.E.: ImageNet classification with deep convolutional neural networks. In: Pereira, F., Burges, C.J., Bottou, L., Weinberger, K.Q. (eds.) Advances in Neural Information Processing Systems, vol. 25 (NIPS 2012) (2012)
9. Slyusar, V., et al.: Improvement of the object recognition model on aerophotos using deep convolutional neural network. Eastern-Eur. J. Enterprise Technol. **5**(2) (113) (2021). https://doi.org/10.15587/1729-4061.2021.243094
10. Keras vs Tensorflow vs Pytorch: Key Differences Among the Deep Learning Framework. https://www.simplilearn.com/keras-vs-tensorflow-vs-pytorch-article. Accessed 30 Aug 2022
11. Mohammad-Ali A'râbi, Viktoria Schwarz. General Constraints in Embedded Machine Learning and How to Overcome Them – A Survey Paper (2019). https://doi.org/10.13140/RG.2.2.14747.21280
12. Andrade, L., Prost-Boucle, A., Pétrot, F.: Overview of the state of the art in embedded machine learning. In: 2018 Design, Automation & Test in Europe Conference & Exhibition (DATE), pp. 1033–1038 (2018). https://doi.org/10.23919/DATE.2018.8342164
13. Zhao, Z., Barijough, K.M., Gerstlauer, A.: DeepThings: distributed adaptive deep learning inference on resource-constrained IoT edge clusters. IEEE Trans. Comput. Aided Des. Integr. Circuits Syst. **37**(11), 2348–2359 (2018). https://doi.org/10.1109/TCAD.2018.2858384
14. Li, E., Zhou, Z., Chen, X.: Edge Intelligence: On-Demand Deep Learning Model. https://arxiv.org/abs/1806.07840. Accessed 21 Aug 2022
15. Vreča, J., et al.: Accelerating deep learning inference in constrained embedded devices using hardware loops and a dot product unit. IEEE Access **8**, 165913–165926 (2020). https://doi.org/10.1109/ACCESS.2020.3022824
16. Simonyan, K., Zisserman, A.: Very deep convolutional networks for large-scale image recognition, arXiv e-prints (2014)
17. Rastegari, M., Ordonez, V., et al.: XNOR-NET: ImageNet classification using binary convolutional neural networks. In: European Conference on Computer Vision, pp. 525–542. Springer, Cham (2016)
18. Murari, M.M., Kumar, L.K., Vipparthi, S.K.: MOR-UAV: a benchmark dataset and baselines for moving object recognition in UAV videos. In: MM 2020: Proceedings of the 28th ACM

International Conference on Multimedia, pp. 2626–2635 (2020). https://doi.org/10.1145/3394171.3413934
19. Caron, M., Bojanowski, P., Joulin, A., Douze, M.: Deep clustering for unsupervised learning of visual features. In: Ferrari, V., Hebert, M., Sminchisescu, C., Weiss, Y. (eds.) Computer Vision – ECCV 2018. ECCV 2018. LNCS, vol. 11218. Springer, Cham (2018)
20. Robotics, C.P.: Vision and Control. Fundamental Algorithms in MATLAB. Springer, Cham (2017)
21. Voitenko, V.P.: Algorithm stages of quasi-optimal regulation in system with a pulse converter. Tech. Electrodynamics (3), 125–126 (2012)
22. Navon, D.: Forest before the tree: The precedence of global features in visual perception. Cogn. Psychol. **9**, 353–383 (1977)
23. Deruelle, C., Fagot, J.: Visual search for global/local stimulus features in humans and baboons. Psychon. Bull. Rev. **5**, 476–481 (1998). https://doi.org/10.3758/BF03208825
24. Howitt, S., Richards, D.: The human machine interface for airborne control of UAVs. In: 2nd AIAA "Unmanned Unlimited" Conference and Workshop & Exhibit (2003). https://doi.org/10.2514/6.2003-6593
25. Saarinen, J.: Visual search for global and local stimulus features. Perception **23**, 237–243 (1994)
26. Kun, A.L.: Human-machine interaction for vehicles: review and outlook. Found. Trends® Hum. Comput. Interact. **11**(4), 201–293 (2018). https://doi.org/10.1561/1100000069

Modeling of Special Equipment Test System: Distribution of Personnel between Test Stages

Ihor Korniienko, Svitlana Korniienko, Volodymyr Dmytriiev, Anatolii Pavlenko, and Dmytro Kamak

Abstract The war of the russian federation against Ukraine has once again confirmed the axiom that in today's world nations must have a highly trained army equipped with modern weapons and other military equipment to defend itself. Ukraine is one of the developers in the world of modern models of weapons and equipment which are currently showing their combat effectiveness. During the process of developing special equipment, the issue of conducting high-quality and on schedule tests is very important. Given the wide range and scope of modern developments, it is necessary to have an effective test system.

The search for ways of optimal planning of all processes concerning the operation of the test organization, in the conditions of an intensive request flow for special equipment testing, is carried out by modeling. Using the analytical apparatus of queuing systems, a model of servicing incoming test requests at the preparatory and practical stages of testing has been developed. The structure of the model includes elements for modeling different productivities of service channels, which characterize the test teams with different staff sizes.

The simulation was performed using real statistical parameters for the incoming flow of test requests, statistical estimates for the expectation of request flows during the test preparations and the service intensity during the practical tests. The test system capacity has been studied while varying the parameters for the test teams

I. Korniienko (✉) · S. Korniienko
Chernihiv Polytechnic National University, Shevchenka 95, Chernihiv 14035, Ukraine
e-mail: cornelukr@gmail.com

S. Korniienko
e-mail: cornel@ukr.net

V. Dmytriiev · A. Pavlenko · D. Kamak
State Scientific Research Institute of Armament and Military Equipment Testing and Certification, Strilets'ka 1, Chernihiv 14033, Ukraine
e-mail: vadmitriev@ukr.net

A. Pavlenko
e-mail: agpav@ukr.net

D. Kamak
e-mail: dkam@meta.ua

S. Shkarlet et al. (eds.), *Mathematical Modeling and Simulation of Systems*, Lecture Notes in Networks and Systems 667,
https://doi.org/10.1007/978-3-031-30251-0_24

sizes and the total number of personnel in the test stages. Limiting the number of staff and the slow change in the intensity of the incoming flow of requests, which is usually observed over the period of a calendar year, has been analyzed. The issue of distribution of personnel between stages of testing in case of limitation of the total staff number is investigated.

Further research into the model includes the introduction of the concept of priority and urgency of requests for testing samples of weapons and military equipment. It is assumed that the simulation results will be used to form decision-making algorithms in the test support information system.

Keywords Testing · Queuing · Special equipment · Input flow · Service requests

1 Introduction

The military aggression of the russian federation and the presence of a Ukrainian developed military-industrial complex in Ukraine contributed to a significant increase in the number and range of the latest defense developments. This has led to an overall increase in the flow of requests for testing weapons and military equipment (further referred to as special equipment) conducted by a specialized state testing institute.

Special equipment sample testing is a complex process which includes several stages: preparation for testing, practical tests, processing of the results and the preparation of a test results report. Test delays can occur at any stage, but studies have shown [1] that major time losses occur during the test preparation and practical tests stages. This necessitated the study of processes and work at these stages and the search for ways and opportunities to reduce time costs. One of the possible ways to increase the testing efficiency is to optimally distribute the available resources, such as: human resources (scientists, engineers and testing personnel) and material resources (laboratories, proving grounds and testing facilities). According to the study, the most critical resource that can be influenced by the test organization is human resources. Expanding the test institute structure and having reserve personnel in the test teams, to some extent, solved the problem of a variable request flow, but did not eliminate the risk for delays arising during the testing process due to overload. In addition, excessive expansion of the structure and staff of the institute leads to unjustified costs for its financing, which has impact on the state budget.

2 Background and Related Work

This article is a continuation of a series of publications covering the issues of optimizing the weapons and military equipment testing system. A lot of studies are devoted to the problems of optimization of test systems: cost reduction issues in testing are considered in [2], the organization of the procurement of weapons and military equipment and its testing is discussed in [3], the procedure for organizing and conducting research of military equipment is described in [4] and studies on the organization of the test system for air forces military equipment are given in [5]. In our case, the study of the test system is carried out using: the well-known analytical apparatus of queuing systems, the main theory is set out in [6, 7] (examples of which for solving military-special problems are given in [8]), modeling of military-technical systems [9, 10], a study of the effectiveness of combat operations [11], in the comparative evaluation of military equipment [12].

In [13], a model of the preparatory stage of the test system is proposed and a study of its behavior and adequacy is conducted. Using the apparatus of mathematical statistics [14, 15] and probability theory [16], a general array of statistical data on the input flow of testing samples of weapons and military equipment, which is presented in [1], was processed.

3 Research Method

3.1 Description of the queueing model

General Representation of the Test Organization Model. The structure of the queuing system of the test organization in general can be represented as follows (Fig. 1) [1].

The main studied statistical parameters of the model are given in [1]. According to the defined characteristics of the input flow and the service device, the model of the test institute has the classification $M/M/n/\infty$—multichannel queuing system, with Poisson's distribution of input flow intensity, Poisson's distribution of service intensity, with n service channels and unlimited queues. Service channels (Fig. 1) are formed by test teams which can have any number of specialists available. Theoretically, the queue with requests is endless, due to the fact that there are no factors that can exclude a request from service. The general representation of the queuing system of the test organization includes two queuing systems connected in series: the queuing system of the preparatory stage and the queuing system of the practical test stage. Let's consider these models in detail.

Model of Preparatory Stage Queues. To model the service processes of requests at the preparatory stage, we use the scheme of Markov processes (Fig. 2):

In Fig. 2, the preparatory stage of the tests is depicted in the form of a diagram of state transitions, where each state S^b characterizes the number of occupied channels b of the total B.

In practice, different numbers of specialists may be involved in testing samples, which is determined by the complexity of the sample and the availability of human resources. This in turn leads to different performance of service channels, which is

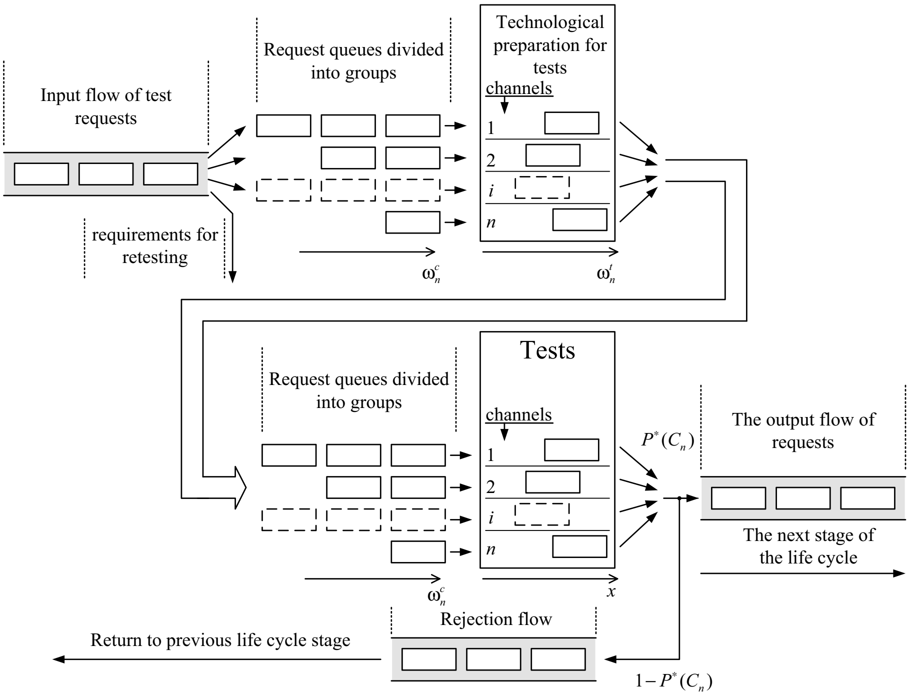


Fig. 1 Diagram of the testing institution queuing system

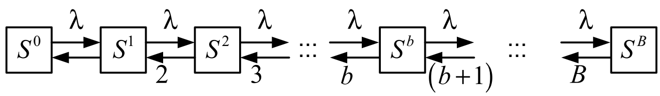


Fig. 2 Presentation of the preparatory stage of tests according to the scheme of Markov processes

characterized by the intensity of service μ. In this case, the set of teams will form the overall intensity of service

$$\mu_p = k_1\mu + k_2\mu + ...k_i\mu + ... + k_n\mu, \tag{1}$$

where μ_p—the total intensity of service requests at the preparatory stage, k_i—number of testers in a team i, n—total number of teams, μ—productivity of one tester. Considering the possibilityof differences in test team size, for modeling the intensity of service requests, expression (1) can be converted into $\mu_p = n\overline{k}\mu$, where $\overline{k}$—mathematical expectation valuefor the number of testers in a team. Then, the state transition diagram can be represented as follows (Fig. 3).

The theoretical number of channels in the queuing system of the preparatory stage is limited to $B^* = \left[K/\overline{k}\right]+1$, where K—the total number of personnel participating

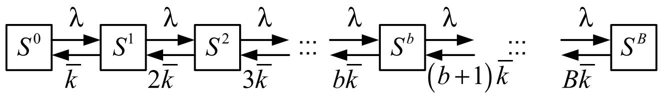


Fig. 3 Representation of the preparatory stage for testing according to the scheme of Markov processes with the average number of staff in the test teams

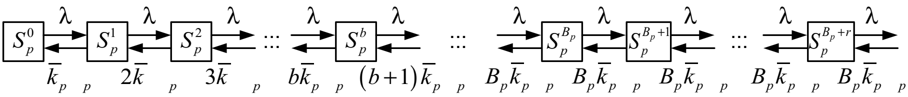


Fig. 4 Representation of the preparatory stage for testing according to the scheme of Markov processes with the average number of staff in test teams and an unlimited queue

in the tests. The last team can have a number of specialists: $k_B \leq \overline{k}$, then the intensity of service requests in the state S^B will be: $\mu_{S^B} = B\left(K - \left[K/\overline{k}\right] + \overline{k}\right)\mu$. However, in practice this situation is impossible due to the involvement of the staff at other stages of the test, so the possibility with the number of staff in the team $k_B \leq \overline{k}$ could be neglected.

Note that the queue of requests of the preparatory stage of the test can be considered theoretically unlimited. Therefore, the model in Fig. 3 can be represented as follows (Fig. 4), in which for further representation of the queuing system of the preparatory stage of the test we enter the lower index "p".

At present, we do not have statistical information on the intensity of work at the preparatory stage, both in terms of the intensity of service of one request and the productivity of an individual test engineer. As an output, one can choose the mathematical expectation value for the service time at the preparatory stage $M^*[\omega]$. Assuming that the average number of staff in the team at the preparatory stage is known, the productivity of one tester in the preparatory stage (taking into account that the request is present in two queues) will be

$$\mu_p = \frac{1}{M^*[\omega] \cdot \overline{k}^*}, \tag{2}$$

where $\overline{k}^*$—mathematical expectation value for the number of personnel in the test team at the test preparatory stage. Obviously, the obtained value μ_p will be under estimated due to the waiting time for requests in the queue. It is estimated that using (2), the performance of an individual test engineer is within $\mu_p \in [0.01,\ 0.05]$.

Let's use the known relations for the formation of service performance indicators of queuing multichannel systems with an unlimited queue [13, 17–19]. Stationary probabilities of queuing states are as follows:

- stationary probability of finding the preparatory stage queuing system of the test in the state S_p^0 (Fig. 4): $P_p^0 = \left(\sum_{b=0}^{B_p} \frac{\rho_p^b}{b!} + \frac{\rho_p^{B_p+1}}{B_p!(B_p - \rho_p)} \right)^{-1}$, where $\rho_p = \frac{\lambda}{\overline{k}_p \mu_p}$—queuing system load;
- stationary probability of being in a state S_p^b: $P_p^b = P_p^0 \left(\rho_p^b / b! \right)$;
- stationary probability of being in a state $S_p^{B_p+r}$: $P_p^{B_p+r} = P_p^0 \left(\rho_p^{B_p+r} \Big/ B_p! B_p^r \right)$;
- stationary probability of getting the request in the queue: $P_p^0 \left(\rho_p^{B_p} \Big/ B_p! \right)$;
- stationary probability of uninterrupted service of the request: $P_p^S = 1 - P_p^B$;
- absolute capacity of the preparatory stage of the test (for systems with unlimited queue $Q_p = 1$ is accepted, however, we are more interested in the request flow intensity which occurs without queuing): $Q_p^* = \lambda P_p^S$;
- average number of service channels (service requests): $L_p^S = \lambda / \overline{k}_p \mu_p = \rho_p$;
- load factor: $\eta_p = \sum_{b=1}^{B_p} S_p^b P_p^b = \rho_p \left(1 - P_p^0 \right)$;
- average number of occupied specialists:

$$\overline{N}_p = \overline{k}_p \rho_p = \lambda / \mu_p; \tag{3}$$

- the average number of requests in the queue:

$$L_p^q = \frac{\rho_p^{B_p+1} B_p}{B_p! \left(B_p - \rho_p \right)^2} P_p^0; \tag{4}$$

- the average number of requests at the preparatory stage: $L_p = L_p^S + L_p^q$;
- average downtime: $t_p^0 = P_p^0 T$, where: T—system analysis period;
- the average waiting time for the request in the queue: $t_p^q = L_p^q / \lambda$;
- stationary average time of the request at the preparatory stage of the test: $t_p = L_p / \lambda$;
- stationary average intensity of request advancement:

$$\lambda_p^i = 1 / t_p. \tag{5}$$

The Queue Model of the Practical Test Stage. Similarly, we present the process of practical tests according to the scheme of Markov processes (Fig. 5):

Index "t" determines the correspondence to the practical test stage. The input flow intensity at the stage of practical testing (Fig. 1) is determined by the test request flow intensity and the special equipment sample flow intensity which were previously removed from the testing stage and sent for revision to the manufacturer: $\lambda_t = \lambda + \lambda(1 - P^*(C_n))$. Further modeling will require the ratio coefficient for the

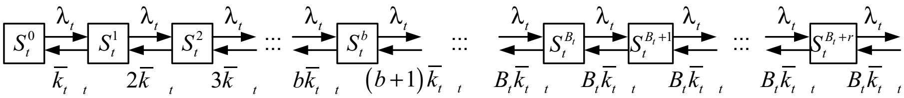


Fig. 5 Representation of the practical test stage according to the scheme of Markov processes with the average number of staff in test teams and an unlimited queue

distribution of personnel between the stages of the test: $\overline{k}_p b_p(t) + \overline{k}_t b_t(t) \leq K$, where $b_p(t)$ and $b_t(t)$—instantaneous values for the number of channels at the appropriate stages.

In practical tests of a single sample the most important factors are the number of specimens of the test sample and their degree of preparation for testing, as practice has shown. However, these factors are external to the organization's testing system and cannot be influenced. On the other hand, the lack of staff in test teams can negatively affect the timeliness of practical tests.

The calculated value for the statistical intensity of practical tests of the sample is $\mu = 0.29$. Taking into account the mathematical expectation value for the number of staff in the test team, the productivity of one specialist can be expressed as $\mu_t = \mu / \overline{k}_t$. If the statistical mathematical expectation value is $\overline{k}_t^* = 11.2$, then the productivity per person is $\mu_t = 0.025$. For further modeling, it should be considered that increasing the intensity of practical tests is often not possible due to the limited number of samples sent for testing.

Given the homogeneity of the queuing system models of the preparatory stage (Fig. 3) and the practical test stage (Fig. 4), the performance indicators will also be determined in a similar way by replacing the index "p" with index "t".

3.2 Modeling and analysis of the performance of the test organization

Objectives and Limitations of Modeling. In order to achieve the optimal operation mode of the test organization, a balance in the distribution of limited staff resources has to be found, based on statistical indicators for the incoming of test request flow. Using a queuing system will help answer the following questions:

- what is the minimum number of staff that will provide service for the test request flow;
- how personnel constraints will affect the throughput of the test system;
- what is the optimal plan for the distribution of personnel between the stages of testing with a limited staff number;
- which impact on the performance of the test system has the number of test teams;
- how periodic fluctuations in the intensity of the incoming request flow affect.

Limitations and assumptions in the modeling:

- statistical indicators of the service intensity at the test stages are in the range from the calculated minimum value to 3–fivefold increase;
- at the same time the specialists of the test team can serve only one request at one stage of the test;
- test requests are considered to have equivalent priority.

Minimum Requirements for Servicing the Test Request Flow. The stationary intensity of incoming test request flow and the accepted minimum productivity of specialists at the preparatory stage are determined in [1] to be $\mu_p = 0.01$, the average performance of the test system at the practical test stage is $\mu_t = 0.25$. Therefore, the stationary average number of specialists servicing the flow of requests in accordance with (3) will be $\overline{N}_p = 194$ and $\overline{N}_t = 89$. Disagreements with the calculations of the number of persons determined in [1] $N_{researcher} \geq 110$ and $N_{tester} \geq 60$ can be explained by assumption that the personnel of the test teams perform tasks related exclusively to the tests and considering losses in the testing stages. However, current data are obtained from statistical indicators that take into account the actual time spent on testing and other activities. In addition, the value of $\mu_p = 0.01$ is selected as the minimum of the specified range. In the case staff productivity at the preparatory stage lies within $\mu_p \in [0.01,\ 0.05]$, the staff stationary average need is shown in Fig. 6:

Less staff at the test stages at given intensities will lead to unlimited queue growth. This can ultimately lead to a reduction in the quality of tests due to attempts to artificially increase the system bandwidth.

System Bandwidth $M/M/n/\infty$. For further modeling, we choose for the average value of staff productivity at the preparatory stage $\mu_p = 0.03$ (minimum personnel need - 64 people) and defined $\mu_t = 0.025$ at the practical stage of testing (minimum personnel need - 85 people). According to the modeling conditions, the queuing

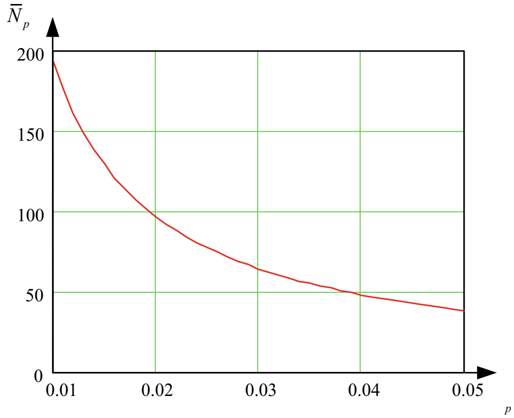


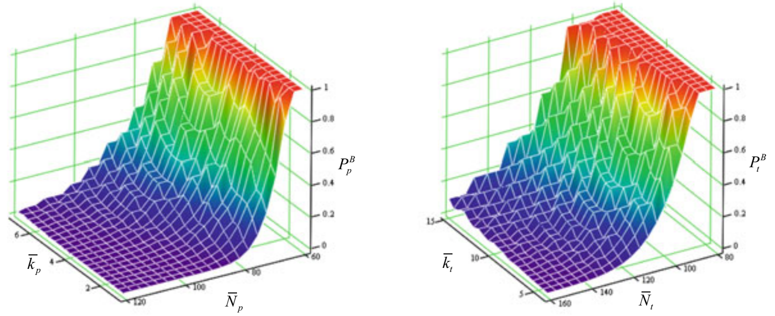
Fig. 6 Average stationary need of personnel at the preparatory stage of the test $\overline{N}_p$ with the productivity of the separate expert $\mu_p \in [0.01,\ 0.05]$

system has theoretically unlimited queues at both stages. Accordingly, the only condition for servicing the flow of requests is the following inequality [13]:

$$\lambda / \overline{k}\mu < B, \tag{6}$$

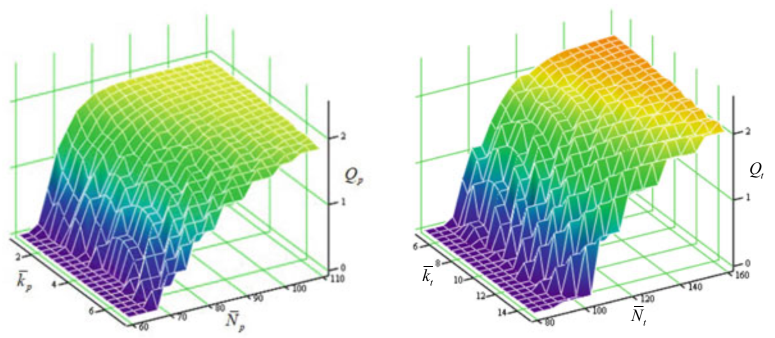
when this inequality is satisfied, there is no unlimited growth of the queue. The results of modeling the main indicators of the efficiency of the test system are shown in Fig. 7–10. Figure 7 shows the dependency of the stationary probability of the request in the queue for different personnel numbers at the test stages—$\overline{N}_i$ and different mathematical expectations values for the number of personnel in the test team—$\overline{k}_i$.

The graphs show that the increase of $\overline{k}$ leads to a certain increase in the probability of request queuing and in accordance with (4) and (5), we should expect a certain loss of productivity of the entire system (Fig. 8).



a) b)

Fig. 7 Stationary probabilities of the request queuing: a) at the preparatory stage; b) at the practical stage



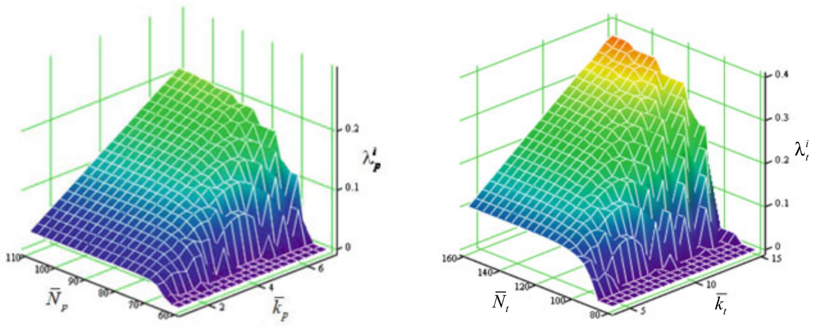
a) b)

Fig. 8 Absolute capacity of the test stages: a) at the preparatory stage; b) at the practical stage

Despite this, there is a certain increase in the intensity of the advancement of the test request (Fig. 9). This can be used to develop rules for the formation of test teams in the presence of priority of the incoming requests.

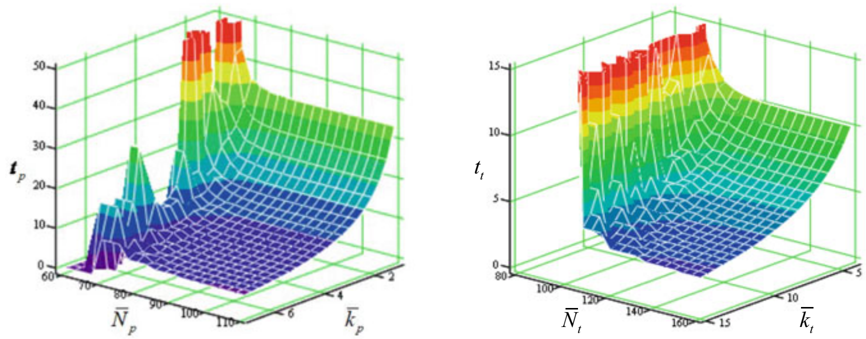
Zero absolute bandwidth of the system on the graphs (Fig. 8) is determined by the number of staff less than the allowable, with an unlimited increase in the queue, according to (6). This is reflected in the time characteristics of requests being in the test system (Fig. 10).

Modeling the Behavior of the Test System with a Limited Number of Personnel. The study of system performance was conducted with a limited number of personnel and slow fluctuations in the intensity of the input flow. For the static characteristics of the number of test teams, the statistical mathematical



a) b)

Fig. 9 Intensity of advancement of a separate request: a) at the preparatory stage; b) at the practical stage



a) b)

Fig. 10 The average length of time a single request is tested: a) at the preparatory stage; b) at the practical stage expectation values $\overline{k}_p^* = 4.3$ and $\overline{k}_t^* = 11.2$ are taken. The total number of staff is limited to the minimum number $N = 150$. The distribution of personnel over the test stages $N_p + N_t = N$. Variation in the intensity of the incoming request flow can be done within $\lambda_p \in [0.5,\ 4]$, which corresponds to the periodic change of intensity shown in Fig. 11 [1]. According to these initial data, the simulation results are shown in Figs. 12–14.

The simulations were performed assuming the worst-case scenario, namely the in-phase peak load at the stages of preparation and practical tests. The distribution of the stationary probability of the request queueing for different values of the intensity of the input flow and the number of personnel at different test stages is shown in the graphs of Figs. 12.

Analysis of the Results. The obtained minimum number of personnel for a certain intensity of the input flow (Fig. 6) will ensure the normal functioning of the test

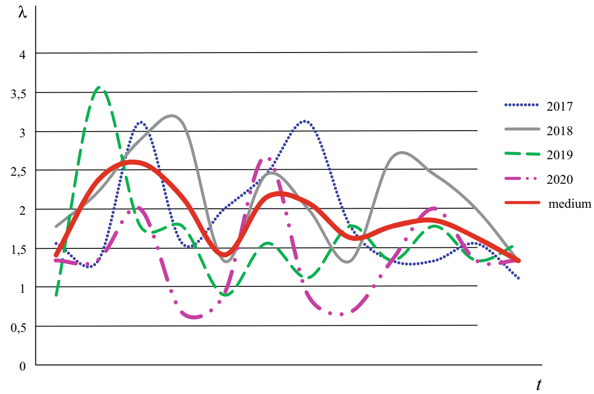


Fig. 11 Smoothed graphs of the annual dynamics of the intensity of the total input test request input flow

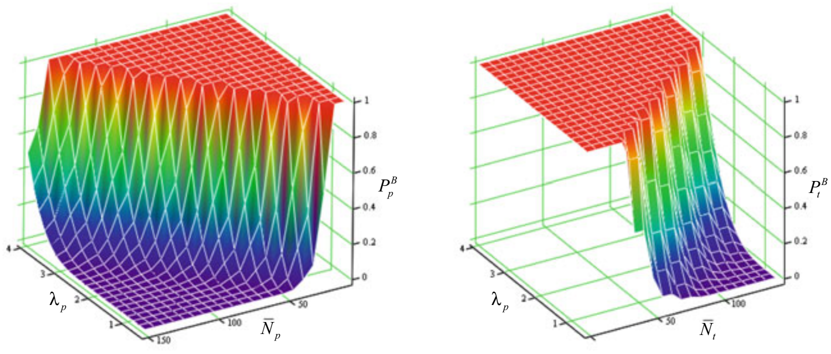


Fig. 12 Stationary probabilities of requestqueuing with a limited number of staff: a) at the preparatory stage; b) at the practical stage

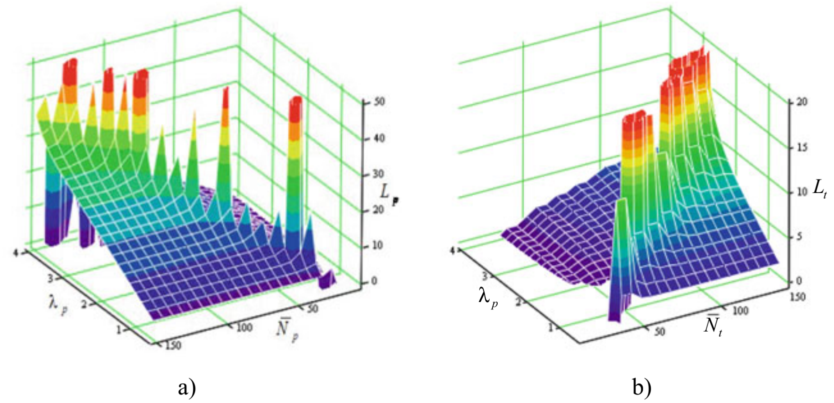


Fig. 13 Average number of requests in the test system with a limited number of staff: a) at the preparatory stage; b) at the practical stage

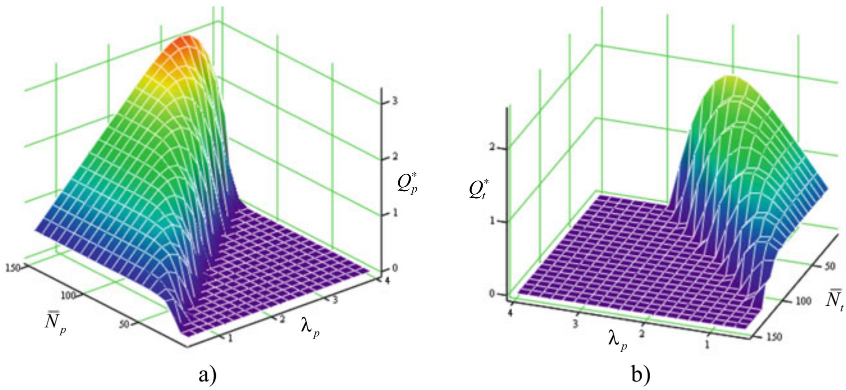


Fig. 14 Absolute capacity of the test system with a limited number of staff: a) at the preparatory stage; b) at the practical stage

system under the condition of a stationary flow. However, as noted in [1], the intensity of the test request flow has a certain variability (Fig. 11). Taking into account the mathematical expectation value for the time the request is in the preparatory stage $M^*[\omega] = 20.27$, and "slow fluctuations" in the intensity of the request flow, the period of which, on average, is 13–18 weeks (Fig. 11), there may be an overlapping of peak loads at different stages. This may require the maximum need of specialists at all stages of testing at the same time and finding ways to significantly increase their productivity.

Figure 7 shows that the stationary probability of getting the request into the queue increases slightly when forming test teams with more personnel in each team. Guaranteed entry of requests into the queue is observed with the number of staff at both test stages $\overline{N}_p < 64$ and $\overline{N}_t < 85$. This in turn is reflected in the "zero" capacity of the test stages (Fig. 8). Although unlimited queuing systems, according to [6, 7], have an absolute capacity $Q = 1$ since in any case they will be serviced. However, we are interested in the flow of requests which will be serviced per unit of time for a given number of personnel. There is also a certain increase in bandwidth during service at both test stages with small sized teams.

However, it is obvious (Fig. 9 and Fig. 10) that the speed of advancement of individual requests at both stages increases significantly when serviced by teams with large sized teams. At the same time, a significant increase in the number of staff at the testing stages above the minimum threshold does not affect the time of requests being in the system (Fig. 10).

Limiting the number of staff can significantly affect the capacity of the test system (Fig. 12–14). The fluctuations in the intensity of the incoming test request flow are critically important. A graphical representation of the optimal distribution of limited personnel between the test stages, under the condition of overlapping peak intensities of the input test request flow, is shown in Fig. 15 (a). The intersection of the constructed surfaces will determine the best option for the distribution of personnel, for which there will be a minimum probability of queuing at the maximum possible capacity.

Graphical representation of the capacity maximization problem with a limited number of staff at different intensities of the incoming request flow is shown in Fig. 15 (b). Figure 15 (b) shows that at low intensity of the request flow, the optimal distribution of personnel tends towards the practical phase of testing until the intensity of the flow reaches the value $\lambda_p \approx 1.94$, after which the optimal distribution is almost constant.

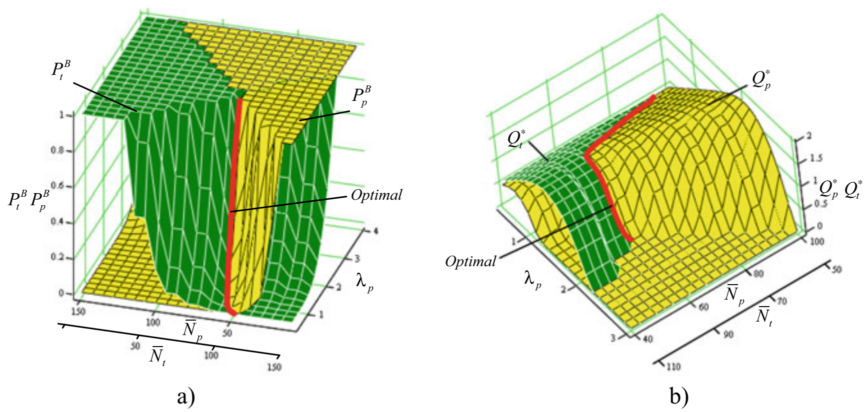


Fig. 15 Optimal distribution of personnel by testing stages at a given intensity of service according: a) to the probability of getting a service request into the queue; b) to the capacity of the test system

4 Discussion

The conducted modeling and graphical interpretation of the results allows to make some major remarks on the functioning of the test system.

The minimum number of staff is calculated according to the intensity of the incoming test request flow. However, in the practical application of the model, it should be considered that test personnel also perform other tasks that are directly or indirectly related to the activities of the test organization. Therefore, the calculated values should be adjusted by a certain factor which will take into account other types of work and statistical time spent on vacations, sick leave, etc. The calculated intensity of the request flow for practical tests is based on the statistical value of the previously failed test. Therefore, this indicator cannot be excluded from the parameters necessary for the control during practical application of the model as part of the test support information system.

The number of test teams does not actually have a significant effect on the capacity of the test system. The consequences of small sized test teams are a slight increase in the capacity of the test system, but also an increase in simultaneous test requests. On the other hand, the intensity of the advancement of individual requests is much higher for larger teams which should be used to service urgent requests and requests with high priority. Important here is the ability to form a large number of test teams when receiving requests with high priority. In this case, it is interesting to investigate the balance between the number of staff that handles requests and staff reserves for serving high priority requests based on the statistics of receipt of such requests.

The distribution of the number of staff between the testing stages under the condition of a stationary request flow should be approximately a constant value. However, in our conditions there are slow changes in the intensity of the inputtest request flow (Fig. 11). According to Figs. 14–15, it can be seen that the optimal distribution will be determined by the balance of flow intensities at a particular time for the preparatory and practical testing stages. Such a balance can be established by calculating the mathematical expectation value for the time requests remain at the preparatory stage $M^*[\omega] = 20.27$ (time from the date of the test request to the beginning of the practical test phase). In this case, there may be undesirable in-phase peak loads for the preparatory and practical testing stages.

5 Conclusions and Future Work

5.1 Conclusions

Based on the apparatus of queuing systems, the analytical model of the special equipment test system is suggested. In the structure of the multichannel queuing system of the preparatory and practical stages of testing, channel elements with different testing team sizes and, accordingly, different service intensities were introduced.

The modeling was carried out based on previously obtained statistical characteristics of the test organization. Estimates of the statistical parameters of the test system were made based on the general data set obtained during 2017–2020. The final probabilistic characteristics of the test system based on the statistics of 2021 could not be obtained due to the war of the russian federation against Ukraine.

The obtained result from analytic modeling allows for the determination of the test system behavior in terms of staffing test teams at the stages of preparation and practical tests. The influence of staff limitation on the capacity of the test system is determined. The basic indicators for the efficiency of the test organization at periodic fluctuations of special equipment test request input flows are investigated.

5.2 Future Work

Investigate the behavior of the test system when receiving samples of special equipment with higher priority and / or limited time for testing. Develop an algorithm for the distribution of personnel by testing stages and the formation of a test personnel reserve for emergency testing.

The developed model of the queuing system of the test system is planned to be implemented in the decision-making algorithms of the automated information system of test support.

References

1. Korniienko, I., Korniienko, S., Dmytriiev, V., Pavlenko, A., Kamak, D.: Modeling and analysis of the main statistical parameters of the testing system of special equipment. In: Korniienko, I., Korniienko, S., Dmytriiev, V., Pavlenko, A., Kamak, D., et al. (eds.) Mathematical Modeling and Simulation of Systems. MODS 2021. Lecture Notes in Networks and Systems, vol. 344, pp. 465–476. Springer, Cham (2022). https://doi.org/10.1007/978-3-030-89902-8_34
2. Fox, B., Boito, M., Graser, J.C., Younossi, O.: Test and evaluation trends and costs for aircraft and guided weapons (No. RAND/MG-109-AF). RAND Corp (2004)
3. Claxton, J.D., Cavoli, C., Johnson, C.: Test and evaluation management guide. Defense Acquisition Univ Ft Belvoir VA (2005)
4. Kass, R.A.: The logic of warfighting experiments. Assistant Secretary of Defense (C3I/Command Control Research Program) (2006). https://doi.org/10.21236/ADA457164
5. McShea, R.: Test and Evaluation of Aircraft Avionics and Weapon Systems (Radar, Sonar and Navigation) 2nd Edition (2014). ISBN-13: 978–1613531761
6. Kleinrock, L.: Queueing Systems—Volume I: Theory. John Wiley & Sons, New York (1975)
7. Kleinrock, L., Gail, R.: Queueing Systems : Problems and Solutions. A Wiley-Interscience publication, New York (1996)
8. Kazachinsky, V.Z., Levitsky, G.E.: Mathematical methods for solving military special tasks, p. 292. Kiev Higher Air Force Academy (1980). (in Russian).
9. Nikolić, N.: Monte Carlo Modeling of military queueing systems -challenge of the initial transience. Belgrade: Andrejevic Endowment/Strategic Research Institute (2008)

10. Jenkins, P.: Using Markov Decision Process with Heterogeneous Queueing System to Examine Military MEDEWAC Dispatching Policies. MSc thesis in Operations Research. Air Force Institute of Technology—Wright-Patterson Air Foce Base, Ohio, USA (2017)
11. Li, L., Liu, F., Long, G., Zhao, H., Mei, Y.: Performance analysis and optimal allocation of layered defense M/M/N queueing systems. Math. Probl. Eng. 1–21 (2016). https://doi.org/10.1155/2016/5915918. Accessed 14 Mar 2021
12. Mao, Z., Yu-juan, W., Chao, W., Sheng, H.: Analysis on aircraft sortie generation rate based on multi-class closed queueing network. In: Proceedings 2nd International Conference on Computer Science and Electronic Engineering, ICCSEE, pp.1877–1880 (2013)
13. Korniienko I., Korniienko S., Dmytriiev V., Pavlenko A., Kamak D. Development of a model of the preparatory phase of test tests: a model of productivity of the test division. Sci. Technol. Air Force Ukraine **4** (45), 124–134 (2021). (in Ukranian)
14. Allen, A.: (1990), Probability, Statistics and Queuing Theory with Compute Science Applications, 2nd edn. Academic Press Inc., Cambridge (1990)
15. Cramer, H.: Mathematical Methods of Statistics. Princeton University Press, Princeton (1946)
16. Ventsel, E.S.: The Probability Theory, 3rd edn. Nauka, Moscow (1969). (in russian)
17. Matveev, V.F., Ushakov, V.G.: Queuing systems: manual on a special course “applied mathematics”. Moscow State Univ. 239 (1984). (in russian).
18. Bocharov, P.P., Pechinkin, A.V.: Queueing Theory: The textbook, p. 529. University under P. Lumumba (1995) (in Russian)
19. Saaty, T.L.: Stochastic network flows: advances in networks of queues. In: Smith, W.L., Wilkinson, W.E. (eds.) Congestion Theory, pp. 86–107. University of North Carolina, Chapel Hill (1964)

Author Index

S. Shkarlet et al. (eds.), *Mathematical Modeling and Simulation of Systems*, Lecture Notes in Networks and Systems 667,
https://doi.org/10.1007/978-3-031-30251-0